

SIEMENS

MAGNETOM

MR

TSG

Troubleshooting Guide

Harmony/Symphony with Turbo/Ultra gradients
Symphony with Quantum gradients
Sonata

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Print No.: M1-010.840.01.03.02

Replaces: M1-010.840.01.02.02

English

Doc. Gen. 12/00

0

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Introduction

Parallel to the advent of a whole new generation of Siemens MR systems is the dawning of a totally new service support philosophy matching and tailored to the modularity and efficiency of the latest hardware technologies incorporated into these advanced line of machines. In keeping with this progress of newer and enhanced designs, it has become not only feasible but also necessary to adapt to a service strategy making the failure isolation and repair as economical as possible. In lieu of these trends, we have attained a level of service friendliness never before achieved. This introduction to the TSG documentation is designed to give you an overview into this new trouble shooting concept for the MAGNETOM Harmony/Symphony.

General

This register contains an introduction to the TSG: what the TSG contains and how it is best used. It will also be attempted to present how the TSG and FUN are coupled together and complement one another. Here you will find a general explanation of the concept structure, the various sections comprising the TSG, and how they can be most effectively used towards trouble shooting.

Repair Instructions can be found in the Manual Replacement of Parts.

Information on the new page format and on-line documentation are covered in the manual FUN, part Prefix, section "Documentation Intro".

The contents of the TSG will support service technicians during on-site trouble shooting and faulty component isolation. While the instructions are not a substitute for an in-depth knowledge of the system, its components or the underlying physics, they will facilitate,

on the one hand, trouble shooting procedures for the less experienced technician and on the other hand, support the more experienced technician in performing essential measurements and tests. The instructions will also explain standard procedures such as measurements of power supplies, etc.

The TSG should not and do not take the place of an error catalogue. In most cases hardware faults are easily and quickly located. Errors which are intermittent require more time... and patience.

It is not within the scope of the TSG to list all possible tests. Its purpose is to support the search for frequently-occurring errors, to allow the technician to limit the error search to certain functional groups, and to propose measurements for localizing a faulty component.

We are aware that the first versions of this manual do not provide the reader with the entire spectrum of information. The manual will be updated accordingly in the future. Any feedback is strongly encouraged and will be warmly welcomed To provide you with the tools you need, we need your help!

Topics

- Categories: Strategy, Procedures and Index.
- Suggestions for use
- Distribution

The goal is to distribute the Online-documentation as well as the paper version to the MR FSEs via a VESPA-distribution list. This means that in the future this documentation will no longer be delivered with the system.

TSG structure

The Troubleshooting Guide is structured as listed below:

- Table of Contents
- Introduction/Safety
- System
- Software
- Host
- SMI
- MMC
- RF system
- Patient handling
- Gradient system
- Cooling
- Line power
- Changes
- Index

The information for the different system components has been subdivided in:

- Test Strategy
- Test Procedures

Strategy

The flowchart is designed as a general guideline for selecting the appropriate trouble shooting procedures.

The objective of the accompanying descriptions is not to give detailed step-by-step instructions into the finer art of trouble shooting, but to offer a means of guiding you, given a particular problem, to the right test procedure in all brevity and simplicity. If more detailed instructions are necessary, then the FSE will be referred to the appropriate procedures in the section "Test Procedures".

Procedures

Here the FSE will find the detailed test instructions. You will also find descriptions for "standard procedures" - for example starting the SeSo, connecting the laptop to the Host, reading log files, etc. This section also contains descriptions of interpreting the test results. Furthermore you will find general information pertaining to parts location, fuse ratings, power supply information, measuring points, descriptions of LEDs and potentiometers in this section.

Index

An index at the end of the book provides you, in addition to the table of contents, with cross references to information on topics you may wish to look into for further information. It is a helpful source of information when looking for related topics or when wanting to concentrate on items of interest. For example, if you would like to know more about the 'adjust frequency' you could either look under 'A' for 'adjust' or 'F' for 'frequency'. Many topics listings are redundant to help make your search more successful.

Safety

All safety guidelines necessary for troubleshooting are listed in this chapter.

TSG specific safety guidelines

General

Note the special safety guidelines indicated in all instructions.
Read these instructions completely prior beginning any work.

NOTICE	Follow all safety guidelines in binder Installation Vol. 1, Register 2, Safety Instructions.
---------------	--

NOTICE	Follow ESD guidelines when handling the components (anti-static wristband, ground cable, grounded pad).
---------------	--

Safety Notes

Protective conductor measurement

NOTICE	The protective conductor resistance must be $\leq 200\text{m}\Omega$. (measured against the LPD protective conductor terminal).
---------------	---

For more detailed information refer to “Protective conductor measurement” in binder: *Installation Vol. 1*; register: *MR-specific safety information*; chapter: *Safety guidelines*.

Ferromagnetic materials

Introduction of ferromagnetic materials or tools in proximity to the magnet results in such objects becoming projectiles which could present a lethal risk to personnel working in the vicinity of the magnet.

Deenergization during a quench

WARNING	When the OR24/OR70 magnet quenches, the magnetic field ramps down to a field strength of 20 mT within 20 seconds.
----------------	--

Working in the magnetic field

Performing installation and service procedures under field is just allowed to be done by SIEMENS personnel. The necessary service tools like none magnetic tools have to be used.

Precaution has to be taken by using electronic measurement devices inside the magnet room when the field is on. Refer to the operators manual of the measurement device whether it can be used in magnetic fields.

CAUTION The magnetic field has to be ramped down when during installation or service any kind of magnetic service aids will be needed, or, related to the documentation, the ramping down of the magnet is required.

WARNING It is prohibited to ramp the magnet with any-one in the bore.

Phantom fluids

For all safety information about handling phantoms and phantom fluid refer to **Phantom fluids** in binder: *Installation Vol. 1*; register: *MR-specific safety information*; chapter: *Safety guidelines*.

Handling heavy loads

For all safety information about handling heavy loads refer to **General requirements for handling heavy loads** in manual: *Installation Vol. 1*; register: *MR-specific safety information*; chapter: *Safety guidelines*.

Laser beams

For all safety information about handling laser beams refer to **Laser beams** in binder: *Installation Vol. 1*; register: *MR-specific safety information*; chapter: *Safety guidelines*.

WARNING Laser products are not allowed to be operated without the appropriate housing. Do not look into the laser beam directly or at its reflection on smooth, mirror-like surfaces.

Image orientation check

CAUTION Perform the "Image Orientation Check" after all repair procedures when you have to disconnect or open the gradient coil power circuit, e.g. replacement of:

- Current sensors
- Gradient filters
- Gradient coil
- Gradient cables

Eddy Current Compensation (ECC)

For all information about executing ECC refer to **Eddy Current Compensation** in binder: *Installation Vol. 1*; register: *General Tune-up*; chapter: *General Tune-up*.

System

In this part you will find test strategies, procedures and appendix instructions encompassing the complete system.

Strategy

In this part you will find test strategies encompassing the complete system.

Test strategy for Host, SMI or SW

To make troubleshooting easier for Host, SMI or SW-problems an overall flowchart table is introduced. By following the questions in this table you are leaded in a fast way to the responsible unit of your problem.

If the question can be answered with **YES** continue with the next question.

Problem case	If No
Any outputs on the console monitor ?	Goto page 59
Booting successful ?	Goto page 29
NUMARIS start-up ok ?	Goto page 29
Image data acquisition ok ?	Goto page 122
3D calculation ok ?	Goto page 123
Measure, archiving, exposure and net transfer ok ?	Goto page 29
Single NUMARIS functions, performance or sporadic problems?	Goto page 29

Procedures

This part is designed to give the FSE an in-depth description of trouble shooting procedures.

In this chapter you will find descriptions for standard common procedures - for example starting the SeSo, connecting the laptop to the Host, etc.

And general information pertaining to parts location, fuse ratings, power supply information, measuring points, descriptions of LEDs and potentiometers can be found.

Starting service platform (SeSo)

Prerequisites

NUMARIS must be up and running

- Select the service function in the system menu:

Output		Input MRC	Notes
Menu	Selection		
		<R> *	Click on image area
Main Menu	System	<R> *	Click on System
System	Service	<R> *	Click on Service

*. Right mouse key

Access to Service Software Platforms

When the service platform is activated, a window is opened and the user is asked to enter the variable password.

The input for the variable password is visible with exception of the last three characters.

When the Service Software platform is accessed for the first time after software installation, no default variable password is displayed.

After the first successful access, the password is stored and displayed as the default in all subsequent dialogues with exception of the last three characters, which you have to type in.

Upper or lower case characters will be accepted.

After typing in the variable password, press <Return>.

The main platform of the SeSo appears in the standard service platform segment.

For further information refer to ["Variable password", page 26](#).

Laptop connection

The service terminal can be connected to the running system or to the non booted system.

Different connection possibilities

For connecting a laptop the Sparc offers two serial interfaces, port A (ttya) and port B (ttyb). This 2 ports are combined to one a common connector (X23 on Sparc).

Depending of the performed test types the laptop has to be connected to either ttya or ttyb.

Serial interface	SW-Settings after booting (Numaris)	
	MRC	MRSC
Port A (ttya)	Modem for remote diagnostics	Terminal
Port B (ttyb)	Terminal	optional Printer

Tab. 1 Sparc serial interface settings

Main console MRC

Port A is only used for testing in the monitor mode. Refer to [“Testing in Monitor mode”](#), page 23.

Port B is used for:

- Testing in Unix level (e.g. HTE). Refer to [“Testing under Unix”](#), page 24.
- Testing in the service platform.

Satellite console MRSC

- **Port A is always used** for connecting the laptop (default setting).

Required tools

If a laptop is used as a service terminal, the following tools are needed:

- Laptop PC
- Terminal emulator program
- Cable to connect the laptop to the SMI5 (Standard cable delivered with the modem 25 pin to 9 pin)
- Null- Modem adapter (P.n. 46 80 773)

NOTICE

The last two listed items are part of the delivered service tool set.

Terminal settings

Before connecting the laptop ensure proper setting of the used emulator program. In the following table the required settings are listed.

Parameter	Setting	
	Monitor mode	Unix level Service platform
Communication type	Asynchron Packet	Asynchron Packet
I/O port	Com 1	Com 1
Speed	9600 bds	38400 bds
Parity	None	None
Stop bit	1	1
Data bits	Ascii-8	Ascii-8
Flow Control	None	None
Packet size length	128	128

Testing in Monitor mode

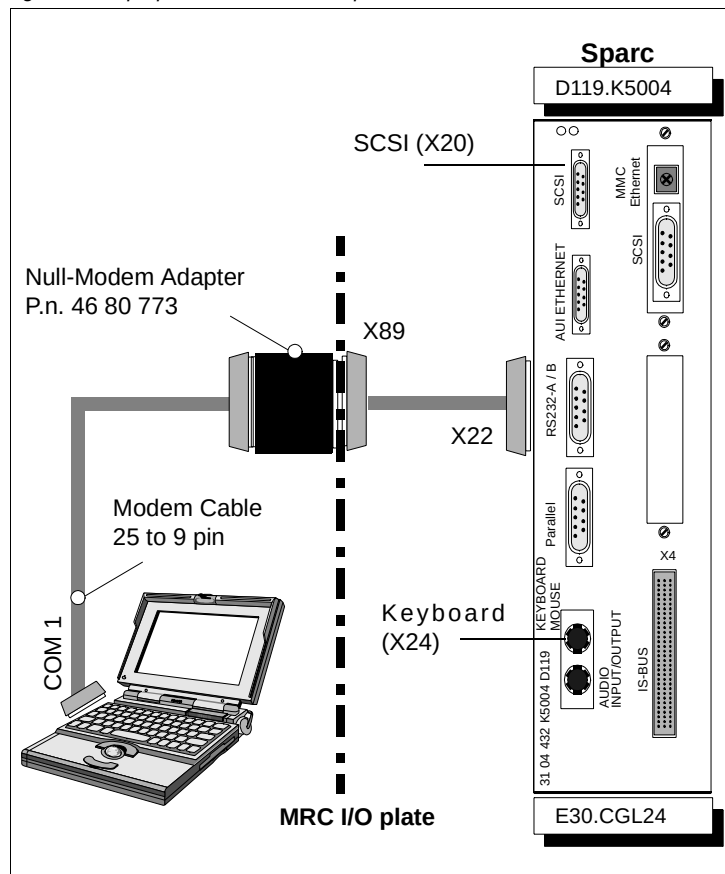
For testing in the **monitor mode** the laptop has to be connected to the serial **port A** (ttya). This applies to an MRC and an MRSC.

Procedure

- Halt the host and switch off the MR(S)C container (Rocker switch S1).
- Pull the keyboard cable (X30) at the rear connector plate of the MR(S)C container (or X24 on Sparc).
- Pull the Y-cable (X89) at the rear connector plate of the **MRC** container (X23 on Sparc).
- Connect the laptop according to [Fig. 1, page 23](#).
- Start terminal program on the laptop.
Refer to "[Terminal settings](#)", page 22 for correct PCLINK set-up.
- Switch on the MR(S)C container (Rocker switch S1).
- Use the Break function (e.g. <CNTRL-Pause>) of your laptop to stop booting.
- Select the new command mode by entering **n ↵**.

NOTICE Refer to Host trouble shooting, "[PROM Diagnostic](#)", [page 81](#), for more information about testing in the monitor mode.

Fig. 1 Laptop onnection to serial port A



Restoring the system setup

NOTICE After testing is finished it is necessary to set the host to its original settings. To do this execute the following steps.

- Halt the host and switch off the MR(S)C container (Rocker switch S1).
- Disconnect the service cable and reconnect the Y-cable (X89) at the rear connector plate of the **MRC** container (X23 on Sparc).
- Reconnect the keyboard cable to X30 at the rear connector plate of the MR(S)C container (X24 on Sparc).
- Switch on the MR(S)C container (Rocker switch S1).
- Execute a normal system booting.

Testing under Unix

MRC

For testing under UNIX (e.g. **HTE**, **SUNDIAG**) the laptop has to be connected to the serial **port B** (ttyb) (X23 on Sparc).

Procedure

- Connect the serial port 1 of the laptop to the 9 pin SUP-D connector of the Y-cable, refer to [Fig. 2, page 25](#).
- Start terminal program on the laptop.
Refer to "[Terminal settings](#)", page 22 for correct terminal set-up.
- For running the tests go back to the respective test description.

MRSC

For testing under UNIX (e.g. **HTE**, **SUNDIAG**) the laptop has to be connected to the serial **port A** (ttya) (X23 on Sparc).

Procedure

- Connect the laptop according to [Fig. 1, page 23](#) to the MRSC.
- Start terminal program on the laptop.
Refer to "[Terminal settings](#)", page 22 for correct terminal set-up.

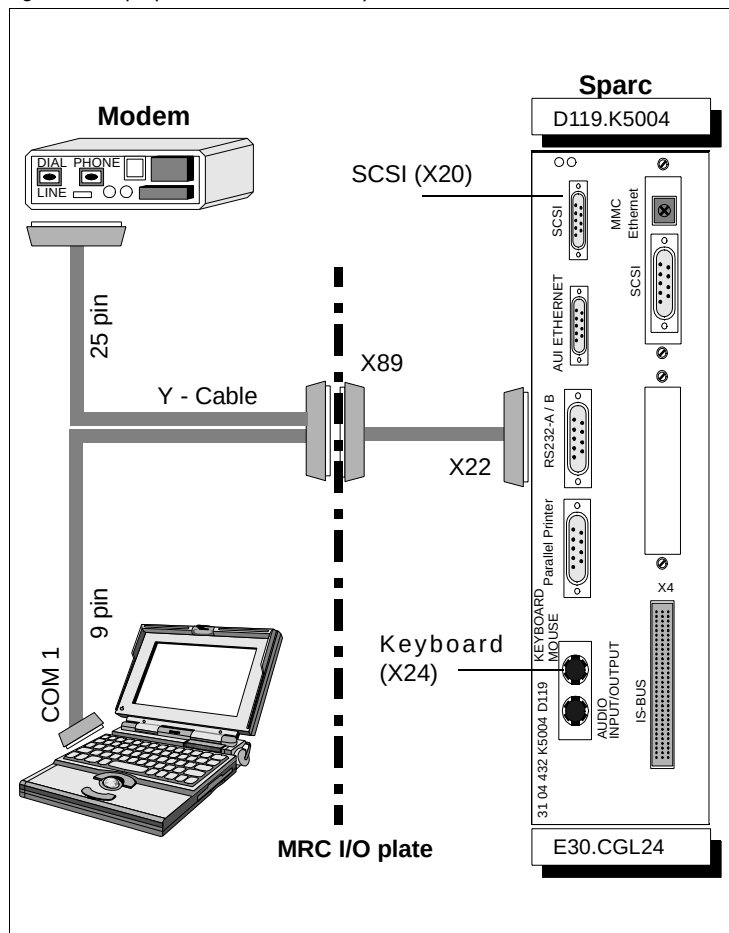
CAUTION For the MRSC the baudrate is set to 9600 bds.

- For running the tests go back to the respective test description.

Restoring the system setup

- Logout by typing <CNTRL-D>.
- Disconnect the service laptop from MR(S)C container.

Fig. 2 Laptop connection to serial port B



Using the service platform

For running the service platform on the **MRC** from the laptop (e.g. shimming, remote diagnostic), the laptop has to be connected to the serial **port B** (ttyb) (X23 on Sparc).

NOTICE

Here is only described how to connect the laptop physically to the system. Which software is required and how to install it is described in the help information "Service-PC Information" of the [SeSo](#). You can find this online help on the MR Service CD-ROM or on the MR system itself.

Procedure

- Connect the serial port 1 of the laptop to the 9 pin SUB-D connector of the Y-cable, refer to [Fig. 2, page 25](#).
- To establish a communication to the MR system refer to the [SeSo](#) help information "Establishing a Direct Cable Connection" on the MR Service CD-ROM or on the MR system itself.

Variable password

General

The variable password protects against unauthorized access. The variable password is generated from a number of parameters, such as host ID, expiration date, serial number and system type.

The variable password has the following properties:

- It is specific to one system.
- It is only valid for a certain time period, as specified by the district.
- The use of a new variable password invalidates the old one.
- It is checked for validity when entered.
- It is entered only once and will then be visible with exception of the last three characters as long as it is valid. The last three characters are not visible (asterisks) and must be typed in.
- When entering a variable password, the typed in characters are displayed on the screen to allow for correction (except for the last three characters).
- Upper and lower case letters may be used when typing in the variable password.
- The expiration date of the password is displayed on the monitor if a valid password has been entered previously.
- You will be informed explicitly when the variable password is about to expire.

Special case: defective CPU

Each host CPU is identified through a unique ID code. This code is required for verification in conjunction with the variable password. If the NVRAM becomes defective and the NVRAM must be replaced,

the current key continues to be valid for a maximum of 4 days, in order to guarantee access to the service software.

Within this time, you should request the ID of the new Host-CPU (type in UNIX command "hostid") and give it to your supervisor to generate a new variable password.

Special case: changing the time

As before, it is possible to set the system time, for example to Daylight Saving Time or to Standard Time.

NOTICE

Imprecise timing can lead to problems for the service engineer.

Back-dating:

Inadvertent back dating (relative to the actual date and time) is neither an advantage nor a disadvantage. However, an expired password cannot be made valid again by setting the time back.

Pre-dating:

It is possible to mistakenly enter a date later than the current date. If this occurs, the user will "loose" the period of time from the current date to the date entered from the period of validity for the variable password. For security reasons, this "error" cannot be corrected except by re-installing the system software.

Generation and Delivery

The software tool for generating the variable password (required to access the advanced diagnostics) is only installed at the headquarters in Erlangen/Germany and Iselin/USA.

This generator is only accessible via the OPSIS facility. Access is restricted to:

- Erlangen headquarters
- Iselin headquarters
- authorized district

NOTICE If you need a password for your site, request it from your district office.

Software

This part of the TSG deals with problems regarding the software. Software strategy, procedures and repair provide the necessary informations to successfully maintain the software system NUMARIS/3.5!

Strategy

Software repair is a difficult task due to the complexity of today's software.

In most cases the first question coming up is: "Is the problem caused by hardware or software"!

Reinstalling the software will answer this question quite reliably, but eventually will cause other problems like loss of data (e.g. patient images, customer protocols, or system specific dynamic data that have not been saved).

To make software troubleshooting as effective as possible here are some procedures to assist you in dealing with software problems!

General

NOTICE

For troubleshooting you should always start with "[Test strategy for Host, SMI or SW](#)", page 19. By following this flowchart table you are led in a fast way to the responsible unit of your problem.

The "Software Strategy" contains a flowchart in table form, the included software subjects are listed in the table "[Subject list](#)" in the table to the right.

Necessary tools

- Password list, refer to the latest Speedinfo, Password List folder.

Variable password (Servicekey).

Definition of fonts used in the documentation

command ↵ All input by the user is indicated in bold and followed by a carriage return sign.

Subject list

No	Problem
1	" System boot ", page 31
2	" Database ", page 32
3	" NUMARIS start-up ", page 32
4	" Single NUMARIS functions ", page 32
5	" Measurement ", page 33
6	" Camera / Exposure ", page 33
7	" Network ", page 33
8	" DICOM ", page 33
9	" Archive ", page 33
10	" Determining problem type HW/SW ", page 33

Software strategy

Remember to return to the strategy page in use, after performing a partial repair/procedure action.

No	Test Step	Status	Service Action	Reference
1.	System boot	Boot hangs	Boot system as single user: >b -sw Ø Password: (Enter root password)	"Single user Boot", TSG, page 34.
		Single user boot hangs	Check host environment, create the boot block or reinstall the kernel (vmunix) manually.	Replacement of Parts, Software, "Boot problems",
		Single user boot still hangs	"Save images from defective system disk"? Ask the customer if the system contains un-archived patient images.	Replacement of Parts, Software, "System disk is not booting"
			Re-install Software	Refer to the Installation Manual, Register 6, Chapter 2 "Getting started and configuration pages", Flowchart: SW reinstallation.
		Single user boot successful	Perform "File System Check"	Replacement of Parts, Software, "File system consistency check (fsck)",
		File system check ran successfully	Reboot the system, in case of problems see system messages.	"System-messages", TSG, page 34.
		Boot still hangs (multi user boot)	"Save images from defective system disk"? Ask the customer if the system contains un-archived patient images.	Replacement of Parts, Software, "Single user boot possible".
			Re-install Software	Refer to the Installation Manual, Register 6, Chapter 2 "Getting started and configuration pages", Flowchart: SW reinstallation.

Software strategy

Remember to return to the strategy page in use, after performing a partial repair/procedure action.

No	Test Step	Status	Service Action	Reference
2.	Database	Database errors.	If the system consists of two consoles check the network connection.	Replacement of Parts, Software, "Net problems"
			Use the "db_repair" tool.	Replacement of Parts, Software, "DATABASE"
3.	NUMARIS start-up	NUMARIS start-up hangs	Check used space on the system disk.	Replacement of Parts, Software, "File system full"
			Check if Database, Firmware server processes, and if Network are available.	Replacement of Parts, Software, "NUMARIS start-up hangs".
			Check at what point NUMARIS hangs.	"SV_MONI", TSG, page 40.
			"Save images from defective system disk"? Ask the customer if the system contains un-archived patient images.	Replacement of Parts, Software, "Single user boot possible"
			Re-install Software	Refer to the Installation Manual, Register 6, Chapter 2 "Getting started and configuration pages", Flowchart: SW reinstallation.
4.	Single NUMARIS functions	One / some NUMARIS function is / are not accessible.	Search for error messages in the NUMARIS error logger.	"Event Log", TSG, page 37.
			Check status files.	Replacement of Parts, Software, "Single NUMARIS functions are failing"
			Use Trouble Reset.	Replacement of Parts, Software, "Trouble Reset"

Software strategy

Remember to return to the strategy page in use, after performing a partial repair/procedure action.

No	Test Step	Status	Service Action	Reference
5.	Measurement	Measurement problems	Measurement hang no images calculated / Not possible to start a measurement	Replacement of Parts, Software, "Measurement problems"
			Coil related measurement problems	Replacement of Parts, Software, "Coil-specific problems"
6.	Camera / Exposure	Exposure problems.	Camera troubleshooting.	Replacement of Parts, Software, "Camera"
7.	Network	Network problems	Test network.	Replacement of Parts, Software, "Net problems"
8.	DICOM	DICOM problems	Check addresses, portnumbers, AETs	Replacement of Parts, Software, "DICOM problems"
9.	Archive	Archiving problems, MOD.	Use "Trouble Reset", option RFIL	Replacement of Parts, Software, "Trouble Reset"
			Perform device installation in INSTALL	Refer to the Installation Manual, Register 6, Chapter 8 "Configuration".
			Check hardware	Host Strategy " Host troubleshooting flow-chart ", page 70
10.	Determining problem type HW/SW	Non classifiable problem.	Determine if problem is caused by software	" Determining problem type HW/SW ", TSG, page 41

Procedures

The following procedures are provided for search of system related error messages (UNIX/NUMARIS messages), displaying system limits, how to determine if a problem is hardware or software-related.

Disabling/Enabling NUMARIS Password

You can disable / enable the NUMARIS password with one of the following procedures. E.g. if the customer has set a new NUMARIS password that is unknown and you need access to the NUMARIS application for service, you can use the **disable_mr_password** to disable the NUMARIS password temporarily.

Output	Disable mr password	Enable mr password
mrc_work% Password:	su install ↵ (enter root password)	su install ↵ (enter root password)
mrc_install%	disable_mr_password ↵	enable_mr_password ↵
mrc_install%	exit	exit

Single user Boot

Output	Input	Note
> Resetting ...	b -sw ↵	
Password:	***** ↵	Enter root password.

System-messages

The **"/var/adm/messages"** file contains general system error messages. If you would like to search effectively for error messages, then use the "egrep" command to specify lines matching search words.

Bad	Bad block: Most likely the internal disk controller is defective, therefore replace the system disk. (The bad block administration is mostly handled internally by the disk without logging into "/var/adm/messages")
failed	Lines containing failed are listed. (E.g SCSI transport failed. In this case the system recognized timing problems on the SCSI-bus. The reason can be incorrect termination, or a defect on the single mode converter.)
fatal	This refers to a fatal error in the hardware. Immediate repair is necessary.
memory or module	This refers to a memory error. The error is combined with the defective memory chip location.
panic	This refers to a system crash. The reason can be inconsistency in the file system, a software failure, or a hardware failure. Please use manual pages for more information "man panic".
reducing	The SCSI bus reduced the transfer speed. Check for defective devices, power supply, and mechanical defects on the SCSI cables.

How to use "grep"

For more information regarding the command `egrep`, use manual pages "man `egrep`".

```
egrep -i "(panic|memory|bad|module)" messages ↵
Apr  5 12:01:46   mrc vmunix: panic on cpu 0: zero
Apr  5 12:01:46   mrc vmunix: 01576 low-memory static kernel pages
```

In the above example we found a panic message, subsequently you can use the `more` command to take a closer look at what happened.

How to use "more"

For the next page, press the **space bar**. For one page backwards, **press b**, to search for a string e.g. "panic" **/panic**, and to quit more, press **q**.

```

#####┐
--More-- (6%

Now enter lpanic (Searches for the string panic)

Apr  5 12:01:46 mrc vmunix: panic on cpu 0: zero
Apr  5 12:01:46 mrc vmunix: syncing file systems... done
Apr  5 12:01:46 mrc vmunix: 01576 low-memory static kernel pages
Apr  5 12:01:46 mrc vmunix: 01214 additional static and sysmap kernel pages
Apr  5 12:01:46 mrc vmunix: 00000 dynamic kernel data pages
Apr  5 12:01:46 mrc vmunix: 00368 additional user structure pages

--More--(86%
```

How to use "tail"

E.g. to list the last 10 lines of a file, refer to the following example:
(you can list any number of lines with the tail command, for more information use "man tail")

```

Mar 7 15:51:06 mrc startup.csh: Starting SMI Firmware server (fws)
Mar 7 15:51:07 mrc startup.csh: Starting CCU firmware server (ccufws)
Mar 7 15:51:08 mrc startup.csh: PPP driver
Mar 7 15:51:14 mrc startup.csh: Remote Diagnostics
Mar 7 15:51:16 mrc mr_startup.csh: running
Mar 7 15:53:15 mrc mr_startup.csh: INGRES system is up
Mar 7 15:53:17 mrc mr_startup.csh: mounts to database host succeeded
Mar 7 15:53:20 mrc mr_startup.csh: starting sv_ralock
Mar 7 15:53:21 mrc mr_startup.csh: install or update dni installation
Mar 7 15:53:22 mrc mr_startup.csh: starting PACSNET CTMRNET DICOMNET
Mar 7 15:53:23 mrc mr_startup.csh: starting sv_patlock
Mar 7 15:53:24 mrc service_startup.csh: starting httpd
Mar 7 15:53:29 mrc service_startup.csh: starting wd.exe (Service Watchdog)
Mar 7 15:53:30 mrc mr_startup.csh: starting protocol conversion
Mar 7 15:53:33 mrc mr_startup.csh: finished

```

How to use "head"

The head command lists the file from the beginning, for more information regarding "head", use manual pages ("man head").

.....↵

Searching for files

find "filename"

If you know only part of a file name e.g. elo1, and you don't have any idea where it's located, use the find command to search for the file.

```

mrc{root}1: find / -name '*elo1*' -print
/usr/appl/diag/ui_melo1.out
/usr/appl/diag/ui_melo1.err
/usr/appl/diag/melo1.out
/usr/appl/diag/melo1.err
mrc{root}2:

```

find combined with mtime

If you would like to list all files created in the last 2 day's under the /usr path, exclude images and database files using the following syntax.

```

mrc{root}2: find /usr/ -type f -name '*' -mtime -2 -exec
ls -la {} \; | egrep -v '(.ima|ingres)' |more
...
-rwxrwxr-x 1 mr 528 Oct 7 09:53 /usr/appl/diag/ui_.out
-rw-rw-rw- 1 root 209715200 Oct 7 01:22 /usr/appl/work/phys/research.tar
-rw-rw-r-- 1 mr 12932 Oct 7 15:46 /usr/appl/anl/stat.anl

```

NUMARIS logfiles

Reports

- Select "Reports" on the top selection bar in the Service Platform.

Site Info

Imprint
Installation_Protocol

Tune-up, QA, Tests

Session_History
Shim_Results
Tune-up_Results
Quality_Results
Test_Results
Export_Data

UNIX

Unix_Error
Unix_Startup

NUMARIS

Process_Diagnostics

Remote Diagnostics

Customer_Log_Status
Authentication_Log
UUCP_Status
UUCP_Queue
UUCP_Process_Status
UUCP_Target_Accessibility
Mail_Status

The options listed on the left hand side are available in the "Report".

Event Log

Select "Event Log" on the top selection bar in the Service Platform

Date: Click "Start" on the list bottom right to
Begin: create the report.

End:

Format:

brif
report
full

Show first:

top
tail

```
glt: Searching for logg entries between 12:28-13:28
glt: 376 logg entries found until 13:28
glt: 95 logg entries found between 12:28-13:28
6718: logg_message      = DTS_SEND_SEM Send semaphore not available
6718: logg_date         = 06-05-1997 13:23:35.442
6718: logg_status       = 192807402
6718: logg_add_status   = 0
6718: logg_info_status  = 0
6718: logg_hexval       = 0x0
6718: logg_module       = free at pos 30
6718: logg_cmd,prg,prc   = -1, -1, -1
6718: logg_fac, class    = 2942, 2
6718: logg_prio         = 0
6718: logg_parnum,text   = 0, 0x0
6718: logg_par1         = [                ]
6718: logg_par2         = [                ]
6718: logg_par3         = [                ]
6718: logg_par4         = [                ]
6718: logg_par5         = [                ]
next entry (brief,full,go,report,reverse,ex):
```

GLT

General Error Logging Tool.

- Open the console window

```
mrc_work% glt ↵
```

```
      General Error Logging Tool
```

```
glt: Enter logg entries day      (ex,[06-05-1997]):
```

```
glt: Enter logg entries begintime (ex,[12:28]):
```

```
glt: Enter logg entries endtime   (ex,[13:28]):
```

```
glt: Enter display format        (ex,brief,report,[full]):
```

```
glt: Stop after each message      (ex,no,[yes]):
```

```
glt: Display last entry first     (ex,no,[yes]):
```

Process Status

To get process status information, use the command. `ps -ax` (Display the status of current processes, include information about processes owned by others and processes with no controlling terminal.) For more information use the manual pages.

The **first** letter in the 3rd column indicates the runnability of the process.

mrc_work% `ps -ax`

```
0 ?  D    0:02 swapper
1 ?  S    0:00 /sbin/init -
2 ?  D    0:00 pagedaemon
53 ? S    0:02 portmap
58 ? IW   0:00 keyserv
67 ? S    0:04 in.routed
```

R		Runnable processes.
T		Stopped processes.
P		Processes in page wait.
D		Processes in non-interruptible waits, typically short-term waits for disk or NFS,I/O.
S		Processes sleeping for less than about 20 seconds.
I		Processes that are idle (sleeping longer than 20 seconds).
Z		Processes that have terminated and waits for their parent process to do, a wait (2V) ("zombie" processes).

The **second** letter in the 3rd column indicates whether a process is swapped out.

```
19261 co  I   0:00      sv_adjman 5 32
25450 co  I   0:00      ui_register 5 34
25483 co  I   0:02      ui_external 5 33
25587 co  I   0:00      ui_patset 5 35
25674 co  IW  0:00      ui_window 5 36
```

	W	Process is swapped out.
		(Blank) Process is loaded in memory.

The **third** letter in the 3rd column indicates whether a process is running with an altered CPU scheduling priority **nice(1)**.

```
26012 co  S N  0:00      sed sdV.*$dd
26013 co  R N  0:00      pstat -T
26014 co  S N  0:00      head -1
```

		N	The process priority is reduced.
		<	The process priority has been raised artificially.
			(Blank) Represented as a SPACE character in this position indicates that the process is running without special treatment

System Environment

Environment settings

To set the environment variables (NVRAM) as set in factory, use following script.

- Open the console window.

Output	Input	Note
mrc_work% Password:	su ↵ ***** ↵	
mrc{root}1:	/install/admin/numi_eeeprom.csh ↵	

Disk space on file systems

With df you get the disk space of all mounted file systems.

```
mrc_work% df ↵
```

Filesystem	kbytes	used	avail	capacity	Mounted on
/dev/sd0a	34343	13510	17399	44%	/
/dev/sd0g	1812101	756421	1037559	42%	/usr
/dev/sd1a	2066117	1475393	384113	79%	/disk2

Checking swap space

pstat (print system facts). With the qualifier -s the information about the actual swap space usage is displayed.

```
work% pstat -s ↵
```

58088k allocated + 55756k reserved = 113844k used, 86792k available

Display system limits

```
mrc_work% limit ↵
```

cputime	unlimited
filesize	40000 kbytes
datasize	unlimited
stacksize	8192 kbytes
coredumpsize	unlimited
memoryuse	unlimited
descriptors	256

NUMARIS monitoring

Show measurement options

Check installed (measurent) options

- To check installed options you can use "meas in", open the console window or login as user work.

```
work% 00000000↵
```

```
024 system          yes
047 system          no
EPI system option    system option
SPEC system option   no
HIGH system option   no
Array receiving option yes
Broadband RFPA option no
Gradient adjustment inline no
PALI not exist       yes
RFSWD no abort       yes
Autoscout allowed    yes
Register renew default no
Calibration of RF chara. no
Power switch not possible no
MAP Shim adjustment  no
MR tuning inline     no
Flexible coil handling no
Confirm window after Pat/Pos no
```

NUMARIS Start-up

Display start-up messages in console window

In case of start-up problems, it is necessary to display the process outputs in the console window. This way you see which processes are started and where the system hangs.

Output	Input	Note
mrc_work% Password:	su root ↵ ***	Enter root password.
mrc{root}1:	proc_outs_to_console ↵	
proc_outs_to_console : Changed moni_prctab.tab to /dev/console output . proc_outs_to_console : Please restart application.		

- Start the NUMARIS application (Login as user mr).

SV_MONI

During login of MR, the file /usr/appl/exe/.login will be read. In this file the graphic user interface Open Windows is started. During initialization it reads the file /usr/appl/exe/.openwin-init. This file starts the process **sv_moni** (first started NUMARIS process). The sv_moni starts all other NUMARIS processes.

To see in which order the NUMARIS processes are started in your actual software version, list the file with the cat command:

"cat moni_prctab.tab ↵" . For detailed information, list the file "moni_prctab.ref", for each process the **-o**, and **-e** tells where the output and error messages are logged in.

For detailed information regarding the NUMARIS Start-up refer to "NUMARIS Startup", FUN, page 64.

Determining problem type HW/SW

To eliminate software as reason for a problem, proceed as follows:

- Perform a Ptree-Installation ->will take about 10-20 minutes (depends on the speed of the CD-ROM drive)
 - login under INSTALL (password required)
 - do in the commandtool
 - cd /usr/appl/install ↵ #change directory
 - ptree_installation.csh ↵ #start ptree_installation
 - wait for the script to terminate
 - cd / ↵
 - ll ↵ #check the owner of disk2 in the printout
 - chown mr disk2 #if the owner is not 'mr' change it
- System reboots automatically
- Check if the problem is still present

With updating the PTREE most of the entire software is replaced. If the software remains suspicious proceed as follows:

- Save dynamic data to **avoid loss of customer protocols** (save the actual tune-up data), label the MOD with **(D1)**.
- Archive all patient images.
- Reinstall Software (Actual version), **do not restore dynamic data!**
- Perform full configuration (Install).
- Perform system Tune-up (Coil tune-up, when coil related problem).

- Check if the problem can be reproduced.

Problem remains	Problem is solved	
Restore the dynamic data labeled " D1 ". Most likely the problem is due to defective hardware. Hardware Trouble Shooting! END	Save dynamic data on a new medium, label this side of the MOD with (D2) . (Don't overwrite the set labeled with "D1")	
	Restore the set of dynamic data that was created in the first step, labeled with (D1) .	
	Check if the problem can be reproduced since the restore of the origin set of dynamic data.	
	Problem returned (Parts of the D1 set is corrupted)	Problem is solved (Software was most likely corrupted)
Restore dynamic data, labeled with D2 .		
Perform coil tune-up or restore the coil files from the D1 save set, Refer to Replacement of Parts, Software, "Restore parts of dynamic data from MOD".		END
If the problem is solved, save dynamic data! (This must be performed to have a complete save set including customer protocols)		

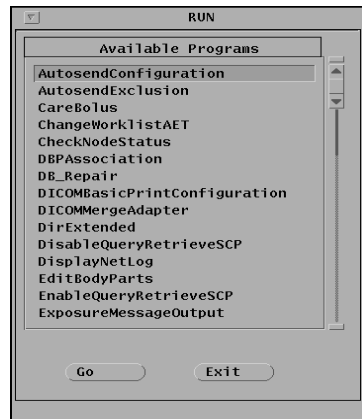
SYSTEM/RUN

Via the SYSTEM/RUN menu it is possible to start

- application programs
- programs, procedures which can be used to get informations about the system,
- programs, procedures to adapt the system for changed demands, environment and
- programs, procedures to maintain system performance.

These programs and procedures are accessible via SYSTEM/RUN and generally avoid the rundown of the complete application. Just the relevant tasks will be stopped for the modifications and restarted afterwards. (This is NOT valid for ALL processes!)

Menu



On the following pages mainly those programs, procedures are described which are potentially helpful for troubleshooting.

For programs, procedures which are described in a different (part of this) manual just the reference is listed.

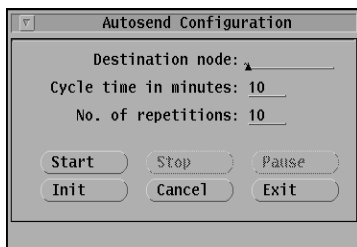
List of programs, procedures accessible via System/Run:

Bold printed programs, procedures are not available for all system types!

AutosendConfiguration	Mobile_Update
AutosendExclusion	NetCopyControl
CareBolus	NetworkInstallation
ChangeWorklistAET	ProtectPatientByUser
CheckNodeStatus	QueryRetrieveSecurity
DBPAssociation	RelocateImages2MRSC
DB_Repair	RemovePatientProtections
DICOMBasicPrintConfiguration	RightHandMouse
DICOMMergeAdapter	
DirExtended	SaveMPCUTrace
DisableQueryRetrieveSCP	Series_CreMod_Behaviour
DisplayNetlog	SetMagnifySharp
EditBodyParts	SetMagnifySmooth
EnableQueryRetrieveSCP	SetSystemLanguage
ExposureMessageOutput	ShowCMSInteractive
GarbageCollection	ShowCoilInformation

GetImageFilename	ShowPatientProtections
Ima=	SortPatients
LeftHandMouse	StartMPCUTrace
StopMPCUTrace	TroubleReset
removeAllCores	set_mr_password
set_window_color	show_printer_info

AutosendConfiguration



Allows to assign one node for the Autosend feature and to control the status for this function.

- Init to set up Autosend for one destination node.
- Start to set Autosend to active.
- Stop to terminate Autosend.
- Pause to stop Autosend temporarily (->heavy netload).
- Cancel to leave the menu without changing anything.
- Exit to exit the menu and activating the changes.

AutosendExclusion

Autosend Exclusion Utility

When using the Autosend functionality to send patient images automatically to another network node (DICOM, PACSNET) each patient is a candidate for Autosend.

If you want to prevent the system from sending certain patients due to any reason (e.g. experimental data, large amount of images, teaching data, VIP data,...), you can enter via this utility those patients. Already excluded patients are marked with a 'X' in the last column. If you enter such a patient the exclusion will be removed again.

UID	PatientName	PatientId	RegistrationDate	Excluded
101	MR13141		11-DEC-1998 09:47:14:000	
100	MR9021	?	14-JUN-1999 08:54:21:000	

Enter UID to change Autosend Exclusion flag or 'e' for exit : 100
Update done.

UID	PatientName	PatientId	RegistrationDate	Excluded
101	MR13141		11-DEC-1998 09:47:14:000	
100	MR9021	?	14-JUN-1999 08:54:21:000	X

Enter UID to change Autosend Exclusion flag or 'e' for exit :

CareBolus

Application -> see also NUMARIS handbook for details!

ChangeWorklistAET



When set to 'Yes' only worklists for the own node are used. Any worklist is selectable when set to 'No'!

CheckNodeStatus

This tool allows to check the reachability of remote nodes known by the system! With the help of 'ping' and 'mc3echo' the most important

parameters regarding the known nodes (listed in /etc/hosts) are determined and listed in a table.

```
CheckNodeStatus.sh; exit
```

```
Check Node Status:
=====
```

```
Check all Nodes      ..... A
Check selected Nodes ..... S
```

```
Exit ..... [EX]
```

```
Your Choice: a
```

```
Running this tool may take a couple of minutes!
Do you really want to run Check all Nodes? (y/[n]):y
```

```
Logging Information
=====
```

```
Date: Fri Jul 2 18:19:08 MET DST 1999
```

```
Check all Nodes
-----
```

```
18:19:15: Checking all the nodes
Please wait ...
18:19:52: Building results table
```

Node Name	Node Type	Node Address	PACS - PLA DICOM - AE	Port Number	Host Type	Node Status	
						Internet	DICOM
						Connection	
td2mrc	PACS STORAGE	1.987	049SA1MR990000	50026	Own	online	not applicable
rdmr	PACS STORAGE	185.2.16.207	049SA1DC74RCV0	50026	Remote	online	not applicable
rdmrnt	PACS STORAGE	185.2.48.196	049SA1DCCRRVC0	50026	Remote	online	not applicable
isidec	PACS STORAGE	185.2.46.118	049SA1OT72FOS0	50026	Remote	online	not applicable
werner	DICOM STORAGE	185.2.3.30	werner	104	Remote	offline*	offline **

tdmrc	DICOM STORAGE	185.2.19.210	049SA1MR27DIC1	50082	Remote	online	online
arc29	DICOM STORAGE	185.2.61.29	049SA1OT32DRSP	50082	Remote	offline*	offline **
testpr	DICOM BASIC PRINT	185.2.2.189	TESTPR	2002	Remote	offline*	offline **
mike	DICOM WORKLIST SCP	185.2.47.213	mike	60000	Remote	offline*	offline **
arc29	DICOM QUERY SCU	185.2.61.29	049SA1OT32DQRY	50090	Remote	offline*	offline **
mike	DICOM QUERY SCU	185.2.47.213	mike	104	Remote	offline*	offline **

*) Possible reasons: host down, IP Address incorrect, Router missing, Netmask incorrect.

**) Possible reasons: incompatible protocols (both nodes must have the same protocol), DICOM-AE incorrect, DICOM application down.

No purging necessary
(logfile length:3.26758, max.length:50 kBytes)
Please press RETURN to continue ...

DBPAssociation

Define DICOM Basic Print Association

Usage:

The DBPAssociation tool allows the user to switch the network connection from permanent association on to permanent association off.

-Permanent association on: means the connection to the Basic Print Camera exists from the time when the first job is started to the selected Camera till the time when a new job to a different Camera is started.
-Permanent association off: means the connection to the Basic Print Camera exists while sequential jobs to this Camera are in the job queue.

Permanent association on

Permanent association off? (Y/N): ☐

Important at sites where a dicom camera is shared by multiple modalities. This parameter effects how and when the camera can be accessed by the different modalities. Observe also:

- type of camera -> max. number of open associations

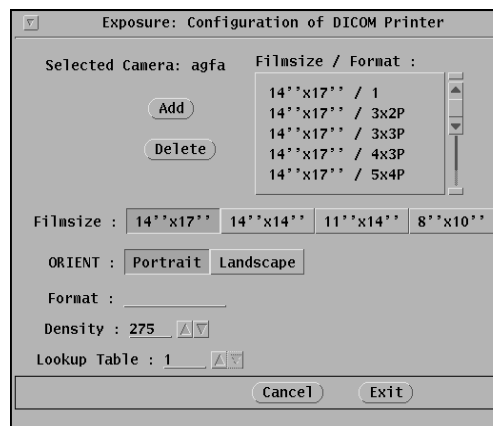
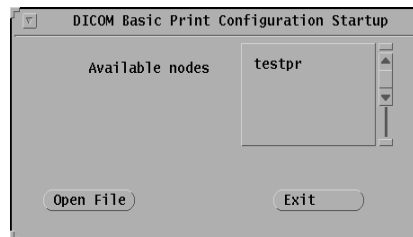
DB_Repair

Provides direct access to the db-repair tool. For details refer to the software troubleshooting guide.

DICOMBasicPrintConfiguration

Clicking on this item will display a popup-menu for the selection of installed DICOM cameras.

Selecting one of the camera files (in this case just one DICOM camera is installed) and clicking on 'Open file' opens the menu to set different parameters of the selected camera.



To confirm selected settings and the selection of the dicom camera exit both menus via 'Exit'!

DICOMMergeAdapter

Dicom Merge Adapter

This tool serves as a DICOM communication adapter between the Numaris System 3B33A and other Numaris Systems (35A11A, 3B33A, 3B31x).
The available VR_CODES are:
'UN' as the official DICOM VR CODE for 3B33A <-> 3B33A , 3B31x

'OB' as a workaround VR_CODE for 3B33A <-> 35A11A

Please press RETURN to continue ...

Dicom Merge Adapter:

=====

Set VR_CODE = UN (Official VR_CODE) UN
Set VR_CODE = OB (Workaround VR_CODE) OB

Exit [EX]

Your Choice: ob

Now please run Trouble Reset with Option 'RSIN'
or boot the System.

Please press RETURN to continue ...

DirExtended

This program lists information (located in each image header) about patient, measurement, etc. of the image in the actual segment. Click the button(s) to select the composition of data you need!

See also TSG System, chapter 5, 'Retrieval of image header information

Information about:		Show:
Measurement	Installation	☒ Comment
Sequence	Adjustment	☐ NEMA
	Patient	☐ Go
	Evaluation	☐ Config ☐ Help ☐ Exit

G21.Re12.Mr.NumberOfSlicesCur = 11 current Number of Slices	
G21.Re12.Mr.CurrentSliceNumber = 9 Current Slice Number	
G21.Re12.Mr.CurrentGroupNumber = 1 Current Group Number	
G21.Re12.Mr.SignalMask = 0 Signal Mask	
G21.Re12.Mr.NumberOfEchoes = 1 total Number of Echoes	
G21.Re12.Mr.CurrentSliceDistanceFactor = 0.1999980000000005 Current Slice Distance Factor	
G51.Txt.SaturationRegions = "1 SAT"	
Patient	

G08.Ide.AdmittingDiagnosis = " " Admitting Diagnosis	
G09.Ide.LastMoveDate.Year = 1998 Last Move Date	
G09.Ide.LastMoveDate.Month = 10	
G09.Ide.LastMoveDate.Day = 16	
G09.Ide.LastMoveTime.Hour = 9 Last Move Time	
G09.Ide.LastMoveTime.Minute = 25	
G09.Ide.LastMoveTime.Second = 4	
G09.Ide.LastMoveTime.Fraction = 0	
G10.Pat.PatientName = "Tf1-P32"	" Patient Name
G10.Pat.PatientId = "12345678"	" Patient Id
G10.Pat.PatientBirthdate.Year = 1967	Patient Birthdate

DisableQueryRetrieveSCP

DICOM Query/Retrieve enables foreign nodes (if also equipped with this feature and properly configured at both ends) to directly list patient data and retrieve images so they can further processed at that node.

Use this item of System/Run to disable the access via DICOM Query/Retrieve from 'foreign nodes' to this system!

DisplayNetLog

Since imagetransfers CAN fail and the according messages of the right screen server are easily missed, (**also the customer can**) use this tools to ensure the proper transmission of images to a foreign node.

The program lists all patients for which images were transfered. An error count is added. If patients exist with error count > 0 the detailed receive - or send - log can be displayed to find out which image transmission failed! (**The customer can resend these images to**

complete the original job!)

The percentage of failed jobs may serve as an indication for the quality of the net connection.

Use the timestamps of records with a 'transmission failed' message to classify the load of the network during the day.

List of patients with NetLog files

UID	PatientName	PatientId	SendLog	ErrCnt	RecLog	ErrCnt	MaxDate

0	Dummy patient 0 - Receive errors -		X	1			13-nov-1998
103	ASCK42dicom2310,okt,3.53.5 1		X	-		-	13-nov-1998
101	Josey tk		X	-	X	-	13-nov-1998
104	MPR tester CSA,ASC,diff,mo 2		X	-		-	13-nov-1998
105	Mustermann,Corina 250		X	-		-	13-nov-1998
102	TF1-P32	12345678	X	3		-	13-nov-1998

Enter UID or 'e' for exit :

101

Enter 'S' for SendLog or 'R' for ReceiveLog : s

Display of file /usr/appl/common/db/dev0/NL0101NetSnd.log

[BOF]

***** 13-nov-1998 09:56:29.576: Start (PACS) Transfer to K46: (OK)

*Elapsed*Bytes*

[09:56:29.781] (PACS) Transfer Image 1 to K46: (OK) | 1.890|137278

.

[09:56:42.112] (PACS) Transfer Image 10 to K46: (OK) | 1.186|137280

[09:56:43.299] (PACS) Transfer Image 11 to K46: (OK) | 1.224|137280

[09:56:44.524] (PACS) Transfer Image 12 to K46: (OK) | 1.410|137280

[09:56:45.935] (PACS) Transfer Image 13 to K46: (OK) | 1.386|137280

[09:56:47.322] (PACS) Transfer Image 14 to K46: (OK) | 1.381|137280

[09:56:48.703] (PACS) Transfer Image 15 to K46: (OK) | 1.310|137280

[09:56:50.014] (PACS) Transfer Image 16 to K46: (OK) | 1.308|137280

.

[10:04:21.832] (PACS) Transfer Image 305 to K46: (OK) | 1.471|140212

[10:04:23.322] (PACS) Transfer Image 306 to K46: (OK) | 1.432|140212

***** 13-nov-1998 10:04:24.928: End (PACS) Transfer to K46: (OK)

*Elapsed*Bytes*

[EOF]

Enter 'e' for exit or any other character to continue: r

List of patients with NetLog files

=====

UID	PatientName	PatientId	SendLog	ErrCnt	RecLog	ErrCnt	MaxDate

0	Dummy patient 0 - Receive errors			-	X	1	13-nov-1998
103	ASCK42dicom2310,okt,3.53.5 1		X	-		-	13-nov-1998
101	Josey tk		X	-	X	-	13-nov-1998
104	MPR tester CSA,ASC,diff,mo 2		X	-		-	13-nov-1998
105	Mustermann,Corina 250		X	-		-	13-nov-1998

102 TF1-P32 12345678 X 3 - 13-nov-1998

Enter UID or 'e' for exit : 101

Enter 'S' for SendLog or 'R' for ReceiveLog : r

Display of file /usr/appl/common/db/dev0/NL0101NetRec.log

[BOF]

***** 13-nov-1998 09:06:51.787: Start (DICOM) Transfer from K46: (OK)

*Elapsed*Bytes*

[09:06:59.370] (DICOM) Transfer Image 1 from K46: (OK) | 2.063|137026

[09:07:03.469] (DICOM) Transfer Image 2 from K46: (OK) | 0.805|137028

.

[09:14:51.033] (DICOM) Transfer Image 283 from K46: (OK) | 0.589|530606

[09:14:53.608] (DICOM) Transfer Image 284 from K46: (OK) | 0.605|530606

[09:14:56.208] (DICOM) Transfer Image 285 from K46: (OK) | 0.571|530606

[EOF]

Enter 'e' for exit or any other character to continue:

EditBodyParts

Application -> see also NUMARIS handbook for details!

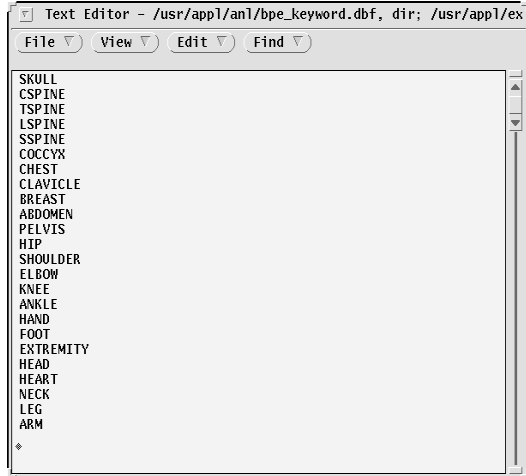
The text editor is called now on the file containing the values for Body Part Examined.

You can restore the original Siemens delivery by removing file /usr/appl/anl/bpe_keyword.dbf.

You should know how to use the text editor.

Don't forget to save your changes via the 'File' button !

Use CTRL^C to abort.



EnableQueryRetrieveSCP

DICOM Query/Retrieve enables foreign nodes (if also equipped with this feature and properly configured at both nodes) to directly list patient data and retrieve images so they can further be processed at that node.

Use this item of System/Run to enable the access from 'foreign nodes' to this system via DICOM Query/Retrieve!

ExposureMessageOutput

Switch on/off the output of EXPO messages at this console.
At least at one console the output of the EXPO messages must be ON!

Message Output: On Off

Exit

(In previous versions this setting was done via the EXPOSURE menu!)

GarbageCollection

Run this program frequently to clean the image file systems from garbage in order to keep system performance up. Follow the advices printed in the preamble.

```
Garbage Collection Tool
=====
This tool checks all files on the DBDATADIR* directories, including the
DBDATADIR directory. A filename check is done whether it is a valid image file.
The garbage files will be displayed and additionally written to a script file.
The filename of this script is
```

```
'/tmp/remove_the_garbage'.
```

The script can be started directly from this tool. It can also be started later from a command tool, but be careful when using the editor on this file.

This job takes some seconds for each filesystem depending on the number of files on the disk.

Don't do too much in parallel when running this tool.

Do you want to start this job ? (Y,[N])

GetImageName

In some cases it is necessary to know the 'name' of a certain image.

Display the image in the actual segment and invoke GetImageName.

```
Active Segment No.      : 2
Active Segment Name     : TF1-P32
Active Image ID         : 10200131
Active Study ID         : 10200011
Active SegPatientID     : 102
Active SegImageNumber   : 131
```

Physical name: 102-11-131.ima stored on DBDATADIR1_1
File path : /usr/appl/data/k49/disk1/102-11-131.ima

Press RETURN to continue :

Ima=

Application -> see NUMARIS handbook for details!

LeftHandMouse

Selfexplaining program. Make the mouse a lefthand one!

MobileUpdate

Besides the possibility to select a certain DICOM camera via system/run also the local camera can be changed without the need to invoke 'install'!

(Mostly used with mobile MR scanners which need to be connected to a different SPCI/SPDI - camera (-> MAXICAM) when moved to the 'next' Hospital!)

Special Mobile Update [June 1996 (v2.1)]

... opening the database

=====
General Installation
=====

Name of Institution [k49] :

Station ID [k49] :

+-----+
| Camera Installation |
+-----+

Birthdate / Age ([Birthdate]/Age) ([0]/1):

Localizer ([First]/All/None) ([0]/1/2):

Maximum Number of Copies (old) : 9
Your Input (max. value 999) :

Maximum Number of Copies (remains) : 9

Number of local cameras connected to system (old) : 1
Your Input (max. value 1) :
Number of local cameras connected to system (remains) : 1

MENU : Camera connected to console

k49 (Main) 0
k49a (Satellite) 1

the Camera is connected to console : has been : k49a (Satellite)
.

Your Choice [(k49a (Satellite)
.) = 1] :

the Camera is connected to console : remains : k49a (Satellite)
.

MENU : VCR for MAXICAM

PAL 0
NTSC 1

videonorm for MAXICAM is has been : PAL

Your Choice [(PAL) = 0] :

videonorm for MAXICAM is remains : PAL

MENU : camera name

3M_DryView_8700 0
3M_M952_SHCI 1
3M_M952_SHPT 2
3M_M969_HQ_SHPT 3
3M_P831_SHCI 4
3M_P831_SHPT 5
3M_XL_SHCI 6
3M_XL_SHPT 7
Agfa_LR3300 8
Agfa_MCL 9
Digicam_LHP 10
Dupont_LP400 11
Fuji_3543_Landscape 12
Fuji_3543_Portrait 13
Fuji_FL-IMD 14
Kodak_1120 15
Kodak_2180 16
Kodak_KELP100 17
Kodak_XLP 18
Konica_LI10 19
Konica_LI21 20
Polaroid_Helios_1417 21

```

connected camera has been : Polaroid_Helios_1417

Your Choice [(Polaroid_Helios_1417) = 21] :

connected camera remains : Polaroid_Helios_1417

Remove old files

=====
Update of Site name and Camera configuration is complete
=====

/usr/appl/install/initstc starting with option(s): s

Installation of SMI5 (Version 5.17 97/08/08)
-----

starting IRS and booting SIAM

starting IRS and booting GIPSY-Kernels

Previous boot file renamed from /usr/appl/smi_load/smi_mr/irsboot.boo
to /usr/appl/smi_load/smi_mr/irsboot.boo%

5 boards found in SMI5:
slot_no=0000: Board D0020 SIAM 16 MB
slot_no=0001: Board D0062 IMAGER
slot_no=0002: Board D0072 LISA
slot_no=0003: Board D0055 MAP4 160 MB
slot_no=0004: Board D0030 DATA-IO 8 MB

Is the hardware correctly identified ? ([y]/n)

installing SMI5 for OPEN

checking piggyback boards of DATA-IO

module_no = 0002: Board D0024

Are the piggyback boards correctly identified ? ([y]/n)
SMI5 installation completed

===== Initialize SMI at satellite console (k49a) =====

/usr/appl/install/initstc starting with option(s): s

Installation of SMI5 (Version 5.17 97/08/08)
-----

```

```

starting IRS and booting SIAM

starting IRS and booting GIPSY-Kernels

Previous boot file renamed from /usr/appl/smi_load/smi_mr/irsboot.boo
to /usr/appl/smi_load/smi_mr/irsboot.boo%

5 boards found in SMI5:
slot_no=0000: Board D0020 SIAM 16 MB
slot_no=0001: Board D0062 IMAGER
slot_no=0002: Board D0072 LISA
slot_no=0003: Board D0055 MAP4 160 MB
slot_no=0004: Board D0066 MAXICAM

Is the hardware correctly identified ? ([y]/n)

```

Note: MobileUpdate provides the possibility to initialize the SMI without needing the password for install!

NetCopyControl

Allows to control the jobs set up by the Autosend feature.

```

List of jobs
=====
ncc_joblist>>No NetJob files available !

NetCopyControl Menu
=====
Pause -> 1 (A Running job will finish normally)
Continue -> 2
List of jobs -> 3
Recover job -> 4
Cancel job -> 5
Remove all jobs -> 6 (A running job will finish normally)

Enter selection (0=exit) :3

List of jobs
=====
ncc_joblist>>No NetJob files available !

```

```

NetCopyControl Menu
=====
Pause -> 1 (A Running job will finish normally)
Continue -> 2
List of jobs -> 3
Recover job -> 4
Cancel job -> 5
Remove all jobs -> 6 (A running job will finish normally)

Enter selection (0=exit) :4

```

No NetJob files available !

NetCopyControl Menu

```
=====
Pause          -> 1      (A Running job will finish normally)
Continue       -> 2
List of jobs   -> 3
Recover job    -> 4
Cancel job     -> 5
Remove all jobs -> 6      (A running job will finish normally)
```

Enter selection (0=exit) :1

```
*****
* NetCopy Send process in state 'pause' !!!! *
*                                             *
* Don't forget to activate the NetCopy Send *
* process with 'Continue'.                  *
* 'pause' will be automatically reset by reboot. *
* Running jobs will finish normally.         *
*****
```

```
*****
* NetCopy Send process in state 'pause' !!!! *
*                                             *
* Don't forget to activate the NetCopy Send *
* process with 'Continue'.                  *
* 'pause' will be automatically reset by reboot. *
*****
```

NetCopyControl Menu

```
=====
Pause          -> 1      (A Running job will finish normally)
Continue       -> 2
List of jobs   -> 3
Recover job    -> 4
Cancel job     -> 5
Remove all jobs -> 6      (A running job will finish normally)
```

Enter selection (0=exit) :2

Activating Net Copy server...

NetCopyControl Menu

```
=====
Pause          -> 1      (A Running job will finish normally)
Continue       -> 2
List of jobs   -> 3
Recover job    -> 4
Cancel job     -> 5
Remove all jobs -> 6      (A running job will finish normally)
```

Enter selection (0=exit) :

NetworkInstallation

This is another function of 'install' for the MRC accessible via System/Run. Numaris needs not to be ended. Just the relevant processes are stopped and restarted after the modification is done. (This is not valid for the MRSC because the related net processes will not be restarted by this tool!)

NetworkInstallation

Networkinstallation will be activated

Please note that active archive or copy jobs will be stopped
Make sure that no archive or copy job is active

Please note that active network transfers will be stopped
Make sure that no network transfer ist active

Continue: y/[n]

Your Choice: y

LOGIN now as user install. You have to enter the password.
Abort the login with CTRL^Z if you don't know the password.
Password:
Last login: Mon Jan 25 14:45:51 on console
SunOS Release 4.1.4 (MR_014_mrc_sun4m) #2: Wed Jan 21 22:39:11 MET 1998

Common Installation [October 1993]
Target Version: __STC_91E11_VERSION

... opening the database

Structure of Status File is up-to-date

Table of Nodes								
Node	Node Type	Node address	Node Name	Logical Name	Node active	SUN	Port number	PLA/AE number
OWN	SIEnet	1.906	tdmrc	TDMRC	Online	YES	50026	049SA1MR270000
"	DICOM StU						50081	049SA1MR27DFOS

"	DICOM StP						50082	049SA1MR27DIC1
"	DICOM PrU						50084	049SA1MR27DBPR
"	DICOM QuP						50089	049SA1MR27DQRY
"	DICOM WU						60011	049SA1MR27DBWM
REMOTE	SiEnet	185.2.16.207	rdmr	RDMR	Online	YES	50026	049SA1DC74RCV0
REMOTE	SiEnet	185.20.13.22	k49	K49	Online	YES	50026	049SA1MR490000
REMOTE	SiEnet	185.2.48.196	rdmmt	RDMRNT	Online	YES	50026	049SA1DCCRRCV0
REMOTE	SiEnet	185.2.8.3	server	SERVER	Online	YES	50026	unknown
REMOTE	DICOM StP	185.2.3.30	werner	WERNER	Online	NO	104	werner
REMOTE	DICOM StP	185.2.47.37	td2mrc	TD2MRC	Online	YES	50082	049SA1MR99DIC1
REMOTE	DICOM StP	185.2.48.101	h6	H6	Online	YES	50082	049SA1MR55DIC1

MENU : Installation of Nodes

Delete D
Create C
Modify M
Exit the Installation of Nodes E

Your Choice : m

Which node should be modified? Your Choice (0-7) : 2

If you want to change a parameter which isn't in dialog:
Use subsequent 'delete' and 'create'

** Info : ... TCP/IP installation is running **

Node name [k49] :

Protocol Type (DECnet/[TCP/IP]) (0/[1]):

Is node a SUN? Your Choice ([Y]/N) :

PLA [049SA1MR490000] :

Port Number [50026] :

** Info : ... TCP/IP installation is running **

Table of Nodes								
Node	Node Type	Node address	Node Name	Logical Name	Node active	SUN	Port number	PLA/AE number
OWN	SiEnet	1.906	tdmrc	TDMRC	Online	YES	50026	049SA1MR270000
"	DICOM StU						50081	049SA1MR27DFOS
"	DICOM StP						50082	049SA1MR27DIC1
"	DICOM PrU						50084	049SA1MR27DBPR
"	DICOM QuP						50089	049SA1MR27DQRY
"	DICOM WU						60011	049SA1MR27DBWM
REMOTE	SiEnet	185.2.16.207	rdmr	RDMR	Online	YES	50026	049SA1DC74RCV0
REMOTE	SiEnet	185.20.13.22	k49	K49	Online	YES	50026	049SA1MR490000
REMOTE	SiEnet	185.2.48.196	rdmmt	RDMRNT	Online	YES	50026	049SA1DCCRRCV0
REMOTE	SiEnet	185.2.8.3	server	SERVER	Online	YES	50026	unknown

REMOTE	DICOM SiP	185.2.3.30	werner	WERNER	Online	NO	104	werner
REMOTE	DICOM SiP	185.2.47.37	td2mrc	TD2MRC	Online	YES	50082	049SA1MR99DIC1
REMOTE	DICOM SiP	185.2.48.101	h6	H6	Online	YES	50082	049SA1MR55DIC1

```
-----
MENU : Installation of Nodes
-----

Delete . . . . . D
Create . . . . . C
Modify . . . . . M
Exit the Installation of Nodes . . . . . E

                                Your Choice : e
Connection closed.
Killing and restarting now the network processes and sv_file ...
9768 co S N   0:00 sv_netcp_2 6 23
9763 p2 S N   0:00 /usr/appl/exe/sv_netcp_2
9789 p2 S N   0:00 grep netcp_2
9779 p2 S N   0:00 /usr/appl/dtk/dircv/dtk_dircv
9791 p2 S N   0:00 grep dircv
9788 co S     0:00 sv_file 6 18
9793 p2 S N   0:00 grep sv_file

2 sv_netcp_2 processes, DICOM ToolKit server and sv_file restarted.

Do System/End/Application/'no restart' on satellite
(if available) in order that the modified nodes should
be accessible also on satellite console

Please press RETURN to continue ...
```

ProtectPatientByUser

```
User Protection of patients
-----
The user may protect patients. This type of protection cannot
be released by any available program, but only by the user
itself. The effect is that FILE/DELETE cannot delete such a
protected patient.

Enter pat_uid (if unknown enter 0, enter -1 for abort) :
```

QueryRetrieveSecurity

```
Query/Retrieve Security
=====
```

DICOM Query/Retrieve users may query your database and retrieve images from remote workstations. To protect your system from unauthorized access you have the possibility of

- o Disabling Q/R
- o Removing critical workstations inside the Network Installation
- o Protecting the patients

When disabling Q/R you will loose the benfits of Q/R.
It is recommended to install only 'reliable' Q/R nodes.
Patients inside NUMARIS can be protected from the usage of remote Q/R.
Depending on your system environment you can define the default setting to 'Patients are protected automatically and have to be released for Q/R' or 'Patients are not protected and can be protected if necessary'
Contact your service technician for information how to define this default.

Protected patients are marked with a 'X' in the second column. By entering the unique UID the protection is set/unset (toggle).

UID	Protected	PatientName	PatientId	RegistrationDate
101		MR13141		11-DEC-1998 09:47:14:000
100		MR9021	?	14-JUN-1999 08:54:21:000

Enter UID to change Protection flag or 'e' for exit : 100
Update done.

UID	Protected	PatientName	PatientId	RegistrationDate
101		MR13141		11-DEC-1998 09:47:14:000
100	X	MR9021	?	14-JUN-1999 08:54:21:000

Enter UID to change Protection flag or 'e' for exit :

RelocateImages2MRSC

Relocate images from MRC to MRSC

Relocating images from MRC to MRSC improves the MRC performance in the case that the local MRC disks are full and remote storage is applied.

As you don't have a satellite console this image relocation tool cannot work. The application will display some more information about the NUMARIS disk partitions and will stop before user interaction start

You will now see a list of all filesystems.
Based on the content you can decide whether image relocation might be reasonable. You will be prompted for the amount of disk space to relocate from the MRC disk(s) to MRSC disk(s).

You are currently working at the main console MRC. Relocating images is only possible from the satellite console MRSC due to performance reasons.

Device	Hostname	FS	Index	Status	Fr.Bytes	All.bytes	256	Filesystem
11	td2mrc	1	1	1	1465009120	114614720	10677	/usr/appl/data/td2mrc/disk1
19	td2mrc	1	9	1	1534882368	62185920	11185	/usr/appl/data/td2mrc/disk9

MRC disk(s) available disk space: 21862 images (256*256)/ 5655 images (512*512)
MRSC disk(s) available disk space: 0 images (256*256)/ 0 images (512*512)

```
-----
Total available disk space: 21862 images (256*256)/ 5655 images (512*512)
```

Note: If there is no contiguous disk space available for the measurement on MRC disks (filesystems), the MRSC disk(s) will be used. This slows down the system.

Can't start without connected MRSC.
Thanks for calling RelocateImages2MRSC

Enter 'e' for exit or any other character to continue:

RemovePatientProtections

The operator can intentionally protect patient images from being deleted. Due to system inconsistencies this also can happen accidentally. With this program the protections can be removed for all or a single patient.

```
[A]ll : delete all patient protections !
[S]ingle: delete only a single patient from list !
Enter : s
```

```
Patient Protection List
=====
pat_name : Mustermann,Corina
pat_id   : 250
pat_uid  : 105
EXPOSURE          9 time(s) on k49, mode 1
```

---Enter pat_uid:

RightHandMouse

Selfexplaining program. Make the mouse a righthand one!

SaveMPCUTrace

Saves the MPCU trace files for further processing.

Series_CreMod_Behaviour

- Output when set to 'strict validation of the FrameOfReference'!

Configuration Series/Create and Series/Modify

```
=====
Series/Create and Series/Modify check the 'FrameOfReference' and the
'StudyInstanceUID' before images are copied together. According to
DICOM standard version 3.0 the 'FrameOfReference' ensures spatial
relationship of images within a series. Additionally images of one
```

series should not have alternating study parameters (e.g. Study date, diagnosis,...) In NUMARIS the 'FrameOfReference' is generated based on the table move date & time. Disabling the check of the 'FrameOfReference' and 'StudyInstanceUID' means, that these DICOM rules can be violated. Inside of NUMARIS you will not have any problems, but you might get problems from other network nodes.

You have been informed about the possible errors due to the missing validation checks of the 'FrameOfReference' and 'StudyInstanceUID'.

Do you agree, that Series/Create and Series/Modify may process images with non unique 'FrameOfReference' or 'StudyInstanceUID' ? (Y,N) :y

Your answer has been stored on the system. Series/Create and Series/Modify does now allow a mix of images with different 'FrameOfReference' or 'StudyInstanceUID'.

You can return to the original behaviour by calling this program once more.

Please enter any character to continue.

- Output when **NOT** set to 'strict validation of the FrameOfReference'!

Configuration Series/Create and Series/Modify

```
=====
You already confirmed that Series/Create and Series/Modify
does not insist on a strict validation of the 'FrameOfReference'
and 'StudyInstanceUID'.
```

Do you want to have a strict validation of the 'FrameOfReference' and 'StudyInstanceUID' in Series/Create and Series?modify ? (Y,N) :

SetMagnifySharp

```
-----
----->Please Restart Application
-----> to activate cubic-spline
-----> algo for MAGNIFY
-----
```

-----> Please press RETURN

Set cubic-spline algorithm for magnify!

SetMagnifySmooth

```
-----
----->Please Restart Application
-----> to install the default
```

```
----->      algo for MAGNIFY
-----
```

```
----->      Please press RETURN
```

Set the default filter (smooth) for magnify!

SetSystemLanguage

Provides the choice to set the systems language either to english or german.

```
=====
System Language
=====

Please enter desired system language:
english/german . . . . . ([0]/1):

System Language hasn't changed.
Actual System Language is english.

Press <return> to continue:
```

ShowCMSInteractive

Provides access to the image header variables! Use (e.g.) GetImageFilename to determine the full filename (with path) for the input of ShowCMSInteractive

SHOWCMSINTERACTIVE AVAILABLE COMMANDS:

```
h[elp]          display all available commands
d[isplay]       display the contents of all header variables
d[isplay] var   display the contents of a header variable
e[nter] var expr set the contents of a header variable
dump           dump the contents of an image file to stdout
dump file_name dump contents of an image file to file 'file_name'
use image       switch to another image file
f[ile]         display name of an image file in use

ex[it], 'Return' leave ShowCMSInteractive
```

SHOWCMSINTERACTIVE: Please enter image file name: 122-6-125.ima

```
STC: Cannot access status file "122-6-125.ima"
      : No such file or directory
SHOWCMSINTERACTIVE: file "122-6-125.ima" cannot be mapped.
Create (y/n): n
SHOWCMSINTERACTIVE> f
      status file currently in use: "122-6-125.ima"
SHOWCMSINTERACTIVE> f /usr/appl/data/td_mrc/disk7/122-6-124.ima
```

```
unrecognized command/syntax "f"
(Type 'help' for help)
SHOWCMSINTERACTIVE> use
```

```
SHOWCMSINTERACTIVE> f
      status file currently in use: "122-6-125.ima"
SHOWCMSINTERACTIVE> use /usr/appl/data/td_mrc/disk7/122-6-124.ima
STC: Cannot access status file "/usr/appl/data/td_mrc/disk7/122-6-124.ima"
      : No such file or directory
SHOWCMSINTERACTIVE> use /usr/appl/data/td_mrc/disk7/122-6-125.ima
SHOWCMSINTERACTIVE: switched to file /usr/appl/data/td_mrc/disk7/122-6-125.ima
SHOWCMSINTERACTIVE> f
      status file currently in use: "/usr/appl/data/td_mrc/disk7/122-6-125.ima"
SHOWCMSINTERACTIVE> d
G08.Ide_buf = ""
G08.Ide.StudyDate.Year = 1994;          /* Study Date (0008,0020) */
G08.Ide.StudyDate.Month = 11
G08.Ide.StudyDate.Day = 9
G08.Ide.AcquisitionDate.Year = 1994;    /* Acquisition Date
      * (0008,0022) */
G08.Ide.AcquisitionDate.Month = 11
G08.Ide.AcquisitionDate.Day = 9
G08.Ide.ImageDate.Year = 1994;          /* Image Date (0008,0023) */
.
.
.
G08.Ide.AdmittingDiagnosis = "still unknown
33: Terminated literal
"; /* Admitting Diagnosis (0008,1080) */
34: Terminated literal
G08.Ide.ManufacturerModel = "MAGNETOM EXPERT
35: Terminated literal
"; /* Manufacturer Model (0008,1090) */
G08.Ide.AttendingPhysician = "?";          /* Attending Physician
      * (0008,1050) */
G08.Ide.Radiologist = "?"; /* Radiologist (0008,1060) */
G08.Ide.OperatorIdentification = "?"; /* Operator Identification
      * (0008,1070) */
G09.Ide_buf = ""
G09.Ide.DataObjectSubtype.M = Object_m_MRU; /* Data Object Subtype
.
.
.
t.PatientMaidenName = ""; /* Patient Maiden Name (0010,1005) */
G10.Pat.PatientAge = "028Y"; /* Patient Age (0010,1010) in Years | Months
      * | Days */
G10.Pat.PatientSize = -19222.0; /* Patient Size (0010,1020) in meters */
G10.Pat.PatientWeight = 60; /* Patient Weight (0010,1030) in kilograms */
G10.Pat.FrameOfReferenceUID =
"1.3.12.2.1107.5.2.6.13001.19941109084147000"; /* Frame of Reference
      * UID (0020,0052) */
G51.Txt.InstallationName = ""
625: Terminated literal
G51.Txt.SoftwareVersion = "TVB21A
626: Terminated literal
"
G51.Txt.Matrix = ""
G51.Txt.TypeOfMeasurement = " tse2-5 180"
G51.Txt.ScanNumber = ""
G51.Txt.RepetitionTime = "STR 4000.0"
G51.Txt.EchoTime = ""
G51.Txt.GatingAndTrigger = "T "
G51.Txt.TubeCurrent = ""
```

```

G51.Txt.TubeVoltage = "???"
G51.Txt.SliceThickness = ""
G51.Txt.SlicePosition = "SSP      14.3"
G51.Txt.SliceOrientationNo1 = ""
G51.Txt.SliceOrientationNo2 = "T      "
G51.Txt.FieldOfView = ""
G51.Txt.ZoomCenter = "F?"
G51.Txt.GantryTilt = ""
G51.Txt.TablePosition = "?TP      0"
G51.Txt.MipHeadLine = ""
G51.Txt.MipLine = "???"
G51.Txt.MipColumn = ""
G51.Txt.MipSlice = "???"
G51.Txt.StudyNumber = ""
G51.Txt.Contrast = "S      "
G51.Txt.PatientBirthdate = ""
G51.Txt.SequenceInformation = "0 R D      "
G51.Txt.SaturationRegions = ""
G51.Txt.DataSetId = "STU/IMA 6/125"
G51.Txt.MagnificationFactor = "      "
654: Unterminated literal
G51.Txt.ManufacturerModel = "MAGNETOM EXPERT
655: Unterminated literal
"
G51.Txt.PatientName = ""
G51.Txt.TimeOfScanning = "D09:42:56"
SHOWCMSINTERACTIVE> h
SHOWCMSINTERACTIVE> AVAILABLE COMMANDS:

```

```

h[elp]          display all available commands
d[isplay]      display the contents of all header variables
d[isplay] var  display the contents of a header variable
e[nter] var expr set the contents of a header variable
dump           dump the contents of an image file to stdout
dump file_name dump contents of an image file to file 'file_name'
use image      switch to another image file
f[ile]         display name of an image file in use

ex[it], 'Return' leave ShowCMSInteractive

```

```

SHOWCMSINTERACTIVE> d G51.Txt.ManufacturerModel
G51.Txt.ManufacturerModel = "MAGNETOM EXPERT
SHOWCMSINTERACTIVE>

```

This example above just shows a small part of all image header variables!

ShowCoilInformation

```

Active Segment No. : 2
Active Segment Name : MR13141
Active Image ID : 10100013
Active Study ID : 10100002
Active SegPatientID : 101
Active SegImageNumber: 13

```

```

Physical name: 101-2-13.ima stored on DBDADIR1_9
File : /usr/appl/data/td2mrc/disk9/101-2-13.1maHeader values:

```

```

HE
G19.Acq2.Mr.ACElementToADCConnect[00] = 1 -> HE

```

Enter 'e' for exit or any other character to continue:

ShowPatientProtections

Lists patients with currently protected images!

```

Patient Protection List
=====
pat_name : Mustermann, Corina
pat_id : 250
pat_uid : 105
EXPOSURE          9 time(s) on k49, mode 1

db_PrintPatientProtection - Duration:      0.026 sec

```

Press RETURN to continue :

SortPatients

Patient display order change utility

=====

The order of patients in the patient list can be changed dynamically for the STORE device.

- 1.) Register date sorting: the patients are displayed in decreasing order by the register date
- 2.) Name sorting: the patients are displayed in ascending order by the patient name and patient id

Enter type of sorting (Register Date sorting=1, Name sorting=2, Abort=0) :

StartMPCUTrace

Start to log information via MPCUTrace.

StopMPCUTrace

Stop to log information from the MPCUTrace tool. Prerequisite for doing SaveMPCUTrace.

TroubleReset

Invokes TroubleReset. See the Software Troubleshooting Guide for details.

RemoveAllCores

Run this procedure in case you suspect that disk space is wasted by numerous core files. This procedure looks for corefiles in all directories and deletes them subsequently!

set_mr_password

Run this procedure to protect the account "mr" by a password.

set_window_color

Adjust the background of the B/W-monitor. Type in one of the eight possible grey levels. Restart the system!

Current color is gray level 6

```
*****
* "set_window_color" allows setting of the window background color. *
* 8 gray levels are available: level 2-9 (2 is dark, 9 is bright) *
* After changing the color it is necessary to execute *
* "End Application No Restart" and logging into MR *
* to get proper colors (else you get a mixture). *
* WARNING: You have to exit all windows for proper operation *
*****
```

```
Enter either gray level      2-9
or restore factory settings  factory
or help for getting a long help help
or quit                     [q] :
```

show_printer_info

Lists details regarding the installed printer along with an example for its usage.

```
printer tools:

printer is installed on:   td2mrc
printer name is:          ps_p
printer connection is: parallel

use
  lpr -Pps_p <file>
for printing Postscript files
  lpr -Pps_pa <file>
for printing ASCII files

Press RETURN to continue:
```


Host

In this part you will find test strategies, procedures, repair instructions encompassing the complete host and peripherals.

Strategy

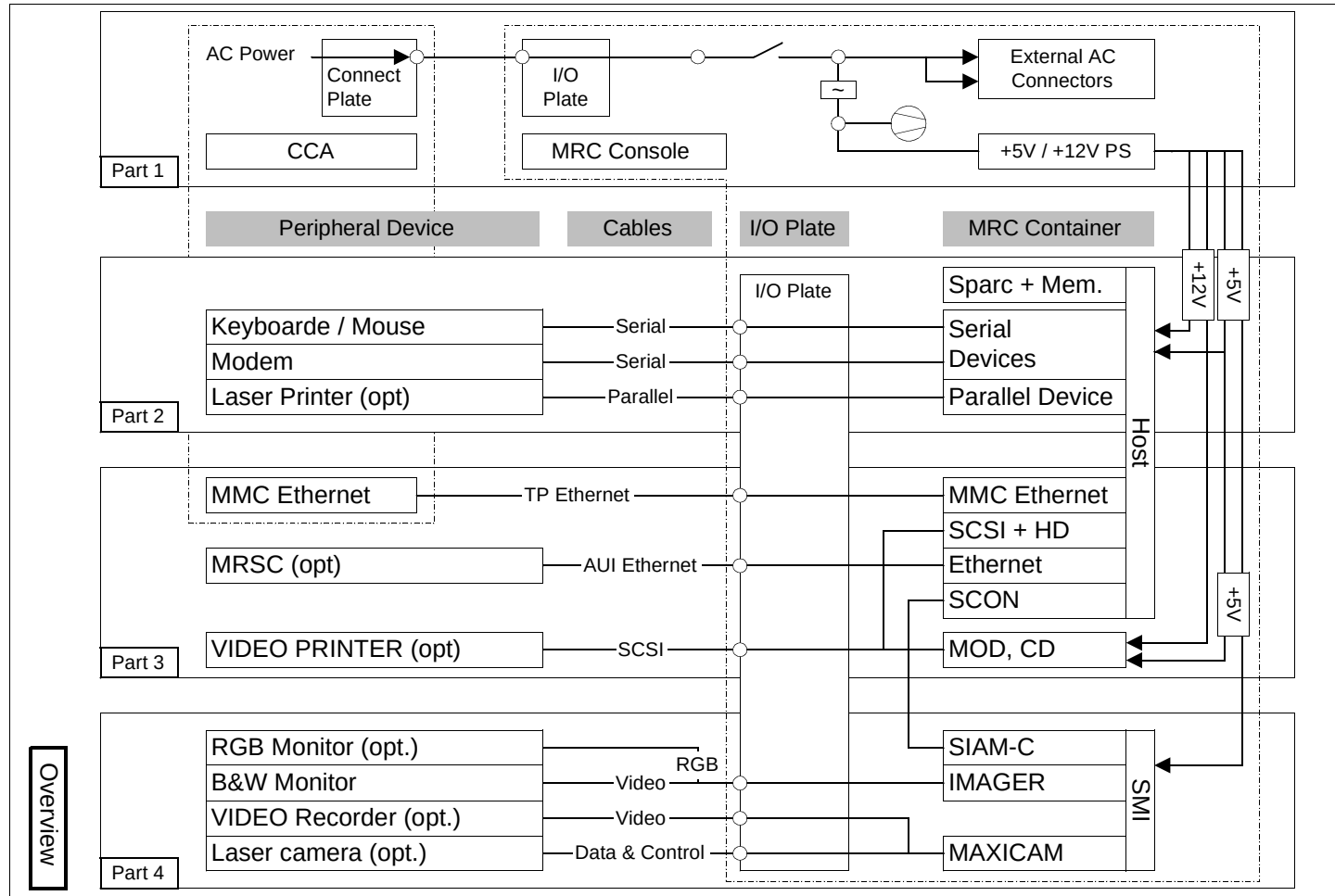
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MR(S)C system overview (schematics)

- The following diagrams (**overview** and **part 1 - 4**) show schematic the MR(S)C and the internal and peripheral components.
- Every peripheral and internal component is connected to the MR(S)C via cables and connectors (data cables and for power cables - also fuses).
- On the right hand side of the schematic diagrams you will find reference pages, where additional information can be received. You can also add your own remarks.

Fig. 3 System overview



Goto

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Fig. 4 Part 1

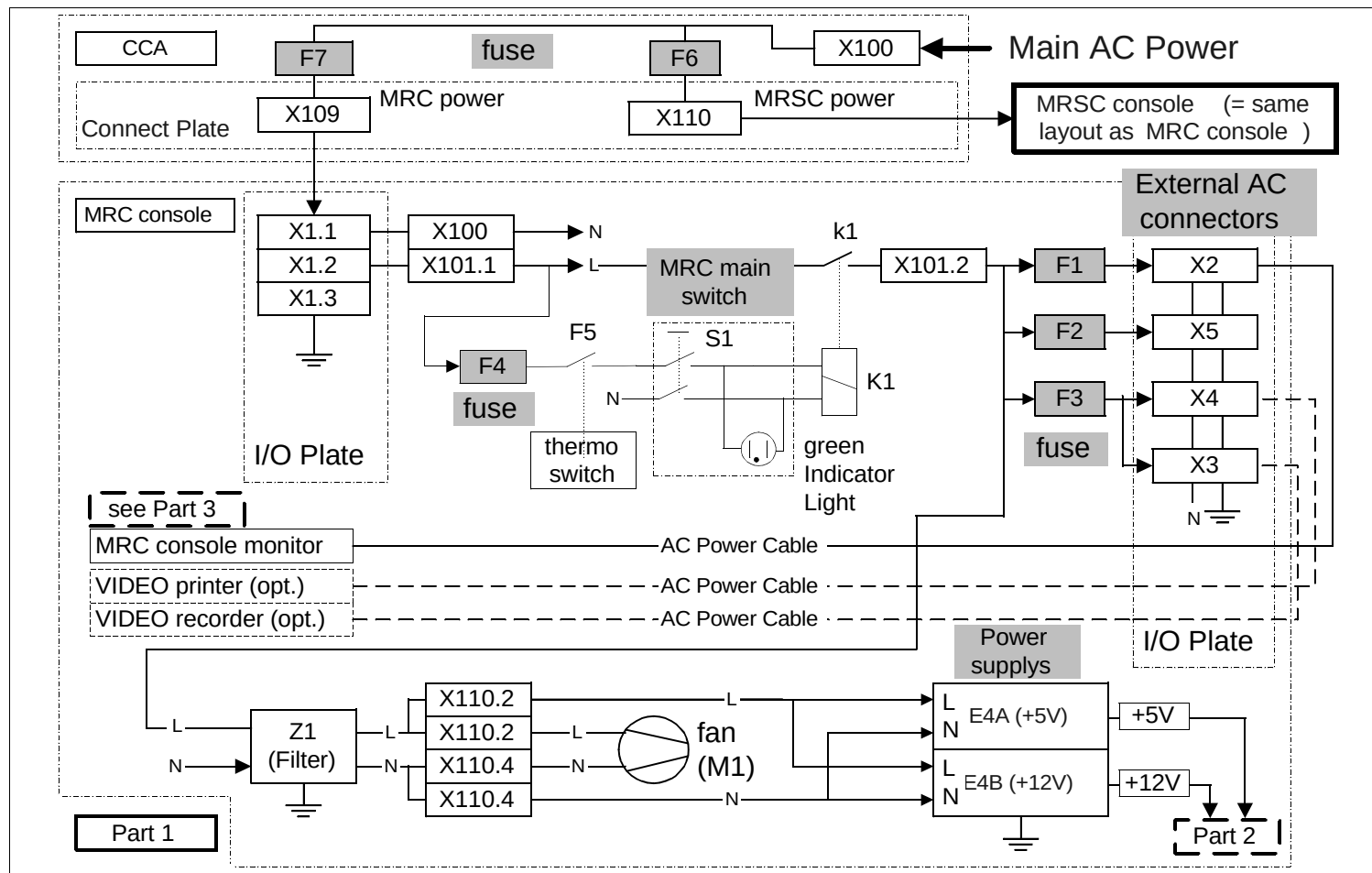


Fig. 5 Part 2

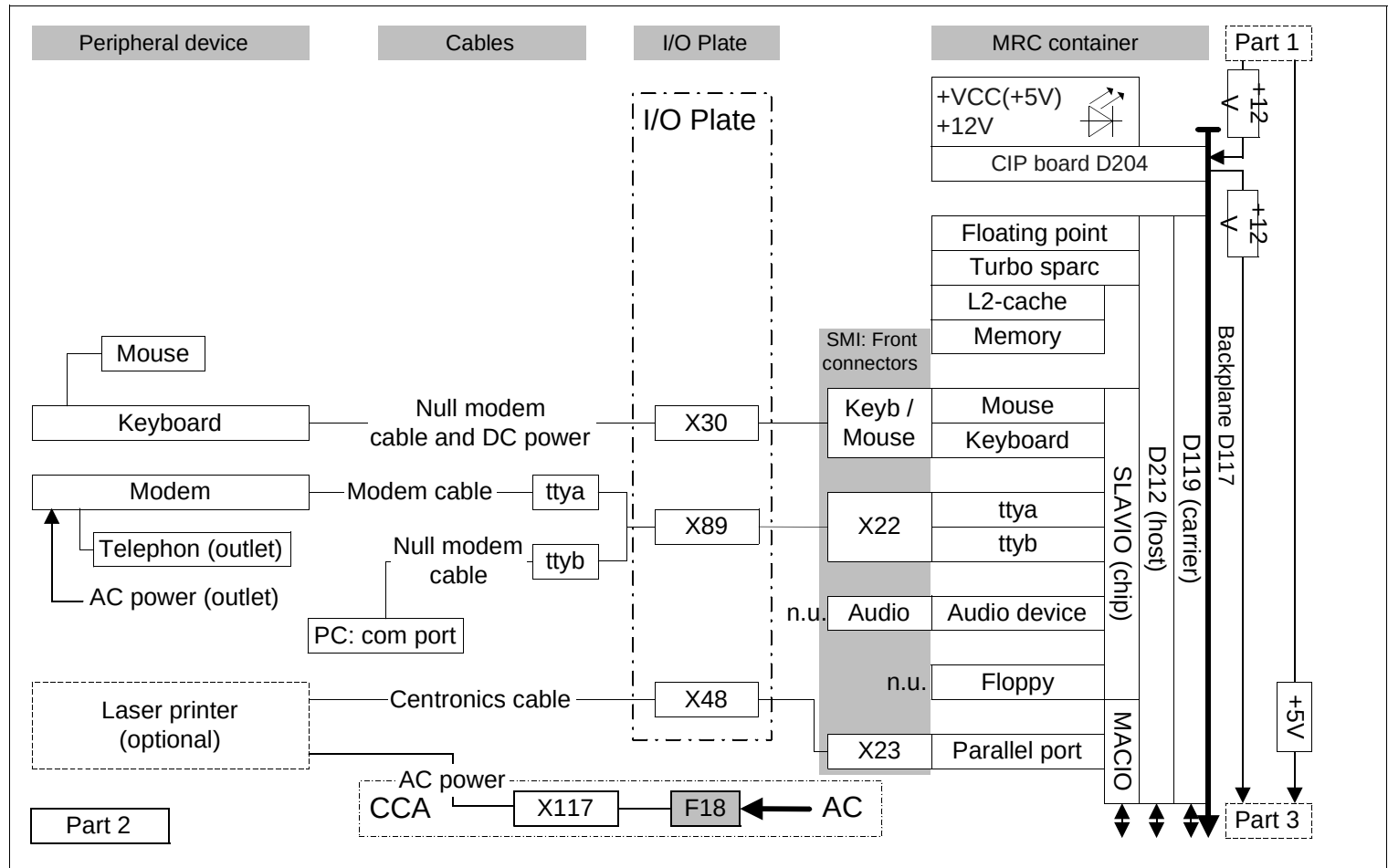


Fig. 6 Part 3

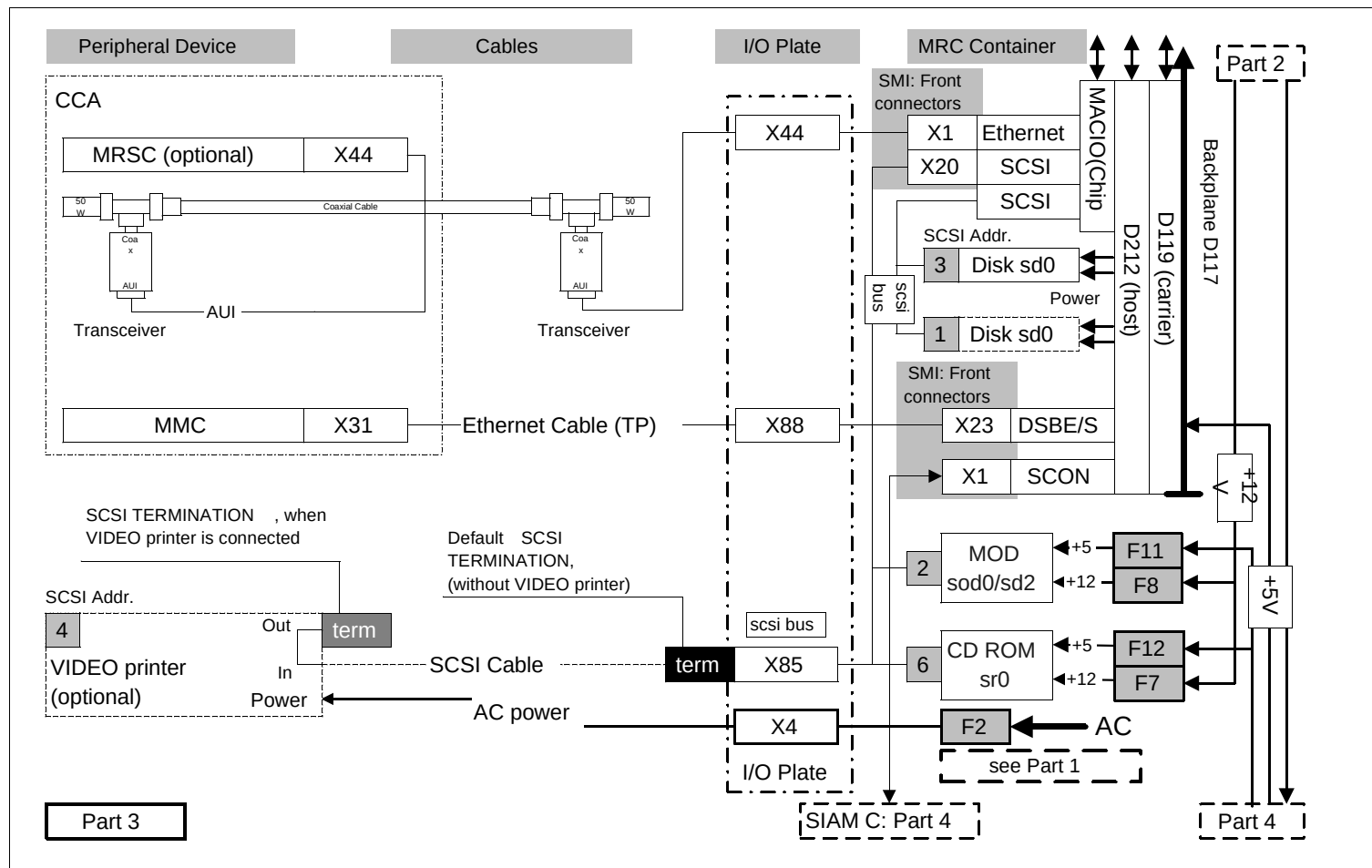
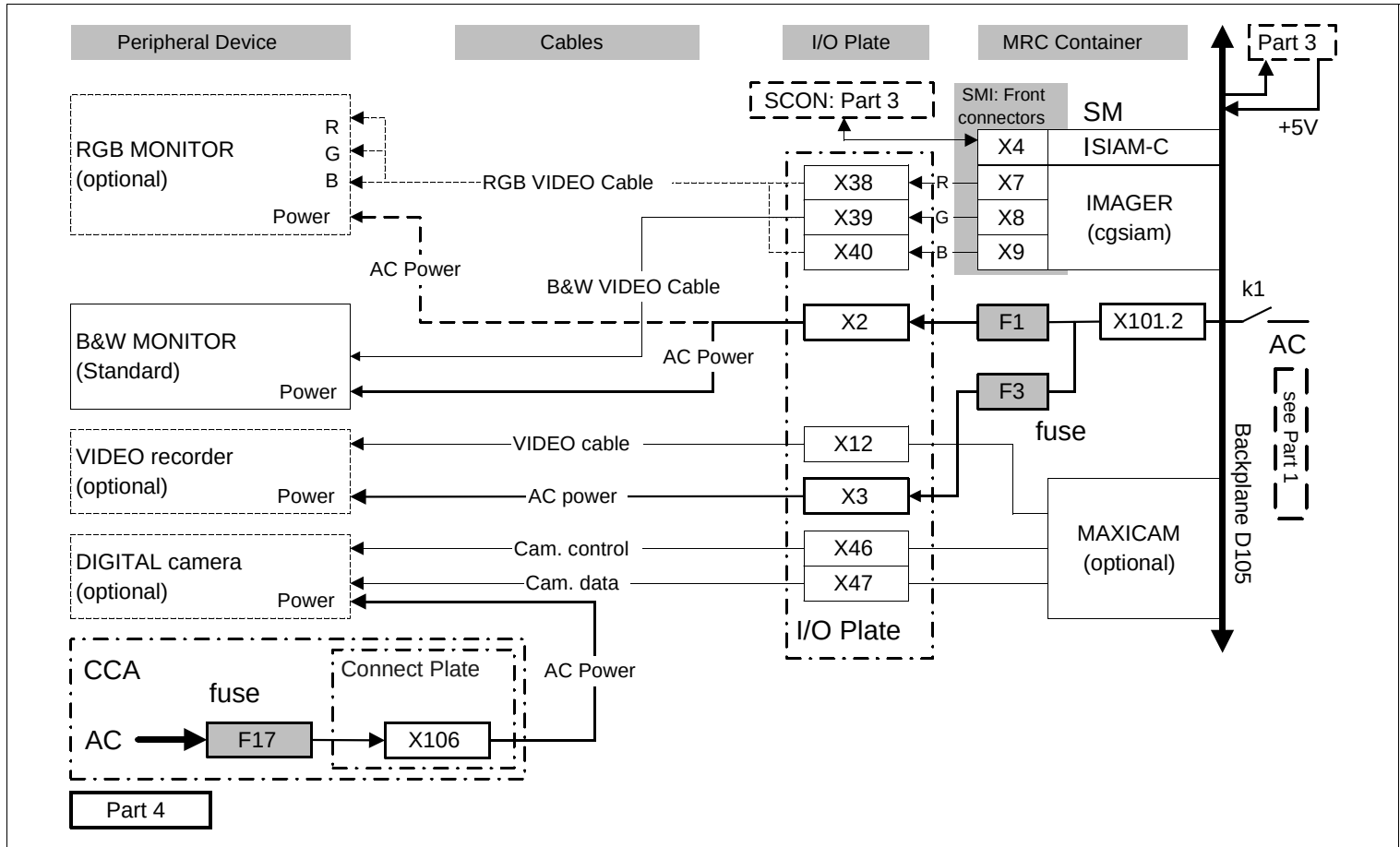
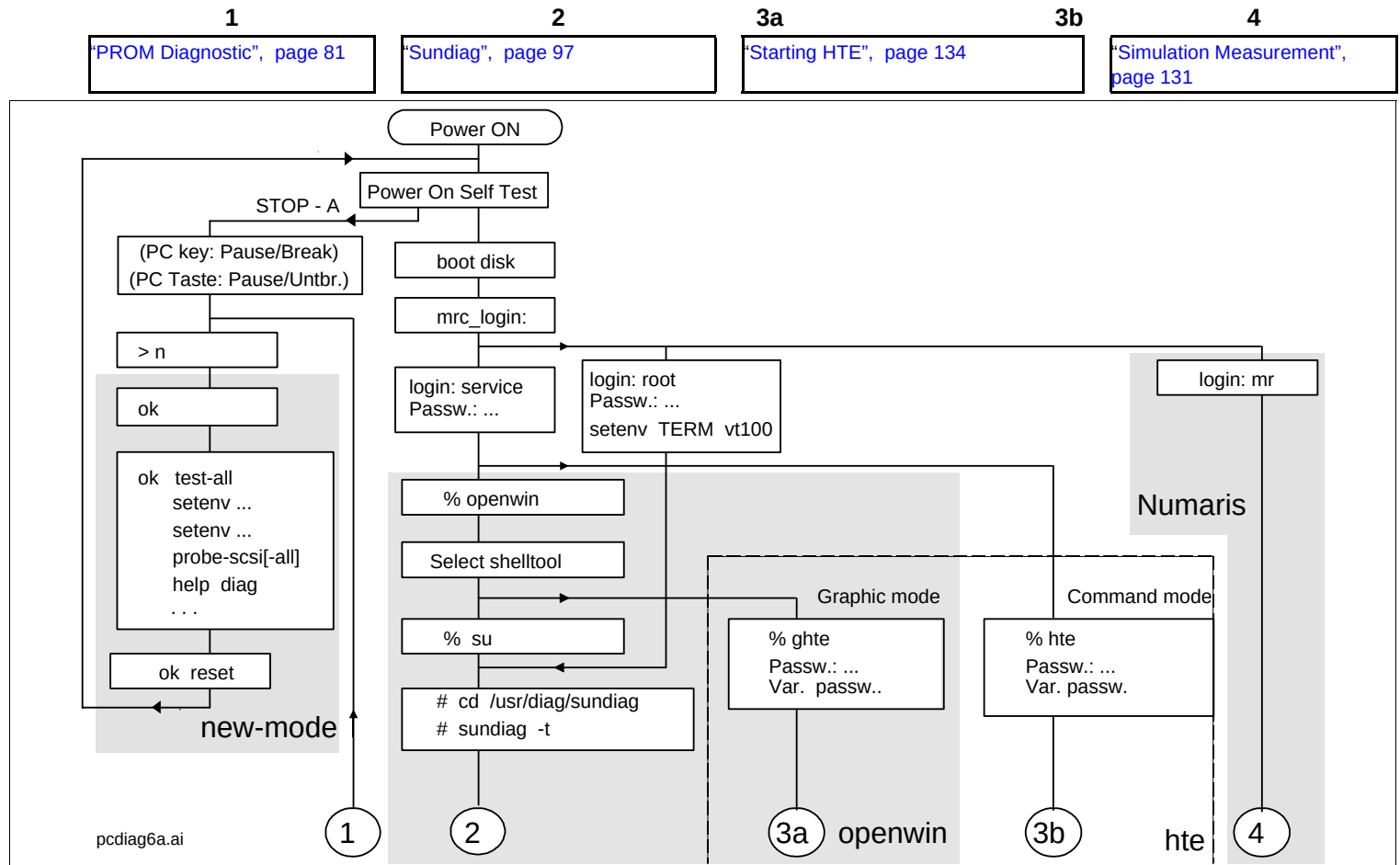
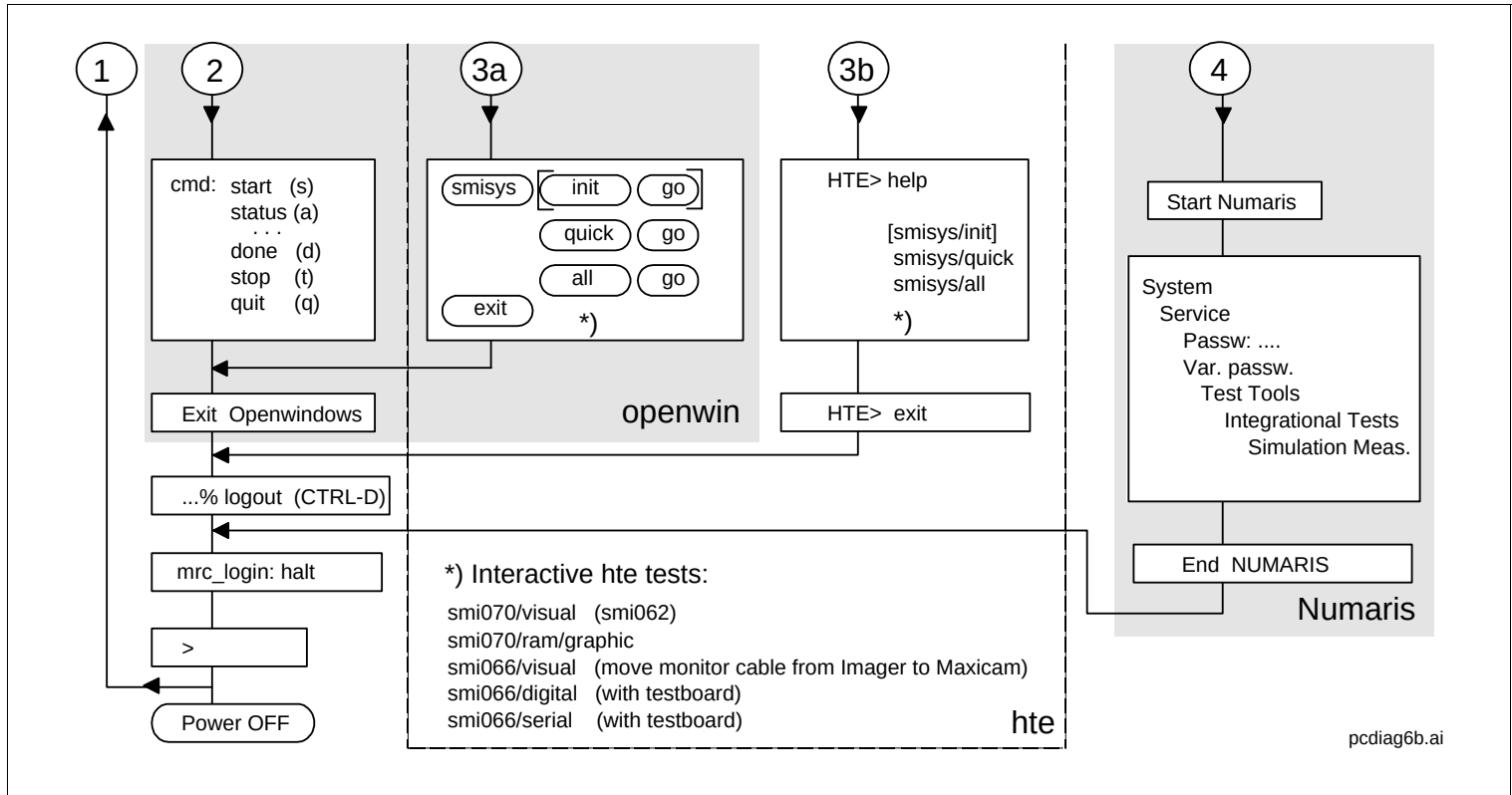


Fig. 7 Part 4

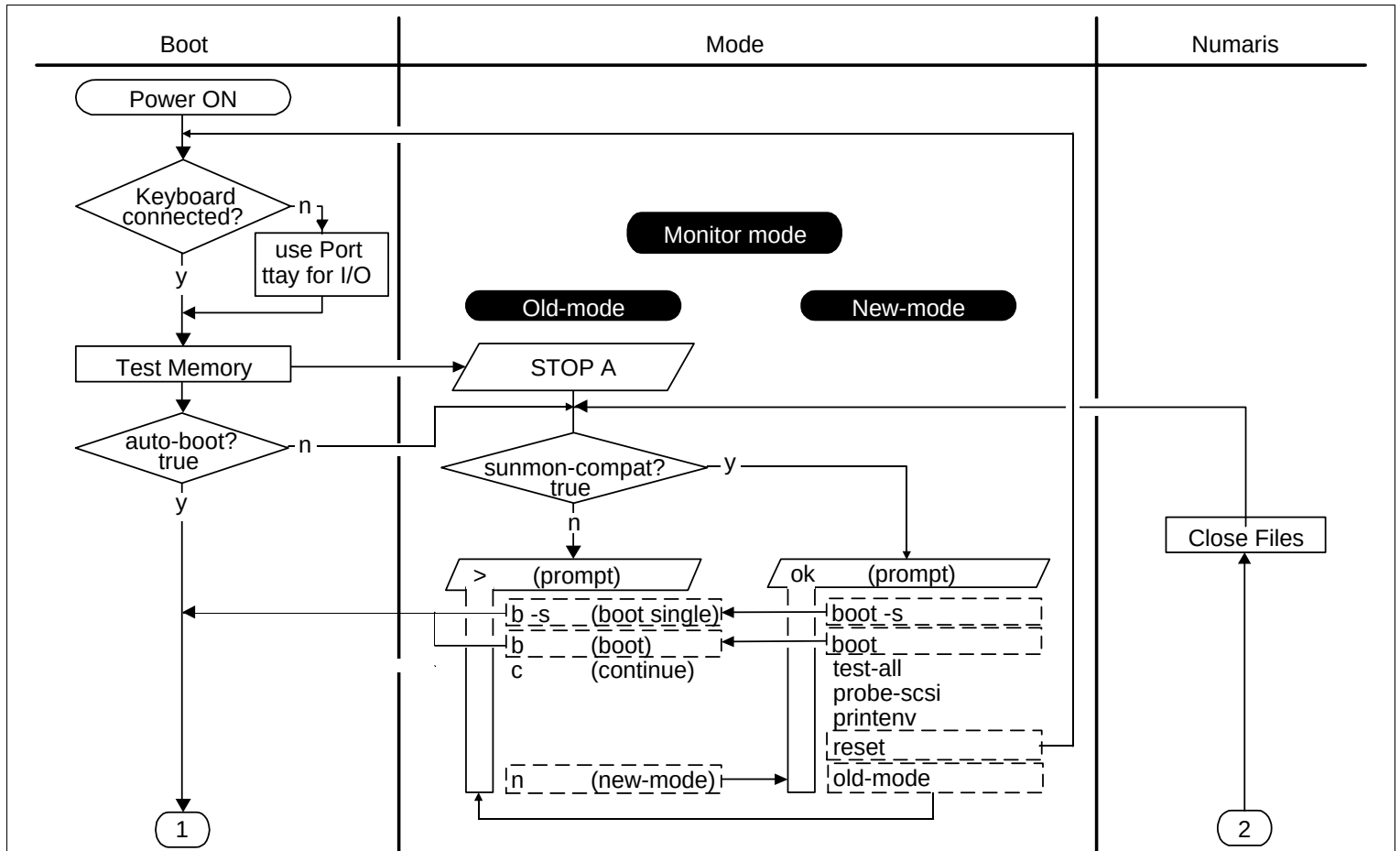


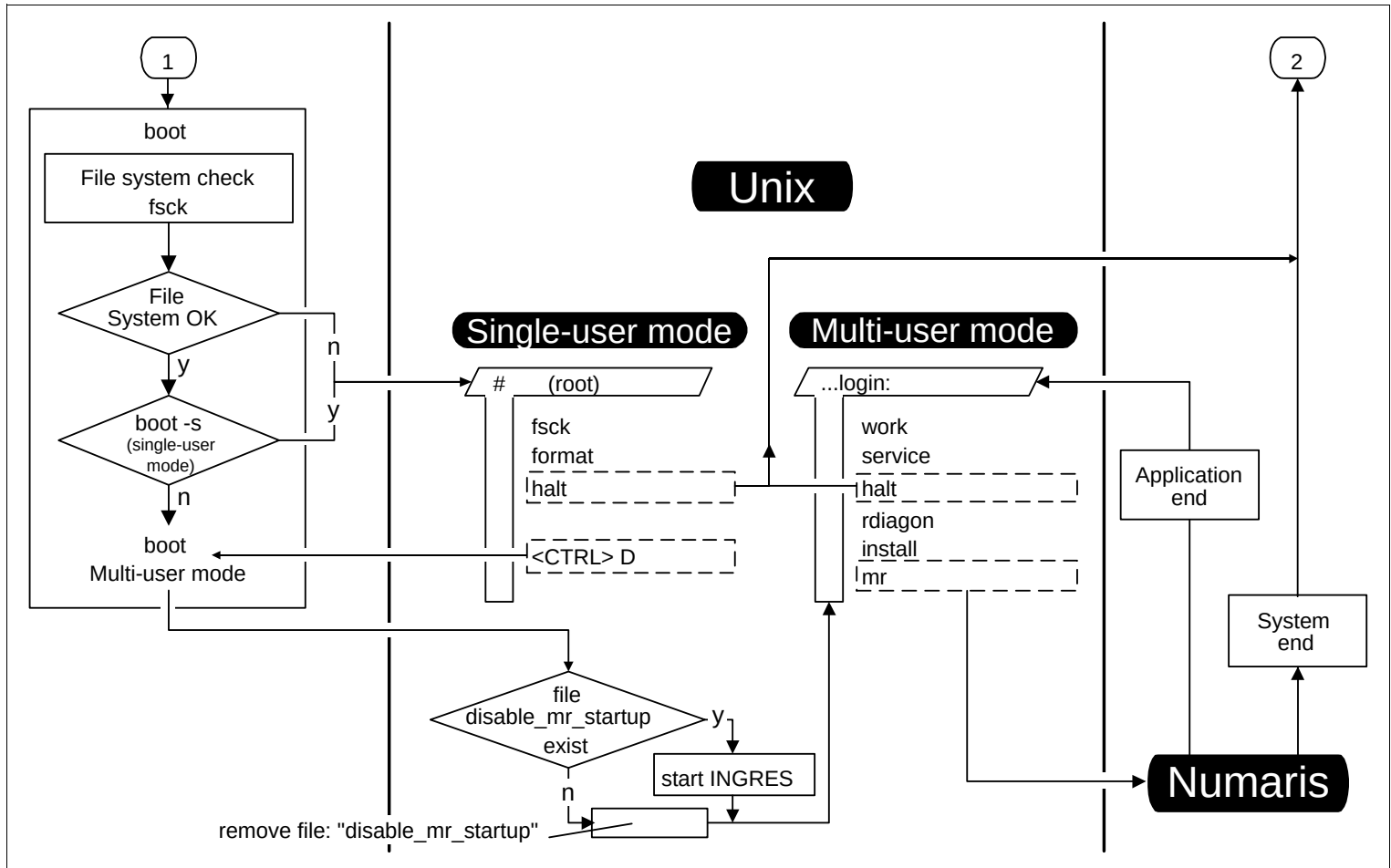
Diagnostics level overview





Boot flowchart





See also "[Boot procedure: Overview](#)", page 109

Explanation of the "Main flowchart"

The tables at the next pages are flow charts.

- The table below shows, how to use this flow charts.

START

Follow the table, line by line, beginning at **START**

CONCERNING PART: Check "CONCERNING PART".

- Preparing step(s): What has to be done before the next step or after the test.
- ...

<i>y(n)</i>	<i>n(y)</i>	TEST STEP SUCCESSFUL? (branch <i>Y/N</i>): Branch □□□ or □□□□□ depending on the answer.	Notes (concerning test step)
	PROBABLE CAUSE Check up □□□□□□□□□□□□□□" by following the rows.	• SERVICE ACTION A bullet says: □□□□□□□□□□□□□□.	⇒ Repair/Replace PARTS or follow REFERENCE ...this is the meaning of an arrow.
	...	...	...

(GOTO next **CONCERNING PART**)

Host troubleshooting flowchart

See also schematic overview at [Fig. 3](#), page 60 ff.

START

POWER

- Open MRC double top cover
 - Pull first cover upward (catch locked).
 - Turn the three clamping bolts (90 deg) at front of 2nd cover and lift it off.
- Switch ON the system and look at power switch in back of the MR(S)C

<i>y</i>	<i>n</i>	Is the green indicator light ON?	The green indicator light is an indicator that the MRC AC Power is OK.	
		Main AC POWER missing	⇔ Check main fuse and power	• See diagrams
		Circuit breaker is OFF	⇔ Check Breakers and power	• See Fig. 4 , page 61
		POWER cable is not connected or bad	⇔ Check cables and power	• Replace defective parts
		CONNECTOR terminal(s) bad	⇔ Check connectors and power	• Replace defective parts
		Room temperature too high?	⇔ Check air condition etc.	

KEYBOARD

- System is ON
- Hit CAPS LOCK then SHIFT several times

<i>y</i>	<i>n</i>	Does the CAPS LOCK LED toggle?	CAPS LOCK LED toggles as an indicator that the KEYBOARD Power is OK.
----------	----------	--------------------------------	--

KEYBOARD defective (spill, damage, break, drop etc.?)	Visually check KEYBOARD (how does it look).	⇒ Replace KEYBOARD if defective
KEYBOARD Power Supply defective?	Check +5V Power in MR(S)C. If Power Supply is defective, the	INTERCOM, host/SMI and drives are also not working.

- Switch off the MRC(S)C

KEYBOARD connectors or KEYBOARD cables defective? Check connectors and cables.	- Pull and plug in again KEYBOARD connectors: "KEYB. /MOUSE" conn. at SUN - MRC I/O Panel connector X30 - KEYBOARD: Cable connector	Check cable if damaged, twisted or bent. ⇒ Replace connectors along with cables if necessary.
--	--	---

MONITOR

- Switch on and boot the system

y	n	Are there normal boot Messages on the Screen after a while?	If the KEYBOARD is defective or not connected, Input/Output is forced to ttya
		(MONITOR is completely dark) MONITOR LINE Power is missing VIDEO Signal is missing MONITOR is defective SMI Imager is defective With a bad MONITOR, a normal boot can be performed via PC Terminal, while Input/Output is switched to ttya.	- Check Circuit Breaker, Power cable, -connectors - Check Video cable, -connectors - Check Monitor with an external Video Source. ⇒ Replace cable ⇒ Replace Monitor ⇒ GOTO SMI test " Blank screen test ", page 124

HOST interface "cgsiam" not visible The connection to the SMI is done via SIAM board D21 (cgsiam). The SIAM is connected to s-bus slot 3 and works together with the SMI as a color graphic system interface and memory (cgsiam). For details (boot log) see also "SMI-5 Imager: SIAM-C is connected via the SCON board to sbus slot 2", page 112	Watch SMI start-up log.messages during boot: "Assuming Imager Type D70 SMED, cgsiam0 at SBus slot 3 0x20000 pri 9 ... simema0 at SBus slot 3 0x10000 ... sidedr0 at SBus slot 3 0x10000 pri 9 ... iopuls0 at SBus slot 3 0x1080c0 ... iointctl10 at SBus slot 3 0x1080d4 pri 9 ... scon0 at SBus slot 3 0x7f80 pri 13"	In the above messages, the SMI was accessible via SIAM. If the Sparc cannot access the SMI, the actual, listed lines are just missing. No special error message is printed out here, but later on there will be problems ! In case of problems ⇔ exchange SIAM or SPARC
KEYBOARD connected, but defective The KEYBOARD has to be connected, to detect its defect. A defective KEYBOARD can force Input/Output to ttya (like no keyboard is connected).	The unusual Message appears at the MONITOR: "Can't open input device Keyboard not present..." - Connect PC Terminal to HOST serial Port ttya - Switch OFF system. - Wait 10sec.	- Switch ON system - Watch PC-Terminal Display: If the boot Messages, displayed at the PC Terminal, are fine, there is a serious KEYBOARD problem. - Trace the Fault in the KEYBOARD Section above.
KEYBOARD is OK, but Connection is bad. Bad cable or connectors have the same effect and force Input/Output also to ttya.		

HOST

- Shutdown system or
- Switch on and HALT with the system "**STOP A**" (Break), (see "["STOP A BREAK \(at PC\)"](#)", TSG, page 79)

y	n	Is BOOT Prompt ">" or "ok" visible?	
		POWER Problem?	⇔ GOTO POWER
		KEYBOARD Problem?	⇔ GOTO KEYBOARD
		MONITOR Problem?	⇔ GOTO MONITOR

NVRAM

- When you turn on the system and your output looks something like ...

...

```
ID PROM invalid.  
Testing 0 Megabytes of Memory ... Completed.  
ERROR: missing or invalid ID prom  
Requesting Internet address for 0:0:0:0:0:0
```

□□

```
Ethernet address ff:ff:ff:ff:ff:ff, Host ID ffffffff  
Invalid format type in NVRAM  
The IDPROM contents are invalid
```

□□

The clock on the system has stopped working. When you turn off the machine it loses track of the time and/or complains about invalid TOD when you turn it on.

□□

You get warnings about by NVRAM battery being low when you boot but otherwise it works nicely.

□□

When your system boots you get warnings about the IDPROM checksum being incorrect and/or Invalid format type in NVRAM.'

□□

The hostid and/or ethernet address on your system have been corrupted.

...

- ... order and replace the NVRAM.

NOTICE

To be able to run the system without a new NVRAM (for a short period of time only) see the following procedure.

- Shutdown (end) the system.
- At ">" Prompt enter "n" to switch to new-mode ("ok" prompt)
- Type **printenv** to display the "**NVRAM parameter table**", page 89

<i>y</i>	<i>n</i>	PARAMETERS ok?	BOOT procedure is controlled by Parameters. See " NVRAM parameters, that affect boot ", TSG, page 92
		NVRAM battery is empty Order and replace the NVRAM The NVRAM Battery can not be replaced, because it is sealed.	⇒ GOTO " Setting the MR specific parameter values ", page 89
		PARAMETER changed by FAULT or was changed to wrong value. Check parameter values refer to "NVRAM parameter", page 89 and following parameter table.	⇒ GOTO " MR specific NVRAM parameter ", page 91
		Something is still wrong with the PARAMETER setting - If manual correction didn't help: - Order and replace the NVRAM	

Boot problems caused by NVRAM

<i>n</i>	<i>y</i>	Problems during boot?	BOOT procedure is controlled by Parameters. See " NVRAM parameters, that affect boot ", TSG, page 92
		Severe problems during boot To set the NVRAM parameters to their default values can solve the problem. MR specific values have to be entered manually if Unix is not running.	⇒ " set-defaults ", TSG, page 89 Numaris will work with "default" settings.
		Unix is running	⇒ GOTO " Setting the MR specific parameter values ", page 89

DRIVES

- Drive power must be OK (check if you have power problems or if you are not sure) refer to Replacement of Parts, Host, "SCSI Target address".
- Halt system. At ">" Prompt enter "n" to switch to new-mode ("ok" prompt)
- First check termination refer to "[SCSI-Termination](#)", page 86 (Per default no terminator on drives). Terminate on X85 via plug.
- For trouble shooting use one drive after the other and terminate the drive, if you need information about the necessary termination refer to:
Replacement of Parts, Host, "Replacing the Hard Disks",
Replacement of Parts, Host, "Replacing the MOD", or
Replacement of Parts, Host, "Replacing the CD-ROM (MRC only)"

NOTICE Don't forget to disable/remove termination after testing.

- Then check whether it is visible for the SCSI host adapter or not via the monitor mode command ok "[probe-scsi](#)", TSG, page 85.

y	n	System disk (target 3) is visible.	SCSI Bus layout and SCSI-Termination see page 86
		• For trouble shooting use one drive after the other. Then check whether it is visible for the SCSI host adapter or not via the monitor mode command ok "probe-scsi", TSG, page 85.	⇒ For testing disks via SunDiag, see " Test of SCSI Fixed Disks ", page 100
y	n	Optional data disk (target 1) visible?	SCSI Bus layout and SCSI-Termination see page 86
		• For trouble shooting use one drive after the other. Then check whether it is visible for the SCSI host adapter or not via the monitor mode command ok "probe-scsi", TSG, page 85.	⇒ For testing disks via SunDiag, see " Test of SCSI Fixed Disks ", page 100
y	n	Is the CD-ROM (target 6) visible?	SCSI Bus layout and SCSI-Termination see page 86

- For trouble shooting use one drive after the other. Then check whether it is visible for the SCSI host adapter or not via the monitor mode command ok "[probe-scsi](#)", TSG, page 85.

y	n	Is the MOD (target 2) visible?	SCSI Bus layout and SCSI-Termination see page 86
		• For trouble shooting use one drive after the other. Then check whether it is visible for the SCSI host adapter or not via the monitor mode command ok "probe-scsi", TSG, page 85.	

MOUSE

y	n	Was MOUSE Cursor always OK?	If MOUSE did work fine till now, MOUSE seems to be OK (branch Y)
		MOUSE connectors and Interface cable • Pull and plug in again MOUSE and KEYBOARD connectors - KEYB. /MOUSE Connect at SUN - MRC.X30	- Keyboard: Cable connector ⇒ Replace cables along with connectors if necessary.
		MOUSE or MOUSE cable defective? • Check cable if damaged, twisted or bent? • Check if electrical or mechanical-	⇒ Replace cable ⇒ Replace MOUSE

CONCLUSION

- If the problem still remains, it's not covered by this flowchart.
- CHECK operational faults
- Perform regular troubleshooting
- Call experienced colleague or Hotline.

END

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Tools and auxiliary material

- AUI Ethernet loopback connector p/n 94 52 384
- TP Ethernet loopback connector (local vendor)

Safety notes

CAUTION	Follow the C-MOS guidelines when handling components.
----------------	---

CAUTION	Perform a protective conductor measurement if you removed ground-protected parts such as covers etc. Protective conductor resistance ≤ 200mOhm.
----------------	--

Power On Self Test (POST)

What does POST test?

The POST test includes a complete memory test. See also “[Diagnostics level overview](#)”, page 65.

- ⇒ POST is executed from the boot PROM.
- ⇒ It runs after turning on the power (power cycle) or after simulating the power cycle, triggered by the reset command.
- ⇒ POST checks the integrity of the CPU, FPU and memory. It is basically a go/no-go test and checks about 85% of the sparc board logic.
- ⇒ If the test is successful, no messages are displayed.
- ⇒ Booting follows (if "auto-boot? true").

Monitor prompt (only) ">" / "ok" is up

The system doesn't boot. What is working?

- ⇒ CPU basic functions
- ⇒ Floating Point Unit (FPU) Basic functions
- ⇒ The tested memory area is OK

Start POST automatically

- ⇒ By default, POST tests 1 MB of memory.

NOTICE

To perform a complete memory Test*) follow the Instruction: “[Complete memory test \(manually\)](#)”, page 78.

- ⇒ After switch on, POST runs automatically.

Start of “POST”

```
1 Imagers available
32 MB Mailbox Memory available          (Memory on SIAM)
Harmony VA11A 10999
Initializing Memory |/-\|/-\|/-\|/-\...
```

End of “POST”

- If the NVRAM parameter is "auto-boot **true**", boot starts automatically.

Start of boot *)

```
Boot device: /iommu/sbus/espdma@f,400000 esp@f,800000/sd@3,0
File and args:
root on /iommu@f,e0000000/sbus@f,e0001000/espdma@f,400000
esp@f,800000/sd@3,0:a fstype 4.2
Boot: vmunix
Size: 1654784+515768+206912 bytes
```

- *) For a detailed description of the boot process, see “[Boot procedure: Overview](#)”, page 109.

Complete memory test (manually)

- ok "reset" will start POST manually
- or
- Switch on the system.
 - Wait until the monitor displays the first messages.:

```
1 Imagers available
32 MB Mailbox Memory available
...
```

NOTICE

- Mailbox memory is not system RAM.
- Mailbox memory resides on SIAM.
- System RAM is located on processor board, see Replacement of Parts, Host.

- Type “**STOP A BREAK (at PC)**” to interrupt system boot.

“ STOP A BREAK (at PC) ”	Halt System *)
> n	go to new-mode**)
ok diag-switch? = true	Set Diag Mode for extended tests
ok ok	Simulate Power Cycle

*) STOP ... escapes, see “**STOP A BREAK (at PC)**”, page 80.

) “Help (new-mode)**”, page 93.

- When the memory test is finished, boot messages will appear.

```
1 Imagers available
32 MB Mailbox Memory available

Harmony VA11A 10999      (e.g.: oem-banner)
Boot device: /iommu/sbus/...(default device: disk)
...
```

- Type “**STOP A BREAK (at PC)**” to interrupt system boot.

“ STOP A BREAK (at PC) ” ok	HALT System *)
ok diag-switch? = false	Unset Diag Mode

*) STOP ... escapes, see: “**STOP A BREAK (at PC)**”, page 80.

Watchdog Reset

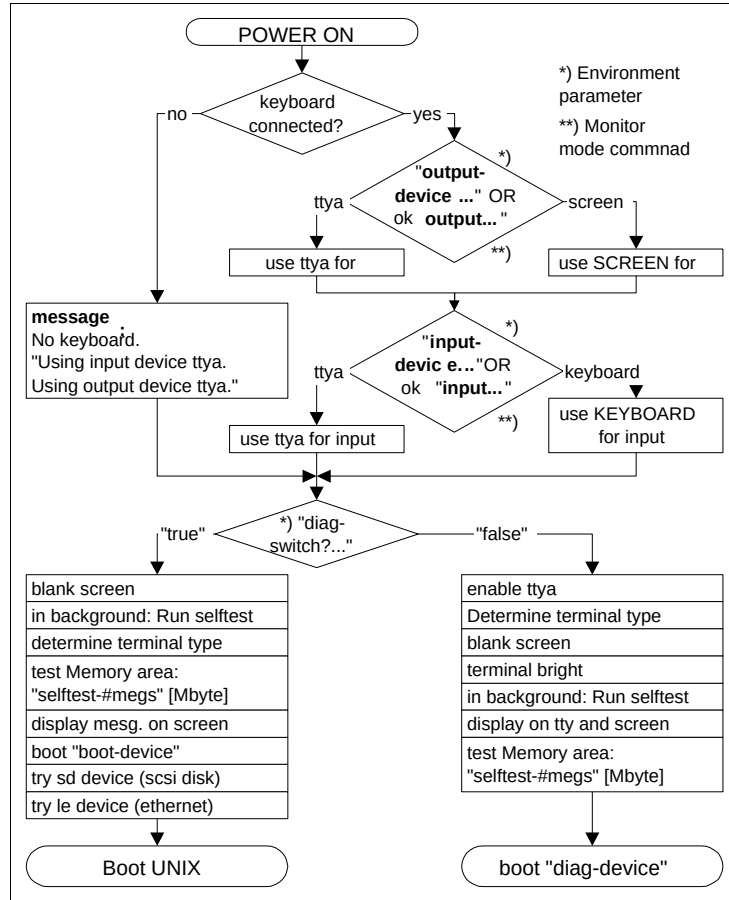
This fatal error usually indicates some kind of hardware problem. Data corruption on the system is possible. Look for some other message that might help diagnose the problem. A watchdog reset doesn't provide enough information in itself because **traps** are disabled and all information has been lost.

Watchdog Reset, Rebooting

See the message “**Start POST automatically**” for details. This rebooting message occurs under the same conditions, but when the EEPROM's **watchdog-reboot?** variable is set to **true**, causing the machine to automatically reboot itself. Data corruption on the system is possible.

Power cycle

Fig. 8 Power cycle and diag mode



STOP (A, N ...) escape hatches

POST reacts to certain "escape hatches"

STOP A BREAK (at PC)	The normal procedure used to stop a sparc system, after all other procedures to correctly stop the system (such as login: halt or <system shutdown>) have failed.
STOP F	Force Input and Output to ttya port *) Connect terminal to ttya and display POST output messages. *) Or select " ttya: Temporary used ", page 82
STOP N	Force NVRAM settings to be changed to their original (factory) default settings. Used only, when boot is not possible. The MR-specific Environment Variables must be set manually. To set the Environment, see " MR specific NVRAM parameter ", page 91
STOP D	Full memory test Force "diag-switch?" setting to be set to "true" and all memory is tested which is defined by default "selftest-#megs" (in MB).

A power cycle is initiated by switching the system on or activating "ok reset".

PROM Diagnostic

Monitor mode "ok" - prompt: Extended commands

Monitor mode / new mode

CAUTION Do not interrupt the computer if shutdown proceeds correctly. When any access to the system fails, interrupt the system with "STOP A BREAK (at PC)" on the MR(S)C - on the laptop use "STOP A BREAK (at PC)".

If the system is running:	
Shutdown the NUMARIS system: <Main menu> <System> <Shutdown>	
or	
halt the system by typing	
login: halt	Login halt account

- Select **new - mode** (ok prompt).

Type b(boot), c(continue), or n(new command mode)
> n
ok

"new-mode" options

- **boot** from an alternate device, like cdrom (see command "**de-valias**", page 88" to display available devices).
- Change **CONSOLE input / output to ttya**, see [page 82](#)).
- Display **system information**, see commands from [page 87](#)).
- **Diagnostics**, see [page 83](#).

"old-mode" options

- **b (boot)** from an alternate device (disk, cdrom, tape, net).
- **c (continue)** - after "**STOP A BREAK (at PC)**" continues as if nothing happened.
- **n (new command mode)** goes to new-mode.

"new-mode" <-> "old-mode"

Toggle between new-mode and old-mode
Enter new-mode > n ok Return to old-mode ok old-mode0 Type b (boot),c(continue), or n (new command mode) >

Perform the first tests

ok " test-all "	(Quick Test of devices)	page 83
---------------------------------	---------------------------	-------------------------

NOTICE

- Use the following with care.
- Do not set NVRAM parameters to default settings, if the system boot is OK.

NVRAM: low battery

See "[NVRAM](#)", page 73.

Numaris will work with "default" settings.

ok " set-defaults "	(Force NVRAM default settings)	page 89
or		page 80
type " STOP N "	during Power cycle	

ttya: Temporary used

Temporary tty access is controlled by commands.

Switch on the system Disconnect the Keyboard from sparc
After the CONSOLE Message:
No Keyboard. Using Input device ttya. Using output device ttya.
ttya is used for Communication.

OR

Monitor-Mode (ok-prompt), type:

ok ttya output0	temporary output to ttya
ok ttya input0	temporary input to ttya
ttya is used for Communication.	

OR

Monitor-Mode (ok-prompt), type:	
ok ttya io0	temporary input/output to ttya
ttya is used for Communication.	

ttya: Permanent use

Permanent tty access is controlled by Environment Variables

The System is in Monitor-Mode (ok-prompt). Type:	
ok setenv output-device ttya0 ok setenv input-device ttya0	output to ttya input to ttya or pull keyboard connector instead
ok reset	Simulate Power Cycle
ttya is used for Communication.	

PROM diagnostic commands

Commands	Notes	FRU
<u>test-all</u>	Quick Test of devices (see: □□□□□□□□")	
ok setenv diag-switch? true diag-switch?= true test-all ok	Diag Mode for extended test Start Test	MEM
Testing /memory@0,0 Testing /obio/SUNW,fdtwo@0,700000 Testing floppy disk system. A formatted disk should be in the drive. Recalibrate failed. The floppy drive is either missing, improperly connected, or defective. Selftest failed. Return code = -1Testing /obio/zs@0,0 Keyboard Present (Testing /obio/zs@0,0 No Keyboard Selftest failed. Return code = -1) Testing /obio/zs@0,100000 !"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNPQRSTUVWXYZ[\]^_`abcde ghijklmnopqrstuvwxyz{ }~ Testing /iommu@f,e0000000/sbus@f,e0001000/SMED,cgsiam@2,20000 Testing /iommu@f,e0000000/sbus@f,e0001000/ledma@f,400010/le@f,c00000 Using AUI Ethernet Interface Internal loopback test -- succeeded. External loopback test -- succeeded. Using TP Ethernet Interface Internal loopback test -- succeeded. External loopback test -- Lost Carrier (transceiver cable problem?) send failed. Testing /iommu@f,e0000000/sbus@f,e0001000/espdma@f,400000/esp@f,800000	Floppy test must fail because no floppy is installed Test of working keyboard is OK (If keyboard is not working, not connected or cable is defective) Test Character generator test interface to the IMAGER SIAM-C Test AUI Ethernet Adapter Test the TP Ethernet Adapter Test Sbus Interface	Keyboard Keyboard SIAM-C CPU CPU CPU
ok setenv diag-switch? false diag-switch?= false		

Commands	Notes	FRU
<u>test net</u>	Test net access to Ethernet.	
ok test net Using AUI Ethernet Interface Internal loopback test -- succeeded. External loopback test -- succeeded.*) *) a loopback connector (p/n 94 52 384) or a connected network is required.	Test succeeds This test consists of two parts. The first part checks the path to the network-controller of the host board. The second part checks the path to the connector. To complete the second part of the test successfully,	HOST NET CABLE
ok test net Using AUI Ethernet Interface Internal loopback test -- succeeded. *) External loopback test -- Lost Carrier (transceiver cable problem? send failed.	Test fails *)Always successful if the internal network controller is working.	HOST NET CABLE
<u>test net: On MRC and MRSC</u>	"TP Ethernet Interface" between MRC - MRSC	
Preparing the MRC:		HOST NET CABLE
ok tpe-link-test? true tpe-link-test?= true Preparing the MRSC:		
ok tpe-link-test? true tpe-link-test?= true Performing on the MRC or MRSC:		
ok test net Using AUI Ethernet Interface Internal loopback test -- succeeded. External loopback test -- succeeded.*)		
	*) If AUI cable is connected	

Testing SCSI bus

The SCSI bus type, used in MR(S)C:

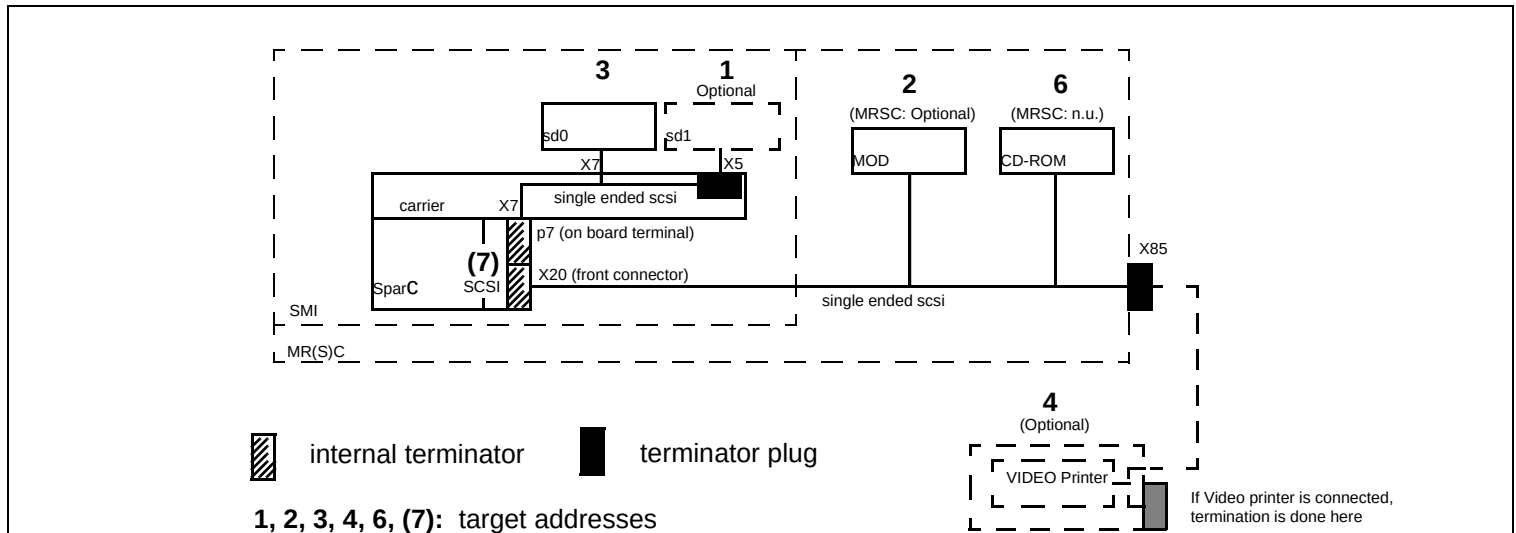
- Single-ended SCSI bus

Commands	Notes	FRU
<u>probe-scsi</u>		
<pre>ok probe-scsi Target 1 Unit 0 Disk SEAGATE ST34371N ... (optional) Target 2 Unit 0 Removable Disk PIONEER DE-C7001 0620 Target 3 Unit 0 Disk SEAGATE ST32171N ... Target 6 Unit 0 Removable Read Only device PIONEER (CD ROM)</pre> ok If one or more devices do not answer, the basic functions of the bus are still operational but you should check the defective device. If it is impossible to identify any device, the SCSI bus and cable itself must be included when tracing the fault.	Displays all visible SCSI devices This test is important for tracing a fault if boot problems occur. The SCSI controller probes the bus with inquiry commands for connected devices. On a faultless bus system (Address and Termination is perfect), the devices answer with special information, permitting identification of devices by the SCSI controller (see the example below). Some Devices change their response on certain Jumper Settings (e.g. PIONEER MOD Mode Jumpers (default: OFF))	HARD-DISK HARD-DISK-JUMPER SCSI-CABLES, -CONNECTORS, -PLUGS POWER-CABLE SCSI-ADAPTER (SPARC)
<u>probe-scsi (if no device is responding)</u>	.. if no device is responding	
<pre>ok probe-scsi ok</pre>		SCSI bus

SCSI-Termination

- SCSI-TERMINATION is done externally (visibly and with easier access), via a terminator plug at the outermost end(s), or at the last drive(s) at the SCSI bus. The SCSI host adapter is terminated internally.
- The hard disk drive must not be terminated, if they are attached to the carrier board (**D119**). The termination for the hard disk drives is done on the carrier board.
- If an additional drive is connected to the SCSI bus, ensure that the terminators are **removed** or termination is disabled via jumpers.

Fig. 9 SCSI Termination



Refer also to "[MR\(S\)C system overview \(schematics\)](#)", page 59.

PROM information commands

<u>test /memory</u>	Test memory area	
<pre> (Preparation) ok setenv selftest-#megs 64 selftest-#megs = 64 ok test /memory (Test Memory /-\\/-\\ ...) (Conclusion) ok setenv selftest-#megs 1 selftest-#megs = 1 ok</pre>	The memory area to be tested is predefined in the NVRAM Variable selftest-#megs. Changes default value to 64 MB. Tests complete memory. Don't forget to restore the previous value (1 MB) after the test to provide faster boot performance. The default tested area of 1MB is a compromise between test time and tested memory used for loading the operating system.	MEMORY

Commands	Notes	FRU
<u>watch-clock</u>	Tests basic function of the timing system.	
<pre> ok watch-clock Watching the 'seconds' register of the real time clock chip. It should be 'ticking' once per second. Type any key to stop. 10 <- (Second counter) <CTRL C> <- (to stop the test)</pre>	Every second a counter increments on the screen.	

Commands	Notes	FRU
<u>printenv</u>	Displays the NVRAM Parameters.	
<pre> ok printenv Parameter Name Value Default Value ...</pre>	Infos about System Configuration see " NVRAM parameter ", page 89.	NVRAM

<u>show-devs</u>		Show devices	
ok	<pre> show-devs... /obio/SUNW,fdtwo@0,700000 (FLOPPY, n.u.) ... /obio/zs@0,0 /obio/zs@0,100000 /iommu@f,e0000000/sbus@f,e0001000 /iommu@f,e0000000/sbus@f,e0001000/SMED,cgsiam@2,20000 ... /iommu@f,e0000000/sbus@f,e0001000/ledma@f,400010 /iommu@f,e0000000/sbus@f,e0001000/espdma@f,400000 ... /iommu@f,e0000000/sbus@f,e0001000/ledma@f,400010/le@f,c00000 /iommu@f,e0000000/sbus@f,e0001000/espdma@f,400000/esp@f,800000 /iommu@f,e0000000/sbus@f,e0001000/espdma@f,400000/esp@f,800000/st (TAPE) /iommu@f,e0000000/sbus@f,e0001000/espdma@f,400000/esp@f,800000/sd (DISK) </pre>	All ".../esp@f, ..." and ".../le@f, ..." devices are bootable devices.	NVRAM

<u>devalias</u>		Show device aliases	
ok	<pre> devalias screen /iommu@f,e0000000/sbus@f,e0001000/SMED,cgsiam@2,20000 floppy /obio/SUNW,fdtwo scsi /iommu/sbus/espdma@f,400000/esp@f,800000 net-aui /iommu/sbus/ledma@f,400010:aui/le@f,c00000 net /iommu/sbus/ledma@f,400010/le@f,c00000 disk /iommu/sbus/espdma@f,400000/esp@f,800000/sd@3,0 cdrom /iommu/sbus/espdma@f,400000/esp@f,800000/sd@6,0:d tape /iommu/sbus/espdma@f,400000/esp@f,800000/st@4,0 ... disk3 /iommu/sbus/espdma@f,400000/esp@f,800000/sd@0,0 disk2 /iommu/sbus/espdma@f,400000/esp@f,800000/sd@2,0 disk1 /iommu/sbus/espdma@f,400000/esp@f,800000/sd@1,0 disk0 /iommu/sbus/espdma@f,400000/esp@f,800000/sd@3,0 ttyb /obio/zs@0,100000:b ttya /obio/zs@0,100000:a keyboard! /obio/zs@0,0:forcemode keyboard /obio/zs@0,0 </pre>	The following devices are bootable: floppy net-aui disk cdrom tape	NVRAM

NVRAM parameter

NVRAM parameter = Environment variables

NOTICE MR-specific values are underlined and bold. All other parameters are set to default values. All parameter names should be checked, if problems occur.

Setting the MR specific parameter values

(Run the script “# /install/admin/numi_eeprom.csh” via UNIX):
For details see Replacement of Parts, Host, “Setting the NVRAM parameter”

If the script doesn't work, use procedure “[MR specific NVRAM parameter](#)”, page 91

NVRAM parameter table

(see next page)

Modify or display NVRAM parameters

Commands		Notes	FRU
<u>setenv</u>		Changes the parameter (variable) to the new value;	
ok	setenv <parameter> <value> Examples: ok setenv diag-switch? false diag-switch? = false ok setenv screen-#columns 171 screen-#columns = 171	As a rule: Logical variables display a (?) following them. Numerical variables use the (#) symbol. Put value in quotes ("), if it contains blanks.	NVRAM

Commands		Notes	FRU
<u>set-defaults</u>		Setting NVRAM parameters to default values.	
ok	set-defaults	This could be very helpful, if NVRAM problems occurred during the boot. (In case the battery is low, replace the NVRAM as soon as possible.)	NVRAM
ok	Setting NVRAM parameters to default values. NOTICE STOP N during Power On Self Test (POST) performs the same. See STOP (A, N ...) escape hatches .		
		NOTICE: To enter MR-specific values, refer to the “ Setting the MR specific parameter values ” from below.	

Parameter Name	Value	Default Value	Note	Example
tpe-link-test?	true	true	If true enables Link Integrity Test on TPE	true / false
etq-testarea	0	0		
bootflash-load-base	4294967295	4294967295		
bootflash-#devices	0	0		
bootflash-#megs	0	0		
wd-ena?	0	0		
level-15-intr-ena?	false	false		
output-device	screen	screen	Power-on output device	screen, ttya or ttyb
input-device	keyboard	keyboard	Power-on input device	keyboard, ttya or ttyb
keyboard-click?	false	false	If true, enable keyboard click	true / false
keymap		<no default>	Location for special key tables	.
ttyb-rts-dtr-off	false	false	If true, SunOS does not assert DTR and RTS on ttyb	true / false
ttyb-ignore-cd	true	true	If true, SunOS ignores carrier-detect on ttyb	true / false
ttya-rts-dtr-off	false	false	If true, SunOS does not assert DTR and RTS on ttya	true / false
ttya-ignore-cd	true	true	If true, SunOS ignores carrier-detect on ttya	true / false
ttyb-mode	9600,8,n,1,-	9600,8,n,1,-	ttyb (baud,#bits,parity,#stop,handshake)	baudrate: 75...19200, #bits: 7/8, parity:
ttya-mode	9600,8,n,1,-	9600,8,n,1,-	ttya (baud,#bits,parity,#stop,handshake)	n/y, stopbits: 1/2, handshake: -/+
fcode-debug?	false	false	If true, include name fields for plug-in device fcodes	true / false
local-mac-address?	false	false	The Ethernet address	true / false
screen-#columns	171	80	Number of on-screen columns (characters/line)	range 1 ...
screen-#rows	45	34	Number of on-screen rows (lines) used	range 1 ...
selftest-#megs	1	1	MB of RAM to test on power-up or on test-memory	range 1 ... 64
scsi-initiator-id	7	7	SCSI bus address of host adapter	range 0 ... 7
auto-boot?	true	true	If true, boot automatically after power up	true / false
watchdog-reboot?	false	false	If true, reboot after watchdog reset	true / false
diag-file		<no default>	Name of the file on your [diag-device] to boot from	
diag-device	disk	net	Diagnostic boot source	disk, net, tape ...
boot-file		<no default>	Name of the file on your [boot-device] to boot from	e.g.: vmunix, ...
boot-device	disk	disk	Boot source (default device is disk)	disk, net, tape ...
sbus-probe-list	f0123	f0123	Which s-bus slots are probed and in what order	range: 01234... (= target 0,1,...)
use-nvramrc?	false	false	Utilized for programming needs; do not set true	true / false
nvramrc		<no default>	Location for commands needed at power-up or reset	
sunmon-compat?	true	false	If true, come up with old-style monitor prompt [>]	true / false
security-mode	none	none	System security level	none, command, full
security-password		<no default>	System security password (never displayed)	
security-#badlogins	0	<no default>	Displays count of outside login attempts	
oem-logo		<no default>	Byte array custom oem logo (use oem-logo? true	
oem-logo?	false	false	If true, use custom oem logo (or use Sun logo)	true / false
oem-banner	<Harmony><SW-V><s/n>	<no default>	Custom oem banner (enabled by oem-banner? true)	"Customer specific screen banner"
oem-banner?	true	false	If true, use custom oem banner	true / false
hardware-revision		<no default>	System version information	
last-hardware-update		<no default>	System update information	
testarea	0	0	One-byte field, available for read/write test	range: 0 ...
mfg-switch?	false	false	If true, perform repeated system self-tests	true / false
diag-switch?	false	false	If true, run diagnostic mode	true / false

Commands		Notes	FRU
<u>MR specific NVRAM parameter</u>			
ok	<pre>setenv diag-device disk diag-device = disk setenv screen-#columns 171 screen-#columns = 171 setenv screen-#rows 45 screen-#rows = 45 setenv use-nvramrc? true use-nvramrc? = true setenv security-#badlogins 0 security-#badlogins = 0 setenv oem-banner "Harmony <SW-Version> <Serial number>" oem-banner = (e.g.) Harmony VA11B 10999 setenv oem-banner? true oem-banner? = true</pre>	NOTICE ----- - The UNIX command <code>"/usr/etc/eeeprom"</code> displays the NVRAM environment variables in a <code>"cmdtool"</code>. - The UNIX command <code>"printenv"</code> display the UNIX environment variables. - Both are variables of different environments. NOTICE: (If the boot procedure is OK) To enter MR-specific values via a script, refer to the "Setting the MR specific parameter values", page 89. Put text in quotes <code>"..."</code> !	
<u>.idprom</u>		Displays the 16 byte IDPROM information (+ 16 byte reserved).	
ok	<pre>.idprom (e.g.) Format/type: 1 80 Ethernet: 8 0 20 0 18 46 Date: 0 0 0 0 Serial: 3a 18 46 Checksum: xx Reserved: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0</pre>	The Ethernet address and system serial # and hostid # are originated and permanently stored in the IDPROM (part of the NVRAM) (as long as the lithium battery is in good condition).	NVRAM
<u>.version</u>		Displays the SUN firmware release.	
ok	<pre>.version (Note leading period "." Example:) Release 2.15 Version 1 created 94/03/03 13:46:20</pre>	Release version of firmware (Boot PROM)	BOOT PROM

<u>NVRAM parameters, that affect boot</u>			Boot is controlled by NVRAM Parameters, see also " NVRAM parameter ", page 89
Parameter	Value	Default	Display NVRAM parameters with command " printenv ", page 87.
mfg-switch?	false	false	If "true", the sparc executes self tests continuously. Host is always testing. No boot possible.
diag-switch?	false	false	If set "true", the sparc tries to boot the "diag-device".
diag-device	disk	net	Boot from this device if "diag-switch" is set "true",
auto-boot?	true	true	If set false, stop at boot prompt ">" or "ok" Automatically boots the system - per default - from the device defined by the parameter: "boot-device"
boot-device	disk	disk net	Boots from this device if "diag-switch" is set "false", The "boot-device" determines where to boot from: here, from the disk. "disk", the default boot device, is a " devalias ", page 88.

Help (new-mode)

Commands	Notes	FRU
<u>help</u>	Display new-mode command categories	
ok help Enter 'help command-name' or 'help category-name' for more help (Use ONLY the first word of a category description) Examples: help select -or- help line Main categories are: File download and boot Resume execution Diag (diagnostic routines) Power on reset -prompt Floppy eject Select I/O devices Ethernet System and boot configuration parameters Line editor Tools (memory, numbers, new commands, loops) Assembly debugging (breakpoints, registers, disassembly, symbolic) Sync (synchronize disk data) Nvramrc (making new commands permanent)	Command categories are groups of commands or a single command (e.g: >-Prompt command)	

<u>help [category]</u>	Displays help for a category																			
ok help [category]	[category] refer to the above																			
<u>help diag</u>	Displays the available tests																			
ok help diag Category: Diag (diagnostic routines) <table> <tr><td>test-net</td><td>(--)</td><td>test network interface</td></tr> <tr><td>test-cache</td><td>(--)</td><td>test cache tag and data fields</td></tr> <tr><td>test-memory</td><td>(--)</td><td>test main memory ; uses self-test-#megs option</td></tr> <tr><td>watch-clock</td><td>(--)</td><td>show increments of real-time clock</td></tr> <tr><td>test-control-regs</td><td>(--)</td><td>test system control registers</td></tr> <tr><td>probe-scsi</td><td>(--)</td><td>show attached SCSI devices</td></tr> </table>	test-net	(--)	test network interface	test-cache	(--)	test cache tag and data fields	test-memory	(--)	test main memory ; uses self-test-#megs option	watch-clock	(--)	show increments of real-time clock	test-control-regs	(--)	test system control registers	probe-scsi	(--)	show attached SCSI devices	i.e. category help information The list displayed is not a complete overview of available tests. For additional test, see below.	
test-net	(--)	test network interface																		
test-cache	(--)	test cache tag and data fields																		
test-memory	(--)	test main memory ; uses self-test-#megs option																		
watch-clock	(--)	show increments of real-time clock																		
test-control-regs	(--)	test system control registers																		
probe-scsi	(--)	show attached SCSI devices																		

UNIX commands

SunOS is up and running

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Commands	Notes
<u>dinfo</u>	Report information about a disk's geometry and partitioning
work% /usr/etc/dinfo <disk> [partition]	Syntax
Man page extract (% man dinfo): "dinfo" gives the total number of cylinders, heads, and sectors or tracks on the specified disk, and gives this	information along with the starting cylinder for the specified partition. If no partition is specified on the command line, dinfo reports on all partitions.

Example:

```
work% dinfo sd0
sd0: SCSI CCS controller at addr 78800000, unit #
241864 cylinders 7 heads 80 sectors/track
a: 67200 sectors (120 cyls)
   starting cylinder 0
b: 100800 sectors (180 cyls)
   starting cylinder 120
c: 1043840 sectors (1864 cyls)
```

```
starting cylinder 0
d: No such device or address
e: No such device or address
f: No such device or address
g: 672000 sectors (1200 cyls)
   starting cylinder 300
h: 203840 sectors (364 cyls)
   starting cylinder 1500
```

Commands	Notes
<u>devinfo</u>	Reports all known devices
work% /usr/etc/devinfo [-pv] [-f promdevice]	Syntax
Man page extract (% man devinfo): "devinfo" displays the devices that the system knows about. The output will state the name of the device, its unit number, and whether a system device driver has claimed it. Since the internal system representation of this information is an n-ary tree, indent (tab) is used to denote a parent-child relationship, and devices reported at the same indent level are considered sibling devices.	OPTIONS -f With the -p option, use the argument promdevice as the monitor PROM device, instead of the default /dev/openprom. -p Report the monitor's view of the attached devices, instead of the kernel's view. -v Report hardware specifications such as register addresses and interrupt priorities for each device.

Example

work% devinfo

Node 'FORCE, ESP8', unit #0 (no driver) SuperSPARC module

Node 'packages', unit #0 (no driver)

Node 'options', unit #0 (no driver)

Node 'aliases', unit #0 (no driver)

Node 'openprom', unit #0 (no driver)

Node 'iommu', unit #0

Node 'sbus', unit #0

S-BUS

Node 'espdma', unit #0

Node 'esp', unit #0

SCSI adapter

Node 'sd', unit #3

System Disk

Node 'sd', unit #0

Store Disk

Node 'SUNW,bpp', unit #0

Node 'power-management', unit #0 (no driver)

Node 'cgsix', unit #0

CGSIAM

Node 'obio', unit #0

IO Interface

Node 'zs', unit #0

Serial:keyb,mous

Node 'zs', unit #1

Serial:ttya,ttyb

Node 'eeprom', unit #0 (no driver)

EEPROM

Node 'ledma', unit #0

Node 'le', unit #0

Ethernet

Node 'SUNW,CS4231', unit #0 (no driver)

Node 'power-management', unit #0 (no driver)

Node 'slaviocfg', unit #0 (no driver)

Node 'auxio', unit #0 (no driver)

Node 'counter', unit #0 (no driver)

Node 'interrupt', unit #0 (no driver)

Node 'power', unit #0 (no driver)

Node 'SUNW,fdtwo', unit #0

Node 'memory', unit #0 (no driver)

Node 'virtual-memory', unit #0 (no driver)

Node 'FMI,MB86904', unit #0 (no driver)

work%

<u>df</u>	Report free disk space on file systems
work% <code>df [-a] [-i] [-t type] [filesystem..] [filename..]</code>	Syntax
Man page extract (man df): df displays the amount of disk space occupied by currently mounted file systems, the amount used and available.	space, and how much of the file system's total capacity has been used. Used without arguments, df reports on all mounted file systems, producing something like:

Example

```
work% df
Filesystem kbytes  used  avail  capacity  Mounted on
/dev/ip0a   7445   4714 1986    70%       /
/dev/ip0g  42277  35291 2758    93%      /usr
```

Note: used and available is less than the amount of space in the file system (kbytes); this is because the system reserves a portion of the space for its file system allocation routines.

```
work% df .      (Note period ".")
Filesystem    kbytes   used   avail capacity  Mounted on
/dev/sd0g     1236830  870677 242470    78%      /usr
```

Shows the amount of space on the file system containing the current directory.

<u>fpurel</u>	Floating Point Co-processor Test
root# <code>fpurel [-v] [-p[count]] [-r]</code>	Syntax
Man page extract (% man fpurel): "fpurel" performs a series of functional and computational tests for the Sun Floating Point Co-processor to verify that it is operational and accurate. With no options, fpurel runs one pass silently in the foreground...	-v Verbose. Display the name and results of each test on the console. The default is to run silently. -p[count] Passcount. Specify the number of times to run the test suite. The default is to run one pass.

Example

```
root# /usr/diag/fpurel -p 10
root#
```

fpurel runs 10 passes silently without errors.

Sundiag

Use a "**shelltool**" in OpenWindows to call Sundiag.
(See "**Start Sundiag**", page 97).

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General description

With Sundiag you can investigate the **FUNCTIONS** and the **INTER-FACES** of the host board and **DEVICES** connected.

An auto-configure-routine detects the hardware components and selects those unites, which can be tested without assistance.

If started, **Sundiag** runs all selected tests in multi-testing-mode in an endless loop (default setting).

Start and Exit of Sundiag

Sundiag assumes a properly working UNIX operating system.

Start Sundiag

- Shut down Numaris ('Application End') and choose 1. or 2. :
 1. **MR console**: Call up **Sundiag** as follows.

Commands	Notes
login: service Password: ... *) % openwin % su Password: ... *) # cd /usr/diag/sundiag # sundiag -t	- Login service account - Enter the password - Start OpenWindows - Switch User to root - Enter the root password - Change directory - Start the sundiag

2. **Laptop** via Serial Port **ttyb** (Refer to "**Testing under Unix**", page 24). Call up **Sundiag** as follows:

Commands	Notes
login: root Password:... *) # setenv TERM vt100-nam # cd /usr/diag/sundiag # sundiag -t	- Login root - Enter the root password - Set terminal mode - Change directory - Start the sundiag

*) If password is unknown, refer to the latest Speedinfo, folder: Password List.

NOTICE

- The menu bar shows **upper** case command words in brackets. Use these words, however, in **lower case** or enter **single letter**.
- In the following, **single letter** (=short commands, e.g. "s") and the corresponding command words, e.g. [start] are described.
- Terminate commands with **<Return>**.
- Some commands have "optional" menus (**next screen**). Switch to **next screen** by typing "n" or [next].
- Enter "d" or [done] to return to the **Main Menu**.
- It is not possible to print a screen dump from **tty** mode. To store logfiles see "**I [logfiles]**", page 106.

Recommended procedures**Start Sundiag tests**

- Enter from **Main menu: s** [start]

Stop Sundiag tests

- Enter from **Main menu: t** [stop]

Exit Sundiag

- Enter from **Main menu: q** [quit]

TEST with Sundiag**Main commands (Main Menu)**

For additional information about "Main commands", refer to [page 103](#).

```

-----
| Sundiag 2.3.2  CPU:ESP8S                               System status: idle |
|-----|
| [START] [OPTIONS] [STATUS] [SCHEDULE] [OPTFILES] [STOP] [QUIT] |
| [RESET] [SUSPEND] [RESUME] [EEPROM] [LOGFILES] [HELP] [NEXT] |
| [PROCESSORS] |
|   [INTERVENTION]: Disable                                [TESTS]: DEFAULT [NONE/ALL] |
|
|   MEMORY DEVICES |
|   ->(mem) Physical                                [OPTION] |
|   ->(kmem) Virtual                                [OPTION] |
|   DISK DEVICES |
|   ->(sd3) SCSI Disk #3                                [OPTION] |
|   ->(sd0) SCSI Disk #0                                [OPTION] |
|   ->(fd0) 0B Floppy Disk #0                            [OPTION] |
|   CPU DEVICES |
|   ->(fpu) FPU |
|   ->(le0) Ethernet                                [OPTION] |
|       (zs0) Serial Ports                            [OPTION] |
|   ->(cgsix0) Color FB |
|-----|
| Command: n |
| Message: |
|-----|

```

Test of physical memory

This test locates parity errors, hard and soft ECC errors, memory read errors, and addressing problems. Through the memory device, it maps, then reads a page repeatedly throughout the memory.

To test the physical memory ONLY, enter the following commands in the order shown into the command line of the main display:

Commands	Notes
tests n	disables all tests
mem	enables the physical memory test
mem o	displays the option menu for this memory test.

```
-----
| Sundiag 2.3.2 CPU:ESP8S                               System status: idle |
|-----|
|      << Test Options - Physical Memory >>                |
|      [DEFAULT(f)] [DONE(d)] [HELP(h)]                  |
|                                                         |
|      Configurations:                                     |
|          Amount:  64 MB                                   |
|                                                         |
|      Sub-tests:  None                                     |
|                                                         |
|      Options:    None                                     |
|                                                         |
|-----|
| Command: mem o                                           |
| Message:                                                 |
|-----|
```

- Only the size of the physical memory is displayed.
- The test of physical memory has no subtests or options.

Test of virtual memory

Virtual memory is the combination of **physical** memory and the **swap** partition of the disk.

The virtual memory test allocates, writes and then reads as much virtual memory as possible. The test also detects parity errors, memory read errors, and addressing problems.

To test the virtual memory ONLY, enter the following commands in the order shown into the Main Display:

Commands	Notes
tests n	disables all tests
kmem	enables the virtual memory test
kmem o	displays the option menu

```
-----
| Sundiag 2.3.2 CPU:ESP8S                               System status: idle |
|-----|
|      << Test Options - Virtual Memory >>                |
|      [DEFAULT(f)] [DONE(d)] [HELP(h)]                  |
|      Configurations:                                     |
|          Amount:  94 MB                                   |
|                                                         |
|      Sub-tests:  None                                     |
|                                                         |
|      Options:                                           |
|          [W]Wait Time:  0                                 |
|          [R]Reserve:    0                                 |
|      ...                                                 |
|-----|
| Command: kmem o                                           |
| Message:                                                 |
|-----|
```

Options

You may control the time interval between **kmem** sessions. One kmem session can last.

Because this test puts a significant burden on the operating system, there is a configurable "**wait time**" option.

Enter **w** to cycle the wait time from **zero** to **30**, **60**, or **90** seconds.

w 60	cycle the wait time 60 seconds.
-------------	--

Start test

s	Start test
----------	------------

Test of SCSI Fixed Disks

To test the SCSI (fixed) hard disks **ONLY**, enter in the main display the following commands in the order given:

Commands	Notes
tests n	disables all tests
sd0	enables the test of the SCSI disk with the logical address 0 (= target address 3)
sd0 o	displays the option menu.

```

-----
| Sundiag 2.3.2 CPU:ESP8S                               System status: idle |
|-----|
|                                     |
| << Test Options - SCSI Disk #0 >> |
| [DEFAULT(f)] [DONE(d)] [HELP(h)] |
|                                     |
| Configurations:                     |
| Capacity: 534.4 MB                  |
| Controller: SCSI CCS                |
| Host Adaptor: esp0                  |
|                                     |
| Sub-tests:                         |
| [R]Rawtest : Disable               |
| [W]Filetest: Disable               |
|                                     |
| Options:                           |
| [O]Rawtest Mode:                    Readonly |
| [P]Rawtest Partition:              c |
| [M]Rawtest Size (MB):              1200 |
|                                     |
|-----|
| Command: sd0 o                     |
| Message:                           |
|-----|

```

- Additional drives are tested analogously.

Configuration

The disk capacity and controller type are displayed.

Subtests

You may choose **[R] Rawtest** and/or **[W] Filetest**.

The **Filetest** exercises the disk controller and the drive.

It writes two .5 MB files with random data in */tmp* directory, then reads and compares them. If the system has no */tmp* directory, Sundiag creates a */tmp-disktop-sd* directory to fill with random data, and deletes it when you exit.

To run this test, the disk must be mounted and contain valid file systems.

The **Rawtest** performs read-only and read/write tests on a disk, using generic SunOS routines.

Options

For the **Rawtest**, you may choose **Read-only** or **Read/Write**. What you choose depends on the problem:

Read problem: Choose **Read-only**

Write problem: Choose **Read/Write**

Sundiag - CPU tests

Test of FPU

This test checks the floating point unit. Refer also to “[fpurel](#)”, page 96. It performs some register tests and series of single and double precision calculations, checking the answer after each computation. There are no options associated with this test.

Test of Serial Ports

Before you choose the test of serial ports, you must enable [INTERVENTION] mode.

To test the Serial Ports ONLY, enter the following commands into the main display in the order shown.

Commands	Notes
tests n	disables all tests

i	enables interventional mode
zs0	enables the tty tests
zs0 o	displays the option menu

Configuration:

The existing serial ports are displayed.

Options:

Enter **L** to select one of the following options:

Commands	Notes
L	Select options
a-b	Both ports are tested with a special loopback cable.
a b	Both ports are tested separately. A loopback connector port (Siemens part number 15 97 561) is needed for the selected port.
a or b	Only one port is tested. A loopback connector (Siemens part number 15 97 561) is needed for the selected port.

Test of Network: le0 (ESP8S) and le1 (DSBE/S board)

The Network test checks the function of the Ethernet hardware on the host board. For this test to be meaningful, the host board being tested must be attached to an Ethernet with at least one other system on the network.

Initially, the Ethernet internal driver error counts are checked. After running the tests, the internal driver error counts are checked again for any differences.

Certain subtests send out messages on a valid network interface and expect to hear a reply from other host devices besides the host itself. Due to the low level of protocol, only one Ethernet test at a time can be executed on a system.

The subtests wait for replies for a given period of time. They then count the replies and check to make sure that the data is the same as originally sent. If no replies come from hosts other than the test host, an error condition is recorded.

To test the network ONLY, enter the following commands in the sequence shown into the main display:

Test le0

Commands	Notes
tests n	disables all tests
le0	enables the network test
le0 o	displays the option menu

Test le1

Commands	Notes
tests n	disables all tests
le1	enables the network test
le1 o	displays the option menu

For testing under Unix the Ethernet interface to the MPCU refer to Replacement of Parts, Software, "Communication to MPCU, MPCU_GW".

Configuration:

The characteristics of the node are displayed.

Options menu

You may choose to enable or disable the spray subtests. This subtest stresses the standard transmit and receive paths by sending out packets continuously.

A delay time is predefined between the sending of each packet. A delay increases heavy activity on the net.

Warning messages are printed on the console only if the driver warning is enabled.

Main Commands (Main Menu)

```
-----
| Sundiag 2.3.2 CPU:ESP8S                               System status: idle |
|-----|
| [START] [OPTIONS] [STATUS] [SCHEDULE] [OPTFILES] [STOP] [QUIT] |
| [RESET] [SUSPEND] [RESUME] [EEPROM] [LOGFILES] [HELP] [NEXT] |
| [PROCESSORS] |
| [INTERVENTION]: Disable           [TESTS]: DEFAULT [NONE/ALL] |
|
| MEMORY DEVICES |
|   ->(mem) Physical           [OPTION] |
|   ->(kmem) Virtual           [OPTION] |
| DISK DEVICES |
|   ->(sd3) SCSI Disk #3       [OPTION] |
|   ->(sd0) SCSI Disk #0       [OPTION] |
|   ->(fd0) OB Floppy Disk #0  [OPTION] |
| CPU DEVICES |
|   ->(fpu) FPU |
|   ->(le0) Ethernet           [OPTION] |
|       (zs0) Serial Ports     [OPTION] |
|   ->(cgsix0) Color FB |
|-----|
| Command: n |
| Message: |
|-----|
```

- Type in: **n** [next]
Main Menu (Next Screen)

```
-----
| Sundiag 2.3.2 CPU:ESP8S                               System status: idle |
|-----|
| [START] [OPTIONS] [STATUS] [SCHEDULE] [OPTFILES] [STOP] [QUIT] |
| [RESET] [SUSPEND] [RESUME] [EEPROM] [LOGFILES] [HELP] [NEXT] |
| [PROCESSORS] |
| [INTERVENTION]: Disable           [TESTS]: DEFAULT [NONE/ALL] |
|
| OTHER DEVICES |
|   (bpp0) Parallel Port           [OPTION] |
|-----|
| Command: n |
| Message: |
|-----|
```

s [start]

After selection of tests and entering system level and test level options, you must enter **<start>** or **<s>** to begin testing. Tests begin immediately. "System status: **testing**".

o [options]

Calls up a menu of **global system level options** that affect every test. When you enter **<default>** or **<f>** in the options menu, the pre-selected default system options are enabled. To exit from this display, enter **<done>** or **<d>**. **Options Menu:** Refer to page 58.

a [status]

Brings up an informative display about the tests in progress. The display shows the SunOS device name, the test name for each device being tested, and a running total of test passes and errors. The tests are grouped by device type, and an actively running test is marked with an asterisk **"*"**.

⇌ Results

The results displayed are stored: See **[logfiles]**.

⇌ System passes

In other words: If the **kmem** test has completed **two** passes while all other tests have completed **thirteen** passes, the status panel will show the lowest number: "**System passes: 2**"

```
Status Menu: "System status:         "
```

```
-----
| Sundiag 2.3.2 CPU:ESP8S                               System status: idle |
|-----|
| [NEXT] [DONE] [HELP]                                     |
| System passes: 2      Total errors: 0      Elapsed time: 000:01:39 |
| [PROCESSORS]                                             |
| MEMORY DEVICE TESTS:                                     |
|      *(mem) pmem      passes: 2      errors: 0          |
|      *(kmem) vmem     passes: 2      errors: 0          |
| DISK DEVICE TESTS:                                       |
| ... |
```

- Type in: **n** [next]

```
Status Menu (Next Screen): "System status: □□□□
□□□"
```

```

-----
| Sundiag 2.3.2 CPU:ESP8S                               System status: testing |
|-----|
| [NEXT] [DONE] [HELP]                                   |
| System passes: 0      Total errors: 0      Elapsed time: 000:02:37 |
| [PROCESSORS]                                           |
|      (fd0) rawtest      passes: 3      errors: 0      |
| CPU DEVICE TESTS:                                       |
|      (fpu) fputest      passes: 13      errors: 0      |
|      *(le0) nettest      passes: 3      errors: 0      |
|      *(cgsix0) cg6test   passes: 2      errors: 0      |
| ... |

```

m [schedule]

Calls up a display of scheduling options. The schedule option menu allows you to specify the start and stop times and dates for Sundiag to activate. The test can run overnight, if necessary. This feature is not important for our usage.

```

Sundia 2.3.2 CPU:ESP8S                               System status: idle
-----
<< SCHEDULE Option Menu >>

[DONE(d)] [CANCEL(c)] [HELP(h)]

[SCHEDULER(s)]:           Disable

[RUNTIME (x)]:             hhh:mm:ss
[STARTDATE(r)]:            mm/dd/yy
[STARTTIME(t)]:            hh:mm:ss
[STOPDATE (o)]:            mm/dd/yy
[STOPTIME (m)]:            hh:mm:ss

-----
Command: m
Message:
-----

```

Examples

Sets the test run time for the next **two** hours.

x 02:00:00	Enter runtime 2 h
------------	-------------------

or

start tests at **04-jul-97 00:00:00** and end test **05-jul-97 06:30:00**

r 07/04/97	Start at 04-jul-97
t 00:00:00	Start time 00:00
o 07/05/97	Stop at 05-jul-97
m 06:30:00	Stop time 06:30

f [optfiles]

You may create **option files** with different names. This calls up a display from which you can load and store the global **system options** chosen with the **options** command.

To create a new **option file**, enter: **n** [filename]

To save the new file, enter: **s** [store]

To restore the **option file** enter: **l** [load]

Sundiag 2.3.2 CPU:ESP8S	System status: idle
<< Option File Menu >>	
[DONE(d)] [HELP(h)]	
[LOAD(l)] [STORE(s)] [REMOVE(r)] [NEXT(n)]	
[NAME(m)]: option-01	
Available option files:	
option-01	
Command: s	
Message:	

t [stop]

Stops all testing. When Sundiag stops the tests in progress, the "System status" is "stopping."

When all tests are stopped, the "System status" changes to "idle." To stop an individual test, simply type in the test name in the command line. If a test is already stopped, enter its name to restart the test. Enter the test name again to stop the test once more.

q [quit]

Exits Sundiag

r [reset]

Resets the passes and error count to zero for each test, and resets „system passes and total errors" display. This option is valid only while tests are stopped (<t>).

p [suspend]

When you enter this command "Systemstatus: suspended" all tests are suspended. **p [suspend]** stops all testing until you enter **u [resume]**.

u [resume]

Restarts testing after the **p [suspend]** command. "system status": testing appears.

e [eeeprom]

Not available in this version.

l [logfiles]

View, print or delete log files stored by Sundiag in the directory /usr/adm/sundiaglog.

A **vi** Editor is used to display the files. You may use **vi** commands such as **CTRL f** or **CTRL b** to page forward and backward through the file. To return to the log file menu, enter **ESC** then **": q !" Ø**

1. Error File: /var/adm/sundiaglog/sundiag.err

m [schedule] e	Select (m) file type error (e) to be viewed.
l [display]	Display the error file in the vi editor.
p [print]	Print error file to the system printer.
r [remove]	Attention: Deletes the error file.

2. Info Log: /var/adm/sundiaglog/sundiag.info

m i	Select file type information to be viewed.
l	display the information file in vi editor.
p	Print information file to the system printer.
r	Attention: Deletes the information file.

3. Unix Log:

m u	Select file type unix to be viewed.
l	Displays the information file in vi editor.
Display the unix log file which contains the date, host name, vmunix, the device type, the time and date of the last log in, etc.	
p	- Prints unix file to the system printer.
r	- Caution: deletes the unix file.

d [done] To return to the previous menu.

h [help] Not available in this version.

n [next]

Displays the next screen of information, if one exists. When you reach the end of the display, enter n again to page back to the beginning of the display. You may then enter a new command, selecting one of the commands in brackets at the top of the screen.

i [intervention]

When you enter intervention or i from the main display, the [INTERVENTION]: text item toggles from „Disable“ to „Enable“ or vice versa.

When you enable the intervention mode, you may choose tests that require special test tools (e.g. loopback connector or scratch tape). If the intervention mode is not enabled and you try to run a test that requires it, a message at the bottom of the screen will tell you to enable the intervention mode.

tests

Enter **tests d** [default] to select the default tests.

The -> symbol appears before each default test.
These tests will run when you enter **s** [start].

Enter **tests n** [none] that no tests are enabled.

All the -> symbols disappear from the test names.

Enter **tests a** [all] with [intervention] disabled, to enable all tests that do not require user intervention.

- Enter **tests a** with [intervention] enabled, to enable all tests.

To enable or disable tests individually, simply enter the test name as shown in parentheses. For example, to enable the physical memory test, enter **<mem>**.

An arrow will appear immediately before the test name, indicating that it is presently enabled. To disable the test, enter **<mem>** again. The arrow is removed, and the test is disabled.

Tests that do not have an arrow in front of the test name will not run when you enter **<start>**.

You may also enter the name of a test to stop an individual test in progress, or to start a new test.

Options Menu

The options toggle from "disable" to "enable" when you enter the command:

o [corefile]	t [trace]
s [single pass]	a [auto start]
q [quick test]	r [run on error]
v [verbose]	m [send mail]

Sundiag allows you to enter **maxerrors <number> 0** or simply **<number> 0** to change the maximum number of errors before stopping tests.

Options

```
-----
| Sundiag 2.3.2 CPU:ESP8S                               System status: idle |
|-----|
|                                     << System Option Menu >> |
|                                     [DEFAULT(f)] [DONE(d)] [HELP(h)] |
|                                     | |
| [COREFILE(o)]: Disable |
| [SINGLEPASS(s)]: Disable |
| [QUICKTEST(q)]: Disable |
| [VERBOSE(v)]: Disable |
| [TRACE(t)]: Disable |
| [AUTOSTART(a)]: Disable |
| [RUNERROR(r)]: Disable |
| [SENDMAIL(m)]: Disable |
| [MAXPASSES(x)]: 0 |
| [MAXERRORS(y)]: 1 |
| [CONCURRENT(c)]: 2 |
| [LOGPERIOD(l)]: 0 |
| [EADDRESS(e)]: root |
|-----|
| Command: o |
| Message: |
|-----|
```

What is happening during boot?

- See also “[Boot procedure: Overview](#)”, page 109
- For a detailed graphic scheme, see [Fig. 11](#), page 110
- For a commented bootlog, see [page 110](#).
- See also “[Boot flowchart](#)”, page 67

Switch on the Sparc

After power is switched on, the POST (Power On Self Test) - host-internal hardware diagnostics - are started from the boot prom.

POST can be interrupted by typing **STOP A**. This ends any processor activity and enters the Monitor Mode (“>” Prompt).

After entering ‘b’ in the Monitor Mode (“>” Prompt or ‘boot’ when the ‘ok’ prompt appears (new-mode after entering > “n”), the host computer starts to boot. The boot device can be defined (e.g. disk1, disk2, cdrom, tape etc.) These devices are device alias. Use the monitor mode command ok “**devalias**” to see all device aliases.

- e.g. **disk** or **disk0** are device aliases to
“/iommu/sbus/espdma@f,400000/esp@f,800000/sd@3,0”
The device at **target 3, Unit 0** = sd0

boot device

The default boot device is stored in the NVRAM parameter “**boot-device**” and is per default “**disk**”.

Label (B0)

In order to be able to localize the beginning of the boot-partition, the label which is always located on sector 0 (B0) of the first partition (as a rule partition a) must be interpreted first.

boot block

In order to load the **boot** program “/boot”, the boot block must be loaded first. This boot block is always located on sectors 1 - 15 on the boot partition, which normally corresponds to the a-partition.

/boot

/boot is a program which resides in the root directory of the boot partition, since at that point in time, it cannot yet be accessed.

/boot via the file system. The blocks where **/boot** is located are marked in the **boot block**. After **/boot** is loaded, it will automatically check from which partition it was loaded, and will try to mount this partition and then load the **kernel** “/vmunix” (parameter in the NVRAM) from the same partition.

Kernel

After the loading process, the kernel is given control and numerous tasks are started.

Boot UNIX in multi-user mode

- To boot the UNIX operating system in multi-user mode enter ‘b’ or ‘boot’.

```
Type b (boot), c (continue), or n (new command mode)
> b (or ok boot)
...
Boot device: /iommu/sbus/espdma@f,400000/esp@f,800000/sd@3,0
File and args:
...
mrc_login:
```

The host computer boots to multi-user mode, which is available after the “login” prompt appears.

Booting UNIX in single-user mode

- To boot the UNIX operating in system single - user mode, enter '**b -s**' or '**boot -s**'

```
> b -s (or ok boot -s) (-s = boot string argument for single - user mode)
...
Boot device: /iommu/sbus/espdma@f,400000/esp@f,800000/sd@3,0
File and args: -s
...
# (root prompt for single-user-mode)
```

Shutting down UNIX

NOTICE Use the following procedure for shutdown, whenever possible.

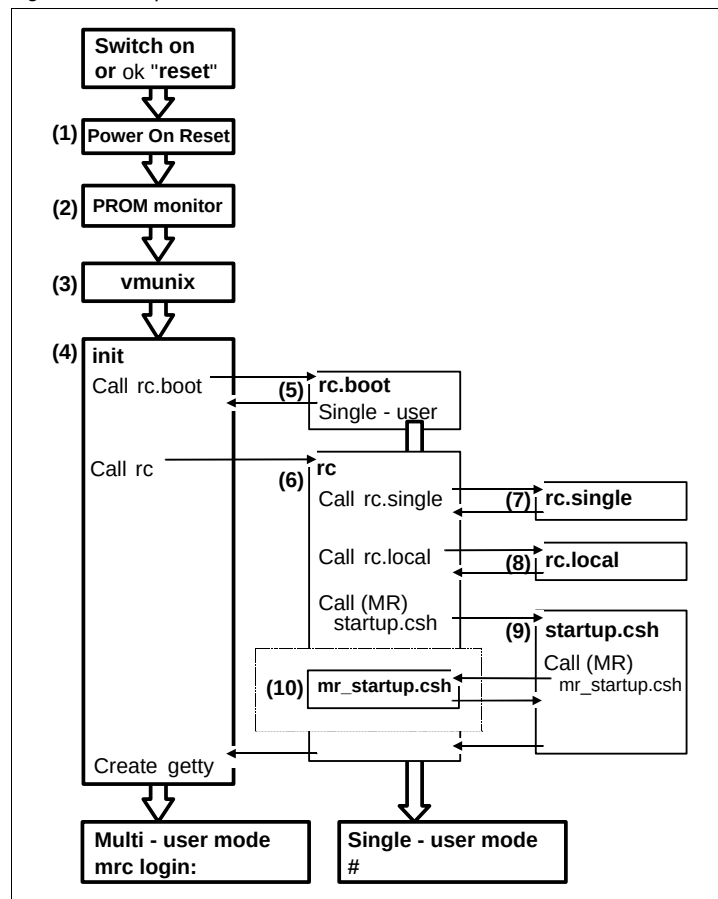
- If NUMARIS is ended already, enter '**halt**' to end the UNIX operating system

```
mrc_login: halt
...
Mar 12 17:17:13 mrc halt: halted by halt
Mar 12 17:17:14 mrc syslogd: going down on signal 15
syncing file systems... done
Halted
>
(ok if NVRAM parameter sunmon-compat? true)
```

- Turn off the system (host computer), when a prompt appears.

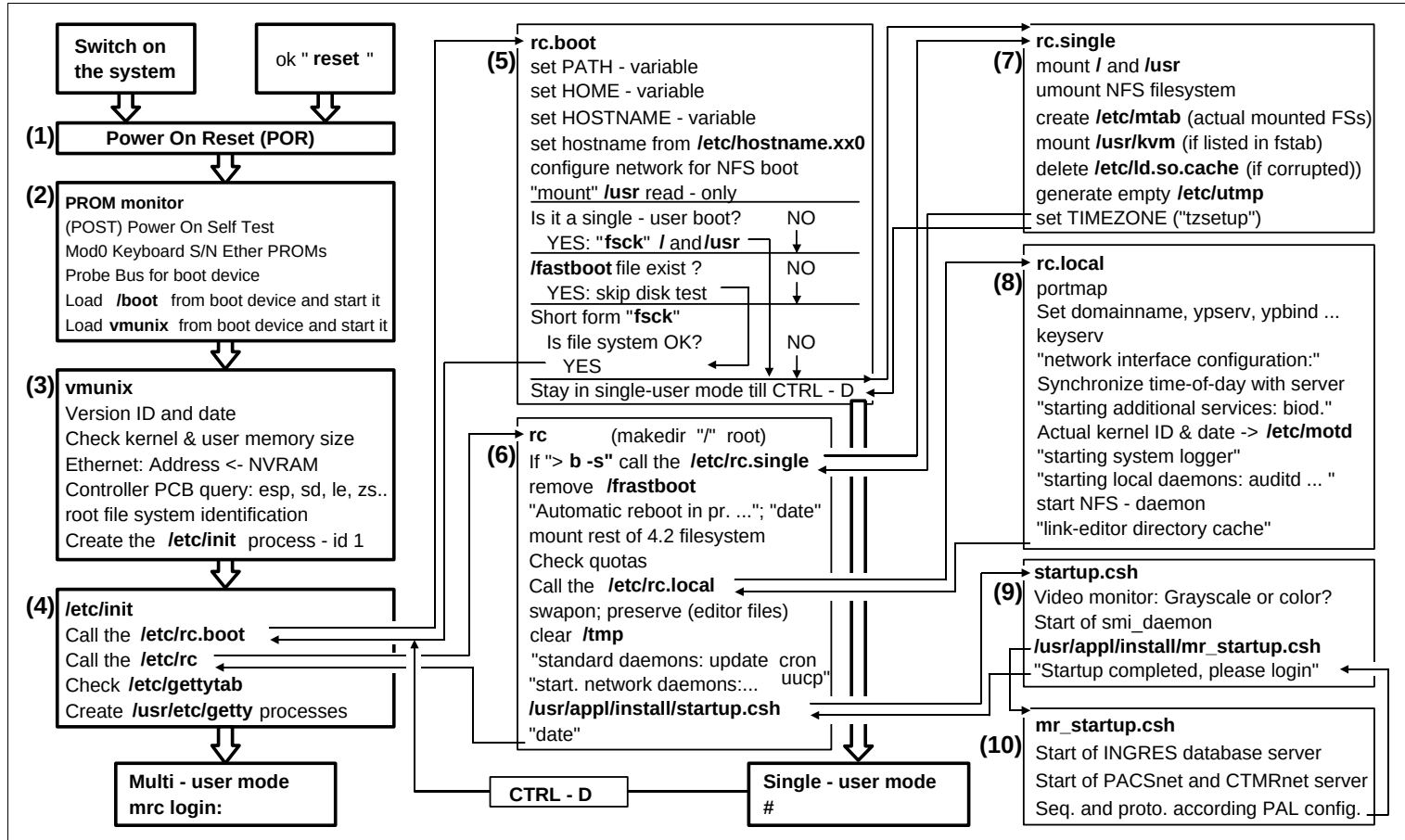
Boot procedure: Overview

Fig. 10 boot procedure overview



Boot procedure: Scheme

Fig. 11 Boot procedure



Bootlog

Explanation

- Lines with a preceding "-" are boot steps from scheme [Fig. 11](#).
- All boot messages are framed.
- Lines with a preceding "#" are "original comments" from the unix shell scripts (executed during boot; e.g. "/etc/rc").
- **Bold** lines are "**comments**", not messages.
- Numbers in front of paragraphs - e.g. "**(1)**" - are numbers in relation to scheme [Fig. 11](#).
- Indent means a "program call" or "exit" and return to the calling procedure.

This "bootlog" is an extended version from the graphic scheme of the boot procedure (see [Fig. 10](#) / [Fig. 11](#)).

Power-On Reset (1)

- Switch on the system (or type: ok "reset").

Call the PROM Monitor program.

PROM monitor (2)

After POR, the PROM monitor program preforms a Power On Self Test (POST), to test the basic functions of the CPU.

- (POST) Power On Self Test
- Mod, Keyboard, S/N (Serial Number), Ethernet address

```
1 Imagers available
32 MB Mailbox Memory available
Harmony VA11A 10999
Initializing Memory I/-\/-\/-\/-\...
```

- Probe the bus for boot device.

If no boot device was defined in the boot string, read the default boot device from NVRAM (parameter: "boot-device" value: "disk") and extract the device alias from the NVRAM and boot it.

The boot device is e.g.:

```
disk =/iommu/sbus/espdma@f,400000/esp@f,800000/sd@3,0
(sd@3,0 = target 3, Unit 0. If not defined, the default boot partition is a) = sd0a
```

Read label (block 0) and then load the boot block program (blocks 1 - 15) and start it.

```
Boot device: /iommu/sbus/espdma@f,400000/esp@f,800000/sd@3,0  File and args:
```

- Load **/boot** program from boot device (sd0a) and start it.

Mount root directory "/" read-only on sd0a as unix file system type 4.2 (to be able to load and run the /vmunix kernel)

```
root on /iommu@f,e0000000/sbus@f,e0001000/espdma@f,400000 esp@f,800000/
sd@3,0:a fstype 4.2
```

- Load **vmunix** from boot device (sd0a) and start it.

```
Boot: vmunix
Size: 1654784+515768+206912 bytes
SuperSPARC: PAC ENABLED
```

vmunix (3)

After the loading process, the kernel is given control and numerous tasks are started.

- Version ID message and date (e.g. kernel "**MR_024_mrc_sun4m**") and **Checking the CPU type**

```
SunOS Release 4.1.4 (MR_024_mrc_sun4m) #3: Tue Nov 12 00:17:32 MET 1996
Copyright (c) 1983-1993, Sun Microsystems, Inc.

cpu = FORCE, ESP8S
mod0 = TI,TMS390Z50 (mid = 8)
```

- Check kernel & user memory size

Checking the size of the available memory, subtract the kernel size and mark the rest as available memory.

```
mem = 65128K (0x3f9a000)
avail mem = 59990016
cpu0 at Mbus 0x8 0x260000
entering uniprocessor mode
```

- Ethernet address <- NVRAM

```
Ethernet address = 8:0:20:1a:e6:41
```

- Controller PCB (Printed Circuit Board) query: esp, sd, le, zs, etc.

esp0 is the on board SCSI controller: Output of all known devices and checking whether responsive peripheral units (disks, tapes, etc.) are available at the respective controllers.

sd0: First hard disk (system disk), SCSI-address 3, lun 0 = logical unit number 0.
The driver displays the label info: <MAGNETOM 05 cyl ... >

sd1: Second hard disk optional for data): SCSI-address 1, lun 0
The driver displays the label info: <MAGNETOM 06 data cyl ...>

sd2: Optical disk is not installed e.g. (LD 1200).

sd2: MOD drive **sd0** = **sd2** (two names for same MOD), target 2

```
espdma0 at SBus slot f 0x400000
esp0 at SBus slot f 0x800000 pri 4 (onboard)

sd0 at esp0 target 3 lun 0
sd0: <MAGNETOM 05 cyl 2733 alt 2 hd 19 sec 80>

sd1 at esp0 target 1 lun 0
sd1: <MAGNETOM 06 data cyl 5179 alt 2 hd 10 sec 164>

sd2: sod_findslave: failed, scsi_slave stat=0x2

sd0: found Direct Access device (tgt2, lun0) on esp0
Vendor/Product ID = PIONEER DE-C7001
sd0 at esp0 target 2 lun 0
sd2 at esp0 target 2 lun 0
sd2: Vendor 'PIONEER', product 'DE-C7001', (unknown capacity)
```

CD-ROM drive sr0 target 6

Ethernet interface

Parallel printer interface

Audio Interface

```
sr0 at esp0 target 6 lun 0

ledma0 at SBus slot f 0x400010
le0 at SBus slot f 0xc00000 pri 6 (onboard)

SUNW,bpp0 at SBus slot f 0x4800000 pri 3 (sbus level 2)
SUNW,DBRle0 at SBus slot f 0x8010000 pri 9 (sbus level 5)

Audio: no Audio device found
```

SMI-5 Imager: SIAM-C is connected via the SCON board to sbus slot 2

The Sparc CPU considers the SIAM-C (cgsiam = Color Graphic System Interface and Memory) as a normal workstation-frame-buffer. The Sparc checks the SMI-5 via SCON (D138) and SIAM-C (D21). In the following bootlog example, the SIAM-C (D21) was accessible.

cgsiam0: SIAM-C board

simema0: Imager Memory

iopals0: PAL bank

ioinctl0: Interrupt controller

scon0: SCON board

```
Assuming Imager Type D70
SMED,cgsiam0 at SBus slot 2 0x20000 pri 9 (sbus level 5)
SMED,simema0 at SBus slot 2 0x10000
SMED,sidedr0 at SBus slot 2 0x10000 pri 9 (sbus level 5)
SMED,iopals0 at SBus slot 2 0x1080c0
SMED,ioinctl0 at SBus slot 2 0x1080d4 pri 9 (sbus level 5)
SMED,scon0 at SBus slot 2 0x7f80 pri 13 (sbus level 7), pri 9 (sbus level 5)
```

If the Sparc cannot access the SMI-5, the lines (listed above) are just missing. No special error message is printed, but later imager problems will occur. In case of communication problems between Sparc and SMI-5, replace SCON (D138), SIAM-C (D21) or check the front cable connection.

For testing the communication to the SIAM-C, use the unix command [“devinfo”](#), [page 95](#).

zs = ZILOG 8530 serial communication controller
zs0 serial communication ports of CPU: Keyboard, Mouse
zs1 serial communication ports ttya, ttyb
fdtwo0: Floppy disk (not used)

```
zs0 at obio 0x100000 pri 12 (onboard)
zs1 at obio 0x0 pri 12 (onboard)
SUNW,fdtwo0 at obio 0x700000 pri 11 (onboard)
```

- root file system identification

The root partition is mounted read-only and the swap and dump partitions are fixed

```
root on sd0a fstype 4.2
swap on sd0b fstype spec size 200640K
dump on sd0b fstype spec size 200628K
```

kernel tasks are finished, kernel is running.

- Create the /etc/init process - id 1.

init (4)

The kernel now start the first process, the "init" process, which gets the first pid = 1 (process identification number).

The init process checks whether the AUI (15 pin ITT cannon plug) is connected to the network.

```
le0: AUI Ethernet
```

- Call the /etc/rc.boot (shell script)

rc.boot (5)

In SunOS 4.x, block devices are no longer consistent with file systems mounted **read-write**. To allow file systems to be reliably **fsck**'ed during system reboots, the **root 4.2** (UFS) file system is

mounted **read-only** as default. **Fsck** uses the **raw** device to check for any file system damage. This way **fsck** can repair any file system damage without the **kernel**, overwriting it with any potentially corrupt "in-core- file-system-control-information" it might have. The file systems will remain **read-only** until a "**remount**" operation by **/etc/rc.single** is done after the disks check out okay. Thus if **fsck** finds and repairs a corrupt file system which is currently **mounted**, the system should be **rebooted** immediately.

For fastboot: The file /fastboot does exist:

```
ifconfig: ioctl (SIOCGIFFLAGS): no such interface
```

Everything is ok when if the file **/fastboot** exists. It is okay to skip **fsck** for the file systems.

Check filesystem with command **fsck** (no /fastboot exist)

```
ifconfig: ioctl (SIOCGIFFLAGS): no such interface
checking filesystems
/dev/rsd0a: is stable.
/dev/rsd0g: is stable.vc
```

Everything looks good. Finish the **single user** set-up which will **remount** the file systems as **read-write** and do other work which can be done only on a **writable root** file system.

- Call the **/etc/rc.single** (if "> b -s")

rc.single (7) if single - user boot

Run interactively from **single-user mode** after any file system problems have been corrected when **read-write** access is needed for the **root** file system.

```
...(Enter single - user mode) ...
#
```

Single - user mode till CTRL - D. If no single - user boot was invoked, or CTRL - D was hit from Single - user mode, exit rc.single.

- Exit rc.single, return to /etc/rc.boot

rc.boot (5)

Check whether rc.single successfully completed. It can fail, for example, if **/etc/fstab** has not successfully run.

If rc.single was successfully completed exit rc.boot.

- Exit rc.boot, return to /etc/init

init (4)

```
smikbdopen():nothing linked below us
```

- Call the **/etc/rc**

rc (6)

The **rc** shell script gets the **console**. All commands are by default protected from tty signals (i.e. are not interruptible). **Interrupts** can be **enabled** by preceding any command line with "**intr**". Console redirection is no longer needed.

Check that the file system is still mounted **read-only**. If so, perform all the remounting and set up work now.

```
... (Every thing is OK) ...
```

- Call the **/etc/rc.single** (if "file system ... still mounted read-only")

rc.single (7) if single - user boot

This function is performed when the file systems look ok. It will remount the **/** and **/usr** file systems (which are probably still mounted

read-only), fix up **/mtab**, clean up the **ld.so** cache, reset **/etc/utmp**, use **tzsetup** to set the kernel's notion of **timezone** for compatibility with old binaries, and use **loadkeys** to load the keyboard translation tables for the current keyboard. This function is invoked from **/etc/rc.boot** on a clean **fsck**, from **/etc/rc** when it detects that the **root** file system is still mounted **read-only**, and can be run interactively from **single-user mode** after any file system problems have been corrected when **read-write** access is needed for the **root** file system.

```
... (Every thing is OK) ...
```

- Exit **rc.single**, return to **/etc/rc**.

rc (6)

Automatic reboot.

Check user disk space (quotas)

```
Automatic reboot in progress...
Thu Jun 19 10:44:13 MET DST 1997
checking quotas: done
```

- Call the **/etc/rc.local**.

rc.local (8)

- Portmap

PORTMAP is a server that converts **TCP/IP** protocol port numbers into **RPC** (remote procedure calls) program numbers.

```
starting rpc port mapper.
```

- Set domain name (if NIS domain), **ypserv**, **ybind**, **yupdate**

... OK ...

- keysevr

KEYSERV is server for storing public and private keys.

starting RPC key server.

- "network interface configuration:"

Routing - system support for local network routing to a specific network or host

Mount NFS (network file systems) listed in /etc/fstab

biod is a daemon that handles client filesystem requests

network interface configuration:
le0: flags=63<UP,BROADCAST,NOTRAILERS,RUNNING>
inet 185.2.47.37 netmask ffff0000 broadcast 185.2.0.0
ether 0:40:dc:0:10:ff
lo0: flags=49<UP,LOOPBACK,RUNNING>
inet 127.0.0.1 netmask ff000000

running routing daemon
mount -vat nfs
starting additional services: biod.

- Actual kernel ID & date ("dmesg" command) -> **/etc/motd**

- "starting system logger"

syslogd forwards system messages to the appropriate log files.

starting system logger

- "starting local daemons: auditd ... "

starting local daemons: auditd sendmail nfsd Jun 19 10:44:27 mrc
vmunix: smikbdopen(): nothing linked below us

rapd bootparamd statd lockd pldauthd.
link-editor directory cache

- Exit **rc.local**, return to **/etc/rc**

rc (6)

Delete all files in directory /tmp

Start network daemon

preserving editor files
clearing /tmp
standard daemons: update cron uucp.
starting network daemons: inetd printer.

Start MR specific startup program "**startup.csh**"

- Call the **/usr/appl/install/startup.csh**

startup.csh (9)

For details, refer to the Software Manual.

Jun 19 10:44:33 mrc startup.csh: **running**

- Video Monitor: Grayscale or color?

Jun 19 10:44:34 mrc startup.csh: Grayscale monitor

- Start of **smi_daemon**

```
Jun 19 10:44:50 mrc disk_cleaner_at_boot_time: running
Jun 19 10:44:51 mrc startup.csh: Starting smi_daemon
Jun 19 10:44:51 mrc startup.csh: define environment with numglob
Database server not running on local host
```

```
Jun 19 10:44:53 mrc startup.csh: Starting SMI Firmware server
(fws)
Jun 19 10:44:54 mrc startup.csh: Starting CCU firmware server
(ccufws)
Jun 19 10:44:54 mrc startup.csh: PPP driver
%FWS-S-START, Firmware Server Version 4.3
module loaded; id = 1
```

- Call the /usr/appl/install/mr_startup.csh.

mr_startup.csh (10)

For details, refer to the [“NUMARIS start-up”, page 32](#), [page 30](#).

```
Jun 19 10:44:59 mrc mr_startup.csh: running
CHECK Raw Partition
CHECK d-partition OK
```

- Start INGRES database server.

```
Starting INGRES soon
Starting INGRES now
Startup messages go to file /usr/ingres/db_startup.log
INGRES logfile not available, configuring whole INGRES
new
```

1. step : Creating logfile on raw-partition
2. step : Installing resources
3. step : Initialize log-file
4. step : Configuration of Database Servers
5. step : Startup of Database Servers
6. step : INGRES configuration + startup completed.

```
7. step : Creating INGRES Database
8. step : Creating NUMARIS Database
9. step : Creating NUMARIS Tables
10. step: Verifying databases
Database >> CONSISTENT
```

```
Database >> STABLE
INGRES startup & check done
Jun 19 10:50:55 mrc mr_startup.csh: INGRES system is
up
```

```
Jun 19 10:50:57 mrc mr_startup.csh: mounts to database
host succeeded
Jun 19 10:50:59 mrc mr_startup.csh: starting sv_ralock
Jun 19 10:50:59 mrc mr_startup.csh: install or update dni
installation
```

- Start PACSNET and CTMRNET server.

```
Jun 19 10:50:59 td2mrc mr_startup.csh: starting PACS-
NET CTMRNET DICOMNET [1] 1901
Jun 19 10:51:00 td2mrc mr_startup.csh: starting
sv_patlock
```

```
Jun 19 10:51:01 td2mrc service_startup.csh: load environ-
ments for file/image transfer functions
Jun 19 10:51:02 td2mrc service_startup.csh: starting httpd
Jun 19 10:51:06 td2mrc service_startup.csh: starting
wd.exe (Service Watchdog)
```

```
Jun 19 10:51:07 td2mrc service_startup.csh: starting
tt_setup.exe (TestTools Configurator)
Jun 19 10:51:08 td2mrc service_startup.csh: Netscape
default files installed
Jun 19 10:51:08 td2mrc service_startup.csh: Remote
Diagnostics
```


- Sequences and protocols according to PAL configuration.

```
ERROR: proto_default.csh: no list of customer protocols
found:
/usr/appl/install/proto_default_m24s.dat
proto_default.csh. Exited.

Jun 19 10:51:10 mrc mr_startup.csh: starting protocol con-
version

Jun 19 10:51:14 mrc mr_startup.csh: finished
```

- Exit mr_startup.csh, return to startup.csh.

startup.csh (9)

```
Jun 19 10:51:14 mrc startup.csh: Startup completed, please login
```

- Exit startup.csh, return to /etc/rc.

rc (6)

```
Thu Jun 19 10:51:14 MET DST 1997
```

- Exit startup.csh, return to /etc/init.

init (4)

- Create **/usr/etc/getty** process.

```
mrc login:
```

Start of network image receiver processes (if Network is installed).

```
Jun 19 10:51:52 mrc start NetReceiver.csh: dnserver not on air: no PACSNET , no
CTMRNET , no DICOM

Jun 19 10:51:52 mrc start NetReceiver.csh: startup of sv_netcp_2 , sv_ctmrnet
anddtk_dirv skipped
```

Start failed: No PACSNET, CTMRNET and DICOM installed.

In this part you will find test strategies, procedures and repair instructions encompassing the complete SMI.

Strategy

General

NOTICE	Always start with “Test strategy for Host, SMI or SW” , page 19, when troubleshooting. Following this flowchart table will help will help facilitate error localization.
---------------	--

NOTICE	If the problem is not corrected after following the flowchart tables, refer to the strategy of the Host , especially the : “Host troubleshooting flowchart” , page 70; “MR(S)C system overview (schematics)” , page 59; “Diagnostics level overview” , page 65.
---------------	--

Necessary tools

- Voltage meter Fluke 8060A
- Fluke Combi Scope PM 3390A or equivalent
- Special chip extractor for PLCC sockets (P.n. 46 73 620)
- Test boards (P.n. 99 00 416 GE999) for testing the serial and parallel interfaces of the MAXICAM D66 to the digital camera.
- 75 Ohm Adapter (P.n. 46 74 230)

Definition of fonts used in the documentation

text	All outputs on the screen is written in Courier font
command ↵	All input by the user is indicated in bold and followed by a carriage return sign.

SMI not accessible

If the SMI is not accessible during booting there is a fast way to find the reason of the problem. Check the MR start-up log for the following entries.

You can check the file `var/adm/messages` for the entries with the Unix command

more /var/adm/messages

or connect the laptop to the serial port of the host and reboot the host. You can check the log for the entries with the help of your terminal emulator.

Bootlog interpretation

SIAM-C is connected via the SCON board to sbus slot 3

The Sparc CPU considers the SIAM-C (cgsiam = Color Graphic System Interface and Memory) as a normal workstation-frame-buffer. The host checks the SMI-5 via SCON (D138) and SIAM-C (D21). In the following bootlog example, the SIAM-C (D21) was accessible.

Device	Explanation
cgsiam0	SIAM-C board
simema0	Imager Memory
iopals0	PAL bank
ioinctl0	Interrupt controller
scon0	SCON board

MR start-up extract

Step	Output
1	SMED,scon0 at SBUS slot 3 0x7f80 pri 13 (sbus level 7), pri 9 (sbus level 5) Assuming Imager Type D70
2	SMED,cgsiam0 at SBUS slot 3 0x20000 pri 9 (sbus level 5) SMED,simema0 at SBUS slot 3 0x10000 SMED,sidedr0 at SBUS slot 3 0x10000 pri 9 (sbus level 5)
3	SMED,iopals0 at SBUS slot 3 0x1080c0
4	SMED,ioinctl0 at SBUS slot 3 0x1080d4 pri 9 (sbus level 5)

If the host cannot access the SMI-5, the lines (listed above) are just missing. No special error message is printed, but later imager problems will occur.

The host checks the connection to the SMI5 in 4 steps.

First step:

Is the S-CON (interface on the Sparc mother board) accessible? In this example the SCON was accessible. If the Sparc cannot access the device, the actual listed line is just missing. No special error message is printed out here, but later problems will occur! In case of problems exchange SCON.

Second step:

Is the SIAM-C accessible via S-CON, front cable and K-Bus?

In this example the SIAM-C was accessible. If the Sparc cannot access the device, the actual listed line is just missing. No special error message is printed out here, but later problems will occur! In case of problems exchange SIAM-C or front cable connection.

Third step:

Is the PAL-bank accessible via S-CON, Carrier, top-backplane, CIB-MR D204 ?

In this example it is ok. If the Sparc cannot access the device, the actual listed line is just missing. No special error message is printed out here, but later problems will occur!

In this case you should replace the S-CON or CIB board.

Fourth step:

Is the Interrupt controller accessible via S-CON, Carrier, top-backplane, CIB-MR D204?

The interrupt controller checks voltages and temperatures on the CIB. In this example everything was ok. If the Sparc cannot access the device, the actual listed line is just missing. No special error message is printed out here, but later problems will occur!

In this case you should replace the CIB board.

Image data acquisition problem

Data acquisition is interrupted during measurement . The following are needed for successfull calculation: Sparc, keybd & mouse, system disk, power supply, SMI, monitor.

No	Test Step	Status	Service Action	Reference
1	Check system report for SMI errors.	SMI error occurred	Do a “Simulation Measurement”. If any error occurs perform the actions suggested in the help text of the failed test step.	Go to page 131
		No SMI error occurred	Execute a Trouble Reset “Reset Measurement Problems” (RMEA), in System / Run.	Go to the manual Replacement of Parts, part Software
			Restart Simulation Measurement	Go to page 131
2	Start HTE test (see page 134)	SMIINIT was not running	Do partial configuration tests.	Go to page 132
		SMIINIT is running	Do the special MAP test.	Go to “ MAP D 55 ”, page 150

Remarks: none

3D reconstruction problem

The 3-D reconstruction is interrupted. The following are needed for successful reconstruction: Sparc, keybd & mouse, system disk, power supply, SMI, monitor.

No	Test Step	Status	Service Action	Reference
1	Check system report for SMI errors.	SMI error occurred	Do a "Simulation Measurement". If any error occurs perform the actions suggested in the help text of the failed test step.	Go to page 131
		No SMI error occurred	- Execute a Trouble Reset "Reset Measurement Problems" (RMEA), in System / Run. - Restart Simulation Measurement	Go to the manual Replacement of Parts, part Software
			Check if any SW process hangs.	Go to the manual Replacement of Parts, part Software "Single NUMARIS functions are failing"
			Execute trouble reset and repeat 3D image reconstruction	Go to the manual Replacement of Parts, part Software
			Restart Simulation Measurement	Go to page 131
2	Start HTE test (see page 134)	SMIINIT was not running	Do partial configuration tests.	Go to page 132
		SMIINIT is running	Do the special MAP test.	Go to "MAP D 55", page 150

Remarks: none

Blank screen test

If you have no display at the console monitor, do the following tests. Boards involved: Imager-ML D70, console monitor and video cable.

No	Test Step	Status	Service Action	Reference
1	Disconnect the video cable from X8 of the D70 and connect a scope to X8 (Fig. 12, page 125)	No video signal present. Refer to Fig. 13, page 125 for a typical video signal.	• Connect your PC-laptop to the serial port of the host.	page 22
			• Start the HTE test	page 134
			• Do the HTE tests for the Imager. smi070/all	page 151
			• If SMINIT was not running, do partial configuration tests.	page 132
		Video signal present. See Fig. 13, page 125	• Reconnect the video cable to X8 on D70.	
2	Disconnect the video cable from the monitor and connect a scope to the video cable	Video signal present. See Fig. 13, page 125	• Check the fuses of the monitor or replace the monitor.	
		No video signal present.	• Exchange the cable between Imager and monitor	

Remarks: none

Fig. 12 Video test for Imager D70

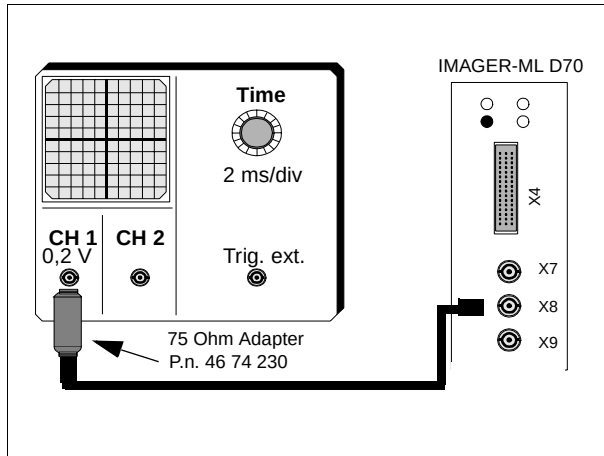
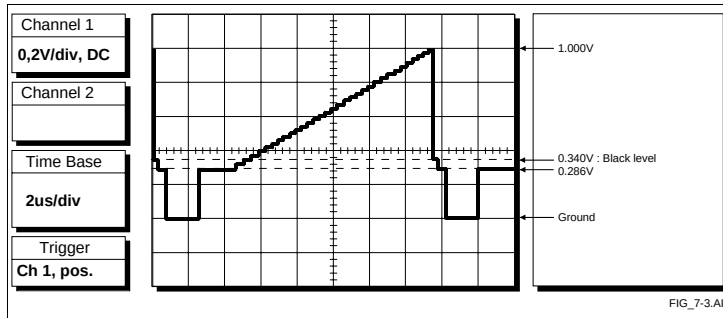


Fig. 13 Typical BAS signal for the Imager output



Recommended tests

The following tests are recommended for testing the SMI. The following are needed for successful boot: Sparc, keybd & mouse, system disk, powersupply, SMI, monitor.

No	Test	Required time	Description	Detailed description
1	Simulation Measurement	approx. 4 min.	The Simulation Measurement performs a quick check of the SMI. The Simulation Measurement includes raw data handling, image calculation and image display. Use the online help text information of the Simulation Measurement to interpret the test results.	Refer to “Simulation Measurement” , page 131
2	SMIINIT	approx. 10 sec.	The SMI initialization is similar to a fast check. The printout of the smiinit can be used for a fast analysis of the SMI status.	Refer to “smiinit” , page 145
3	SMISYS/QUICK	approx. 30 min.	Performs test 'smi0xx/quick' for each board which is found in the real SMI-system. The boards are tested analogous to the order of the plugin cards in the frame.	Refer to “smisys” , page 147
4	SMISYS/ALL	approx. 8 hours.	Tests all boards; all single tests are included except the interactive tests (Refer to “smisys” , page 147).	Refer to “smisys” , page 147
5	SMISYS/PARALLEL/RAM/ALL	approx. 1 h 20 min.	Parallel execution of all GIPSY data RAM and all main RAM tests	Refer to “smiparallel” , page 147

Note: The required test times are given for a SMI configuration with 2 MAP4. The test times can vary depending on the installed options.

Caution: After you interrupt a test with CNTRL-C or after the program ends a test with an error, initialize the SMI with 'smisys/init' or 'smiinit' to be sure that the SMI is in a defined state.

Hints

SMI Boot problems

If NUMARIS start-up is interrupted, the source for error may be booting problems in the SMI.

SMI boot problems occur when the locally recognized SMI HW is different from the content of the SMI bootfile.

In this case, an SMI installation should be performed to correct the problem. The SMI boards recognized by the install procedure should be identical to the boards actually installed.

Refer also "[SMI not accessible](#)", page 120 or to "[HOST interface 'cgsiam' not visible](#)", page 72 for boot log interpretation when the SMI is not booting.

Visual inspection

All slots of the SMI 5 frame should be filled under normal operating conditions. Dummy boards must be inserted in empty spaces to maintain continuity on the system bus

Dummy board part number: 54 53 618 K5004

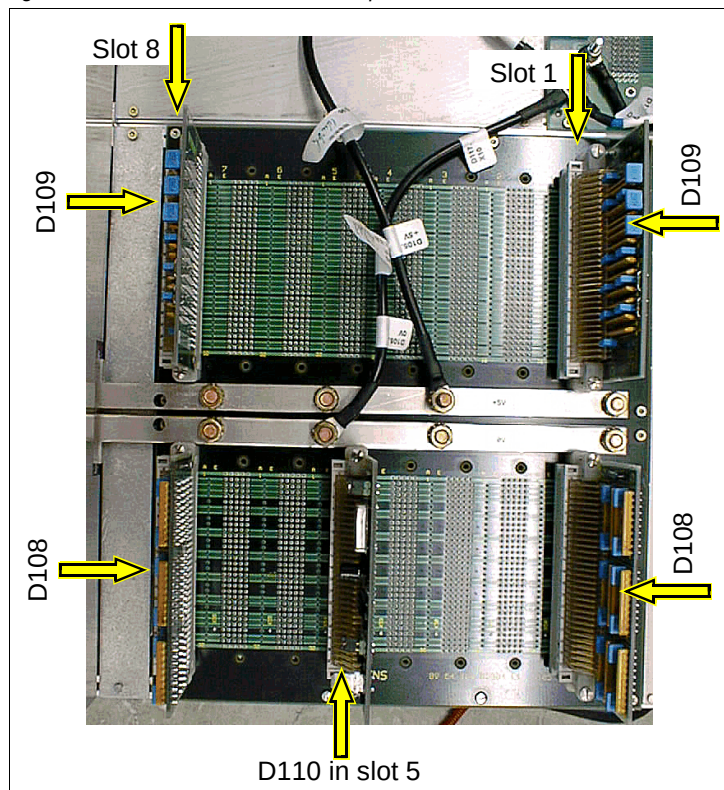
After transporting the SMI 5, the terminator or bus master boards on the backplane of the SMI 5 may be loose (D108, D109 and D110).

If your system shows **MULTIMUX** or **Reset_Power_Fail** errors, check the jumpers on the bus master D110 and verify that all boards are plugged into the backplane. (Refer to the [Fig. 14, page 127](#) for the location of D108, D109 and D110).

Jumpers on bus master D110:

X2	1-2	The oscillator on D110 is switched off
X3	1-2	Internal Power_Fail supervision is switched off
X4	n.used	Power_Fail connection to the power supply)

Fig. 14 Rear view of the SMI 5 backplane



Board swapping possibilities

If you have a second console (MRSC), all boards can be swapped without requiring any changes.

The only exceptions are the SIAM-C, the CIB and the TurboSparc. When swapping the TurboSparc, the NVRAM has to be exchanged.

Necessary procedure for changing the SIAM-C

On the SIAM-C the System PAL is located. Two types of PAL's are used:

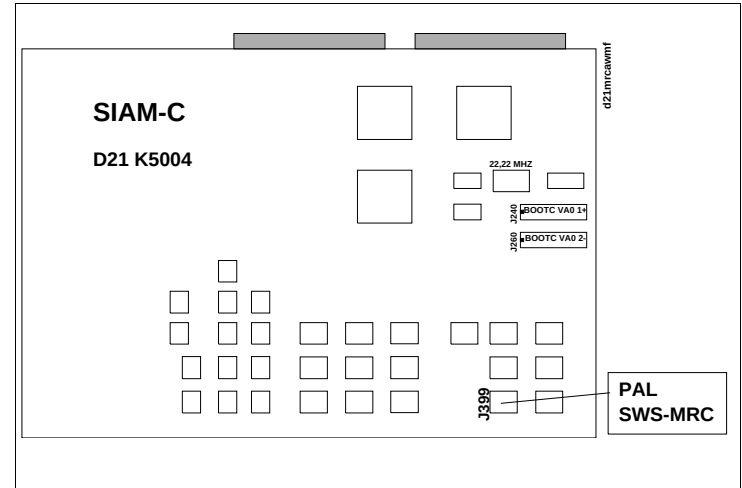
- one for the main console MRC
- and one for the satellite console MRSC

NUMARIS 3 uses the system PAL to distinguish between MRC and MRSC.

- When exchanging the SIAM-C, you have to swap the system PAL (J399). For the location of the system PAL refer to [Fig. 15](#).

NOTICE To exchange the system PAL, you need a special chip extractor for PLCC sockets (P.n. 46 73 620)

Fig. 15 Location of the system PAL on the SIAM-C



Board Lockups

MAP 4

It is possible to run the system with only one MAP 4 until the spare part arrives if one of the MAP4 boards is defective. To use your MRC with one MAP 4, you must do the following:

- Insert a dummy board (54 53 618 K5004) inplace of the defective MAP 4 board.
- Use a D166 M4-XBUS-1 board (P.n. 28 22 927) to connect the two extension busses together on the front of the MAP4. If your system has a satellite console use the D166 board of the MRSC.
- Switch on the system and after booting initialize the SMI. Refer to manual Installation Vol.1, Register Installation SW, Chapter Configuration, Section Initialize SMI.

Procedures

General

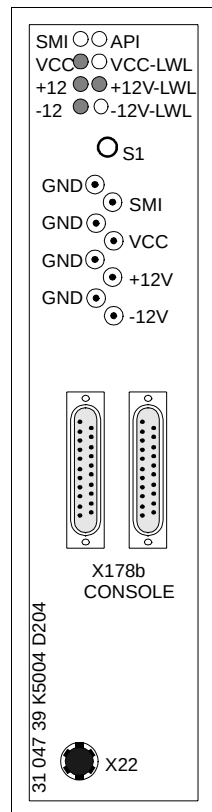
Required Tools

- Anti-static wristband and grounded tabletop pad (TDK equipment Type 8005 from 3M Deutschland GmbH).
- Screwdriver
- Service Laptop
- Terminal emulation program
- Cable for Laptop-Host connection

Power supply voltages

The host and the SMI 5 have a common +5 Volt power supply voltage.

Fig. 16 CIB front view



- The power supply voltage (+ 5V) of the SMI 5 is ok, as long as the VCC LED on the front of the CIB D204 is on.

The power supply voltage (+ 5V) of the SMI 5 can be verified with a voltmeter at the specific service connectors on the front of the CIB D204. (between connectors VCC and GND)

⇨ The +5V supplies should be within the range of 5.005V to 5.045V.

- The power supply voltage (+ 12V) for the different disk drives is ok, as long as the +12 LED on the front of the CIB D204 is on. The power supply voltage (+ 12V) for the different disk drives can be verified with a voltmeter at the specific service connectors on the front of the CIB D204 (between connectors +12 and GND).

⇨ The supply voltage should be within the range of 11.94V to 12.06V.

If the +12 V voltage supply for the drives fails, the corresponding LED on the CIB glows continuously at half intensity.

NOTICE

For the MRSC you have to measure directly at the power supplies.

LED status

After power-up

The following figure shows the LED status after the SMI 5 is switched on.

- ⇒ 'POR' LED lights up on all SMI boards. This means that a power-on reset is active on all boards and the GIPSY is in the boot addressing mode.

Fig. 17 SMI 5 LED status after power-up

16 43 100 K5004 D21	○○ ●○ ○○ ○○ ○○
SIAM-C	
54 70 448 K5004 D70	○○ ●○
IMAGER_ML	
28 19 840 K5004 D55	○○ ○○ ○○ ○○ ○○
MAP4	
28 19 840 K5004 D55	○○ ●○ ○○ ○○ ○○ ○○
MAP4	
89 65 683 K5004 D30	○○ ●○ ○○ ○○ ○○
DATA-IO	
54 55 357 K5004 D66	○○ ○○ ○○ ○○
MAXICAM	

After completion of booting (mrc login)

The following figure shows the LED status when the login prompt appears on the console monitor.

- ⇒ 'MAG' LED on SIAM-C board is on. This means that a power-on was performed and the GIPSY on the SIAM-C is in the Mailbox Access mode.

Fig. 18 SMI 5 LED status after completion of booting

16 43 100 K5004 D21	○○ ○○ ●○ ○○ ○○
SIAM-C	
54 70 448 K5004 D70	○○ ○○
IMAGER_ML	
28 19 840 K5004 D55	○○ ○○ ○○ ○○ ○○ ○○
MAP4	
28 19 840 K5004 D55	○○ ○○ ○○ ○○ ○○ ○○
MAP4	
89 65 683 K5004 D30	○○ ●○ ○○ ○○ ○○
DATA-IO	
54 55 357 K5004 D66	○○ ○○ ○○ ○○
MAXICAM	

After completion of MR start-up

The following figure shows the LED status when the message “Measurement system ready” appears in the Right Screen Server.

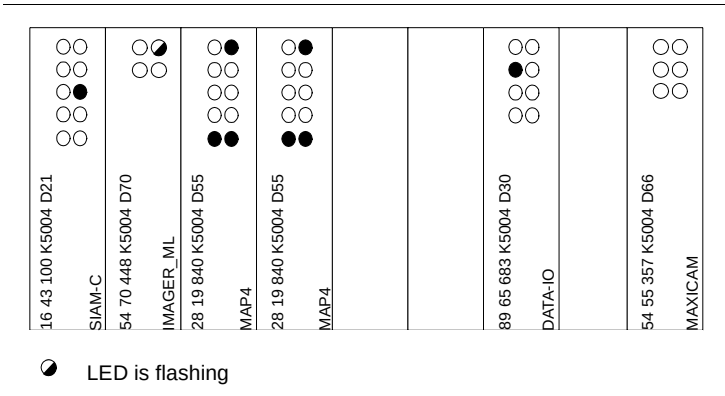
⇒ 'MAG' LED on SIAM-C board is on. This means that a power-on was performed and the GIPSY on the SIAM-C is in the Mailbox Access mode.

'INT' LED on the IMAGER-ML is flashing. This means the GIPSY of the IMAGER-ML is in the interrupt mode.

On the MAP 4 boards, the following LED's are on:

INT	GIPSY is in the interrupt mode
ATA	The processors have access to the address translation RAM
PBA	The processor bus is active (cycles of processors on the processor bus)

Fig. 19 SMI 5 LED status after completion of MR start-up



Simulation Measurement

Prerequisites

- **NUMARIS is up.**
- Service platform is started.
Refer to “[Starting service platform \(SeSo\)](#)”, page 21.

Starting the test

- Click **Test Tools** in the main platform with the **left** mouse key. Select **SMI Tests** with the left mouse button.
- The **Simulation Measurement** will be started as soon as the **Start** button is pressed.

NOTICE

For additional information, refer to the online help texts of the Simulation Measurement. The help texts for each test can be selected by clicking on the Help button with the left mouse key.

Partial configuration

If the system bus or one of the MMX-busses seems to be blocked by a board, or if identical or similar errors occur with several of the tested boards, the bus is probably blocked.

The defective board can be detected by using the partial configuration procedure.

The partial configuration procedure starts by establishing the [Minimal Configuration](#).

After you verify correct functioning, continue the testing by systematically increasing the number of boards inserted in the SMI rack. Always perform the specific board tests (refer to ["Board specific HTE commands"](#), [page 148](#) for the necessary HTE commands).

Minimal Configuration

A minimum of two boards must be inserted. The SIAM-C and another board must always be inserted. If possible, use the IMAGER-ML as the second board, so that you can use the HTE in the graphic mode.

- The front connection between SIAM-C and Imager must be plugged in.
- The BNC cable to the monitor should be connected to X8 of the Imager.
- Remove all other boards from the SMI rack.
- Ensure that the boards necessary for performing tests, i.e. **D21** and **D70**, are functioning properly .
(required HTE tests: **smi021/all** and **smi070/all**)

NOTICE

If minimum configuration is not working correctly or smisys/init does not work, use any other board instead of the D70 and repeat the tests at the PC laptop in command mode.
Refer to ["Testing under Unix"](#), [page 24](#) for connecting the laptop to the host.

NOTICE

If smisys/init still does not work for the partial configuration with any other board in place of the IMAGER-ML D70, the SIAM-C or the host may be defective. Go to the chapter Host strategy, section ["Host troubleshooting flowchart"](#), [page 70](#).

NOTICE

If smisys/init works for the partial configuration with any other board, the IMAGER-ML is defective.

Priority chain

- A Grant card has to be inserted in each slot between the SIAM-C board and the board to the extreme right. The Grant cards (Dummy boards) inserted in the slots to the extreme right may be used for this purpose.

Test order

- Bring down the system. Refer to [“Ending HTE”, page 136](#).
- Switch off the SMI (rocker switch S1 in the SMI container).
- Establish the [Minimal Configuration](#).

NOTICE Observe ESD guidelines.

- Switch on the SMI-5 and boot the system.
- Start HTE. The SMI will initialize automatically and the current configuration will be determined.
- Ensure that the boards necessary for the minimum configuration, i.e. **D21** and **D70**, are functioning properly (HTE test: **smi021/all** and **smi070/all**).

NOTICE If the minimal configuration is not working properly, refer to the additional information under [“Minimal Configuration”, page 132](#).

- Switch off the SMI-5.
- Insert the next board in the next available slot on the right side.
- Switch on the SMI-5 and boot the system.
- Start HTE. The SMI will initialize automatically and the current configuration will be determined.

- Start the test routine for the board to be tested (**smi0xx/all**).

Board	HTE test
MAP4 (D55)	smi055:n/quick n=1 for the first MAP n=2 for the second MAP
DATA-IO (D30)	smi030/all
Data Receiver (D24)	smi024
MAXICAM (D66)	smi066/all

NOTICE If the SMI initialization or the test is not running, the last board inserted is defective.

- Restore the SMI to its original configuration upon completion of all of the tests in the desired configuration.

Note

The extension bus (front connector) for **MAP4** boards must always be terminated with a termination board. Depending on the number of boards installed , different termination boards are required.

No. of MAP's	Labeling	Part number
1	D166 M4-XBUS-1 board	28 22 927
2	D167 M4-XBUS-2 board	28 22 943

Starting HTE

Graphic mode GHTE

Prerequisites

- Unix is booted.

NOTICE During the start-up of the GHTE, the SMI initializes automatically. This normally takes about 10 seconds.

Output	Input	Notes
login:	service ↵	log in
Password	*	Enter password for service.
Last log in:.... SunOS Release 4.1 (MRSTART)...		
%	openwin ↵	Open windows will be loaded after approx. 30 sec. Position the mouse in one of the windows.
%	ghte ↵	Start of GHTE.
Please enter password:	* ↵	Enter HTE password.
Variable password expires: 04-Aug-1998 Please enter/complete variable password: [S3K769M409QZ ...]	. . . ↵	Enter the last three characters of the variable password. GHTE will be loaded after approx. 10 seconds.
HTE>		Refer to "Common HTE-commands" , page 141 for a description of the tests.

-
- *. If password is unknown, refer to the latest Speedinfo, folder: Password List

Command mode HTE

Prerequisites

- Unix is booted.

NOTICE If it is not possible to start the HTE test program in the graphic mode at the MR-console, the command mode may be called up at the MR console or at the laptop PC.

NOTICE During start-up of the HTE, the SMI initializes automatically. This normally takes about 10 seconds.

Output	Input	Notes
login:	service ↵	log in
Password	*	Enter password for service.
Last log in:.... SunOS Release 4.1 (MRSTART)...		
%	hte ↵	HTE command mode is started.

Output	Input	Notes
Please enter password:	* ↵	Enter HTE password.
Variable password expires: 04-Aug-1998 Please enter/complete variable password: [S3K769M409QZ ...]	. . . ↵	Enter the last three characters of the variable password. HTE will be loaded after approx. 10 seconds.
HTE>	help ↵	Help Infos are displayed. Refer to "Common HTE-commands", page 141 for a description of the tests.

-
- *. If password is unknown, refer to the latest Speedinfo, folder: Password List

Example: Starting one of the tests

Output	Input	Notes
HTE>	smi030 ↵	D30 is tested.
/all/quick/magic/mmx/ram		All options are displayed.
HTE>	smi030/quick ↵	Test is started.

Refer to ["HTE Description", page 137](#) for more information about all test commands.

Ending HTE

- To end the HTE test program, proceed as follows:

Output	Input	Notes
HTE>	end ↵	Type in end in window #3 or select the 'Exit' button in window #2.
%	exit ↵	Click outside the fields with the right mouse key and select 'Exit' in the Workspace menu.
%	logout ↵	Exit service mode.
		If you want to switch off the system, type in the following.
login:	halt ↵	UNIX will be ended. Subsequently, switch off the power.
ok		Switch off the system.

Analyzing HTE messages

During testing with the HTE all inputs (HTE commands) and outputs (Messages) regarding test programs are reported in a protocol ring file (default name: hteprt.dat). Each entry has a time stamp. The file is located in the following path:

/usr/export/home/service/hteprt.dat

To analyze this file, use the HTE command '**protocol/show**'. Refer to "[protocol](#)", [page 142](#) for more detailed information.

During testing with the HTE, all error messages detected by test software are saved in an error ring file (default name: hterr.dat). Each message has a time stamp.

The file is located in the following path:

/usr/export/home/service/hterr.dat

To analyze this file, use the HTE command '**errlog/show**'. Refer to "[errlog](#)", [page 141](#), for more detailed information.

NOTICE

If you cannot analyze the files, send them to Erlangen.

HTE Description

Introduction

The HTE is used to test the SMI 5 hardware. It is possible to run the HTE in two different modes:

- command mode
- graphic mode

The basic version of the HTE test frame supports command-driven interactions. The HTE supports the management of hardware test programs. This command mode can be used if it is not possible to start the more comfortable graphic mode. The HTE command mode runs under the SUN/UNIX operating systems. Either a SUN terminal or a laptop (connected to a serial SUN interface) is suitable as a user interface.

In the graphic mode, a graphics interface is used for operating the basic version of the HTE. Each graphic input results in a command string, which is sent to the HTE and processed as if it were entered directly via the keyboard. The graphic mode runs under Openwindows.

Description of graphics user interface

The user interface created for the SMI5 service test should cover the entire monitor screen. In the following graphic, a typical example is given.

[illegible]

The screen can be divided into four segments:

- Window #1
contains operating graphics for calling the individual test commands. It indicates the status of the hardware configuration.
- Window #2
contains operating elements with HTE internal commands.
- Window #3
Terminal window. The basic version of HTE runs under Window #3.
- Window #4
illustrates the command tree corresponding to the command selected in Window #1.

Operation

Window #1

The command to call the test programs of each board is displayed above each board in the corresponding button of the displayed hardware configuration.

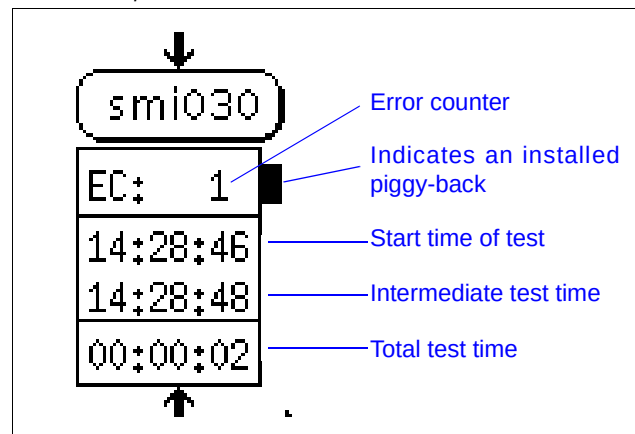
All commands allocated for testing individual boards begin with the prefix “smi” and end with the 3-digit “D” number of the board (for example, D21 - command “smi021”).

Tests higher than board level (for example system test and initialization) are identified by the command “smisys”. A button on the lower left part of window #1 displays this command.

The operating elements used for selecting commands are displayed in the form of buttons.

A command is selected by positioning the mouse cursor within the outline of the button and pressing the left mouse key. The following appears in the rectangle of the individual boards in the order shown below.

Fig. 21 HTE - Representation of a board



The button selected is highlighted. The command tree for this command is displayed in window #4.

If additional boards are plugged into the board selected (“piggy-back”). A “subwindow” is opened with the information described in the following section.

Window #4

A command branch is selected by clicking the left mouse key at the end of the desired branch. The connecting lines and the buttons are highlighted indicating the selected branch.

After a branch is selected, the test is started by clicking the “GO” button. The test program may be started directly using an alternative method, i.e. by double clicking the left mouse key at the end of the branch.

Window #2

Window #2 contains buttons with internal test frame commands.

After a command is selected, various types of subwindows are displayed offering different information.

help

Commands for which a help file exists are available in a “pull-down menu” (entry in the command file).

The command line “help <command>” is transferred to window #3 when a command is selected (right mouse key), and the help file in the “more” mode is displayed. (as with the Unix command ‘more’)

If the help button is selected using the left mouse key, the string “help help” is transmitted to HTE. A short explanation and a list of the parameters for the help command are displayed.

recall

The command “recall” is transmitted to window #3 and the history buffer is displayed.

set

The “global values” are indicated in a subwindow and may be modified. The corresponding command record is sent to window #3 and executed.

go

A command (end point in the command tree) selected in window #4 may be confirmed by pressing the GO button.

exit

This command ends the HTE.

Prior to using this command, any test programs or macros in progress must be ended via the “abort” command.

abort

This command terminates a test program (or a macro) in progress.

With this command, the “^C” is sent to window #3.

Window #3

This is the terminal window in which HTE runs in its basic version.

All graphically generated command records as well as the output from HTE commands are echoed in the terminal window.

- In this window, all HTE test programs can be started by typing in the correct command manually.

Refer to [“Common HTE-commands”, page 141](#), for more detailed information.

Common HTE-commands

NOTICE In this chapter, the most common HTE commands are briefly explained. To get more information on all possible HTE commands, use the HTE help information (e.g. type help smisys).

end

Ends the program normally.

You can also use the command 'exit'.

Type CTRL/C to abort a running test program. The HTE continues to run.

exit

Exits the program normally.

You can also use the command 'end'.

Type CTRL/C to abort a running test program. The HTE continues to run.

help

Entering 'help' displays an overview of the existing commands.

To get more detailed information, type 'help <command>'.

The output is in page mode. After each screen it pauses, shows an information line at the bottom of the screen, and waits for user input:

Possible inputs are:

press <SPACE>	- next page
press <RETURN>	- next line
press h	- get help menu
press b	- previous page
press f	- next page
/<expression>	- search for first occurrence of expression
press q	- end output

errlog

Error messages detected by test software are saved in an error ring file (default name: hterr.dat). Each message has a time stamp.

The terminal output is in page mode. After each screen it pauses, shows an information line at the bottom of the screen, and waits for user input.

Possible inputs are:

press <SPACE>	- next page
press <RETURN>	- next line
press h	- get help menu
press b	- previous page
press f	- next page
/<expression>	- search for first occurrence of expression
press q	- end output

Examples:**HTE>errlog/show**

start-dd-mmm-yyyy-hh-mm >**start-19-JAN-1990-11-00**
 end-dd-mmm-yyyy-hh-mm >

Displays all entries after January 19th 1990 11.00 a.m.

HTE>errlog/show start-today end-today

Displays all of today's entries.

HTE>errlog/show start-yesterday end-today

Displays all entries since yesterday.

HTE>errlog/copy start-today end-today /usr/service/newerr.dat

Saves all of today's entries in a file called "newerr.dat" in path
 "/usr/service".

errlog/init Initializes the error log file. All old entries are lost.

protocol

Inputs (HTE commands) and outputs (Messages) concerning test programs are reported in a protocol ring file. Each entry has a time stamp.

The terminal output is in page mode. After each screen it pauses, shows an information line at the bottom of the screen and waits for user input:

Possible inputs are:

press <SPACE>	- next page
press <RETURN>	- next line
press h	- get help menu
press b	- previous page
press f	- next page
/<expression>	- search for first occurrence of expression
press q	- end output

Examples:**HTE>protocol/show**

start-dd-mmm-yyyy-hh-mm >**start-19-JAN-1990-11-00**
 end-dd-mmm-yyyy-hh-mm >

Displays all entries after January the 19th 1990 11.00 a.m.

HTE>protocol/show start-today end-today

Displays all of today's entries.

HTE>protocol/show start-yesterday end-today

Displays all entries since yesterday.

HTE>protocol/copy start-today end-today /usr/service/newprt.dat

Saves all of today's entries in a file called "newprt.dat" in path "/usr/service".

protocol/init Initializes the protocol file. All old entries are lost.

set

Changes global value settings. Not every global value can be changed at HTE run time.

terminal	on(off)	- Activate(deactivate) terminal output of test programs.
set/printer	(on)off	- Activate(deactivate) printer output of test programs.
set/file	open(close)	- Activate(deactivate) output to error log- and protocol-file.
set/output	full(brief)	- The whole error history is displayed (only the first error is displayed).
set/startmess	yes(no)	- Message (no message) about the start of a test program.
set/endmess	yes(no)	- Message (no message) about the end of a test program.
set/echo	on(off)	- At macro execution the single commands are echoed (not echoed).
set/info	yes(no)	- Display (don't display) information.
set/warning	yes(no)	- Display (don't display) warnings.
set/error	yes(no)	- Display (don't display) errors.
set/fatal	yes(no)	- Display (don't display) fatal errors.
set/trace	yes(no)	- Display (don't display) trace messages of test programs.
set/debug	(yes)no	- Reserved for internal usage.
set/version	yes(no)	- Test programs display (don't display) the version.
set/flog	(yes)no	- Detailed Information about test program execution is protocolled in the HTPFLOG.DAT file (or not protocolled).
set/autoreboot	yes(no)	- After asynchronous test program abortion, (CTRL/C or runtime limit exceeded) an automatic smi-reboot is performed (not performed).
set/protocol[:value]		- Sets the value for maximum number of entries in the protocol file
set/errlog[:value]		- Sets the value for maximum number of entries in the error log file
set/maxerr[:value]		- Sets the value for maximum number of errors that are counted after a subtest is aborted..

show

Lists all global value settings.

show/globals		- Lists the setting of all global values.
show/terminal	on/off	- Commands and messages from test programs are displayed/not displayed on the screen.
show/printr	on/off	- Commands and Messages from test programs are printed/not printed.
show/file	open/close	- Output is written/not written to errorlog- and protocol-file.
show/output	full/brief	- The whole error history is displayed/ only the first error is displayed.
show/startmess	yes/no	- Message/no message about the start of a test program.
show/endmess	yes/no	- Message/no message about the end of a test program.
show/echo	on/off	- At macro execution, the single commands are echoed/not echoed.
show/info	yes/no	- Display/don't display information.
show/warning	yes/no	- Display/don't display warnings.
show/error	yes/no	- Display/don't display errors.
show/fatal	yes/no	- Display/don't display fatal errors.
show/trace	yes/no	- Display/don't display trace messages of test programs.
show/debug	yes/no	- Reserved for internal usage.
show/version	yes/no	- The version of the test program is displayed/not displayed.
show/flog	yes(no)	- Detailed Information about test program execution is protocolled in the file HTPFLOG.DAT (or not protocolled).
show/autoreboot	yes(no)	- After asynchronous test program abortion, (CTRL/C or runtime limit exceeded) an automatic smi-reboot is performed (not performed).
show/maxhistory		- Length of command history buffer. The last 16 commands are saved and can be recalled and executed again.
show/protocol		- Maximum number of entries in protocol file.
show/errlog		- Maximum number of entries in error log file.
show/maxerr		- Maximum number of errors that are accumulated after a subtest is aborted. Default is 4.

smiinit

The printout of the SMI initialization during the HTE can also be used for a quick check of the SMI 5.

During the SMI Initialization, a hardware reset for the entire SMI system is performed and each board is booted. At this time, the existing configuration is tested and the boards the system finds are displayed on the monitor in the order in which they are connected in the SMI (in ascending order from left to right)

During initialization, the following checks will be performed:

- hardware reset of the whole SMI system
- proper sysbus access to each board is tested (boot of each board)
- a GIPSY self-test is performed
- a GIPSY code RAM and data RAM test are executed
- the GIPSY IDPROM and BootPROM are checked (checksum test)
- a check of the kernels to see if they are running or not.

The output shows information about the boards as they are plugged into the frame from left to right.

After "smiinit", all boards should be completely initialized and all GIPSY kernels should be running.

Example of a smiinit printout

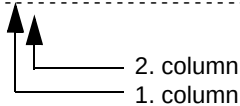
```
Hardware Test Environment (HTE) Version 4.7 (03-SEP-1993)
BEGIN OF STARTFILE
  NAME OF STARTFILE:  tstini.dat
  START-TIME:        Fri Sep 24 11:09:45 1993
%HTE-I-ECHO, set/trace yes
%HTE-I-ECHO, set/version yes
%HTE-I-ECHO, set/autoreboot yes
%HTE-I-ECHO, smisys/init
%HTE-I-STARTMESS, htp (pid 1230) started, max. runtime 120 sec
%SMIHT-I-HTP_, Host test-support-module >>HTP<< VERSION: V4.01 20-JUL-93
%SMIHT-I-HTP_, Date: 24-Sep-93 Time: 11:09:46
%SMIHT-I-BC1, -----
%SMIHT-I-BC1,      smi  Gno  Ver BoardID IDProm CkB DkB EMB Xt      BootProm
%SMIHT-I-BC2,      021:1 0100 V02 1643100 060893 128 126 032 s+ SIAMC_T 141092
%SMIHT-I-BC2,      070:1 0300 V01 5470448 030693 128 112 000 s+ IMP_ML 051192
%SMIHT-I-BC2,      ( 055:1 0500 V02 2819840 050793 128 118 288 s+ SGPR0M 030190
%SMIHT-I-BC2,      ( 055:2 0600 V02 2819840 050793 128 118 288 s+ SGPR0M 030190
%SMIHT-I-BC2,      030:1 0a00 V03 8965683 200991 128 126 008      SGPR0M 030190
%SMIHT-I-BC1, -----
%SMIHT-I-BC5, smi021:1 onboard ram: 032MByte AddrRange:00000000-00ffffff
%SMIHT-I-BC5, smi055:1 onboard ram: 032MByte AddrRange:00000000-00ffffff
%SMIHT-I-BC5, smi055:1 module 1 ram: 256MByte AddrRange:01000000-08ffffff
%SMIHT-I-BC5, smi055:2 onboard ram: 032MByte AddrRange:00000000-00ffffff
%SMIHT-I-BC5, smi055:2 module 1 ram: 256MByte AddrRange:01000000-08ffffff
%SMIHT-I-BC4, smi030:1 module 2 is smi024 V00 BoardID:5454301 IDProm:010390
%SMIHT-I-HTP_, Date: 24-Sep-93 Time: 11:09:54
%HTE-I-ENDMESS, htp (pid 1230) finished, exit status 1
END OF STARTFILE
  NAME OF STARTFILE:  tstini.dat
  END-TIME:          Fri Sep 24 11:09:54 1993
```

NOTICE

The HW recognized has to be identical with the boards physically installed.

Description of the smiinit printout

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
smi	Gno	Ver	BoardID	IDProm	CkB	DkB	EMB	Xt		BootProm
021:1	0100	V02	1643100	060893	128	126	032	s+	SIAMC_T	141092
-070:1	0300	V01	5470448	030693	128	112	000	s+	IMP_ML	051192
(055:1	0500	V02	2819840	050793	128	118	288	s+	SGPROM	030190
(055:2	0600	V02	2819840	050793	128	118	288	s+	SGPROM	030190
-030:1	0a00	V03	8965683	200991	128	126	008		SGPROM	030190
*066:1	0									



1. A "(" in the 1. column indicates that this board is part of a map cluster. A special symbol in the 2. column shows the status of the board after initialization:

symbols	meaning	test working?
(blank)	Succesfully initialized	Yes.
- (hyphen)	Existing board not in boot file	For the most part.
~ (tilde)	Kernel not running	No.
* (asterisk)	Error during boot test	Maybe. Try.
? (question mark)	Invalid version of GIPSY kernel	Maybe. Try.

2. The D-Number of the board is followed by a colon and an ordinal number. This ordinal number counts boards with the same D-Number from left to right.
3. The board's GIPSY-Number (its address). This number is not fixed, but written into the board during the boot sequence.

4. The board's hardware version number.
5. The board's ID (Part number/Sachnummer).
6. The version date of the boards ID Prom (DDMMYY - Day-MonthYear).
7. The length of GIPSY code RAM in kilobytes.
8. The length of GIPSY data RAM in kilobytes.
9. The length of GIPSY external 2 RAM (main memory) in megabytes.
10. "s" means that the board is able to perform multimux secondary lock.
"+" means that the board has a magic+ chip.
11. The GIPSY boot PROM version code.

CAUTION **ATTENTION! This information is only useful if the board's ID PROM is correct.**

Additional information is displayed for the DATA-IO-board (D30), SIAM-C (D21) or the module-MAP-board (D55) :

```
smi021:1 onboard ram: 032MByte Addrance:00000000-00ffffff
smi055:1 onboard ram: 032MByte Addrance:00000000-00ffffff
smi055:1 module 1 ram: 256MByte Addrance:01000000-08ffffff
smi055:2 onboard ram: 032MByte Addrance:00000000-00ffffff
smi055:2 module 1 ram: 256MByte Addrance:01000000-08ffffff
smi030:1 module 2 is smi024 V00 BoardID:5454301 IDProm:010390
```

smisys

Available smi system tests.

smisys/all	Tests the entire SMI, i.e. each existing board is tested with the corresponding program and optionall . The following interactive tests are not done: - smi070/visual - smi070/ram/graphic - smi070/duart - smi066/visual - smi066/digital - smi066/serial
smisys/quick	MI-5 Quick-Test. The quick tests are performed for all boards in the configuration (smi0xx/quick).
smisys/ECinit	Performs the following actions: - error counters are set to zero - inverted boards in the graphic window are lit up - trailing time reports are output in the graphic window
smisys/cmdinit	Generates a new command file for the HTE program that contains only the currently available test commands.

NOTICE	To activate the new command file, you have to exit HTE and restart it.
---------------	--

smisys/errmsg

- n=0 (default) ignore the characters in the message literal up to the second hyphen while searching
- n!=0 use the entire message literal for search

The 'err_lit' command prompts for a message literal for which you want help. If found, the information contains the message literal itself, the hexadecimal message code, the parameterized message text and, if available, an explanation.

Within 'message literal' you may specify the wildcard '*', where the '*' stands for any number of characters (e.g. for the input '*ima*' you'll get information on 'smigt-e-te_imac21', 'smigt-e-te_imamacroa' and 'smigt-i-ti_mi_rimage').

To exit the command, press 'return'. With '? return' you can access online help.

smiparallel

Parallel test of all remote boards. The interface board (SIAM-C) is not tested.

smisys/parallel/all

All gipsy RAM and main RAM tests are executed.

smisys/parallel/quick

Gipsy RAM and main RAM housenumber tests are executed.

Board specific HTE commands

Board	Goto
SIAM-C D21	page 148
Data Receiver D 24	page 149
Data-IO D 30	page 149
MAP D 55	page 150
Imager-ML D 70	page 151
Maxicam D66	page 152

SIAM-C D21

The following HTE commands are suggested to test the board:

smi021/all	Executes the Magic-, MMX-, RAM-tests.
smi021/quick	Full magic-test, quick MMX-test, gipsyram and mainram housenumber test.
smi021/lisa	Performs the ASIC LISA test; a test partner is required. The default test partner for the SIAM-C board is the D70 imager.
smi021/mad	Performs the ASIC MAD test; a test partner is required. The default test partner for the SIAM-C board is the D70 imager.

smi021/mmx

Executes multimux tests, a test partner is required.

As a default test partner the next board on the right is taken. Normally this is the Imager board.

CAUTION

Note that errors in the Multimux test can also be caused by the test partner (Imager). Ensure that the test partner is functioning properly prior to the test.

To eliminate errors from boards not required for tests involving the MMX-BUS, execute the tests using the minimal configuration (refer to **“Partial configuration”**, [page 132](#) in this document).

smi021/ram/all

Performs the RAM test on the board. The gipsyram and the mainram tests are executed.

NOTICE

To get more information on all possible HTE commands for the respective board, use the help information of HTE (e.g. type help smi021).

Data Receiver D 24

The following HTE commands are suggested to test the board:

smi024	Tests the data-IO/MR-receiver interface by using the testmode0 (transfer of random data from data-IO to MR-receiver) and testmode1 (transfer data back from MR-receiver to the extern-2-RAM). The data received are compared with the original data in the gipsy data RAM.
---------------	---

Data-IO D 30

The following HTE commands are suggested to test the board:

smi030/all	Subtests MAGIC, MMX and RAM are performed.
smi030/quick	A quick check, not all subtests are performed. The full magic-test, quick MMX-test, gipsyram and mainram housenumber test are executed.
smi030/magic	Tests all registers of the MAGIC-asics.
smi030/mmx	Tests the Multimux, a test partner is required. the default test partner is the SIAM-C board.

CAUTION

Note that errors in the Multimux test can also be caused by the test partner (SIAM-C). Ensure that the test partner is functioning properly prior to the test.
To eliminate errors from boards not required for tests involving the MMX-BUS, execute the tests using the minimal configuration (refer to [“Partial configuration”](#), page 132 in this document).

smi030/ram/all	Performs the RAM test on the board. The gipsyram and the mainram tests are executed.
-----------------------	--

NOTICE

To get more information on all possible HTE commands for the respective board, use the help information of HTE (e.g. type help smi030).

MAP D 55

The following HTE commands are suggested to test the board:

smi055[:m]/all	Subtests MAGIC, MMX and RAM are performed. If more then one D55 is installed, boards are numbered from left to right. The first one is smi055:1, the second one smi055:2 and so on. If m is omitted, m = 1 is taken by default.
smi055/quick	A quick check, not all subtests are performed. The full magic-test, quick MMX-test, gipsyram and mainram housenumber test are executed.
smi055/magicp/all	All MAGIC+_subtests are performed in succession.
smi055/mmx	Tests the Multimux, a test partner is required. The default test partner is the SIAM-C board.

CAUTION Note that errors in the Multimux test can also be caused by the test partner (SIAM-C). Ensure that the test partner is functioning properly prior to the test.
To eliminate errors from boards not required for tests involving the MMX-BUS, execute the tests using the minimal configuration (refer to chapter [“Partial configuration”](#), page 132 in this document).

smi055/ram/all	Performs the RAM test on the board. The gipsyram and the mainram tests are executed.
smi055/mad	Tests the ASIC MAD. A test partner is required. If no test partner is specified, D21 (SIAM-C) is the default partner.
smi055/mc88110/all	All processors perform all CPU tests in succession. Finally the simultan test is performed.
smi055/mc88110/simul	Simultan test. All 4 processors start at the same time to transfer simultaneously 5 MByte from the same source-buffer to different destination-areas.

NOTICE To get more information on all possible HTE commands for the respective board, use the help information of HTE (e.g. type help smi055).

Imager-ML D 70

The following HTE commands are suggested to test the board:

smi070/all	Subtests MAGIC, MMX and RAM are performed. (except ram/graphic, duart and visual tests)
smi070/quick	A quick check, not all subtests are performed. The full magic-test, quick MMX-test, gipsyram and mainram housenumber test are executed.
smi070/magic	Tests all registers of the MAGIC-asics.
smi070/mmx	Tests the Multimux, a test partner is required. The default test partner is the SIAM-C board.

CAUTION	Note that errors in the Multimux test can also be caused by the test partner (SIAM-C). Ensure that the test partner is functioning properly prior to the test. To eliminate errors from boards not required for tests involving the MMX-BUS, execute the tests using the minimal configuration (refer to chapter “Partial configuration”, page 132 in this document).
----------------	--

smi070/ram/all	Performs the RAM test on the board. The gipsyram and the mainram tests are executed.
smi070/dac	Tests the DAC.

smi070/gsp	Tests the GSP.
smi070/selector	Executes special MMX tests for which a test partner is required. The default partner is SIAM-C D21.
smi070/pmx	Pixelmixer test with image ram.
smi070/visual	Tests image display on the monitor. Different samples can be selected.
smi070/duart	Tests the serial interfaces. Note: Additional test hardware is required. The two serial connectors at the front of the Imager-ML board must be connected together with the test cable (Null-Modem cable) before the 'duart'-test can be executed.

NOTICE	To get more information on all possible HTE commands for the respective board, use the help information of HTE (e.g. type <code>help smi070</code>).
---------------	---

Maxicam D66

The following HTE commands are suggested to test the board:

smi066/all	Subtests MAGIC, MMX and RAM are performed. (except visual, digital and serial tests)
smi066/quick	A quick check, not all subtests are performed. The full magic-test, quick MMX-test, gipsyram and mainram housenumber test are executed.
smi066/magic	Tests all registers of the MAGIC-asics.
smi066/mmx	Tests the Multimux, a test partner is required. the default test partner is the SIAM-C board.

CAUTION Note that errors in the Multimux test can also be caused by the test partner (SIAM-C). Ensure that the test partner is functioning properly prior to the test. To eliminate errors from boards not required for tests involving the MMX-BUS, execute the tests using the minimal configuration (refer to [“Partial configuration”](#), page 132 in this document).

smi066/ram/all	Performs the RAM test on the board. The gipsyram and the mainram tests are executed.
smi066/dac	Tests the DAC.
smi066/gsp	Tests the GSP.

smi066/selector	Executes special MMX tests for which a test partner is required. The default partner is SIAM-C D21.
smi066/pmx	Pixelmixer test with image ram.
smi066/visual	Tests image display on the monitor. Different samples can be selected.
smi066/digital	Test of digital camera interface. The test is divided into two parts: - Part one (static test) checks the control lines. - Part two (dynamic test) does a real data transfer and checks the result.
smi066/serial	Test of the two serial interfaces. The tests assume that the two interface connectors are joined with the assigned cable.

NOTICE For testing the digital and the serial interfaces of the MAXICAM, a special test hardware kit is necessary. You can order this test hardware kit in Erlangen under the following number: P.n. 99 00 416 GE999.
A description is included with the test kit.

NOTICE For camera troubleshooting see also under “Camera”, Software part, in the manual Replacement of Parts.

NOTICE To get more information about cameras, you can order the binder "Service Info, Cameras", print number PA0-00.064.01.
All possible tests for the interface to the camera are described in this binder.

NOTICE To get more information on all possible HTE commands for the respective board, use the help information of HTE (e.g. type help smi066).

Control

Strategy

A description for "**Strategy**" and "**Procedures**" for the MMC is deemed unnecessary since the MMC tests are all inclusive. When confronted with problems concerning the MMC, it is suggested to run the complete battery of tests irrespective of which component may be suspected.

Information about the tests can be found in the online-help of the SeSo test tools. In the event of failures, information to the result and probable cause will be given by the test programs.

RF-System

In this part you will find instructions for the RF-System, containing the **Strategy**, the **Test Procedures** and the **Repair** section.

Strategy

General

One of the essential prerequisites for excellent image quality of a MR-system is a correct and stable RF system.

However, due to the complexity of a MR-system, image quality is influenced by a great variety of components and functions, such as

- Communication systems (e.g. CAN bus, general data transfer,...)
- Control systems (e.g. Host, MMC,...)
- Image generating systems (e.g. RF system, gradient system,...)
- Image acquisition, processing and archiving systems (e.g. MMC, SMI,...)
- Software performance

The main tool for RF troubleshooting are the **RF tests**, available in the service software platform **"Test Tools"**:

The RF test includes communication tests, measurements of the stability, linearity and signal levels of RF components using loop tests, which step by step include more components of the TX/RX path.

In case one of the test fails, a diagnosis will show you the FRU's responsible.

Additional hardware tests are described in the section on **"Procedures"**; the section on **"Repair"** lists the necessary adjustments and final checks when replacing a FRU.

The following is an attempt to provide the FSE with a shortcut that starts with the symptom and ends with identifying the respective FRU in the RF system by using the corresponding test.

Error classification

RF related hardware errors, communication errors

Symptoms	Service action	References
Specific error message Break-down of a specific component (FRU) Adjustments/measurements fail	Check error messages in System/Report/Details for remedy. Check error LED's. Check power supplies, fuses. Consider the possibility of monitoring errors -> check error transmission lines (FOC's, optical transmitter and receiver). Replace defective component; or replace components of equal type (if used in the system) with each other. Perform specific RF component tests.	Functional description of the corresponding FRU's Binder RF system Diagrams RF procedures
Communication error	Run Test Tools : RF: Comm/Status MMC: ISA/CAN, Comm/Status Check communication lines (CAN-units of RF system are: RFSU/TX-module, RFPA, RFAS (BTB,LCCS), RFCI)	Test tools On-line help
Noise image only	Check cables/connections in the RF path Check signal level of RF components with scope.	RF procedures

RF related image quality problems

- Specific QA procedures such as S/N DIP, CalcArtifacts, RF Verify or Stability Check fail and/or general image quality complaints exist

Symptoms	Service action	References
Ghosting in phase encoding direction independent of orientation	Run the RF tests; check the results of stability evaluation	Test Tools RF, on-line help
Low S/N	Run the RF tests; check the results of stability and linearity evaluation. Eliminate possible spike sources.	
Slice profile broadening	Run the RF tests; check the results of linearity evaluation. Check modulator SSB1 with scope	Tune-up, on-line help; RF procedures
IQ problem depends on specific coil	Check the local coil visually for damages. Run the RF tests for the specific RF-path Check the tuning/detuning signals with the help of the service plug (RFCI/LCCS interactive tests). Use another coil of the same type.	RF procedures
IQ problem depends not on specific coil	Run the RF tests especially: Mod/REC1,2; RFPA; RFAS	Test Tools RF, on-line help

Proposed order of RF test procedures

Step	Service action	References
0.	Prerequisites/Preliminary tests	
0.1.	Run Test Tool - Communication Test (MMC/MPCU-, MMC/ISA/CAN-, RF-Comm/Status)	Test Tools On-line help for specific test
0.2.	Run Tests Tools - MMC tests (MC4C40, RX4)	
0.3.	Run Test Tools - SMI tests	
1.	RF tests (a full test takes about 45 minutes)	
1.1.	Run Test Tools RF tests	Test Tools On-line help for specific test
1.2.	Perform specific procedures in case of doubt	RF procedures
2.	Perform Tune-up/QA measurements	
To verify the success of the repair/replace- ment of RF components, specific adjustment must be performed (refer to RF repair proce- dures).		Start up manual: Tuning calibration, coil tune-up Tune-up on-line help QA on-line help
Finally, a complete run of QA is recom- mended.		

Procedures

General

This section describes additional tests for troubleshooting on components (FRU's) of the RF-System, such as voltage level and pulse form measurements at RFSU (modulator, synthesizer), RFPA, RFCI and local coil connectors.

Required Tools

- Functional description RF system (FUN)
- Diagrams RF system
- Set of standard tools
- 50 Ohm/2W BNC Termination resistor (7760457 B1947)
- QLA T-connector (e.g. Suhner 43QLA 01-0-1, or equivalent)
- Service cables QLA-BNC (2× 8733446 K2060)
- Connector BNC-Banana plug (3025749 B1947)
- Digital multimeter, type Fluke 8060A or equivalent
- Oscilloscope, 100MHz bandwidth
- Service plug for LCCS (4795501)

RFSU

Measurement of:

- Modulator SSB1 signal
- RFPA PF (forward wave) signal
- Synthesizer 10 MHz signal

NOTICE

If the MR system is equipped with a second RX_Module, these modules can be swapped for troubleshooting.

SSB1-Signal

Prerequisites

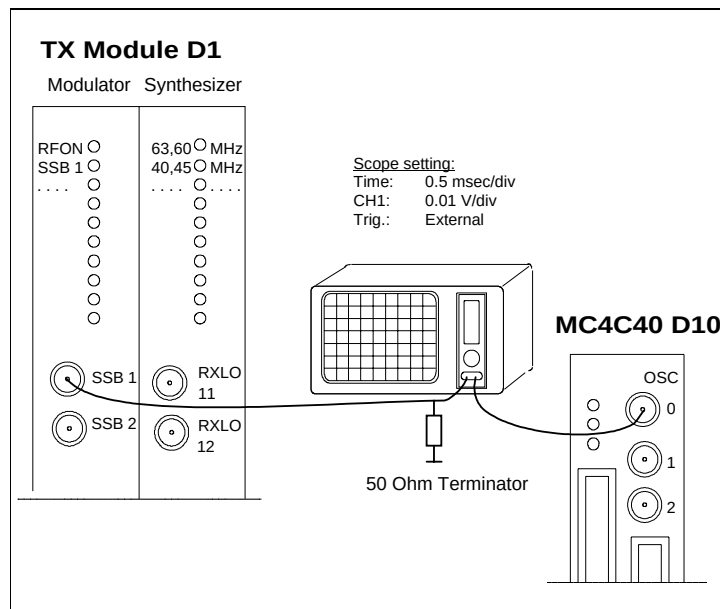
(Measurement of SSB1 and PF/RFAS)

- MR system ready for measurement
- Tuned body phantom with loader positioned in the magnet center.

Procedure

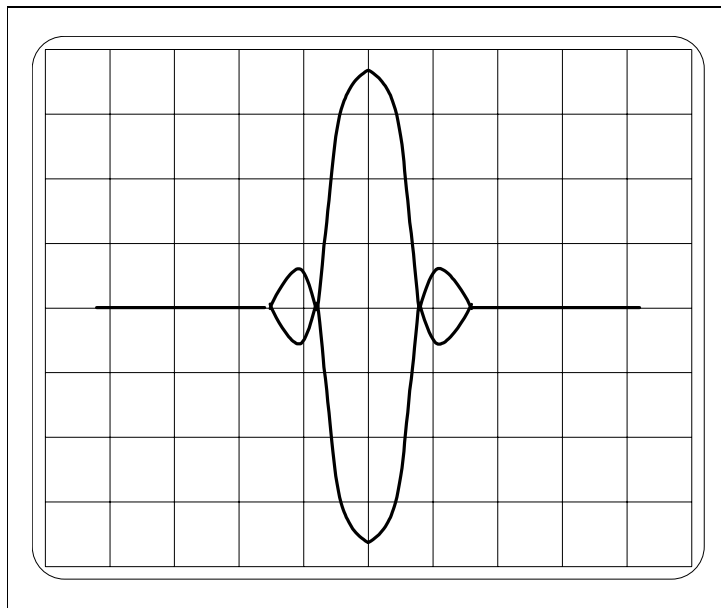
- Start the service software platform "Test Tools"; reduce the platform to an icon.
- Register respectively renew the **"Service Patient"** (System/Patient/Renew).
- Load the sequence: `/usr/appl/service/seq/rf_loop` using the Protocol/Change platform (use the button "Sequence File").
- Set No.Meas = 100; deselect Inline Adj.
- Click on the button "Adjust". In the Adjustment platform click on "Set". It may be necessary to first perform Adjust FRE, TUNE and ADJ/TRA.
- In the platform displayed set RF-amplitudes no.1-6 to 100 V. Leave the platform by clicking on "Apply".
- In the Adjustment platform you can check the setting with "Show actual". Leave the platform with "Cancel".
- Go back to the platform "Protocol/Change" with "Change".
- Remove the cable to RFPA from the SSB1 modulator output. Connect the oscilloscope to modulator output SSB1 via 50 Ohm terminator. Connect the MC4C40 output OSC 0 as external trigger to the oscilloscope.
- Start the sequence.

Fig. 22 Measurement configuration - Signal modulator SSB1



- Oscilloscope setting: CH1: 0.01 V/div, Time: 0.5 ms/div, ext.Trigger.
- Monitor the SSB1 signal (first pulse = constant sinc-pulse) on the oscilloscope.
- A symmetrically-shaped signal must be continuously present, without voltage drop-off or overshoots.
- Typical value: 76 mVpp.

Fig. 23 Typical signal SSB1 (first sinc-pulse of sequence rf_loop)



- Abort the measurement
- Restore the original system configuration.

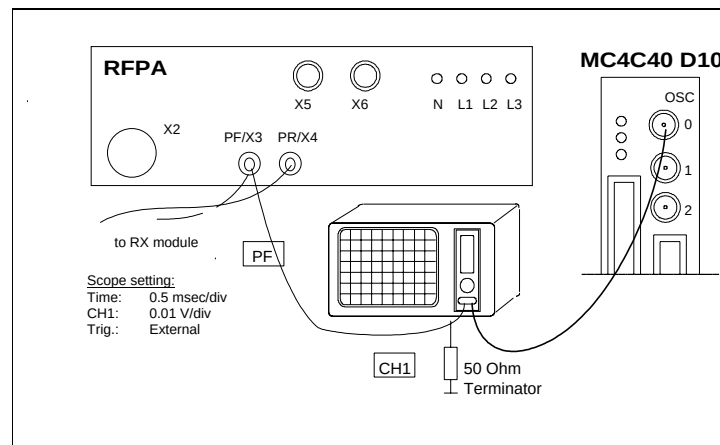
RFPA

Directional coupler signal PF

Procedure

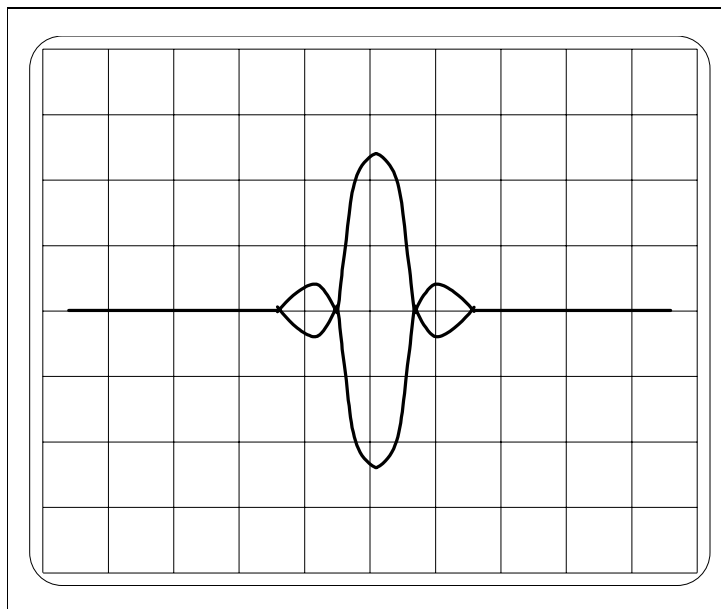
- Remove the cable to the RX module from PF output of RFPA. Connect the CH1 to X3/PF (forward wave of the directional coupler of RFPA) via 50 Ohm terminator. Connect the MC4C40 output OSC 0 as an external trigger to the oscilloscope.
- Start the sequence. The sequence is the same as for the SSB1 signal test.

Fig. 24 Measurement configuration - RFPA signal PF/PR



- Oscilloscope setting: CH1: 0.01 V/div, Time: 0.5 ms/div, ext.Trigger.
- Monitor the signal PF (first pulse = constant sinc-pulse) on the oscilloscope: A symmetrically-shaped signal must be continuously present, without voltage drop-off or overshoots. Refer to [Fig. 25](#).
- Typical value: 48 mVpp (same measurement parameters as SSB1 measurement). The attenuation of the directional coupler is approx. 70 dB at 63 MHz.

Fig. 25 Typical signal PF/RFA (first sinc-pulse of sequence rf_loop)



- Abort the measurement
- Restore the original system configuration.

10 MHz-Signal

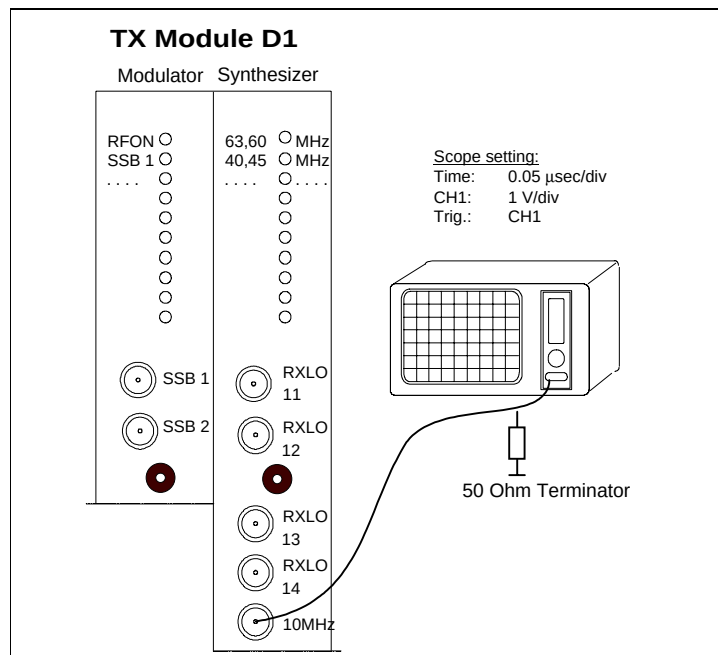
Prerequisite

- MR system is on.

Procedure

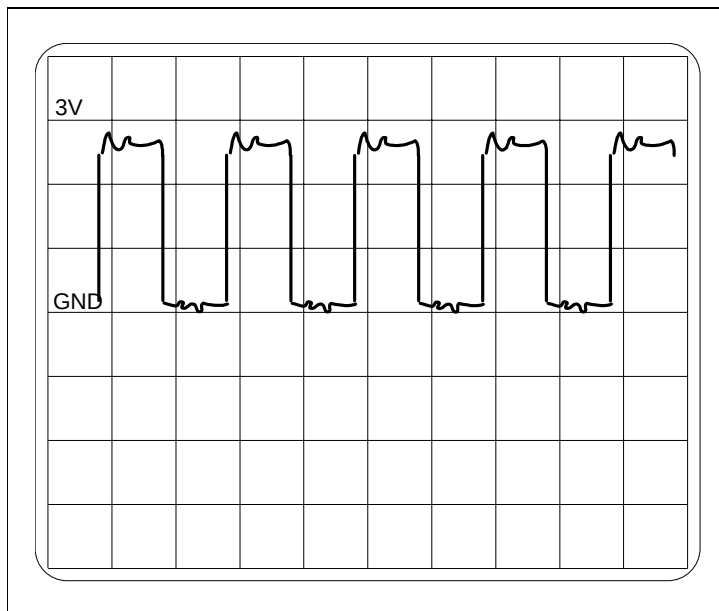
- Connect the oscilloscope to synthesizer output 10MHz via 50 Ohm terminator.
- Oscilloscope setting: CH1: 1V/div, Time: 0.05 μ sec/div, Trigger CH1.

Fig. 26 Measurement configuration - Synthesizer 10 MHz signal



- Monitor the 10 MHz signal on the oscilloscope (TTL-signal)
Nominal value:
High level: 2.6 V +/- 0.2V into 50 Ohm
Low level: max. 0.55 V into 50 Ohm
(rectangular shaped).

Fig. 27 Typical signal 10 MHz



- Restore the original system configuration.

RFAS

500 mV Reference voltage

Prerequisite

MR system is on.

Procedure

- Connect the cable from X7/TALES and a DVM via QLA T-connector to the CV input of Receiver 1/B.
- Switch off the circuit breaker F13 (main power supply for MMC) in the CCA cabinet.
- Switch on F13 again
 - ⇒ During the selftest of the MMC (duration approx. 1 minute, red LED's at RX/TX modules "TEST" light up), the reference voltage of TALES can be measured with the DVM:

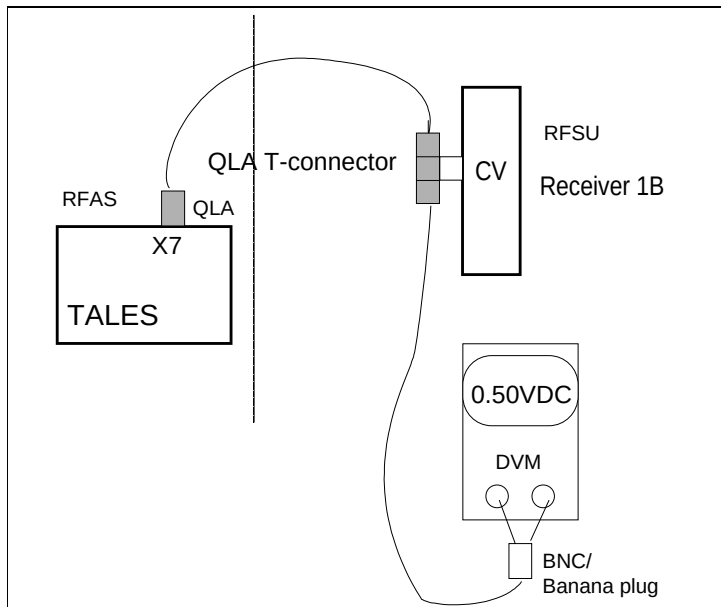
Nominal voltage: 0.475 to 0.525 VDC.

Additional Test of output multiplexer control (TALES)

- Remove the three FOC's U1,2,3 from TALES in the idle state of the MR system.
 - ⇒ All three FOC's are dark
- Switch off/on F13.
 - ⇒ During the selftest of MMC, all three FOC's light up.

(=> the reference voltage is switched to output X7/TALES by the multiplexer control).

Fig. 28 Measurement configuration - TALEs Reference voltage



- Restore the original system configuration.

BTB/RFCI

PIN Diodes control

Prerequisite

- MR system is on.
- Start the service software and select "Test Tools".

Procedure

- Select RFCI
 - ⇒ Follow the instruction of the service program.

After the semi-automatic tests a test for manual DVM measurements is started:

- ⇒ Alternately two voltages will be present at the test points:
 - Reverse voltage of PIN diodes (UR)
 - Forward current voltage drop (IF) at PIN diode (according a forward current of 100 mA for the local coils.

The service plug can be connected to the four plug positions successively without interrupting the test.

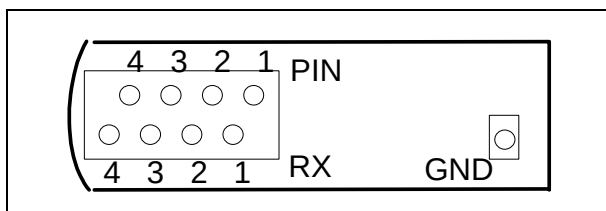
1. Measurements at the service plug:

The service plug is connected to coil connector (x=1...4):

Tolerance range:

- Reverse voltage (UR) at PIN diode: - 27 ... - 33 VDC
- Forward current voltage drop (IF): + 1.9 ... + 2.3 VDC

Fig. 29 Service plug - top view



- Measure the voltages at the four test points (PIN) of the service plug against GND.
- Measure the +10 (+/- 1) VDC RX-preamplifier supply voltage at the test points (RX). The voltage must be stable without drift and hum.
- For an exact measurement of the RX-preamplifier supply voltage, you have to abort the PIN diode test. (During the test, the voltage oscillates corresponding to the PIN diode switching).

2. Measurements at RFCI service board

Test points for local coil PIN diodes:

- J10 left side TP 1-4 --> coil connector 1, PIN diode 1-4
- J10 right side TP 5-8 --> coil connector 2, PIN diode 1-4
- J9 left side TP 9-12 --> coil connector 3, PIN diode 1-4
- J9 right side TP 13-16 --> coil connector 4, PIN diode 1-4
- J7 left top TP --> body coil, PIN diode located in the BTB

Description:

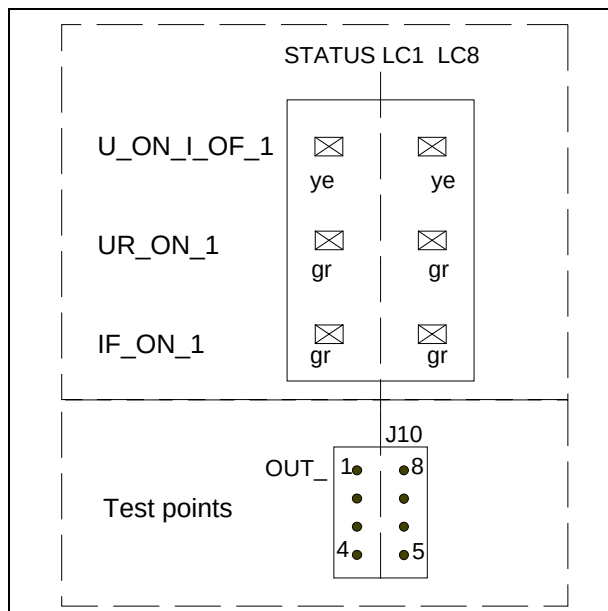
During the RFCI Interactive test the LED's at the RFCI service board alternating light up. There are 4x4 groups of LED's for each of the four PIN diodes connections of every coil plug. Fig. 30 shows the LED's for one PIN diode of two coil connectors.

U_ON_I_OF_1 on: PIN diode control active

UR_ON_x on: UR present at test point x against ground

IF_ON_x on: IF present at test point x against ground

Fig. 30 Service board RFCI: LED's of local coil PIN diode control and test points (partial view).



Tolerance range (local coil) PIN diode control:

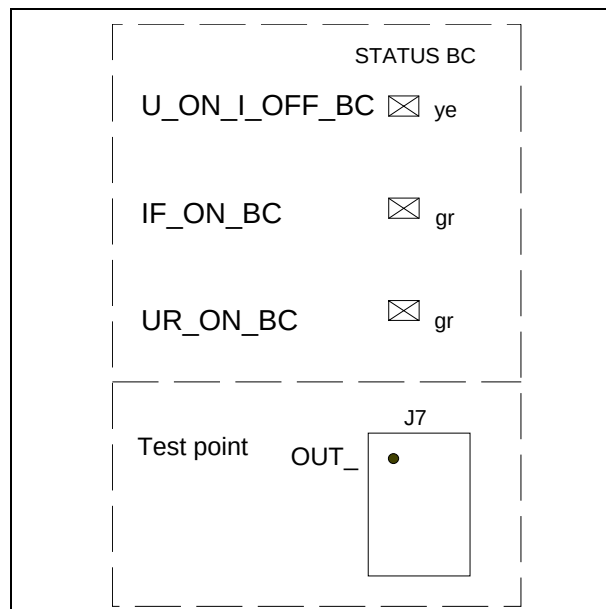
- Reverse voltage at PIN diode (UR): - 27 ... - 33 VDC
- Forward current voltage drop (IF): + 2 ... + 6 VDC
(dependent on the cable length between CCA and patient table).

Error:

Open circuit voltage (IF current regulator at limit):
approx. + 14 VDC

- Leave the voltmeter connected to a test point to see the two alternating voltages

Fig. 31 Service board RFCI: LED's of body coil PIN diode control and test points (partial view).



Tolerance range (Body coil) PIN diode control:

- Reverse voltage (UR) at PIN diode: - 27 ... - 33 VDC
- Forward current voltage drop (IF): + 3 ... + 8 VDC
(according to a forward current of 600 mA and dependent on the cable length between CCA and body tune box).

Error:

- Open circuit voltage (IF current regulator at limit):
approx. + 15 VDC

Conclusions:

1. Measurement at the service plug:

UR	IF	Conclusion
OK	OK	OK
NOT OK	OK	Error RFCI
OK	NOT OK	Error RFCI; PIN diode of service plug may be also defective.
10 VDC (LC preamplifier)		Conclusion
NOT OK		1. Broken or shorted cable between LCCS (X11/12) and coil connector 2. 15V to 10VDC conversion at LCCS defective

2. Measurement at RFCI service board local coil:

UR	IF	Conclusion
NOT OK	OK	Error RFCI
OK	NOT OK	Broken cable between RFCI and LC-connector
NOT OK	NOT OK	Error RFCI or short circuit in the cable between RFCI and LC-connector

3. Measurement at RFCI service board (Body coil):

UR	IF	Conclusion
OK	OK	OK
NOT OK	OK	Error RFCI/ Pin diode in the body coil defective
OK	NOT OK	Defective PIN diode of body coil or broken cable between RFCI and body tune box (BTB). Perform the service program: TestTools/BTB/BC/Functional.
NOT OK	NOT OK	Error RFCI or short circuit in the cable between RFCI and BTB or defective PIN diode of body coil. Perform the service program: TestTools/BTB/BC/Functional.

4. Additionally you can measure several supply voltages at the RFCI service board.

"Excessive reflected RF pulse power was detected by TALES"

This message appears within System/Report and within SeSo -> EventLog.

It is very likely caused by arcing components (e.g., bad cable connection/contact) in the transmit path (RF cables between RFPA and body coil).

Fig. 32

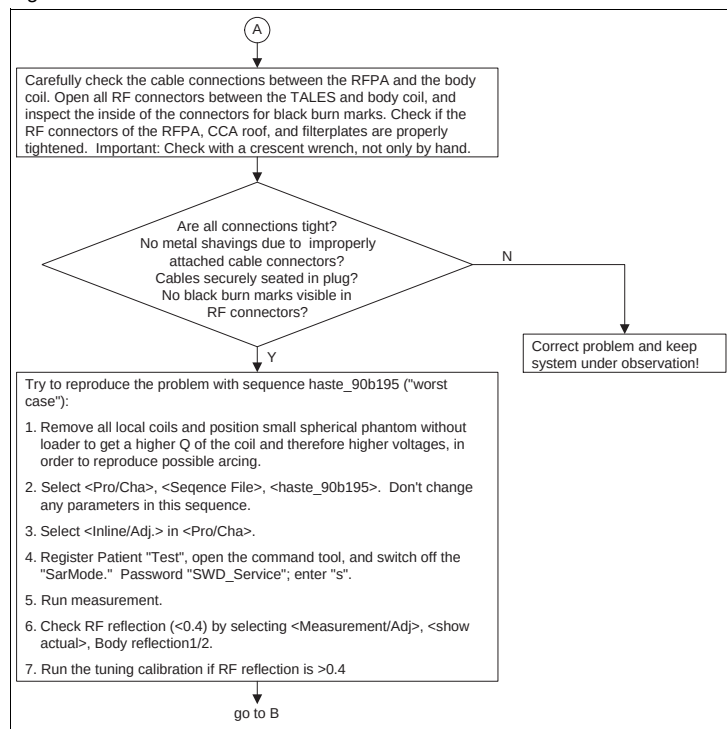


Fig. 33

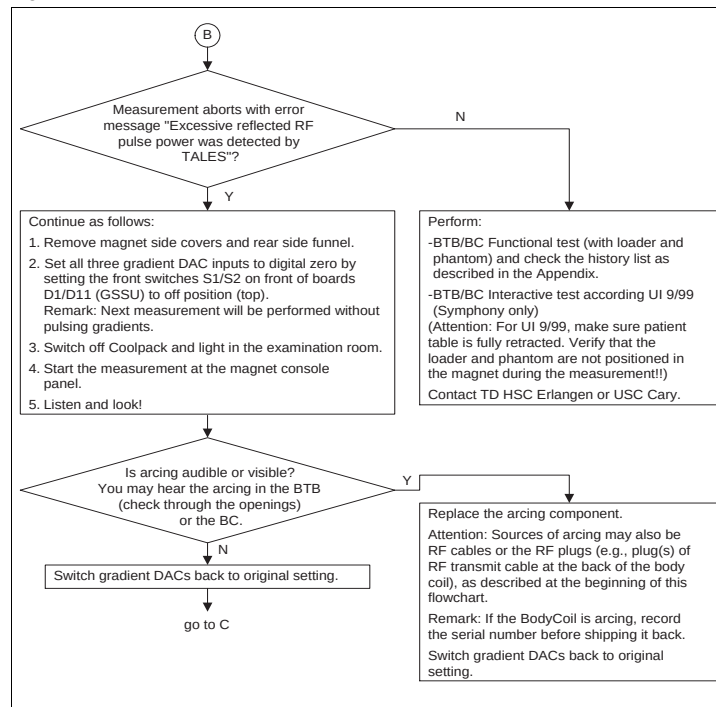


Fig. 34

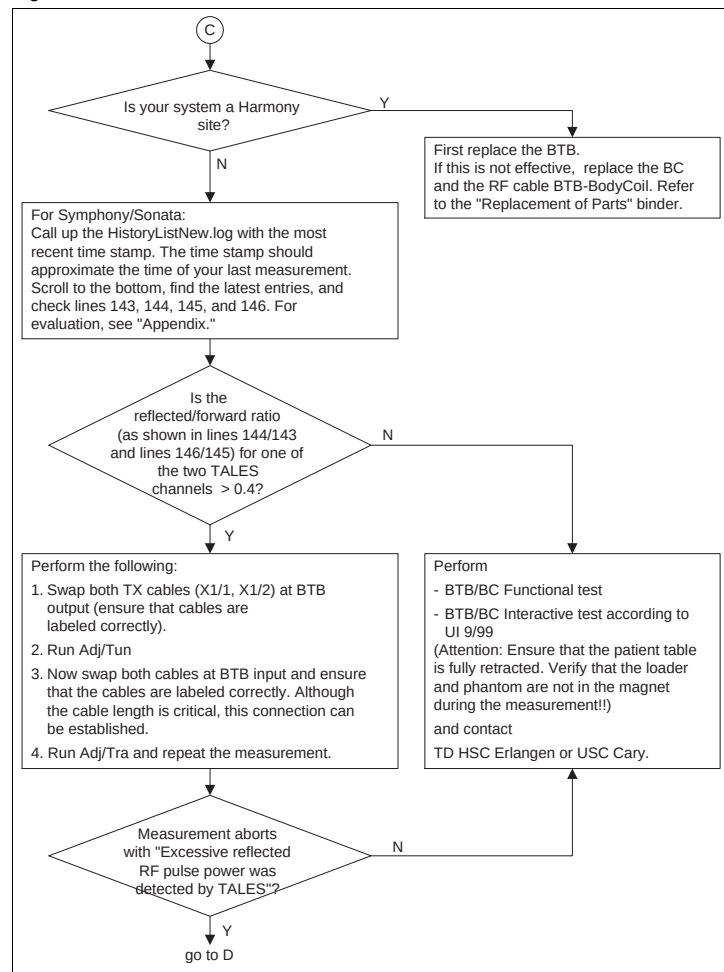
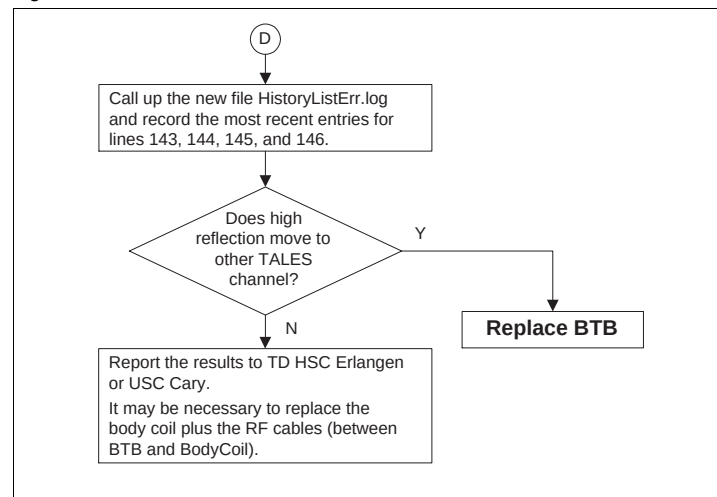


Fig. 35



"RFPA power fault" / "RFPA overcurrent"

Fig. 36

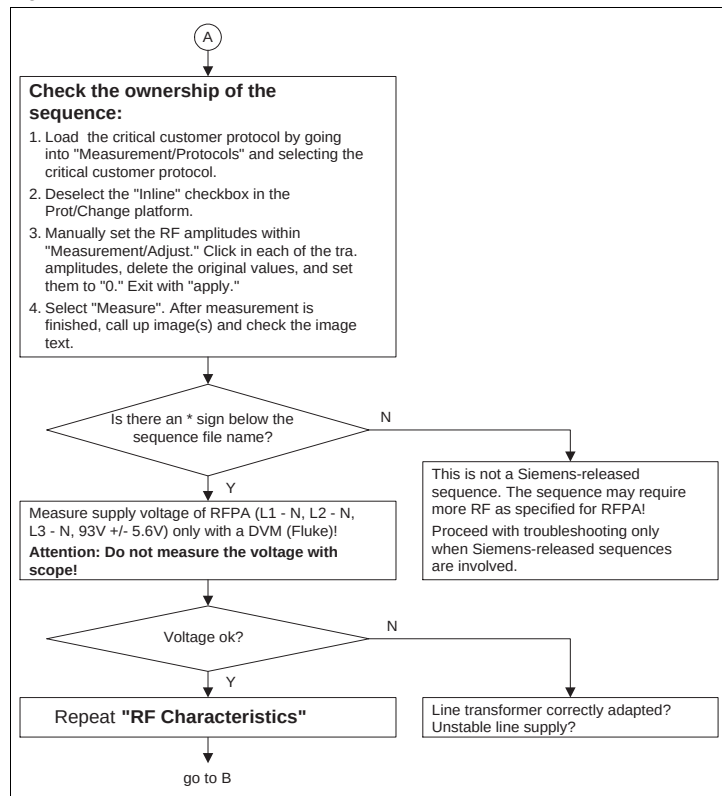
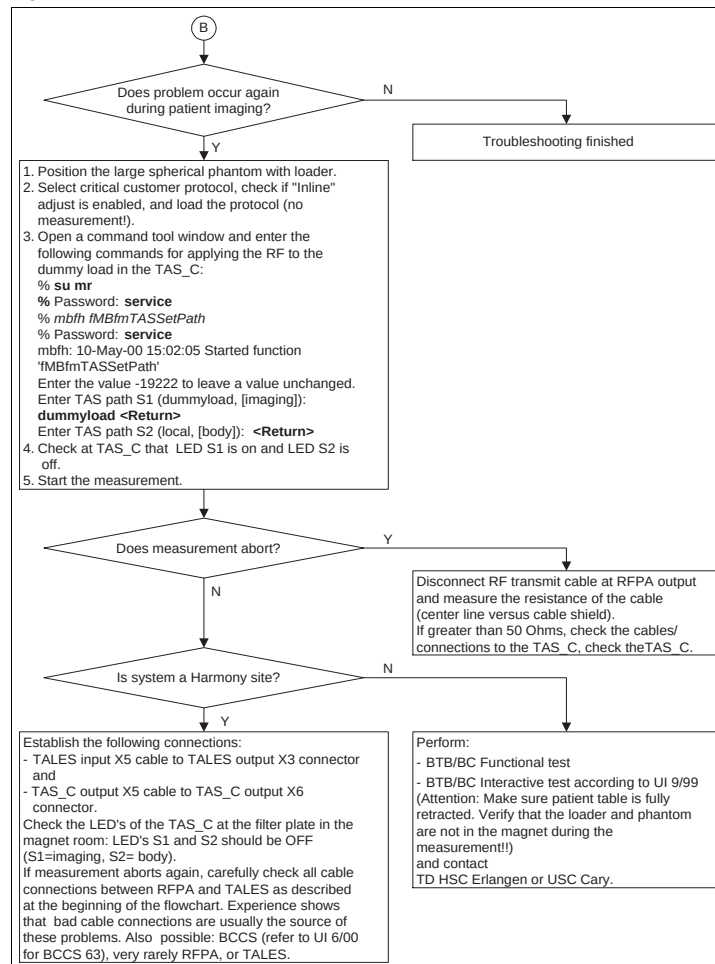
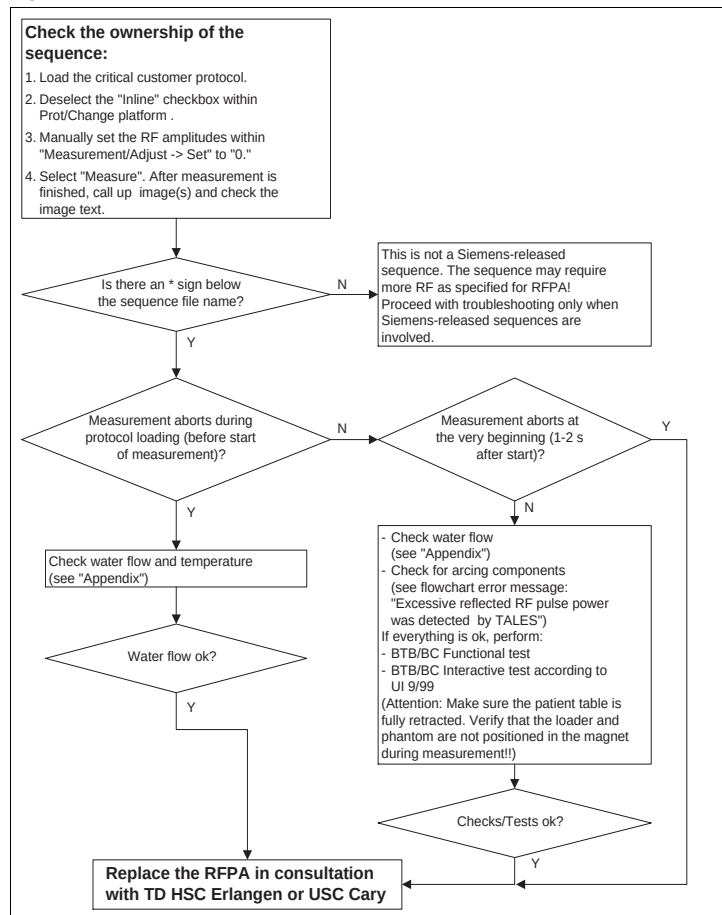


Fig. 37



"RFPA overtemperature"

Fig. 38



HistoryList*.log

This file can be called up with SeSo -> "Reports," menu item "Process diagnostics."

Whenever the patient is changed (Pat/Reg) and the first protocol is started, a new file HistoryListNew.log is generated and the original file is copied to HistoryListOld.log. In the case of high RF reflection, for example, the file is also saved as HistoryListErr.log. Select one of these files and scroll down to the bottom (the file may be quite large). In line 143 and 144, the peak maximum forward and peak maximum reflected power measured by TALES channel 1 are shown in W. In line 145 and 146, the same values measured by TALES channel 2 are shown.

Example (ok condition!):

<TALES Measurement Results>

143 TALESpeakMaxForw_1.....: 1338.660400 [W]

144 TALESpeakMaxRefl_1..... : 84.671265 [W]

145 TALESpeakMaxForw_2.....: 1287.063843 [W]

146 TALESpeakMaxRefl_2..... : 218.698105 [W]

NOTICE

The absolute power values depend on the selected sequence. The forward/reflected ratio of each channel indicates the quality of impedance matching.

Checking secondary water flow through RFPA

Open the water connection at the RFPA output and measure how many liters flow into a bucket in 1/2 min. Given the nominal value of 8 l/min, it should be at least 4 liters.

NOTICE	Two quick-release water couplings are needed to conduct this test: part no. 5940726 (male coupling) and part no. 5940718 (female coupling).
---------------	---

If the water flow is not ok and your system is a Symphony Quantum, refer to UI 2/00.

For Symphony and Sonata systems, refer to UI 11/00.

Patient Handling

In this part you will find test strategies, procedures and repair instructions encompassing the complete Patient Handling.

Strategy

General

The enclosed information applies to all **Magnetom Harmony/Symphony** systems using software version Numaris3.5A.

It contains information about possible errors as well as troubleshooting charts to assist you during troubleshooting.

The main tool for patient handling troubleshooting are:

- PT communication and status test, available in the service software platform "Test Tools" and the Service Software for Patient Table (see PT Procedure).
- PMU communication and status test, available in the service software platform "Test Tools".

The section on "Repair" lists the necessary adjustments and final checks when replacing a FRU.

For specific error messages, check the error message in System/Report/Detail for the corrective action.

NOTICE

If a fatal error occurs, the funnel symbol on the magnet display will begin flashing.

Descriptions of the LED's "[Description of LED's](#)" and fuses "[Description of fuses](#)" are located in the "Procedure" chapter.

Patient Table

Test of CAN communication and PT status

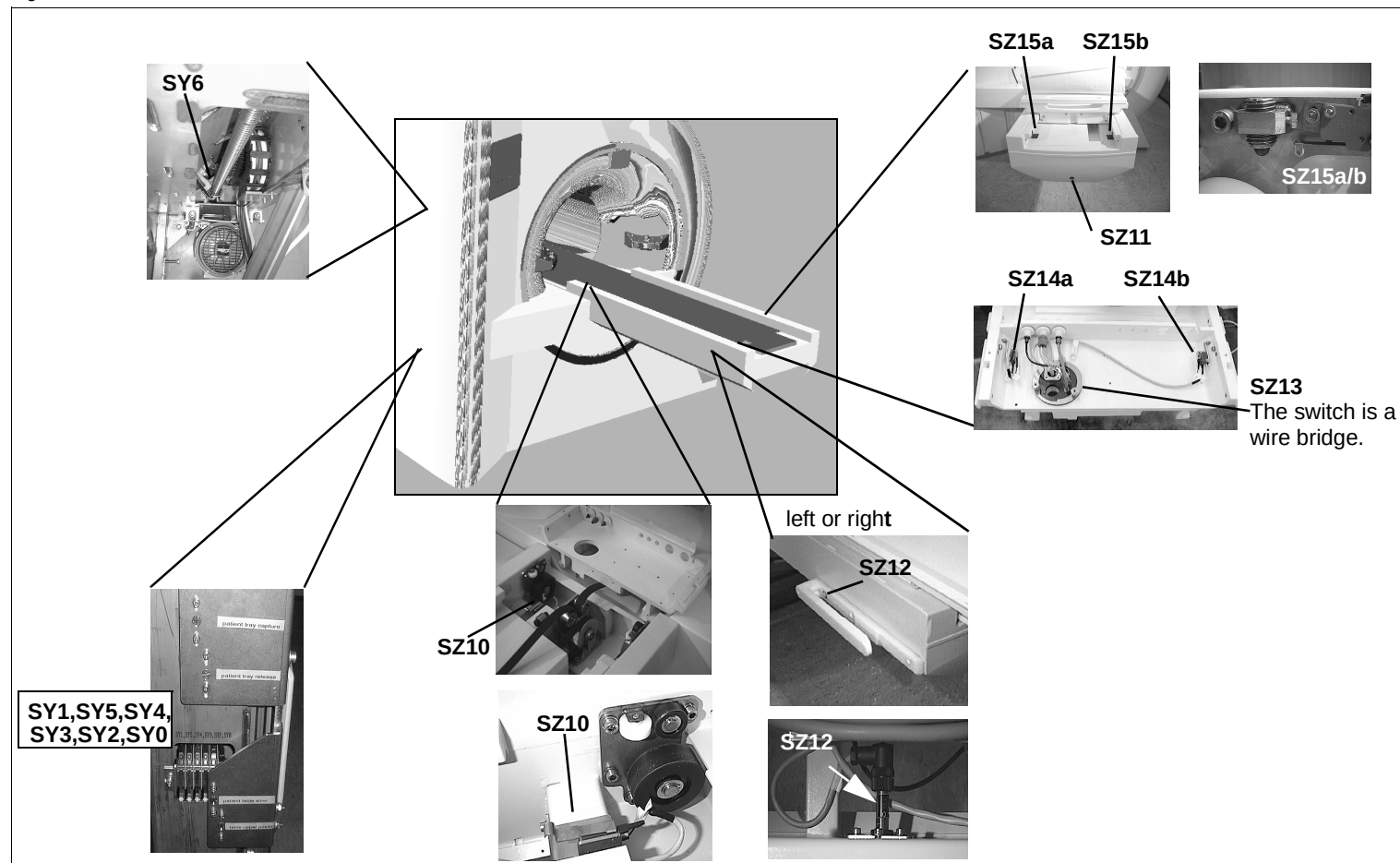
If you select "PT Status" within the platform "Test Tools" of the service software the following two quick tests will be performed:

- Test of the CAN communication.
- A listing of the PT status will be displayed. Error conditions are marked with "!!!".

NOTICE	If no PT status is reported, check the CAN bus connection to the patient table controller.
---------------	--

The table below contains a list of the table status messages and their explanations.

Fig. 39 Switch overview



Status message	Explanation	Status	Hints for troubleshooting
Different sensors	Power supply monitoring for the switches, sensors and table stop circuit	ok	Check fuse F11 on A5132.
+10.5V for cabin UI	Power supply monitor for the control components and display	ok	Check fuses F13 on A5132 and F3 on A5131.
+10.5V for A5181 (PMU)	Power supply monitor for the PMU-converter A5181.	ok	Check fuses F15 on A5132 and F3 on A5131.
+18V for PMU	Power supply monitor for the PMU Front end A5191 and PMU converter A5181.	ok	Check fuses F16 on A5132 and F1 on A5131.
+13.5V for Video	Power supply monitor for the video camera A 5196.	ok	Check fuses F12 on A5132 and F1 on A5131.
-18V for PMU	Power supply monitor for the PMU Front end A5191 and PMU converter A5181.	ok	Check fuses F17 on A5132 and F2 on A5131.
+18V for A5192 (Grad.Tmp.)	Power supply monitor for the gradient temperature supervision on A5192.	ok	Check fuses F18 on A5132 and F1 on A5131.
-18V for A5192 (Grad.Tmp.)	Power supply for the gradient temperature supervision on A5192.	ok	Check fuses F19 on A5132 and F2 on A5131.
For all switch status messages		• Check the functionality of the switches using the service software for patient table “Mode M1, switches and sensors”. • Check the signal path up to the backplane A5150. • Replace controller A5140.	

Status message	Explanation	Status	Hints for troubleshooting
Switch input Y 0_2	Toggle switch, in case the table is in the vertical mode the horizontal mode is locked.	off, when the patient table is in the vertical mode, i.e. the converter is active (the frequency converter display shows 000.0) on, when the patient table is in the horizontal mode	Refer to Patient table in the Adjustment Instructions, Switch adjustment " Switch SY0/SY2 "
Switch input Y 1	table lower position (without trolley connected)	off, when the patient table has reached the lowest position.	
Switch input Y 2	during the table up movement the table speed will be switched to slow speed	off, just before the patient table reaches the highest position (height to enter the magnet).	Refer to patient table in the Adjustment Instructions, Switch adjustment " Switch SY0/SY2 "
Switch input Y 3	If the trolley is connected and the table moves down, the table speed will be switched to slow speed. If the trolley is connected and the table moves up, the table slowly stops	off, when the patient table is above the trolley docking height or at the trolley docking height.	Refer to Patient table in the Abbreviated Adjustment Instructions, Switch Adjustment " Switch SY3/SY4 " Detailed Adjustment Instructions are located in the Installation Volume 1 binder, Register 4 Chapter 6; "Adjusting the trolley height"
Switch input Y 4	If the trolley is connected and the table moves down, the table slowly stops	off, when the patient table is below the trolley docking height or at the trolley docking height.	
Switch input Y 5	Table lower position (with trolley connected)	off, when the patient table has reached the lowest position.	
Switch input Y 6_15	Supervision for the spindle nut (switch SY6) Supervision for the carrier panel (switch SZ15a and SZ15b are in series)	not ok, when the spindle nut is out of the guide and/or when the carrier plate is raised up.	Objects may be located underneath the patient table or the carrier plate. Refer to Patient Table in the Adjustment Instructions, Switch Adjustment " Switch SZ15a/b "

Status message	Explanation	Status	Hints for troubleshooting
Switch input Y 0_1	Toggle switch, in case the table is in the horizontal mode, the vertical mode is locked.	On, when the patient table is in the vertical mode, i.e. the converter is active (the frequency converter display shows 000.0). Off, when the patient table is in the horizontal mode.	Refer to patient table in the Adjustment Instructions, Switch Adjustment " Switch SY0/SY2 "
Switch input Y	Checks whether the switch status of switch SYnn is valid.	ok	Refer to the table " Status of switches "
Switch input Z	Checks whether the switch status of switch SZnn is valid.	ok	Refer to the table " Status of switches "
Switch input Z 10	Locks vertical movement when the table is inside the magnet. Reference position of the Z-axis.	Off, when the table is completely outside the magnet (Home position).	Refer to patient table in the Adjustment Instructions, Switch adjustment " Switch SZ10 "
Switch input Z 11	Sensor; the downward table movement stops, if the trolley is not connected properly or if any obstacle is close to the sensor.	Off, when an obstacle is located within the range of the sensor. The sensor is not always supplied with voltage. <u>Exception:</u> voltage is always present in service mode.	Refer to patient table in the Adjustment Instructions, Switch Adjustment " Switch SZ11 " Check whether obstacles are located underneath the patient table.
Switch input Z 12	Sensor; verifies the trolley connection to the patient table.	On, when the trolley is docked. The sensor is not always supplied with voltage. <u>Exception:</u> voltage is always present in service mode.	Refer to patient table in the Adjustment Instructions, Switch Adjustment " Switch SZ12 "

Status message	Explanation	Status	Hints for troubleshooting
Switch input Z 13	The connection between the carrier panel and the removable tabletop will be verified. The switch is a jumper on connector E543-X2 of the ASV.	Off, when the carrier plate is moved up.	
Switch input Z 14 a&b	Two switches in parallel; the trolley connection to the carrier panel will be verified	Off, when the trolley is connected to the carrier panel (i.e. the carrier panel is positioned on the trolley.)	Check that both trolley latches are locked in.
SW cond. 0 to 7	Current status of the software	Not relevant.	
Cabin Panel 0	Code in front right operating panel	Present, when the respective operating panel is connected. None, corresponding operating panel is not connected.	Check the connection to the operating panel.
Cabin Panel 1	Code in front left operating panel		
Cabin Panel 2	Code in back right operating panel		
Display 0	Code for front display	Present, when the respective display is connected. None, corresponding display is not connected.	Check the connection to the display.
Display 1	Code for back display		
I2C Power Supply	Reports a malfunction of the I2C bus interface of the A5132 power supply	ok	Power-on-Reset via F15 (PCA)

Status message	Explanation	Status	Hints for troubleshooting
I2C Bus	Reports a malfunction of the I2C bus system caused by an incorrect function of the operating panel or display.	ok	Ignore the error, if it occurs only once. Replace operating panel A516 or display A517. Replace power supply A5132 if incorrect error messages regarding power failures are displayed. Replace controller A5140.
Table move	Motorized table movement is enabled.	Not relevant.	
Erroroutput of converter	Internal error of the frequency converter A511.	On	Reload/All. Check whether there are any mechanical problems (table should move smoothly). Overload for table due to non-stop movement or an excessively heavy load, wait approx. 30 min.
Converter help-voltage	Frequency converter auxiliary voltage	On, when the frequency converter is switched on.	Check the frequency converter.
Halt circuit	Table stop circuit output.	Open, if table stop is activated at the operating panel or intercom.	Check the cable connection to the operating panels or dummy plug (X28, on the back-plane) must be present if the rear operating panel is not connected.
Temperature vertical drive	Temperature of the vertical drive A531 is too high.	ok	Reload/All. Overload for table due to non-stop movement or an excessively heavy load. Wait appr. 30 min., then Reload/All.

Status message	Explanation	Status	Hints for troubleshooting
Voltage of intermediate circuit	Voltage is present in the intermediate circuit (A512)	Ok, if the patient table can be moved horizontally.	
Output halt flip flop	Table stop monitor	Low, no table stop. High, table stop.	
Collision detection	Collision vertical	Low	Move the table up, if it is resting on an obstacle. Activate table stop and reset. Check switches SY6, SY15 a/b, Refer to the general information " For all switch status messages. "
Internal error of hor. ampl.	Error in horizontal end stage A512 (over-/undervoltage, overtemperature).	ok	Reset table stop, if the error occurred in conjunction with other errors. Check the patient table supply voltage (195.5V to 253V) on A5131 pin X100.3 and 6. Replace A512.
High current error of hor. ampl.	Shorted output or internal error in horizontal end stage (A512).	ok	Switch off F15 (CCA) and disconnect cable W5421 at the backplane. Switch on F15. If the LED H2 on A512 is glowing, replace A512 and reconnect cable W5421. Cable W5421 or motor A532 are shorted out. Check the cable and motor and replace, if necessary.

Status message	Explanation	Status	Hints for troubleshooting
Antivalence error	Missing antivalence of the signals from the incremental generator of the horizontal drive (A532)	ok	Reload/All and press Home button on the operating panel. Check signal path to the incremental generator (A532). Check voltage supply (+5V; LED H1) at power supply (A5132) and at cable W5422-X2 pin 1 and 6. Replace horizontal drive A532. Replace controller A5140.
No null impulse plausibility	No zero pulse plausibility of the incremental generator of the horizontal drive (A532)	ok	Reload/All and press Home button on the operating panel. Check signal path to the incremental generator (A532). Check voltage supply (+5V; LED H1) at power supply (A5132) and at cable W5422-X2 pin 1 and 6. Replace horizontal drive A532. Replace controller A5140.
Communication to motion controller	Internal or communication error in/to motion controller A5141.	ok	Power-on-Reset via F15 (PCA). Replace the controller A5140.
Reference status	Reference pulse is out of range No reference pulse within the expected range no reference switch found	ok	Check SZ10 and the signal path to SZ10. Refer also general information "For all switch status messages". Check the belt tension.

Status message	Explanation	Status	Hints for troubleshooting
Switch OFF state (Hor. Driver)	Switch-off error at the horizontal driver (A512).	ok	Reset the table stop. Replace horizontal driver A512. Replace controller (A5140).
Switch ON state (Hor. Driver)	Switch-on error at the horizontal driver (A512).	ok	Check the fuses /LED's on the horizontal driver (A512). LED H5 has to glow continuously otherwise: check fuse F31 on board A512 check/replace power supply A5131, transformer T1, cable W5452. LED H4 should glow only when horiz. axis is active (goes off slowly after switching to inactive), if not check/replace F31 and F32 on horizontal driver(A512), controller (A5140).
Maximum current error	Motor current for horizontal axis has been exceeded.	ok	Check for any obstacles and remove them. Check that the table moves smoothly. Reset the table stop. Check the fuses/LED's on the horizontal driver (A512) .
Horizontal movement error	The controller detected that table movement is erratic and stopped table movement.	ok	Check for any obstacles and remove them. Check that the table moves smoothly. Reset the table stop.

Status message	Explanation	Status	Hints for troubleshooting
Switch on error of converter	Vertical end stage (frequency converter A511) could not be switched on	ok	Check fuse F7 on A5131. Check the cables to the frequency converter (W5451, W5453). Replace frequency converter A511. Replace A5131.
Switch off error of converter	Vertical end stage (frequency converter A511) could not be switched off	ok	Reload/All and press Home button on the operating panel. If the error is still present, replace: power supply A5131 controller A5140.
Converter state	Parameter change or communication error from the frequency converter	ok	Patient table in Home position, Reload/All. Reinitialize frequency converter using the service software for patient table, "Mode M7, parameterizing the converter". Replace frequency converter A511.
	Incorrect version of frequency converter firmware		Frequency converter installed is the wrong revision level; should be at least "G". Replace with a new frequency converter.
EEPROM state	Cecksum error or incorrect loadware version	ok	Reload/All (forced). Error appears together with I2C bus. Replace power supply A5132. Replace operating panel A516 or display A517.

Status of switches

With no trolley connected

Patient table position in Z-direction		magnet center	outside (HOME) *)	outside
Patient table position in Y-direction		highest position		lowest position
Toggle switch between vertical mode and horizontal mode	SY 0_2	on	on	off
Lowers table position (without trolley)	SY 1	on	on	off
Switches to slow speed during table upward movement	SY 2	off	off	on
Table speed switches to slow speed if the table moves down and a trolley is connected; table movement stops if the table moves up and a trolley is connected	SY 3	on	on	on
Table movement stops if the switch is activated when the table is moving down and a trolley is connected	SY 4	on	on	on
Lowers table position (with trolley)	SY 5	on	on	off
Supervision for spindle nut (SY6) and for carrier plate (SY15a/b)	SY 6_15	ok	ok	ok
Toggle switch between horizontal mode and vertical mode	SY 0_1	off	off	on
Locks the vertical mode if the table is in the horizontal mode and acts as a reference switch for the horizontal mode	SZ10	off	on	on
Senses any obstacles beneath the table	SZ 11	off	off	on
Sensor; verifies the trolley connection to the patient table	SZ 12	off	off	off
Verifies the connection between the carrier panel and the removable tabletop	SZ 13	on	on	on
Two switches in parallel; verifies the trolley connection to the carrier panel	SZ 14	on	on	on

*) the frequency converter is not activated (the converter display is off)

With trolley connected

Patient table position in Z-direction		magnet center	outside (HOME)*)	outside	
Patient table position in Y-direction		highest position		trolley height**)	lowest position
Toggle switch between vertical mode and horizontal mode	SY 0_2	on	on	off	off
Lowers table position (without trolley)	SY 1	on	on	on	on
Switches to slow speed during table upward movement	SY 2	off	off	on	on
Table speed switches to slow speed if the table moves down and a trolley is connected; table movement stops if the table moves up and a trolley is connected	SY 3	on	on	off	on
Table movement stops if the switch is activated when the table is moving down and a trolley is connected	SY 4	on	on	off	on
Lowers table position (with trolley)	SY 5	on	on	on	off
Supervision for spindle nut (SY6) and for carrier plate (SY15a/b)	SY 6_15	ok	ok	ok	ok
Toggle switch between horizontal mode and vertical mode	SY 0_1	off	off	on	on
Locks the vertical mode if the table is in the horizontal mode and acts as a reference switch for the horizontal mode	SZ10	off	on	on	on
Senses any obstacles beneath the table	SZ 11	off	off	off	off
Sensor; verifies the trolley connection to the patient table	SZ 12	off	off	on	on
Verifies the connection between the carrier panel and the removable tabletop	SZ 13	on	on	on	off
Two switches in parallel; verifies the trolley connection to the carrier panel	SZ 14	on	on	off	off

*) the frequency converter is not activated (the converter display is off)

**) the trolley is connected and looked

PMU

Test of CAN communication and PMU status

If you select "PMU Status" within the platform "Test Tools" of the service software the following two quick tests will be performed:

- Test of the CAN communication.

Status message	Explanation	Status	Hints for trouble shooting
Supply voltages	Monitors the supply voltages at the PMU Frontend	ok	Check the fuse F15 on board A5132 and the fuse F16/F17 on board A5132. Check the cabling between the PMU Converter and the PMU Frontend.
DSP memory test	Tests the DSP memory in the PMU Frontend.	ok	The PMU Frontend should be replaced.
data receiving	Verifies the communication lines of PMU Frontend.	ok	Check the cabling between the PMU Frontend and the PMU Converter A5181 and MPCU.
Other status messages applies only to ECG lead(s), Pulse sensor or active ECG electrodes and have no direct impact for the service. Those are more likely application hints (see the System Guide)			

PMU - Trouble shooting Chart

Symptom	Possible cause	Service action	Reference
No physiological text displayed at the right screen server	Loose cable connection	Check the cabling between the MPCU-Box and the SMI/Imager	
No calibration pulse is displayed at the right screen server	Loose cable connection	Check the cables between the PMU converter A5181 and the PMU Frontend. Check the fibre optical cable between the PMU Converter A5181 and the MPCU. Check the power supply +10.5/ +18V/-18V for the PMU	Use the PT communication & status test
Weak ECG pulse or noise is displayed at the right screen server but a satisfactory calibration pulse is displayed	ECG lead is defective Patient contact with the ECG electrodes is not sufficient.	Check the resistance of the ECG leads Check the proper application of the ECG electrodes (e.g. together with the application specialist, ..)	Between 15kOhm and 30kOhm See System Guide
No pulse from the pulse sensor or respiratory pulse is displayed at the right screen server but a satisfactory calibration pulse is displayed	Pulse sensor or respiration belt is defective PMU Frontend defective	Check the pulse sensor on a volunteer Check the respiration belt by expanding the belt Replacement of the PMU Frontend	

Symptom	Possible cause	Service action	Reference
No beep sound (calibration pulse or ECG signal is visible) at the intercom but the intercom itself is working	The trigger switch is positioned at "0" Loose cable connection	Check the trigger switch on the left side of the intercom Check the cable between MPCU/ISA_CAN D17 and the Intercom	See System Guide
PMU Display on the magnet is off	Power supply not on	Check the F15/RF Room in the CCA Check the fuse F4 on the board A5131 Check the cable between the PMU Converter A5181 and the PMU Display	Supply voltage: appr.13VAC
No calibration pulse is displayed on the PMU Display on the magnet but on the right screen server.	Loose cable connection	Check the fibre optic cable between the PMU Converter A5181 and PMU Display	

Procedures

In this chapter the standard procedures for the patient table are described.

Patient Table

Service-Software for the patient table

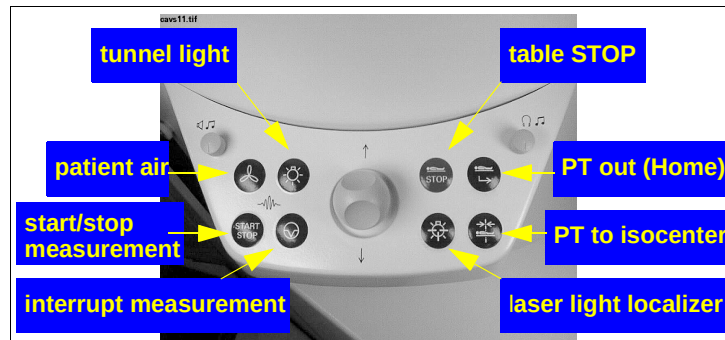
General

The service software for the patient table allows you to test and change the parameters for the patient table. If the patient table electronics is in the service mode, the table functions are limited (no table movement, very limited CAN functionality, no ventilation or lighting control, no table stop reset).

You evoke the service software by simultaneously pressing the UP and DOWN buttons on backplane A5150 (patient table electronics).

The service software for the patient table includes eight different modes, from M1 to M8. Each mode is indicated accordingly on the display together with other mode-specific information.

Fig. 40 Control panel

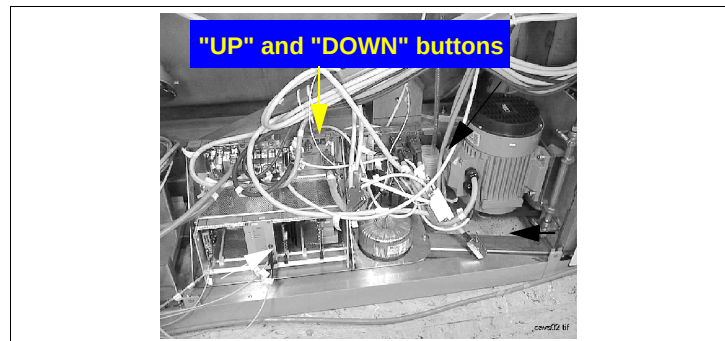


How to operate the service software for the patient table

To evoke the service software:

- Press the "UP" and "DOWN" buttons on Backplane A5150

Fig. 41 "UP" and "DOWN" buttons



- "Start for Servicemode" will appear on the display.

- Press the “measurement start/stop” key at the control panel.
- "M1: . . ." appears on the display.

To switch modes

You can switch between modes M1 through M8 by pressing the “PT out (Home)” or “PT to isocenter” keys through the numbering sequence. The current mode (except for some display text in mode 2) is shown in the upper left of the display.

To exit from the service software

- Press the “PT to isocenter” key at the control panel until the “M8: Leave with START” appears on the display.
- Press the "start/stop measurement" key on the control panel.
- The display switches to standard mode.

Mode M1, switches and sensors

This mode shows the status of the switches and sensors, and provides an easy way to check the condition of the wiring as well as the functions of the switches and sensors. All switches and sensors are supplied via +18V_DC_SENS (board A5132). The HIGH signal means that the respective switch or sensor has been activated, that is +18V_DC_SENS is present at the controller input (board A5140).

Each switch/sensor has been assigned a single digit in the text line of the display (except for SY0 which has two digits). The reference number of the respective switch or sensor is displayed when the controller input has been set to HIGH (switch/sensor is activated). When the controller input is set to LOW, the display shows "-".

For example, the display may show:

M1: SY - 0 1 - 3 4 5 6 SZ 0 1 - 3 4

Details of the display:

Switch	Ref.-No.	Signal	Description of the switch
SY 0_1	0	LOW	Toggle switch, if the table is in the vertical mode, the horizontal mode is locked
SY 0_2	0	HIGH	
SY 1	1	HIGH	Lower table position (without trolley)
SY 2	2	LOW	Switches table speed to slow during the table upward movement
SY 3	3	HIGH	If the trolley is connected and the table moves down the table speed will be switched to slow speed. If the trolley is connected and the table moves up, the table will slowly stop.
SY 4	4	HIGH	If the trolley is connected and the table moves down, the table will slowly stop.
SY 5	5	HIGH	Lower table position (with trolley).
SY 6	6	HIGH	Supervision for the spindle nut.
SZ 10	0	HIGH	Locks the vertical mode if the table is in the horizontal mode, and acts as a reference switch for the horizontal mode.

Switch	Ref.-No.	Signal	Description of the switch
SZ 11	1	HIGH	Sensor; stops table downward movement if the trolley is not connected or if any obstacle is close to the sensor.
SZ 12	2	LOW	Sensor; verifies the trolley connection to the patient table.
SZ 13	3	HIGH	Verifies the connection between the carrier panel and the removable tabletop.
SZ 14	4	HIGH	Two switches in parallel; verifies the trolley connection to the carrier panel.

- The switches for collision detection SY6, SZ15a, and SZ15b are connected in series and data is acquired in common under SY6.
- SY0 gives you the make or break contacts.

"M1: SY - 0 . . ." means that the input SY0_1 (make contact) is LOW and SY0_2 (break contact) is set to HIGH. This means, therefore, that the table is at the correct height to move into the magnet and SY0 is active.

"M1: SY 0 - . ." means that the input SY0_1 (make contact) is HIGH and SY0_2 (break contact) is set to LOW. The table is in the vertical mode.

"M1: SY - - . ." or "M1: SY 0 0 . ." are not acceptable, because they indicate errors.

Mode M2, display

In this mode, the display is provided with four different test images. Use the "measurement start/stop" key and "measurement pause" key to toggle between the images.

The contrast of the display can be changed with the keys for "funnel air" or "funnel light".

Two of the test images will show the currently selected contrast setting.

NOTICE

Use the front control panel to manipulate the display in front. The same allocation applies to the control panel and display located in the back.

For example, the display may show

M2: Display Test (31,31)

Explanation for the example "M2: Display Test (xx,yy)":

Contrast	for Display
xx	Magnet, in front
yy	Magnet, in back

When you exit the service software via mode M8, the contrast setting selected will be saved.

NOTICE

The display should be in operation for at least 20 minutes before you correct the contrast setting.

Mode M3 through M5, control panels

These modes provide you with the status of the respective control panel, if it was recognized at the start of the patient table loadware.

- M3 operating panel PAN_0 (main operating panel, front side, magnet)
- M4 operating panel PAN_1 (auxiliary operating panel, in back of magnet)
- M5 operating panel PAN_2 (auxiliary operating panel, front side, magnet)

For example, the display may show:

M3: PAN_0 KEY: 7F JOY: 126

Key code

The information right after KEY represents the function key pressed on the control panel

Display	Button
7D	patient air
7E	tunnel light
7B	start/stop measurement
77	interrupt measurement
3F	laser light localizer

- "PT out" and "PT to isocenter" keys are tested during the mode changes.
- It is not possible to test the "table STOP" key with the service software. However, you can test its function during normal table operating mode.

Joystick code

The decimal value following JOY represents the joystick position.

Display	Joystick position
127 to 128 (ideal)	Center (not deflected)
117 to 138 (acceptable)	Center (not deflected)
000	Joystick deflected all the way down
255	Joystick deflected all the way up

The table will begin to move at values below 96 or above 180.

If the control panel is not recognized, the display shows, e.g.: "M4: PAN_1 not found"

Mode M6, position acquisition

This mode measures the horizontal table position in increments (inc) and converts them into mm's.

The display may show, for example:
(reference position is known)

M6: Zpos: SZ10+009344inc
0024mm

The display may show, for example:
(reference position is not known)

M6: Zpos: ???+001234inc
0003mm

The position changes immediately in response to tabletop movement.

The entire motorized tabletop travel is 2160 mm or 847,830 inc, that is, 392,514 inc/mm. You can override the table travel limit manually (e.g. after pressing the "table STOP" key).

Mode M7, parameterizing the converter

In this mode, the currently valid set of parameters can be transferred to the converter.

NOTICE Parameters can be changed for the converter only if, prior to starting the service software, the table is in a position where the frequency converter is connected (the converter display shows 000.0).

WARNING The converter should not have been disconnected in the meantime (e.g. by pressing the "start/stop measurement" key).

The display may show
(converter is enabled)

M7: INV-Init with START

The display may show
(converter is not enabled)

M7: INV not available

- Press the "start/stop measurement" key to start the parameterization of the converter.
- The written and stored parameter is shown on the display during automatic transfer of parameters.
- Automatic transfer of parameters is completed in approximately 35 seconds and cannot be interrupted.
- Parameterization completed successfully if the display shows "M7: Success -> CENTER"

- Exit the mode with the “PT to isocenter” key.

Mode M8, Exiting from the Service mode

- Exit the mode with the “start/stop measurement” key.

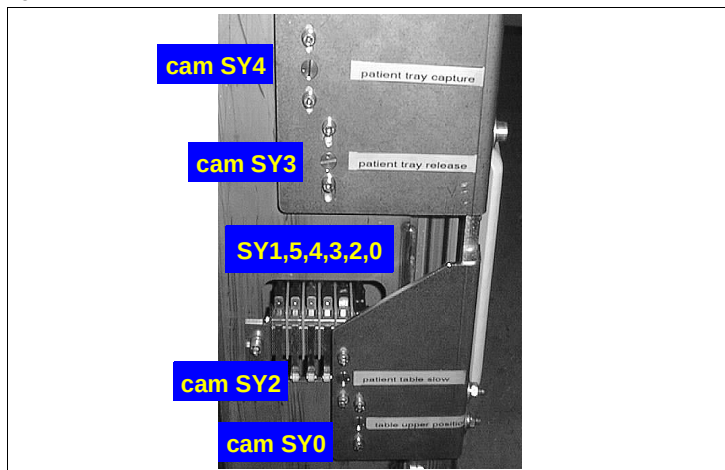
Switch adjustment

Switches SY0, SY... are located on the small cam plate under the left lifting column cover.

If the adjustment range is not sufficient, the small cam plate must be shifted.

The side cover will have to be removed prior to adjusting this switch. Refer to Replacement of Parts, Patient Handling, "Removing the column cover"

Fig. 42 SY switch



Switch SY0/SY2

- Place a weight of approximately 80 kg on the table (use all the phantoms or a volunteer).
- Move the patient table upward and into the magnet.

- The carrier plate rollers should sit exactly on the slideway and should be stable.

Carrier plate position:

carrier plate too high

Release the cam for SY0 and **move it down** by adjusting the eccentric screw. The cam for SY2 should be moved the same distance as for SY0.

carrier plate too low

Release the cam for SY0 and **move it up** by adjusting the eccentric screw. The cam for SY2 should be moved the same distance as for SY0.

- Check the adjustment and repeat, if necessary.

Switch SY3/SY4

The docking heights for upward and downward travel are set separately.

NOTICE

Detailed adjustment instructions are located in the Installation Volume 1 binder, Register 4; "Adjusting the trolley height".

- Position the trolley under the table. Activate the foot brake.
- Place a weight of approximately 80 kg on the table (use all the phantoms or a volunteer).
- Move the table down until it stops above the trolley.

Ideally, there should not be any gap between the removable table and the trolley. You must be able to lock in the tabletop using the foot lever from this position (both latches snap in without rubbing).

Carrier plate position:

The carrier plate stops before it reaches the trolley. Release the cam for SY4 and **move it up** by adjusting the eccentric screw. The cam for SY3 should be moved the same distance as for SY4.

The carrier plate continues to move, even though it is already on the trolley. Release the cam for SY4 and **move it down** by adjusting the eccentric screw. The cam for SY3 should be moved the same distance as for SY4.

- Check the adjustment and repeat it, if necessary.

You have now finished adjusting the docking height from above. Now you can check or adjust the docking height from below.

- Lock in the carrier plate using the foot lever on the trolley.
- Move the carrier plate approximately 40mm downward.
- Move the carrier plate up as far as possible (it should stop automatically at docking height).

The carrier plate must now lie flush on the trolley, and the trolley locking mechanism must be functioning correctly.

The carrier plate should move further upward. Release the cam for SY3 and **move it down** by adjusting the eccentric screw.

- Check the adjustment and repeat it, if necessary.

- Move the carrier plate further up and remove the trolley.

Overlap of switches:

The overlap of both switches should be between 4 mm and 8 mm . Check this using the service software for the patient table **“Mode M1, switches and sensors”**.

Checking the entire docking process

As long as the transfer (with and without patient load) of the carrier plate to the trolley and back functions correctly, the adjustment is complete.

Switch SZ10

NOTICE

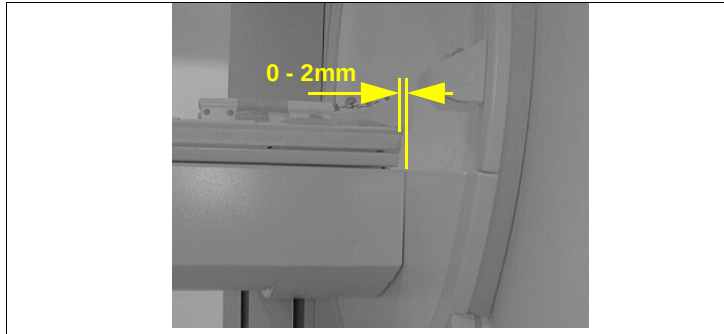
After each adjustment of SZ10 or the cam, the table must be switched briefly off, power on Reset via F15 (PCA).

Setting / checking the reference position for the basic table

- Remove the end cover and the base cover of the support arm. Switch SZ10 is located in the support frame on the magnet side.
- Move the tabletop by approximately 30 cm in the direction of the magnet and then back until it automatically stops.
- Move the table into the “Home-Position”. This initial movement of the table after switching on the voltage is the reference movement.

The head-end of the tabletop must be flush with the table carrier; there must not be any overhang.

Fig. 43 SZ10



The **space** between the tabletop and table carrier should be between 0 and 2mm.

To make the distance smaller Release switch the mounting for switch SZ10 and **move it up**.

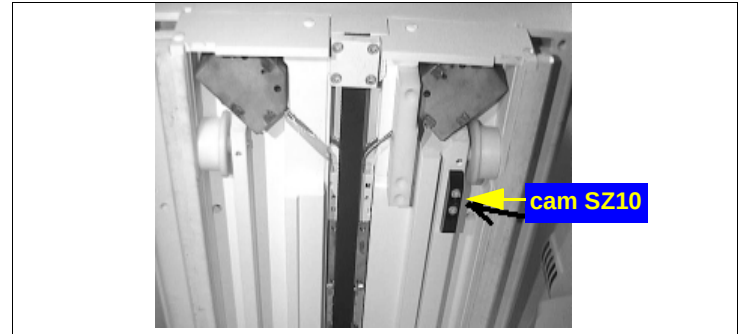
- Switch the system off again briefly and check the adjustment. Repeat the adjustment, if necessary.
- Install the end cover and the base cover of the support arm.

Adjustment/inspection of reference position for trolley operation

- Move the tabletop by approximately 30 cm in the direction of the magnet and then back until it automatically stops.

The head-end of the tabletop must be flush with the table carrier; there must not be any overhang. The **space** between the tabletop and table carrier should be between 0 and 2mm.

Fig. 44 cam for SZ10



The distance must be changed. Move the tabletop far enough into the magnet so that it reaches the switch cam underneath the tabletop carrier.

Adjust the cam by the same distance and in the same direction as the deviation.

- Switch the system off again briefly and check the adjustment.

Switch SZ11

Sensor SZ11 is an optical reflex sensor. It prevents damage to the table and trolley in connection with the mirror, in case the trolley does not dock correctly and is still located beneath the table. It also senses other obstacles as well (but about 10 times worse than mirrors) and switches off downward movement. Since the sensor could interfere with the measurement, voltage to the sensor is switched off at times. In service mode, it is always switched on and can be checked.

Setting the sensitivity

- Remove the covers so that sensor SZ11 is accessible at the foot-end of the carriage frame.
- Rotate the adjustment screw (blue plastic) on the sensor to the right stop (corresponds to maximum sensitivity).
- Using the service software for the patient table “[Mode M1, switches and sensors](#)” check whether sensor SZ11 switches on when an obstacle (e.g. hand) is placed in front of it.
- Perform the same check with the trolley.
- Reinstall all covers.

CAUTION Do not turn the adjustment screw too far.

Switch SZ12

Sensor SZ12 comprises a reflex barrier in conjunction with a small mirror in the docking terminal. It detects whether a trolley is correctly docked. Since the sensor could interfere with the measurement, the supply voltage is switched off for part of the time. In service mode, the sensor is always switched on and can be checked.

Adjusting the sensitivity

- Remove the covers so that switch SZ12 is accessible.
- Turn the adjustment screw (blue plastic) on the sensor to the right until it reaches the end stop (corresponds to maximum sensitivity).
- Position the trolley so that the vertical guide rod is 4 mm before the end stop.
- Shift the sensor laterally so that it switches on 4 mm before the guide rod reaches the end of the docking terminal. Check the adjustment using the service software for the patient table “[Mode M1, switches and sensors](#)”.

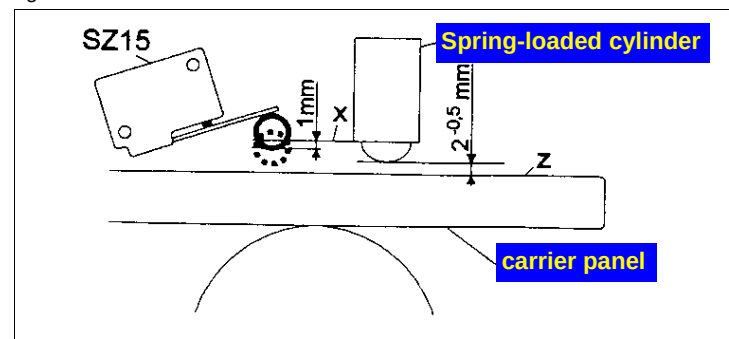
- Reinstall the covers.

Switch SZ15a/b

The spring-loaded cylinder prevents switch SZ15a/b from actuating at too low a pressure. The switch is only actuated when the ball actuator is pressed down into the cylinder.

- Remove the covers so that switch SZ15 a/b is accessible.

Fig. 45 SZ15



- Adjust the switch according to the instructions above.
- Pull the carrier panel slowly upward at the foot-end, first on one side (then on the other). Check the switching action of switch SZ15 a/b using the service software for the patient table “[Mode M1, switches and sensors](#)”.

NOTICE

Check the display for SY6, since switches SZ15 and SY6 are connected in series.

- Reinstall the covers.

Description of LED's

Reference	Name	Function
Power supply		
A5132-H1	+5V_DC_ENC	power on
A5132-H2	+7.5V_DC_CON	power on
A5132-H3	+18V_DC_SENS	power on
A5132-H4	+10.5V_DC_PAN	power on
A5132-H5	+10.5V_DC_PMU	power on
A5132-H6	+18V_DC_PMU	power on
A5132-H7	-18V_DC_PMU	power on
A5132-H8	+13.5V_DC_VID	power on
A5132-H9	+18V_DC_TMP	power on
A5132-H10	-18V_DC_TMP	power on
Lighting		
A5182-H1	POWER or U=	Power supply (intermediate circuit)
A5182-H2	LAMP_1 or UL1	Voltage drop in lamp 1 (on the right when looking at the back of the magnet) - brightness according to light level - off if the lamp shorts out (or 0 level is selected)

A5182-H3	LAMP_2 or UL2	Voltage drop in lamp 2 (on the left when looking at the back of the magnet) - brightness according to light level - off if the lamp shorts out (or 0 level is selected)
A5182-H4	LSB_LEVEL or 2^0	Nominal value for light level (LSB, 2^0)
A5182-H5	MSB-LEVEL or 2^1	Nominal value for light level (MSB, 2^1)
A5182-H6	TEMP_FLT	Overtemperature of heatsink
End stage A512		
A512-H5	POWER	38 V_AC - Power supply for end stage
A512-H4	SUPPLY	Power supply (Intermediate circuit) - off during MR measurement (glows for approx. 30 seconds afterward)
A512-H3	RUN	End stage operating - switched on and enabled and no end stage error
A512-H2	ERROR_C	Short on output circuit (Motor A532, cable W5421, end stage A512)

A512-H1	ERROR_S	Summary error message - on for over or under voltage or overtemperature of the end stage
Controller		
A514-H1	FW	FW LED of the piggy board - flashes, when FW is active - glows (briefly) when parameters are being saved in the EEPROM
A514-H2	LW	LW LED of the piggy board - flashes when the LW is active - on during sleep mode
A514-H3	HORIZ FAULT	Horizontal axis error - on in (Re-) start phase or for Power-On (for approx. 1s) - on for horizontal axis error If the loadware (e.g. invalid) could not initialize the position controller A5141, the controller does not report a horizontal error. - on during sleep mode

A514-H4	HORIZ OK	Horizontal axis ready - on, if there is no horizontal axis error (refer to HORIZ FAULT) - on during sleep mode
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Description of fuses

Assembly Fuse	Value	Name of line	Assembly Fuse	Test points	Units protected
A5131-F1	3,2AT	16.5V_AC_1			Switches, sensors, brake, PMU Front end, Gradient coil monitor (+18V_DC respectively)
A5131-F2	1,6 AT	16.5V_AC_2			PMU Front end, Gradient coil monitor (-18V_DC respectively)
A5131-F3	6,25 AT	10V_AC			Controller, incremental encoder, operating panel, displays, light localizer, PMU Front end and PMU converter
A5131-F4	3,2 AT	13.5V_AC			PMU display
A5131-F5	3,2 AT	-		A5131-X4.1 A5131-X4.2 ! 230 VAC !	Power line transformer A5131-T1, primary
A5131-F6	1 AT	-			Patient air
A5131-F7	15 AM	-			Frequency converter A511
		GND		A5132-X12	
A5132-F11	0,8 AT	+18V_DC_SENS	A5132-H3	A5132-X4	Microswitch/sensors SYx and SZxx, SZ13 (incorporated in the ASV) Brake for drive A532
A5132-F12	0,8 AT	+13.5V_DC_VID	A5132-H8	A5132-X9	Video monitor camera
A5132-F13	3,2 AT	+10.5V_DC_PAN	A5132-H4	A5132-X5	Operating elements, displays and light localizer
A5132-F14	3,2 AT	+7.5V_DC_CON +5V_DC_ENC	A5132-H2 A5132-H1	A5132-X3 A5132-X3	Controller A514 Incremental encoder of drive A532
A5132-F15	0,8 AT	+10.5V_DC_PMU	A5132-H5	A5132-X6	PMU converter A5181, PMU radio

A5132- F16	0,8 AT	+18V_DC_PMU	A5132-H6	A5132-X7	PMU Front end
A5132- F17	0,8 AT	-18V_DC_PMU	A5132-H7	A5132-X8	PMU Front end
A5132- F18	0,2 AT	+18V_DC_TMP	A5132-H9	A5132-X10	Gradient coil monitor D9
A5132- F19	0,2 AT	-18V_DC_TMP	A5132-H10	A5132-X11	Gradient coil monitor D9
A5182- F21	4 AT	18V_AC	A5182-H1	-	Tunnel light
A512- F31	4 AT	38V_AC	A512-H5	-	Horizontal end stage A512
A512- F32	4 AT	-	A5132-H4	-	DC intermediate circuit for the horizontal end stage A512

Gradient System

The enclosed information applies to all **Magnetom Harmony/Symphony/Quantum/Sonata** systems using software version **Numaris3.5A** and equipped with the **gradient cabinet K2209 or K2217 (GPA)**.

In this part you will find test strategies, procedures and repair instructions encompassing the complete gradient system.

Strategy

It contains information about possible errors as well as troubleshooting charts to assist you during troubleshooting.

General

Sources for errors in the gradient system

As compared to most of the other system components, the gradient system contains a number of assemblies that generate and process analog signals. These assemblies, all field replaceable units, (FRU's), form a closed current control circuit, which makes it at times difficult to limit the source of error to a single FRU.

Errors are usually looped across several assemblies. For example the backplane D8/18 of the gradient small signal unit (GSSU) as well as all plug-in connections of the boards (PCB's) may be the source for an error.

Usually, if error messages are sent by the application software (ASW) to the MMC via the CAN-bus, at the same time the modulators will be blocked. The gradient current control is disabled and the **Regulation error message** is generated as a consequence.

Except for image quality errors, the following major types of errors are stored in the system error log (system/report). Several of them also are indicated via LED's on the assemblies of the GSSU:

- **Dynamic errors:**

These errors occur during measurements: They are either reproducible under the same conditions or they appear sporadically. In either case, the measurement is terminated. In most cases, the error message generated corresponds to the actual situation.

- **Static errors:**

These error messages are generated during GPA switch-on. Reset will not remove these errors. The source for error is an actual defect in the component or an error in the monitoring/transfer path (monitoring error).

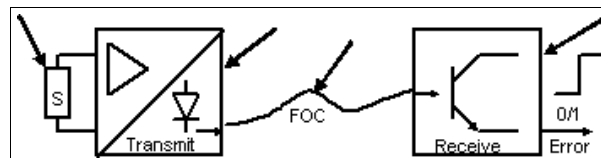
- **Image quality errors:**

Image quality depends on the error-free interactions between all system components, the most important of these are the RF and gradient systems. No specific error message will be generated when image quality is degraded.

Refer to Image Quality - Test Strategy, [page 287](#).

When errors continue to occur, monitoring errors have to be ruled out (refer to [Fig. 46](#), the arrows show the range of error sources). There is a relatively simple test that traces the signal path, e.g. it tests the light emitter and the fiber optic cables (FOC). "Light off" means that "there is an error" (fail safe-principle).

Fig. 46 Error signal sources in a transmission line



Example:

The components involved in the error **COIL_NOT_RDY** (refer to TSG-Chart 01) are:

- NTC in gradient coil
- Coil plug X2
- Line to D9 in patient table electronics
- Plug in location X20 on D9
- D9 monitoring assembly with fiber optic cable transmitter, including voltage supply (from patient table electronics)
- FOC D9-->D7/D17 (in gradient small signal unit, GSSU)
- D7 /D17 FOC assembly in GSSU including voltage supply E4A
- Backplane D8/D18 in the GSSU
- D6D16 Service and CAN assembly in the GSSU
- CAN-Bus transmission path to MPCU

Note:

All error signals could be influenced by a defective backplane D8/ D18 of the GSSU and/or by bad connections between D8/D18 and the relevant board.

Error handling - Error codes

Error messages are forwarded directly or via software-controlled multiplexer (on D4 K2209/D14 K2217, modulator PCB) to the CAN processor.

All occurring changes of the GPA-status are forwarded from the CAN processor on D6 K2209/D16 K2217 to the MPCU via a CAN telegram such as "Alarm", "General Error" or "Warning".

The type of telegram to be used is established in the application software of the GPA CAN unit. The definitions for the three types of telegrams are:

- Alarm:** Fatal error or error status which will terminate a measurement in progress or prevent the start of a measurement. Whenever possible, the error will be indicated independent of the MPCU or MPCU software (e.g. red LED's in the GPC).
- General Error:** Serious error or error status which will terminate a measurement in progress or prevent the start of a measurement.
- Warning:** Error status which neither affects the measurement in progress nor prevents the start of a measurement. However the image quality may be affected !

Different STOP-modes for the GPA K2217:

MOH : All Modulators will be inhibited by loadware.

MOD OFF: All Modulators will be inhibited by hardware.

POW OFF: The GPA will be switched in stand-by mode, i.e. the primary voltage of transformer T1 is switched off by contactor K2, the power stage voltages are off).

The Regulators will be inhibited as a consequence of a modulator inhibit -> green LED "Reg_On" on D12 for X,Y and Z-axis is off.

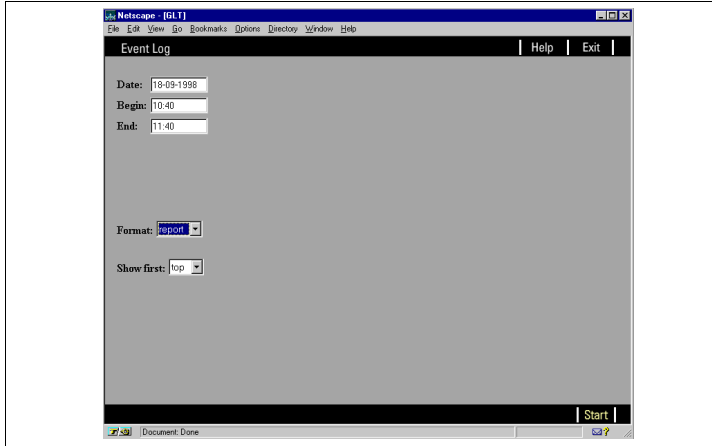
Types of error messages for a CAN Unit

- **Error string**, contains the error message.
- **Error explanation**, contains further information about the error and first aid for the customer.
- **Error remedy**, contains actions for remedy.
- **Error display**, contains the error text displayed on the Right screen server (RSS).

With the exception of the error display text, which is displayed in the RSS, all other information is sent to the error database, the "General Error Logging Tool - GLT", as soon as the error occurs.

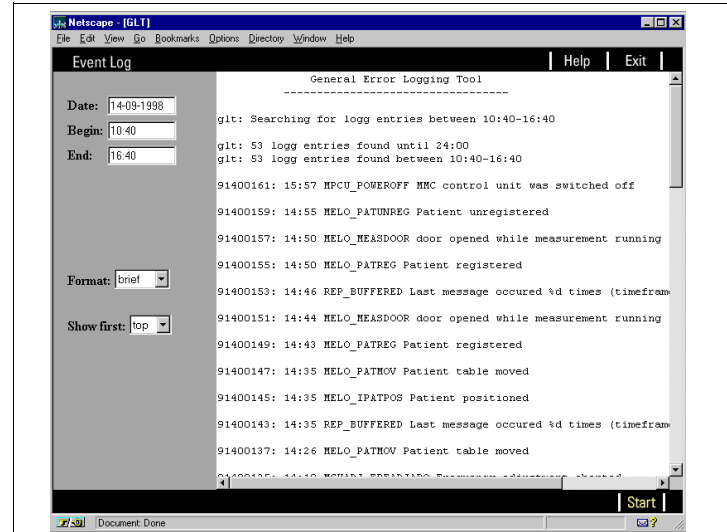
In the Event Log window, available from the Service Software Homepage, you can select more or less detailed information from the error logger. Fig. 47 shows the window with the possible selections.

Fig. 47 Event Log window



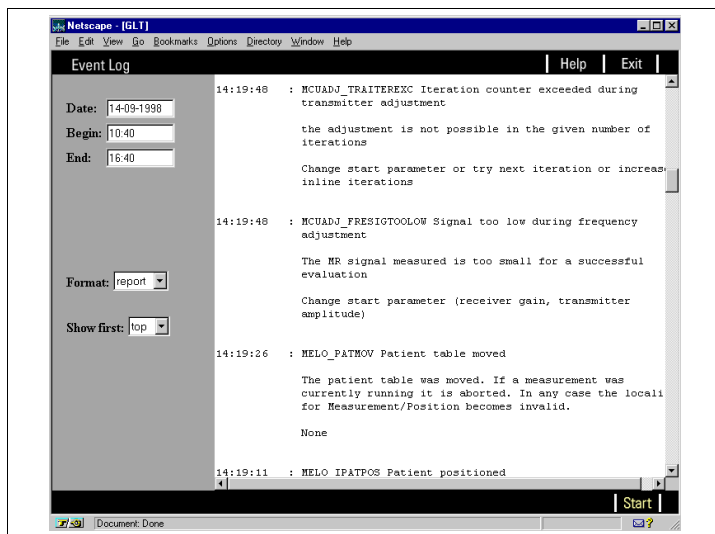
The selection display format "brief" (Fig. 48) gives you a listing of the system messages in the selected day and time interval:

Fig. 48 Example of the Event Log window (display format "brief")



The following window (Fig. 49) shows you more and detailed informations. Select **"report"** as display format.

Fig. 49 Example of the Event Log window (display format "report")



Test of CAN communication and GPA status

If you select "GPA Status" within the platform "Test Tools" of the service software the following two quick tests will be performed:

- Test of the CAN communication between GSSU/D6 with MMC/ISA-CAN.
- A listing of the GPA status will be displayed. Error conditions are marked with "!!!!".

In case of errors the displayed diagnosis will give you hints for further trouble shooting.

Error reset and warm restart of GPA

The latched error signals (displayed by error LED's on the GSSU) can be reset via the "Reset" button on D3 K2209/D17 K2217, except for permanent errors.

Pressing the "CAN reset" button S404 on D6 K2209/D16 K2217 provokes a warm restart, i.e. "Power on" sequence of the GPA. Static errors then will be logged and displayed again.

Troubleshooting Charts

Content of a TSG Chart

- **Error message:**
Name of the error which occurred (System/Report or Event Log)
- **Error Code:**
Name of error signal in the application software
- **Error Description frame:**
Description of the original meaning of the error and routing of the error signal
- **Symptom:**
How you can recognize the error
- **Area of Trouble (Components/function):**
Localization of the error in the system, a component or a system function.
- **Probable Cause:**
Description of probable reasons of the failure
- **Service Action:**
Recommended action for remedy
- **Reference:**
Further support for troubleshooting
- **Remarks:**
Additional information

Comments for using the following troubleshooting charts:

- Check all error messages and the time sequence of their appearance
- Refer to the corresponding TSG charts
- If you do not have a clear idea of the cause of an error, proceed step by step through the recommended service actions of the TSG charts.

Survey of Error messages GPA K2209

GPA K2209 Error Message	Chart
Gradient coil not ready	01/ page 216
Resonant Converter not ready	02/ page 217
Resonant Converter Overtemperature	03/ page 219
Resonant Converter Output in Current Trip	04/ page 220
Resonant Converter defective	05/ page 222
Driver current supply defective	06/ page 223
No Reaction of Modulator	07/ page 225
No Reaction of Resonant Converter	08/ page 226
Synchronization fault	09/ page 227
Triangle voltage error	10/ page 228
Plus 15V/ Minus 15V/ Plus 5V defective	11/ page 229
Upgrade component error	12/ page 230
Power stage defective	13/ page 231
Regulation error on X,Y,Z axis	14/ page 233
Current actual value error on X,Y,Z axis	15/ page 235
High current on X,Y,Z axis	16/ page 237
Duty cycle for power stage X1(2),Y1(2),Z1(2)	17/ page 239

GPA K2209 Error Message	Chart
Controlling error for power stage X1(2),Y1(2),Z1(2)	18/ page 241
Temperature error power stageX1(2),Y1(2),Z1(2)	19/ page 242
High voltage by power stage X1(2),Y1(2),Z1(2)	20/ page 244

CAN-Unit Error Message K2209/2217	Chart
The command is not acknowledged in certain time	21/ page 247
GPA command is not acknowledged in certain time	22/ page 248
GPA "Set loop" error occurred (GPA internal)	23/ page 249
GPA "Set regulator" error occurred (GPA internal)	24/ page 250
GPA "Set mode" error occurred (GPA internal)	25/ page 251
GPA "Set modulator" error occurred (GPA internal)	26/ page 252
GPA power ON/OFF could not be set	27/ page 250

Survey of Error messages GPA K2217

Error Message K2209<->Error Message K2217	Chart
Gradient coil not ready	01a page 254
Resonant Converter not ready <-> Main power supply not ready (MPS_NOTRDY_ERR)	02a page 256
Resonant Converter Overtemperature <-> Rectifier Overtemperature (RECTIFIER_TEMP)	03a page 258
Resonant Converter Output in Current Trip <-> Transformer Overtemperature (TRAFO_TEMP)	04a page 259
No Reaction of Resonant Converter <-> No Reaction of Main power supply (MPS_SET_ERR)	05a page 260
Synchronization error	06a page 262
Triangle voltage error <-> Modulator clock error (MOD_CLK_ERR)	07a page 263
Plus 15V/Minus 15V/Plus 5V defective	08a page 264
Spare error X1, Y1, Z1 <-> Driver supply error X, Y, Z (DRV_VOLT_ERR_X,Y,Z)	09a page 266
Upgrade component error <-> FASTOFF (FAST_OFF)	10a page 268
Regulation error on X, Y, Z-axis (LOOP_ERR_X,Y,Z)	11a page 269

Error Message K2209<->Error Message K2217	Chart
Current actual value error on X, Y, Z-axis (CAV_ERR_X,Y,Z)	12a page 271
Dutycycle/controlling/Temp/OVV/Spare error on X2, Y2, Z2 <-> Overcurrent for X, Y, Z (HIGH_CURR_X,Y,Z)	13a page 272
Dutycycle/controlling/Temp/OVV/Spare error on X3, Y3, Z3 <-> Duty cycle for power stage X, Y, Z	14a page 274
Controlling frequency error for power stage X, Y, Z	15a page 276
No reaction of modulator	16a page 277
Duty cycle for Y1 <-> Modulator defective	17a page 278
Temperature error for power stage X, Y, Z-axis	18a page 279
High voltage error by power stage X, Y, Z-axis	19a page 281
Duty cycle for X1 <-> Pull protection error (PULL_PROT)	20a page 283
Overcurrent for X <-> Danger of stimulation (STIM_DANGER)	21a page 284
Overcurrent for Y <-> FOC outputs to driver inhibited	22a page 286

Note:

If the GPA K2217 is installed on systems running the software NUMARIS 3, some error messages have a different meaning.

The displayed messages on the RSS always correspond to the GPA K2209. In the preceding table and in the following **TSG Charts XXa** the real meanings of the messages for the GPA K2217 are presented.

In the error report of the system the explanation and the remedies for both errors (i.e. GPA 2209 and GPA 2217) are displayed.

Chart 01

Error Message: **Gradient coil not ready**

Error Code: GPA_COIL_NOT_RDY

The gradient coil temperature exceeds 76 degrees C. The temperature is measured with six NTC resistors located in the gradient coil. The monitoring is done on D9 K2209 (patient table) and the error signal is routed via FOC to D7 K2209, D6 K2209 and via CAN Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error during normal operation of the system: FOC from D9 to U41/D7 is glowing weakly; right red LED, V413/D6 is on.	a. Gradient coil temperature too high	Thermal overload caused by specific measurement conditions	Wait 10 minutes until the coil temperature is in safe range. Check sequence parameters.	
		Failure of cooling system	Check cooling system: Pipes, water flow, function of RCA.	TSG Cooling page 369
02. Error exists permanently: FOC from D9 to U41/D7 is glowing weakly; right red LED V413/D6 is on.	a. Monitoring error	NTC/cable defective	Measure the NTC resistance	page 305
		Sensor cable not connected	Check cable and connectors	
		D9 defective or supply voltage is missing if green LED V55 on D9 is off.	Check D9: Red LED V4,12,20 and position of DIP-switches. Replace D9	Test proc. D9 page 304
		FOC from D9 to D7 defective or not connected	Check FOC from U1/D9 to U41/D7 (light off = error)	
		D7 defective (D8 defective)	Replace D7 (Replace D8)	
		D6 defective	Check CAN-Bus communication Replace D6	Test proc. D6 page 300

Remarks: At a coil temperature of about 20°C, the NTC resistance should be approx. 25 - 42 kOhm and at the temperature limit lower than 4 kOhm. A defective NTC can be by-passed by a DIP switch on D9. **There must be at least four working NTC's in the coil to ensure safety.**

Chart 02

Error Message: **Resonant Converter not ready**

Error Code: GPA_RC_NOTRDY_ERR/_WARN

The voltage of the RC does not exceed 60% of the nominal voltage (e.g. 250 VDC_{act} at 415 VDC_{nom}). The voltage is monitored internally in the RC and the error signal is routed via FOC to D7 K2209, forwarded to D6 K2209 and via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. FOC "RDY" from RC to U53/D7 is dark; green LED "RDY" at RC is off. Eventually red LED's "PHASE" and/or "VDIP" and/or "VPRI" and/or "WATER TEMP" and/or "AIR TEMP" and/or "ILIM" at RC are on.	a. Communication error/ no enable signal	Defective FOC signal lines between GSSU/D7/U51; RC or D6 defective	Do "Reload/All", if no success: Press button S402 at GSSU/D6 and check FOC D7/U51; if there is no light, check D6, D7 or D8.	Test proc. D6 page 300
		Defective FOC signal line between RC and GSSU/D7/U53	Check FOC signal line between RC and GSSU/D7/U53.	
	b. 230 VAC aux. supply of RC	Missing "VDIP" AC power supply or VDIP < 185 VAC or > 252VAC or defective RC or control unit	Check AC power supply line; Check green voltage LED's (P15V, N15V, HF15V). Check plug (3-pin) on front panel; replace defective RC	Repair proc. Test proc. RC page 306
	c. Three phase supply and supervision	Three phase AC voltage supply error (missing phase) or "VPRI" supply voltage higher than 528 VAC	Check circuit breaker F9/50A; Provide stable AC supply.	
	d. RC overload	Overtemperature RC or Current limit/trip RC	Check FOC signal lines (light off = error); Check LED's on RC.	TSG Charts 03, 04, 20

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01.continued FOC "RDY" from RC to U53/D7 is dark; green LED "RDY" at RC is off. Eventually red LED's "PHASE" and/or "VDIP" and/or "VPRI" and/or "WATER TEMP" and/or "AIR TEMP" and/or "ILIM" at RC are on.	e. Monitoring error	Defective control board in RC	Replace RC	
02. Service mode is activated: RC is expected to be on; no error message but warning; LED "RDY" on RC lights up	a. Communication error	Defective "RDY" FOC connection	Check FOC at U53/ D7 and RDY/RC: Off: FOC or RC display board defective. Replace defective components.	

Remarks: In **service mode** this error is treated as a **warning** and will not be evaluated, i.e. modulators and regulators will be not disabled. In case of overtemperature, overvoltage or current-limit (primary stage of RC), the ready-signal will become low, when the output-voltage becomes less than 60% of the nominal value. In case of further error messages related to the errors mentioned above, refer to the corresponding Troubleshooting Charts 03/04/20.

Note: After a supply voltage failure for the RC or the GSSU you must switch the GPA into service mode (right switch S1 on D3 in upward position), to reactivate the RC via the button S402 on D6. Afterwards switch back to normal mode and perform "Reload/All").

Chart 03

Error Message: **Resonant Converter Overtemperature**

Error Code: GPA_RC_TEMP

The temperature of the power stages for the RC is too high. The temperature is measured near the IGBT's/diodes in the RC and monitored internally. The error signal is routed via FOC to D7 K2209, forwarded to D6 and via CAN Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error occurs during normal operation FOC "TEMP" from RC to U52/D7 is dark; red LED's "WATER TEMP" and/or "AIR TEMP" at RC is on.	a. Overtemperature of the power stage of the RC	Failure of the cooling system: Chilled ´water and/or fans beside the RC inside the cabinet	Check cooling system: Pipes, water flow/temp., function of RCA, Internal cabinet coolers (fans).	TSG Cooling page 369
		IGBT's/diodes in the RC are not driven in a correct manner (control PCB).	Try to switch on after a few minutes; if no success check all other reasons 01.a,02.a,03a; if negative, replace the RC.	Repair proc. RC
02. Error exists permanently; red LED "TEMP" at RC is off	a. Monitoring error	FOC from RC to D7 defective or not connected	Check FOC from RC to U52/D7 (light off = error).	
03. Error exists permanently; red LED "TEMP" at RC is on		D7 defective	Replace D7.	
		D6 defective	Check CAN-Bus communication; Replace D6.	Test proc. D6 page 300
		Monitoring(sensors) of RC defective	Replace display board at RC if available; Replace RC.	Repair proc. RC

Remarks: If an overtemperature occurs, the RC will be switched off. After the output voltage becomes less than 60% of the nominal voltage, the ready-signal of the RC will become low. The GPA (modulators/regulators) will be inhibited.

Chart 04

Error Message: **Resonant Converter Output in Current Trip**

Error Code: GPA_RC_CURTRIP

One of the six output stages of RC is in current_trip condition. The current is measured internally in RC. The RC will be disabled internally. The error signal is routed via FOC to D7 K2209, forwarded to D6 K2209 and via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error during normal operation: FOC "ITRIP" from RC to U54/D7 becomes dark; a red LED at RC (e.g. IX1) is on, showing the overloaded stage.	a. At least one of the output stages of RC was overloaded	Overload of RC caused by specific measurement conditions, i.e. overload of one of the power stages of the GPA.	Check sequence parameters; Restart the measurement	
		Wrong gradient data from GCTX/MPCU	Check gradient data cable; Run Test DAC/Functional, data transfer between GCTX and DAC	SeSo: Test Tools; Gradient
		D1 (DAC) or D2 (Regulator) defective	Check D1or D2/ Replace defective D1 or D2	Test proc. D1/ D2, page 296
02.(only in Service mode) Error occurs again after an attempt to switch on GPA a second time. (Disable modulators of GPA by switches on D4)	a. Power stage of GPA	Defective power stage (indicated by the corresponding error LED at RC (IX1...IZ2).	Replace the power stage	Repair proc. PS
	b. Secondary rectifier of RC	Defective secondary rectifier of RC	Pull out the corresponding PS and switch on RC; if the error occurs again, replace the RC	Repair proc. RC

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
03. Permanent error: FOC "ITRIP" from RC to U54/D7 is dark; other red LED's at the RC may be on or of	a. Monitoring error	Monitoring(sensors) of RC defective	Replace display board at RC; Replace RC.	Repair proc. RC
		FOC "ITRIP" from RC to U54/D7 defective	Check FOC from RC to U54/D7 (light off = error).	
		D7/D8/D6 defective	Check CAN-Bus communication; Replace D6, Replace D7.	Test proc. D6 page 300

Remarks: If the yellow LED "ILIM" on RC lights up during measurement, the primary power stage of the RC is in current limit. This may be caused if the rating of a measurement sequence is too high. If the current limit remains active for some time, the output voltage of the RC may become less than 60% of the nominal value. Then the ready signal becomes low and switches off the modulators of the GPA; refer to [Chart 02](#). As the current limit signal is not latched within the RC it will normally come back to the OK-status if the overload is removed. The next measurement may successfully restart the GPA.

Refer also to [Chart 13](#), "power stage defective".

Chart 05

Error Message: **Resonant Converter defective**

Error Code: GPA_RESCON_DEF

The three/six signals "High voltage by power stage X1...Z2" (depending on installation of three or six power stages in the gradient system) are present at the same time ($U_{nom} > 425V$). The signals are generated on the RC, routed via FOC's to D7, further via D4 -supervision and error multiplexer to the gradient processor on D6. The gradient processor creates the final error signal which is routed via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. All FOC's (VX1...VZ2) from RC to U153/253/353//163/263/363/D7 are dark. All red error LED's "Overvoltage" at D4/5 are on.	a. RC control	U_{nom} of RC misadjusted or RC control electronic defective	Check the U_{nom} adjustment on the RC (with pot R1 on the left side of the display board); Replace RC	Repair proc. Test proc. RC page 306
		AC supply for RC control UDIP exceeds 252 VAC. (Tolerance range: 207...252 VAC)	Check the correct adaptation of line supply voltage at transformer T1 to the local line voltage.	Tune-up manual; Checking the line voltage
02. RC in standby Error red LED's "Overvoltage" at D4/5 may be on or off	a. Monitoring error	FOC's from RC to U153/253/353/163/263/363/D7 are defective	Check FOC's (light off = error)	
		D7 defective	Replace D7	
		D4/5 (Error multiplexer) defective	Check red error LED's on D4/5; Replace D4/5	Test proc. D4 page 298
		D6 defective	Check CAN-Bus communication; Replace D6	Test proc. D6 page 300

Remarks: The RC is switched into standby mode by the gradient processor, i.e. no output voltage is present. The resonant circuit of the RC will not be inhibited internally.

Chart 06

Error Message: **Driver power supply defective**

Error Code: GPA_DRV_PS_DEF

The driver voltage is not ok. The monitoring is done via the three/six temperature error signals (FOC/T), (depending on installation of three or six power stages in the gradient system). If all temperature error signals are present at the same time (routed from PS via FOC to D7, further to D4 supervision and error multiplexer to gradient processor on D6), the gradient processor creates the final error signal which is routed via CAN Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. All three/six FOC's (T) from PS to D7 are dark; green LED at D10 is off.	a. Synchronization of D10	No synchronization clock, D3/D7 defective, FOC between D7 and D10 is missing or defective. Jumper X4 on D10 is missing	Check red error LED V37/D3 (refer to TSG Chart 09) Replace D3/D7 Check FOC between D7 and D10 Check correct jumper position 1- 2	
	b. GSSU power supply E4A, Slot 2, 15 V supply	E4A defective	Remove the 4- pin connector X1on D10. Measure supply voltage 15V at X1, pin 1- 2 or on terminal of E4A/slot 2; replace E4A	
	c. Power supply connection between E4A and D10	Power line disconnected, bad connector X1	Check power line and connector X1	
	d. D10 Driver supply board	D10 defective	Remove the 24-pin connectors (X2,X3) on D10; if LED (at D10) does not go on, replace D10	

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
02. Green LED lights up after removing the 24-pin connector from D10	a. Power stage driver board	Driver board of power stage defective	Reconnect the 24-pin connector at D10. Remove the 9-pin X1 connectors at the power stages, one after another, until LED on D10 is on. Replace the corresponding power stage.	
03. Green LED at D10 is on.	a. Driver supply connections	Defective driver supply connections to PS	Check at X1 at PS:15V rectangular signal between 1-4, 2-5, 3-6, 7-8. Check X2, X3 at D10.	
	b. Monitoring error	D4/5, D6 defective FOC's (T) between PS and D7 defective	Check CAN-Bus communication; Replace D6/D4/5 Check FOC's (all lights off = error)	Test proc. D6 page 300

Remarks: Check jumper on D10 as well: X4 Pin 1/2 external clock control (normal operation), Pin 2/3 internal clock control.

Chart 07

Error Message: **No Reaction of Modulator**

Error Code: GPA_MOD_SET_ERR/_WARN

The modulators of the three axes x,y,z can be controlled by the service software or by switches on D4, i.e. they can be switched on and off. If after a control command from the MPCU the modulator is not in the desired condition, this error is generated by the gradient processor, using a combination of several control signals (switch signals from D4/5 and control signals from D6). The final error signal will be routed via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Normal operation of MR measurement system or Service software tests running (e.g. gradient tests): Modulator disabled, yellow left LED V1/A2 at D4 is on. (only if modulators disabled by application software of GPA or switches S1/2 on D4). GPA_Status message: X,Y,Z modulator inhibit = yes	a. D4/5	Switch position: S1 left, S2 left and right: Upper position means modulator x,y,z-axis disabled	Set switches in "on" position (downward position) Loading and starting a sequence enables the modulators if no other error condition exists. Eventually do "Reload/All".	SeSo: Test Tools Gradient GPA_Status (Online help)
		D4/D5 defective	Replace D4/5	
	b. D6	D6 defective	Check CAN-Bus communication Replace D6	Test proc. D6 page 300

Remarks: In the **service mode** (right switch S1 on D4 in upper position or service mode enabled by the service software) this error is treated as a **warning** and will not be evaluated, i.e. modulators and regulators will not be disabled (with the exception of manual disabling via switches on D4/5).

Chart 08

Error Message: **No Reaction of Resonant Converter**

Error Code: GPA_RC_SET_ERR

Within five seconds after an "Enable" request (either by NUMARIS or by manually pressing the push button "RC on" S402/D6), the RC does not respond with the "RDY" message. The gradient processor generates the final error message which is routed via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error after GPA "Power On" command	a. Communication between gradient processor and RC	FOC's "EN" and/or "RDY" between RC and D7 missing or defective	Check FOC connections by manually switching the RC on/ off (pressing S402 on D6). The green LED's between S402 and S 403 must be on then.	
		D7 defective	Check FOC connections on D7: U51 (RC_ENABLE_FOC) U53 (RC_READY_FOC) (no light = error)	
	b. Three phase supply and supervision	Three phase AC voltage supply error or missing phase; red LED "PHASE" may be on.	Check circuit breaker F9/50A; Provide stable AC supply.	TSG Chart 02
	c. RC control	RC control defective	Replace the RC	
	d. Monitoring error	D6 defective	Check CAN-Bus communication Replace D6	Test proc. D6 page 300

Remarks: If the RC cannot be switched on by pressing S402 (caused by application software of GPA), proceed as follows:
Switch on the "service mode": right switch S1 at D4 in upward position. Press reset-button S 404 at D6. Now press S402, RC on. If the RC is switched on (green LED "RDY" at RC is on) switch the "service mode" off again (right switch S1 at D4 in downward position). Afterwards do "Reload/All".

Chart 09

Error Message: **Synchronization fault**

Error Code: GPA_SYNC_ERR/_WARN

The synchronized clock signal from MPCU(GCTX) is faulty or missing. This 4MHz clock signal is transmitted via the gradient waveform interface cable from the GCTX/MC4C40 to the D1 K2209. Here the signal is divided into a 50 kHz signal. If the phase detector (PLL) on D3 K2209 detects an error, the error signal will be routed to D6 K2209 and via CAN-Bus to the MPCU; a red LED, V37 "SYNC Error" at D3 will be on.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error during normal operation; red error LED V37 "SYNC Error" at D3 is on. Image quality may be degraded.	a. Communication between MMC and D1	Error on MMC (MC4C40/GCTX)	Run MMC Tests (MC4C40/GCTX)	SeSo: Test Tools MMC
		Gradient data cable and/or connectors defective	Check gradient data cable; Run Test DAC/Functional, data transfer between GCTX and DAC	SeSo: Test Tools Gradient
	b. D1	D1, wrong switch positions (S3/4); D1 defective	Check switch positions on D1: S3-pos.7, S4-pos.2 Replace D1	
02. Red error LED at D3 is on or off.	a. Adjustment error and/or monitoring error	D3 misadjusted or defective	Measure the signal "TRIA"/ "FREQ1"(52 kHz) at D3 X10/12; Adjust D3/R66 (during adjustment remove gradient data cable from MMC to D1); Replace D3	Test proc. D3 page 297
		D6 defective	Check CAN-Bus communication Replace D6	Test proc. D6 page 300

Remarks: In the **service mode** this error is treated as a **warning** and will not be evaluated, i.e. modulators and regulators will be not disabled. Image quality may be degraded.

Chart 10

Error Message: **Triangle voltage error**

Error Code: GPA_TRIA_ERR

The triangle voltage amplitude is out of tolerance or the offset is too high. The error signal is generated on D3 K2209 in combination with the supervision of the signal F_SU; this signal is generated on D4 (Jumper S3) from combination of three clock signals, generated on D3. The final error signal "TRIA ERR" is routed to D6 and via CAN-Bus to the MPCU; a red LED, V38 "TRIA Error" at D3 will be on.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error during normal operation; red error LED V38 "TRIA Error" at D3 is on.	a. D3, generation of triangle voltage (amplitude/offset)	D3 defective	Measure the triangle voltage: "TRIA" (50kHz) at D3/X10 Replace D3	Test proc.D3 page 297
	b. D4, generation of the signal F_SU (25kHz)	D4 defective or incorrect position of DIP switches S3	Check position of DIP switches on D4: S3: 5=off, 7=on. Replace D4	Test proc. D4/5 page 298 and page 299
	c. D2	D2 defective	Replace D2	Repair Proc. GSSU
02. Red error LED at D3 is on or off.	a. Monitoring error	D3 defective	Measure the triangle voltage "TRIA" (50 kHz) at D3/X10; Replace D3	Test proc. D3 page 297
		D6 defective	Check CAN-Bus communication; Replace D6	Test proc. D6 page 300

Chart 11

Error Message: **Plus 15V/ Minus 15V/ Plus 5V defective**

Error Code: GPA_P15/ N15/ VCC_ERR

DC supply voltages from E4A (+5V VCC, Slot1 and +/-15V, Slot3) are out of tolerance (+/- 9%). The voltage is monitored on D3 K2209; the error signals are routed to D6 and via CAN-Bus to the MPCU. In case of a +/-15V failure red LED's V32/31 "+/-15V Error" are on; in case of a +5V failure, green LED V30 "+5V OK" is off.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error after GPA "Power On" Green LED V30 at D3 off Red LED's V32 (+15V) and/or V31 (-15V) are on.	a. Power supply unit E4A	Power supply E4A defective; Fan defective	Measure/adjust the output volt- ages directly at the outputs of the power supply slots. If not adjustable replace the power supply.	Test proc.E4A page 301
	b. Power supply lines	D7,8 defective or not plugged cor- rectly; Cabling between D7 and E4A defective.	Measure the +/-15V and +5V sup- ply voltages at D7/X5; if ok, the backplane D8 is probably defec- tive.	
	c. Boards of GSSU and D10	Defective PCB	Replace the defective board	Remarks
01.continued Error after GPA "Power On" Green LED V30 at D3 off Red LED's V32 (+15V) and/or V31 (-15V) are on.	d. Monitoring error	Supervision on D3 defective	Measure the supply voltages at D3 X16: +15 V, X18: -15V; X20: +5V (+/- 5% tolerance range) if ok, replace D3.	Test proc. D3 page 297
		D6 defective	Check CAN-Bus communication Replace D6	Test proc. D6 page 300

Remarks: In principle each defective board in the GSSU including the D10 and the backplane D8 could be the reason for the supply voltage break down and should be checked. Also check fuse F14/4A in the CCA cabinet. The sum of the supply voltages should be lower than 10 mV_{pp}.

Chart 12

Error Message: **Upgrade component error**

Error Code:

The upgrade component is defective or not available. The error signal is generated on the upgrade component and routed via FOC or wire (D7/X5) to D7 K 2209 forwarded to D6 and via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. FOC D7/U14 is dark. Left red LED V413/D6 is on.	a. Optional upgrade component	Failure of upgrade component	Troubleshooting on upgrade component; Replacement of defective components	
	a. Monitoring error	Defective upgrade interface between D7 (connector X5) and upgrade component	Check connection line	
		D7 defective	Check FOC U14 at D7; (light off = error). Replace D7	
02. FOC D7/U14 lights up		D6 defective	Check CAN-Bus communication Replace D6	Test proc. D6 page 300

Remarks: If the Upgrade component is not installed, a short link must be installed on D7, connector X5, between PIN 8 and 4.
If the error occurs, the RC will be switched to standby mode.

Chart 13

Error Message: **Power stage defective**

Error Code: GPA_PST_DEF

One or more of the six output stages of the RC is in current_trip condition. The current is measured internally in the RC. The RC will be disabled internally. The error signal is routed via FOC to D7 K2209, forwarded to D6 K2209 and via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. The error occurs after the second attempt to switch on the GPA, especially if the modulators are disabled (e.g. in service mode). FOC "ITRIP" from RC to U54/D7 becomes dark; one or more red LED "Current trip" at RC (IX1...IZ2) is on, indicating the overloaded stage.	a. Power stage	Power stage(s) defective	Remove the corresponding power stage (the defective power stage is indicated by a glowing red LED's "Current trip" (IX1 ... IZ2) on the RC) and try to switch on the RC again. If "Current trip" doesn't occur again, replace the gradient power stage.	Repair proc. PS
	b. RC	Secondary Rectifier in RC defective	Remove corresponding gradient power stage and try to switch on the RC again; if "Current trip" occurs again, replace RC.	Repair proc. RC

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
02. No error LED at RC lights up. The RC is disabled, i.e. green "Ready" LED is off.	a. Monitoring error	FOC "ITRIP" defective Supervision of RC defective	Check FOC connection "ITRIP" at RC; if light is off, replace RC	
		FOC "ITRIP" between RC and D7 defective; D7 defective	Check FOC U54 "N_RC_CURTRIP_FOC at D7 (light off = error); replace D7 if FOC "ITRIP" is lighted up at RC.	
		D6 defective	Check CAN-Bus communication Replace D6	Test proc. D6 page 300

Remarks: If the current trip occurs again, even the modulators are disabled (e.g. via service mode switch on D4). This is a strong indication of a defective RC or **defective power stage**.

Chart 14

Error Message: **Regulation error on X,Y,Z-axis**

Error Code: GPA_LOOP_ERR/_WARN_X/_Y/_Z

The error signal and the regulator output signal are monitored in the closed loop error logic on D2 K2209. If one of these signals reaches the upper limit (10V), the yellow LED V27 on D2 lights up and an error signal is generated on D2, routed to D6 and via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Dynamic error while running a measurement; the yellow LED V27 "Regulation Error" on D2 lights up.	a. Measurement sequence	Gradient rise time is too fast	Switch on the Service mode to find out the affected axis. Run the specific sequence again; check the sequence parameters, if other than Siemens-released sequences are used.	
	b. Gradient data transmission	MC4C40/GCTX defective, Data cable and/or connectors defective, D1 K2209 defective	Run MMC Test; run Gradient Test for DAC; check gradient data cable and connectors. Replace D1.	SeSo: Test Tools MMC Gradient
	c. Load circuit gradient coil interrupted	Gradient coil, cable connections, RF filters defective	Check the continuity of the load circuit; check for loose cable connections. Replace defective components.	
	d. Closed control loop	D2, D4/5 defective; D7 defective; FOC's to power stages defective	Perform Regulator Adjust/Check; cyclically replace the D2's; check for other error messages	SeSo: Tune-up/QA and page 296
	e. Measurement of actual current	SENS, D2 defective	If no CAV_ERR_X,Y,Z (refer to TSG Chart 15), cyclically replace D2 (x,y,z); replace defective D2. Otherwise replace defective SENS	Test proc. D2 Repair proc. GSSU TSG Chart 15

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01.continued Dynamic error while running a measurement; the yellow LED V27 "Regulation Error" on D2 lights up.	f. Power stage	Power stage defective	Check for other error messages; cyclically replace power stages	
	g. RC	Too low output voltage	Check for error messages from RC Measure the output voltage of the RC and adjust it if necessary	"RC Test procedure", page 306
02. Error after GPA "Power On"; (in the idle mode), the yellow LED V27 "Regulation Error" on D2 lights up.	a. DAC D1	D1 defective/ high offset	Check voltage at D2, X3 in idle mode: Offset < 1 mV	
	b. Regulator D2	D2 defective	Cyclically Replace the D2's; replace D2	Repair proc. GSSU
	c. Modulator D4/5	D4/5 defective	Replace D4/5	
	d. Current sensors (SENS)	Current sensors (SENS) defective	Replace current sensors (SENS)	Repair proc. SENS
	e. Monitoring error	D6 defective	Check CAN Bus communication Replace D6	Test proc. D6 page 300

Remarks: The regulation error occurs as a consequence of every other error but will be suppressed in this case; you will see only the causal errors in the error report. The regulation error will only be recorded in the system report if it occurs as a primary error. Refer to [Chart 15](#) (Current actual value error) or [Chart 16](#) (High current value error).

In **Service mode**, the regulation error is treated as a **warning** and will not be evaluated, i.e. modulators and regulators will not be disabled.

In case that the SeSo-program "**Regulator Adjust**" **does not work** - overshoot too large and termination of the program with the error message - "Calculated Amp (p) pot is out of allowed range. Setting to default value." - check the U_{nom} adjustment on the RC (with pot R1 on the left side of the display board). Refer to repair proc./Test proc. RC [page 306](#).

Chart 15

Error Message: **Current actual value error on X,Y,Z-axis**

Error Code: GPA_CAV_ERR_X/_Y/_Z

The current sensor signal (actual current value) is not correct. The signal is monitored on D2 K2209 (red LED V24 "SENS Error" lights up) and is forwarded to D6 and routed via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error after GPA "Power On"; the red LED V24 "SENS Error" on D2 lights up.	a. Current sensor supply	Break in signal line or power supply disconnected or supply line shorted	Check supply/signal line from SENS/X1 to D7/X3 for continuity and shorts	
	b. Power line from GPA to the gradient coil	Ground short of a gradient power circuits	Check the gradient coil circuit for ground shorts (gradient coil, gradi- ent cables, gradient filters)	
	c. D2 Regulator	D2 defective/ defective SENS soft start control (the error is detected, if the sen- sors are switched off)	Replace D2	Repair proc. GSSU
	d. D7 FOC board	D7/8 defective	Check connectors; replace D7	

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01.continued Error after GPA "Power On"; the red LED V24 "SENS Error" on D2 lights up.	e.(Type IT600S/LA205-S) Synchronization via FOC	FOC's between D7,U22(X), U24(Y), or U25(Z) defective	Check the FOC's between D7 and SENS	Repair proc. SENS
	f. Current sensors (SENS) Type LB200-S/SP4 paired or Type IT600S combined with LA 205-S	SENS defective: Difference of compensation cur- rents of two paired sensors is higher than 24 mA or in case of the IT600S/LA 205-S combination to high	Measure the sensor signals at D2, X707 and X709 Replace current sensors (SENS)	Repair proc. SENS
No error LED lights up:	g. Monitoring error	D6 defective	Check CAN-Bus communication Replace D6	Test proc.D6 page 300

Chart 16

Error Message: **High current on X,Y,Z-axis**

Error Code: GPA_HI_CURR_X/_Y/_Z

The actual current exceeds the limit of 300A +/- 9A; (red LED's "Overcurrent" for x,y or z axis on D4/5 K2209 light up). The monitoring is done at D4 and an error signal is forwarded to D6 K2209. The final error signal is generated on D6 and routed via CAN Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Dynamic error while running a measurement; the left red LED V4,5 or 6 "Overcurrent X,Y,Z" on D4 lights up	a. Measurement sequence	Sporadic overload	First run the duty_cycle sequence: If ok, check the measurement sequence parameters.	
	b. D2	Regulator D2 oscillates due to wrong parameters; D2 defective	Run the Regulator Adjust/Check; cyclically replace the D2's. Replace defective D2.	SeSo: QA/Tune-up: Regulator Adjust/Check
	c. Gradient load circuit	Short in the gradient load circuit (gradient coil, cables, filters)	Measure the resistance of the load circuit: Short, if only a few mOhm.	
	d. Communication between MMC and D1	Error on MMC (MC4C40/GCTX)	Run MMC Tests	SeSo: Test Tools MMC
		Gradient data cable and/or connectors defective	Check gradient data cable; Run Gradient tests for DAC	SeSo: Test Tools Gradient
		D1 defective	Replace D1	Repair proc. GSSU

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01.continued Dynamic error while running a measurement; the left red LED V4,5 or 6 "Overcurrent X,Y,Z" on D4 lights up	e. D4/5	Defective Modulators	Replace D4/5	Test proc. D4/5 page 298
	f. D7/FOC's to PS's	Defective lightboard and/or FOC's	Replace D7 and or FOC's	
	g. current sensors (SENS)	Measured values incorrect by defective current sensors (SENS)	Replace current sensors (SENS)	TSG chart 15 page 235 Repair proc. SENS
02. Error after GPA "Power On"; the left red LED V4,5 or 6 "Overcurrent X,Y,Z" on D4 lights up	a. Monitoring error	D4 defective	Replace D4	Test proc. D4 page 298
No error LED lights up.		D6 defective	Check CAN-Bus communication Replace D6	Test proc. D6 page 300

Remarks: Check also for accompanying error messages, e.g. "Current actual value error" -> [page 235](#), or "Regulation error on X,Y,Z-axis" -> [page 233](#) and refer to the corresponding TSG-Chart.

Chart 17

Error Message: **Duty cycle for power stage X1(2),Y1(2),Z1(2)**

Error Code: GPA_DUTYCYCLE_X1(2)/_Y1(2)/_Z1(2)

The duty cycle of the actual current is too high; (red LED's "Duty Cycle Error" for x,y or z axis on D4 K2209 light up). The monitoring is done on D4 and an error signal is forwarded to D6 K2209. The final error signal is generated on D6 and routed via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Dynamic error while running a measurement; one or more of the the right red LED V5,6,or 7 "Duty cycle Error X1,Y1,Z1" on D4 lights up.	a. Measurement sequence	Sporadic overload	Check the measurement sequence parameters Check the system by running a high duty cycle sequence.	
	b. Communication between MMC and D1	Error on MMC (MC4C40/GCTX)	Run MMC Tests	SeSo: Test Tools
		Gradient data cable and/or connectors defective	Check gradient data cable; Run Gradient Tests for DAC	SeSo Test Tools: Gradient DAC/ Functional
		D1 defective	Replace D1	
	c. D2	Regulator D2 oscillates due to incorrect parameters; D2 defective	Run Regulator Adjust/Check; Cyclically replace the D2's but remember their original positions. Replace defective D2.	SeSo: QA Tune-up: Regulator Adjust/Check
	d. D4/5	Defective Modulator boards	Replace D4/5	
	e. D7/FOC's to power stages	Defective lightboard D7 and/or FOC's to power stages	Replace D7 and or FOC's to power stages	

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Dynamic error while running a measurement; one or more of the the right red LED V5,6,or 7 "Duty cycle Error X1,Y1,Z1" on D4 lights up.	f. Current sensors (SENS)	Incorrect values measured by defective current sensors (SENS)	Replace current sensors (SENS)	Repair proc. SENS
02. Error after GPA "Power On"; one or more of the right red LED V5,6,or 7 "Duty cycle Error Xi,Yi,Zi" on D4/5 lights up	a. Monitoring error	D4/5 defective, especially if a X,Y,Z_2/_3 error occurs.	Replace D4/5	Test proc. D4/5 page 298
No error LED lights up.		D6 defective	Check CAN-Bus communication Replace D6	Test proc. D6 page 300

Remarks: Check also for accompanying error messages, e.g. "Current actual value error" -> [page 235](#), and refer to the corresponding TSG-Chart.

Chart 18

Error Message: **Controlling error for power stage X1(2),Y1(2),Z1(2)**

Error Code: GPA_CTRL_ERR_X1(2)/_Y1(2)/_Z1(2)

The control signals for the power transistors of a specific power stage are not ok. The monitoring is done on D4/5 K2209 (one red LED "Drive Error" lights up) and an error signal is forwarded to D6 K2209. The final error signal is generated on D6 and routed via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Dynamic error while running a measurement; the right red LED V2/A2 "Drive Error" on D4/5 lights up.	a. D4/D5 (attempt to switch the power stage transistors without dead time; refer to FUN/Gradient system)	D4/D5 defective	Replace the D2 of affected axis with another D2 Replace D2/4/D5	Test proc. D4/5 page 298
02. Error after GPA "Power On"; the right red LED V2/A2 "Drive Error" on D4/5 lights up.	a. Monitoring error	D2/4/5 defective	Replace the D2 of affected axis with another D2 Replace D2/D4/5	Test proc. D4/5 page 298
03. Error after GPA "Power On"; the right red LED V2/A2 "Drive Error" on D4/5 are off		D6 defective	Check CAN-Bus communication Replace D6	Test proc. D6 page 300

Remarks: Because of the possibility of D4/5 errors (i.e. the modulators cannot be disabled), the RC will be switched to standby mode. Check also for accompanying error messages, e.g. **"High current error on...-axis"** -> [page 237](#), and refer to the corresponding TSG-Chart.

Chart 19

Error Message: **Temperature error for power stage X1(2),Y1(2),Z1(2)**

Error Code: GPA_TEMP_ERR_X1(2)/_Y1(2)/_Z1(2)

The power stage temperature exceeds the limit of 56°C. The temperature is measured inside the power stages (at power stage transistors) and an error signal is sent via FOC to D7 K2209. The error signal is monitored on D4/5 (one red LED "Overtemp" lights up) and forwarded to D6 K2209. The final error signal is generated on D6 and routed via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Dynamic error while running a measurement; the left red LED V2/A1 "Overtemperature" on D4/5 lights up.	a. Measurement sequence	Load of GPA by measurement parameters or measurement conditions are too high	Check the sequence parameters; Run an approved high duty cycle sequence	
	b. Cooling system	No water flow or water flow in the cooling system is too low or cabinet cooling system (RCA) defective	Check for other error messages from RCA or MMC (cabinet temperature control) Check RCA, water pipes, flow rate; Check the cabinet cooling system (CCS)	TSG Cooling page 369
	c. GSSU of GPA	Any component of the GSSU defective	Check for other error messages	Corresponding TSG chart
	d. D7	D7 defective	Replace D7	
	e. Power stage	"Drive" FOC's from D7 to power stage defective	Check and replace FOC's	

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
02. Continuous error, trying GPA "Power On" after wait- ing period of approx. 30 sec; the left red LED's V2/A1 "Overtemperature" on D4/5 lights up. One or more of the FOC's "T" from the power stages to D7 are dark.	a. Monitoring error	FOC's "T" from power stage to D7 defective	Check the FOC at power stage "T" and at D7/U152, 252, 352 and D7/U161, 262, 362 (Light off = error)	Test proc. D7 page 301
		Temperature sensors of power stage defective	Replace power stage if "T" is con- tinuously off in idle state	Repair proc. PS
		D7 defective	Check FOC's at D7/U152, 252, 352 and D7/U162, 262, 362 (Light off = error) Replace D7 if light is on	Test proc. D7 page 301
		D4/D5 defective, especially if a X,Y,Z_3 error occurs	Replace D4/5	Test proc. D4/5 page 298
No error LED lights up.		D6 defective	Check CAN-Bus communication Replace D6	Test proc. D6 page 300

Remarks: If the FOC's "T" of all power stages are dark, then probably the driver supply board D10 K2209 supplies no output power; refer to TSG Chart 6, [page 223](#).

If the **duty cycle error for the same axis** (refer to TSG Chart 17, [page 239](#)) **occurs at the same time**, check at first the used **measurement parameters**, otherwise check at first the **cooling system**.

Chart 20

Error Message: **High voltage by power stage X1(2),Y1(2),Z1(2)**

Error Code: GPA_HIVOLT_X1(2)/_Y1(2)/_Z1(2)

The output voltage of one channel of the RC exceeds the upper limit 425V. The voltages are measured on the RC. The error signal is sent via FOC to D7 K2209 and is monitored at D4/5 K2209 (dependent on the axis a red LED, e.g. "Overvoltage X" lights up) and is forwarded to D6 K2209. The final error signal is generated on D6 and routed via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Dynamic error while running a measurement; one or more of the the right red LED's V3,4 "Overvoltage Xi,Yi,Zi" on D4/5 lights up.	a. Power stage	A single defective power stage produces overvoltage spikes.	Swap the power stage with another one. Replace defective power stage.	Repair proc.PS
	b. Drive control (only when D5 installed)	D4/5 defective (possible only if a upgrade component is installed)	Replace D4/5	Test proc. D4/5 page 298
	c. RC	RC defective or misadjusted	Measure the threshold voltage for the overvoltage supervision (6 Testpoints behind the front panel of the RC). Replace RC.	Test proc.RC page 306 Repair proc. RC

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
02. Simultaneous error messages "High voltage" from all power stages	a. RC	RC defective or misadjusted	Adjust the actual output voltage of the RC. Replace RC	Test proc.RC page 306 TSG chart 05 Repair proc. RC
	b. Line voltage out of tolerance or wrong jumpering of primary transformer	Line voltage out of tolerance range	Provide correct line input voltage	

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
03. Error after GPA "Power On"; one or more of the right red LED's V3,4 "Overvoltage Xi,Yi,Zi" on D4/5 lights up. One or more of the FOC's VXi,Yi,Zi from RC to D7 are dark. (i = 1,2)	a. Monitoring error	RC supervision defective	Measure the RC output voltages Check FOC outputs at RC: VX1...VZ2 (Light off = error, i.e. voltage too high) Replace RC	Test proc. RC page 306
		FOC's between RC and D7 defective	Check FOC's at D7/U153, 253, 353 and D7/U163, 263, 363 (Light on = ok)	Test proc. D7 page 301
		D7 defective	Check FOC's at D7/U153, 253, 353 and D7/U163, 263, 363 (Light off = error) Replace D7 if light on	Test proc. D7 page 301
		D4/5 defective	Replace D4/5	Test proc. D4/5 page 298
04. No error LED lights up.		D6 defective	Check CAN-Bus communication Replace D6	Test proc.D6 page 300

Remarks: In this error status, the RC will be switched to standby mode within 0.1 ms by the gradient processor which disables the RC ("Enable"- signal -> off). If all RC stages signal overvoltage at the same time, the RC is probably defective -> refer also to TSG Chart 05, [page 222](#).

Chart 21

Error Message: **The command is not acknowledged in certain time**

Error Code: TIMEOUT

The upgrade component does not respond or is not available/ connected.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01.K2209 The GPA is switched off FOC D7/U14 is dark	a. D7/X5	Jumper of X5, Pin 8 to ground (Pin 3,4,5) is missing	Install the Jumper of X5, Pin 8 to ground (Pin 3,4,5)	Test proc. D7 page 301
	b. CAN unit D1 on D6 K2209	Defective D1/D6 (CAN module)	Replace D6 K2209	SeSo: Test Tools: Gradient/ Communica- tion/Status

Chart 22

Error Message: **GPA command is not acknowledged in certain time**

Error Code: GPATIMEOUT

After a command from the MPSU, the GPA CAN-unit does not send the command acknowledge within the time specified.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01.K2209 The GPA CAN-unit does not respond to any command from the MPSU. The RC is in standby condition.	a. GPA CAN communication to MPCU	Missing or defective CAN-cable or connectors	Check CAN-cable connections to the MPCU at D6 K2209, connector X402.	SeSo: Test Tools: Gradient/ Communication/Status
	b. CAN unit D1 on D6 K2209	Software hang-up; Defective D1 (CAN module)	Try System/Reload all; Check/Replace D6 K2209	Test proc.D6 page 300
02.K2217 The GPA CAN-unit does not respond to any command from the MPSU. The main power supply is in standby condition.	a. GPA CAN communication to MPCU	Missing or defective CAN-cable or connectors	Check CAN-cable connections to the MPCU at D16, connector X402.	SeSo: Test Tools Gradient/ Communication/Status
	b. CAN unit D1 on D16	Software hang-up; Defective D1 (CAN module)	Try System/Reload all; Check/Replace defective D16.	Test proc.D16 page 318

Chart 23

Error Message: **GPA set loop error occurred (GPA internal)**

Error Code: GPALOOPERROR

The gradient measurement loops (stability/linearity) controlled by the SeSo (Test Tools), i.e. DAC-value loop/ Actual-value loop (via Loop MUX on D6/D16) cannot be set correctly. The GPA CAN-unit sends the message "loop setting not possible" to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error message "GPA "Set loop" error occured" at RSS The RC/main power supply is in standby condition.	a. D1/D6/D16	D1/D6/D16 defective	Check/Replace D6 K2209/ D16 K2217	Test proc. D6 page 300 Test proc.D16 (Ref)
	b. Communication/ GPA hardware	Wrong message code was sent back from GPA to MPCU Hardware error GPA	Check CAN-Bus cable Check manual loop setting Perform Regulator Check Check GPA: switch and jumper settings. Check switch setting on D4/D17. Perform Reload/All and try again.	SeSo: Test Tools: Gradient/ Communica- tion/Status SeSo: QA: Regulator Check

Remarks: In the service mode the GPA will be run up to normal operation conditions; however, in this case the regulator setting may originate from the default S1/2 DIP-switch settings on D12 if an error occurs. This may result in suboptimal adjustment of the regulators. A deterioration of the image quality is possible.

Chart 24

Error Message: **GPA set regulator error occurred (GPA internal)**

Error Code: GPAREGERROR

The regulator setting (of P- and I-part of the x,y,z-regulators D2 K2209/D12 K2217) is not successful.
The GPA CAN-unit sends the message "loop setting not possible" to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error message "GPA set regul. err. occurred" at RSS The RC/main power supply is in standby condition.	a. D1/D6/D16	D1/D6/D16 defective	Check/Replace D6 K2209/D16 K2217	Test proc. D6 page 300 Test proc. D16 (Ref)
	b. Communication/ GPA hardware	Wrong message code was sent back from GPA to MPCU Hardware error GPA	Check CAN-Bus cable Perform Regulator Check Check GPA: switch and jumper settings Check switch setting on D4D17 Perform Reload/All and try again	SeSo: Test Tools: Gradient/ Communica- tion/Status SeSo: QA: Regulator Check

Remarks: In the service mode the GPA will be run up to normal operation conditions; however, in this case the regulator setting may originate from the default S1/2 DIP-switch settings on D2 K2209/D12 K2217 if an error occurs. This may result in suboptimal adjustment of the regulators. A deterioration of the image quality is possible.

Chart 25

Error Message: **GPA set mode error occurred (GPA internal) or GPA mode could not be set** Error Code: GPAMODEERROR or GPAMODE-NOTSET

The mode setting - **service/normal mode** - initiated by the MPCU (Software) is not successful.
The GPA CAN-unit sends the message "mode setting not possible" to the MPCU

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error messages "GPA set mode err. occured" or "GPA mode could not be set" at RSS The RC/main power supply is in standby condition.	a. Modulator D4/D14	Right switch S1/D4 or service switch S2/2 on D17 is in upward (i.e. "Service mode")position. (On D4 the left upper yellow LED lights up).	Switch S1 on D4 K2209 or S2/2 on D17 D2217 in downward position.	
	b. D1/D6/D16	D1/D6/D16 defective	Check/Replace D6 K2209/ d16 K2217	Test proc. D6 page 300 Test proc. D16 page 318
	c. Communication/ GPA hardware	Wrong message code was sent back from GPA to MPCU Hardware error GPA	Check CAN-Bus cable Check GPA: Switch and jumper settings Check switch setting S1 on D4 or S2/2 on D17.Perform Reload/All and try again.	

Chart 26

Error Message: **GPA set modulator error occurred (GPA internal)**

Error Code: GPAMODULATORERR

The modulators (D4/5 K2209/D14 K2217) cannot be controlled by the MPCU (Software), i.e. any of the modulators of an axis (x,y,z) cannot be enabled/disabled by a software command. The GPA CAN-unit sends the message "modulator not settable" to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error message "GPA set mod. err. occurred" at RSS The RC/main power supply is in standby condition.	a. Modulator D4/D14	Modulator control switches on D4 or D17 are in upward (i.e. "OFF") position.	Switch S1/2 on D14 or S1/1, S1/2, S2/1 on D17 in downward position	
	b. D1/D6/D16	D1/D6/16 defective	Check/Replace D6 K2209/D16 K2217	Test proc. D6 page 300 Test proc. D16 page 318
	c. Communication/ GPA hardware	Wrong message code was sent back from GPA to MPCU Hardware error GPA	Check CAN-Bus cable Check GPA: switch and jumper settings Check function of control switches S1/2 on D4 or S1/1, S1/2, S2/1 on D17 Perform Reload/All and try again	

Chart 27

Error Message: **GPA power ON/OFF could not be set**

Error Code: GPAPOWERNOTSET

The GPA power cannot be controlled by the MPCU (Software), i.e. the RC/main power supply cannot be switched on or off. If the RC is on, the input voltage of the power stages is approx. 400 VDC. The GPA CAN-unit sends the message "power on/off not settable" to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error message "GPA power ON/OFF not set" at RSS The RC is on or in standby condition.	a. D1/D6/D16	D1/D6/D16 defective	Check/Replace D6 K2209/D16 K2217	Test proc. D6 page 300 Test proc. D16 page 318
	b. Communication/ GPA hardware	Wrong message code was sent back from GPA to MPCU Hardware error GPA	Check CAN-Bus cable Check GPA: switch and jumper settings Perform Regulator Check Perform Reload/All and try again	SeSo: Test Tools: Gradient/ Communication/Status SeSo: QA: Regulator Check

Chart 01a

Error Message: **Gradient coil not ready**

Error Code: GPA_COIL_NOT_RDY

The gradient coil temperature exceeds 76 °C. The temperature is measured with six NTC resistors located in the gradient coil. The monitoring is done on D9 K2209 (patient table) and the error signal is routed via FOC to U7/D17 K2217, through D18 to D16 K2217 and via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error during normal operation of the system: FOC from U1/D9 to U7/D17 is glowing weakly; red LED, V413/D16 lights up. The red LED V38/A1 on D17 lights up.	a. Gradient coil temperature too high	Thermal overload caused by specific measurement conditions	Wait 10 minutes until the coil temperature is in safe range. Check sequence parameters.	
		Failure of cooling system	Check cooling system: Pipes, water flow, function of RCA.	TSG Cooling page 369
02. Error exists permanently: FOC from U1/D9 to U7/D17 is glowing weakly; right red LED V413/D16 "COILERR" is on or off. The red LED V38/A1 on D17 is off.	a. Monitoring error	NTC/cable defective	Measure the NTC resistance.	Test proc. D9 page 304
		Sensor cable not connected	Check cable and connectors.	
		D9 defective or supply voltage is missing if green LED V55 on D9 is off	Check D9: Red LED V4,12,20 and position of DIP-switches. Replace defective D9.	Test proc. D9 page 304
		FOC from D9 to D17 defective or not connected	Check FOC from U1/D9 to U7/D17 (light off = error).	
		D17 defective (D18 defective)	Replace defective D17, (D18).	

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
02. Error exists permanently: FOC from U1/D9 to U7/ D17 is glowing weakly; right red LED V413/D16 "COILERR" lights up. The red LED V38/A1 on D17 is on or off.	a. Monitoring error	D16 defective	Check CAN-Bus communication; replace defective D16.	Test proc. D16 page 318
03. The right red LED V413/D16 "COILERR" lights up, the red LED V38/A1 on D17 is off.		D18 and/or D16 defective	Check D18 and D16.	

Remarks: All modulators will be stopped by the loadware (MOH).

At a coil temperature of about 20°C, the NTC resistance should be approx. 25 - 42 kOhm and at the temperature limit lower than 4 kOhm. A defective NTC can be by-passed by a DIP switch on D9. **There must be at least four working NTC's in the coil to ensure safety.**

Chart 02a

Error Message: **Resonant Converter not ready**
-> Main power supply not ready

Error Code: GPA_RC_NOTRDY_ERR/_WARN

The 400 VDC supply voltage for the power stages is switched off. The error signal is generated by opening of relay contact K3 on D31 K2217(open circuit) as a consequence of not closed main contactor K2 (main supply voltage for GPA transformer). The error signal is routed via cable to D17 K2217, through D18 to D16 and via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. The two green LED's V407 on D16 are off.	a. Communication error: No enable signal from D16 for D31, or no acknowledge signal from D31	Defective cable connection between GSSU X3/D17 and X1/D31	Do "Reload/All", if no success: Check cable between X3/D17 and X1/D31.	Test proc. D16 page 318 Test proc. D17 page 320
		D31 or D16 or D17 defective	Check D16, D17/(D18) and D31.	
	b. Soft Start Unit D31	Cable connection between K2 and D31 defective Defective contactor K1 or K2 on D31	Check K1. Check K2 coil voltage (A1/A2): 24 VDC. Check cable between X2/D31 and K2 contacts 13/14. Replace D31.	Test proc. D31 page 322 Repair proc. D31
	c. Power supply of D31 from GSSU power supply	Line power supply for GSSU power supply is missing or fuse F14 in the CCA is blown. Cable between GSSU/D17 and D31 defective	Check voltage LED's on D14. Check AC power supply line to GSSU power supply via fuse F14 in the CCA. Check cable between GSSU/D17 and D31/X1.	TSG Charts 08a/09a, page 264/ page 266

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01.continued The two green LED's V407 on D16 are off.	d. Monitoring error	Defective relay K3 on D31 or cable between K2 and D31	Check cable between X2/D31 and K2 contacts 13/14. Replace D31.	Test proc. D31 page 322
		Defective D16	Check CAN-Bus communication; replace defective D16.	Test proc. D16 page 318

Remarks: In **service mode** (Switch S2/2 on D17 in upward position or service mode enabled by the service software) this error is treated as a **warning** and will not be evaluated, i.e. modulators and regulators will not be disabled.

In case of further error messages related to the errors mentioned above, refer to the corresponding Troubleshooting Charts (e.g. 03a, 04a, 05a).

In case of this error during normal operation (contactor K2 or K3 opens), the primary voltage of the main transformer T1 will be switched off by contactor K2 (POW OFF).

Chart 03a

Error Message: **Resonant Converter Overtemperature <**
-> Rectifier Overtemp.

Error Code: GPA_RC_TEMP

The temperature of the rectifier assembly on D32 K2217 is too high. The temperature is measured at the cooling plate of the rectifiers on D32 by a NTC-resistor. The monitoring is done on D31 K2217. The error signal is routed via cable to from X1/D32 to X8/D31, from X1/D31 to X3/D17 K2217, forwarded through D18 to D16 and via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error occurs during normal operation Red LED V27/A1 on D17 "RC_TEMP" lights up	a. Cooling system	Failure of the cooling system: Chilled water and/or fans beside the D32 inside the GPA cabinet	Check cooling system: Hoses to D32, water flow/temp., function of RCA/CCS, internal cabinet fans.	TSG Cooling page 369
	b. Rectifier board D32	Defective rectifier on D32	Try to switch on after a few minutes; if no success check all other reasons 01.a,02.a; if negative, replace the D32.	Repair proc. D32
02. Error exists permanently Red LED V27/A1 on D17 "RC_TEMP" lights up	a. Monitoring error	Cable connection between X1/D32 and X8/D31 and/or between X1/ D31 and X3/D17 defective.	Check the cable connections. Check flat cable between D31 and D32.	Test proc.D17 page 324
		NTC-resistor on D32 defective	Measure the NTC-resistance. Replace defective D32.	Test proc. D32 page 324
		D17 defective (D18 defective)	Replace defective D17, (D18)	
		D16 defective	Check CAN-Bus communication; replace defective D16.	Test proc. D16 page 318

Remarks: All modulators will be stopped by the loadware (MOH).

Chart 04a

Error Message: **Res. Conv. Output in Curr. Trip**
-> Transform. overtemperature

Error Code: GPA_RC_CURTRIP

The main supply transformer T1 is in overtemperature condition. The monitoring is done on D31. The error signal is routed via cable to X5/D31, from X1/D31 to X3/D17, forwarded through D18 to D16 and via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error occurs during normal operation Red LED V27/A2 on D17 "TR_TEMP" lights up	a. Main transformer T1 in the GPA cabinet	Transformer T1 thermal overload caused by specific measurement conditions	Wait for cool down.	
		Failure of the CCS air cooling system or fans inside the GPA cabinet	Check function of the CCS and internal cabinet fans.	TSG Cooling page 369
		Main transformer T1 in the GPA cabinet is defective	Try to go on after a few minutes; if no success, check all other reasons 01.a, 02.a; if negative, replace the transformer.	
02. Error exists permanently Red LED V27/A2 on D17 "TR_TEMP" lights up	a. Monitoring error	Temperature sensor in the transformer defective	Check the temperature sensor.	Test.proc.D17 page 320
		Cable connection between temperature sensor and X1/D31 and/or X1/D31 and X3/D17 defective	Check cable connections.	
		D17 defective (D18 defective)	Replace defective D17, (D18).	
		D16 defective	Check CAN-Bus communication; replace defective D16.	Test proc. D16 page 318

Remarks: All modulators will be stopped by the loadware (MOH).

Chart 05a

Error Message: **No Reaction of Resonant Converter**
 -> **No react. of power supp.**

Error Code: GPA_RC_SET_ERR

Within 3 seconds after an "Enable" request (either by NUMARIS or by manually pressing the push-button "Power on" S402/D6), the main power supply does not respond with the "RDY" message (refer to TSG Chart 02). The gradient processor on D16 K2217 generates the final error message which is routed via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error after GPA "Power on" command The two green LED's V407 on D16 stay off.	a. Communication error: No enable signal from D16 for D31, or no acknowl- edge signal from D31	Defective cable connection between GSSU X3/D17 and X1/D31 D31 or D16 defective.	Do "Reload/All", if no success: Check cable between X3/D17 and X1/D31. Check D16, D18/D17 and D31. Replace defective components.	Test proc. D16 page 318 Test proc. D17 page 320
	b. Soft Start Unit D31	Cable connection between K2 and D31 defective Defective contactor K1 or K2 on D31	Check K1. Check K2 coil voltage (A1/A2): 24 VDC. Check cable between X2/D31 and K2 contacts 13/14. Replace D31.	Test proc. D31 page 322
	c. Power supply of D31 from GSSU power supply	Cable between GSSU/D17 and D31 defective	Check cable between GSSU/D17 and D31/X1.	

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error after GPA "Power on" command The two green LED's V407 on D16 stay off.	d. Monitoring error	Defective contactor K3 on D31 or cable between K2 and D31	Check cable between X2/D31 and K2 contacts 13/14. Replace D31.	Test proc. D31 page 322
		Defective D16	Check CAN-Bus communication; replace defective D16.	Test proc. D16 page 318

Remarks: As a consequence of this error the GPA will directly switched or stays in stand-by mode by hardware. (-> K2 off, i.e. primary voltage of transformer T1 off).

Refer also to TSG Chart 02a, ([page 256](#)) and 10a, ([page 268](#)).

Chart 06a

Error Message: **Synchronization fault**

Error Code: GPA_SYNC_ERR/_WARN

The synchronizing clock signal from MPCU(GCTX) is faulty or missing. This 4MHz clock signal is transmitted via the gradient waveform interface cable from the GCTX/MC4C40 to the D11 K2217. Here the signal is divided into a 20kHz signal. If the supervision on D14 K2217 detects an error, the error signal will be routed through D18 to D16 and via CAN-Bus to the MPCU; a yellow LED, V59/A1 "SYNC Error" at D14 will be on.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error during normal operation; red error LED V59/A1 "SYNC Error" at D14 is on. Image quality may be degraded.	a. Communication between MMC and D11	Error on MMC (MC4C40/GCTX)	Run MMC Tests (MC4C40/GCTX).	SeSo: Test Tools MMC
		Gradient data cable and/or connectors defective	Check gradient data cable; Run Test DAC/Functional, data transfer between GCTX and DAC.	SeSo: Test Tools Gradient
	b. DAC board D11	D11, wrong switch position/ (S3/4) D11 defective	Check switch positions on D11: S3-pos.7, S4-pos.2 Replace defective D11.	Test proc. D11 page 313
02. Yellow error LED V59/A1 "SYNC Error" at D14 is on or off.	a. Monitoring error	D14 defective	Replace defective D14.	
		D16 defective	Check CAN-Bus communication; replace defective D16.	Test proc. D16 page 318

Remarks: All modulators will be inhibited by hardware (MOH).

In the **service mode** (Switch S2/2 on D17 in upward position or service mode enabled by the service software) this error is treated as a **warning** and will not be evaluated, i.e. modulators and regulators will be not disabled.

The image quality may be degraded.

Chart 07a

Error Message: **Triangle voltage error -> Modulator clock error**

Error Code: GPA_TRIA_ERR

The internal modulator clock is faulty. The error signal is generated on D14 K2217 in combination with the supervision of Power-up condition and the signals CONFIG_DONE and READY of the programmable D14 chips. The final error signal "CLOCK_ERR" is routed through D18 to D16 and via CAN-Bus to the MPCU; a red LED, V57/A1 "Clock error" at D14 will be on.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error during normal operation; red error LED V57/A1 "Clock Error" at D14 lights up.	a. Modulator board D14	D14 defective	Replace defective D14.	
02. Red error LED V57/A1 "Clock Error" at D14 is on or off.	a. Monitoring error	D14 defective	Replace defective D14.	
		D16 defective	Check CAN-Bus communication; replace defective D16.	Test proc. D16 page 318

Remarks: All modulators will be inhibited by hardware (MOH). The control signals to the switches of the power stages will be inhibited.

Chart 08a

Error Message: **Plus 15V/ Minus 15V/ Plus 5V defective**

Error Code: GPA_P15/ N15/ VCC_ERR

DC supply voltages from E4A (+5V VCC, Slot1 and +/-15V, Slot3) are out of tolerance (+/- 9%). The voltage is monitored on D14 K2217; the error signals are routed through D18 to D16 and via CAN-Bus to the MPCU. In case of a +/-15V failure red LED's on D14, V58/A1/A2 "+/-15V Error" are on; in case of a +5V failure, green LED on D14, V59/A2 "+5V OK" is off.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error after GPA "Power On" Green LED V59/A2 at D14 off and/or Red LED's at D14, V58/A1 (+15V) and/or V58/A2 (-15V) is on.	a. Power supply unit E4A	Power supply E4A defective; Fan defective	Measure/adjust the output volt- ages directly at the outputs of the power supply slots. If not adjustable replace the power supply.	Test proc.E4A page 321
	b. Power supply lines	D17,(D18) defective or not plugged correctly; Cabling between D17 and E4A defective	Measure the +/-15V and +5V sup- ply voltages at D17/X4 and Test pins at front of D17; if ok, the back- plane D18 is probably defective.	Test proc. D17 page 320
	c. Boards of GSSU, D31, D32	Defective PCB	Replace the defective board.	Remarks

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01.continued Error after GPA "Power On"	d. Monitoring error	Supervision on D14 defective	Measure the supply voltages at D17 (+/- 5% tolerance range) if ok, replace defective D14.	Test proc. D17 page 320
Green LED V59/A2 at D14 off and/or Red LED's at D14, V58/A1 (+15V) and/or V58/A2 (-15V) is on.		D16 defective	Check CAN-Bus communication; replace defective D16.	Test proc. D16 page 318

Remarks: In principle each defective board in the GSSU including the D31/32 and the backplane D18 could be the reason for the supply voltage break down and should be checked. The sum of the supply voltages should be lower than 10 mV_{pp}. In case of this error all modulators will be stopped by the loadware (MOH).As a consequence of this error the FOC driver outputs will be inhibited and **overcurrent Y** error is indicated.

Chart 09a

Error Message: **Spare error X1, Y1, Z1**
-> Driver supply error X, Y, Z

Error Code: GPA_RES_X1/_Y1/_Z1_ERR/
 _WARN

The driver voltage of some axis is not ok. The monitoring is done at power stage/D21; (green LED V15/21/38/39 on -> ok). The error signals are routed via cable to D14 K2217 (supervision and error multiplexer), through D18 to D16 and via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. The red LED V64/A2 on D14 is on during state 1,2 and/or 3, according to axis X, Y, or Z. The green LED V86 on power stage/D21 is off. If the green LED V5 is off, check cable connections between E4A/D31 and power stage.	a. Driver supply/D21 inside power stage	D21/driver board (inside PS) defective	V6/7 and V86 on D21 off: Replace the corresponding power stage.	Repair proc. Power stages
	b. Power supply lines	Cable connections between E4A and X10/D31 and/or X12,13,14/D31 and power stages (V5=off)	Check cable connections. Measure the voltage on connector X10,X12,X13,X14 on D31 and X2 on PS (16.5 VDC).	Test proc. E4A page 321
	c. Synchronization of driver supply voltage	D14 defective	Replace defective D14.	
	d. Cable connection D14 -> power stage	Cable connection between power stage and D14 defective	Check/replace the cable.	

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
02. Error LED's on D14 in arbitrary conditions.	a. Monitoring error	Supervision on power stage (D21) defective	Replace the defective power stage.	Repair proc. Power stages
		Cable connection between power stage and D14 defective	Check/replace the cable.	
		D14 defective	Check D14. Replace D14, if green LED's V15 on power stages are on.	Test proc. D14 page 316
		D16 defective	Check CAN-Bus communication; replace defective D16.	Test proc. D16 page 318

Remarks: All modulators will be inhibited by hardware (MOD OFF).

Note:

The driver supply error always is coupled with a temperature error of the corresponding power stage. The reason for this behavior is that the driver supply is also the power supply for the temperature supervision of the power stage transistors.

In the **service mode** (Switch S1/2 on D17 in upward position or service mode enabled by the service software) this error is treated as a **warning** and will not be evaluated, i.e. modulators and regulators will not be disabled.

Meaning of green LED's on D21 (LED is on during normal operation):

V5: DC supply voltage (from GSSU) ok; (LED above connector X2)

V6/7: primary AC supply voltage ok

V86: secondary AC supply voltage ok; (LED on right side below of connector X1)

Chart 10a

Error Message: **Upgrade component error -> FASTOFF**

Error Code: GPA_UPGRADE-COMP_ERR

The voltage across the damping resistors during the "Switch on" sequence of GPA does not decrease in a certain time interval (after closing of contactor K2) and the contactor K2 will drop out again. The voltage is monitored on D31 and the error signal is routed via cable to D17 K2217, through D18 to D16 and via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error after GPA "Power On"	a. Power stages	Power stages short circuit	Check power stages. Replace defective power stage. Check fuses between T1 and D32.	Rep. proc. Power stages
	b. D32 Rectifier board	D32 defective (as a consequence of PS short circuit)	Check D32 Replace defective D32. Check fuses between T1 and D32.	Test proc. D32 page 324
	c. Soft Start Unit D31	Defective/broken damping resistors	Check D31; replace defective D31; check power stages.	Test proc. D31 page 322
	d. Monitoring error	Monitoring circuit on D31 defective	Replace defective D31.	
		Defective error signal path, cable connection between D31 and D17.	Check error signal path: D31->D17-D18->D16.	
		D16 defective	Check CAN-Bus communication; replace defective D16.	Test proc. D16 page 318

Remarks: If a power stage has a short circuit, a high input current could destroy the damping resistors.

As a consequence of this error the GPA will directly switched in stand-by mode by hardware. (-> K2 off, i.e. primary voltage of transformer T1 off: POW OFF).

Chart 11a

Error Message: **Regulation error on X,Y,Z-axis**

Error Code: GPA_LOOP_ERR/_WARN_X/_Y/_Z

The error signal and the regulator output signal are monitored in the closed loop error logic on D12 K2217. If one of these signals reaches the upper limit (10V), an error signal is generated on D12, routed through D18 to D16 and via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Dynamic error while running a measurement; the yellow LED's V135/A1, V135/A2, V136/A1 for X, Y, Z-axis "Loop Error" on D12 light up.	a. Measurement sequence	Gradient rise time is too fast	Switch on the Service mode to find out the affected axis. Run the specific sequence again; check the sequence parameters, if other than Siemens-released sequences are used.	
	b. Gradient data transmission	MC4C40/GCTX defective, Data cable and/or connectors defective, D11 K2209 defective (Gradient rise time is too fast)	Run MMC Test; run Gradient Test for DAC; check gradient data cable and connectors. Replace def. D11.	SeSo: Test Tools MMC Gradient
	c. Load circuit gradient coil interrupted	Gradient coil, cable connections, RF filters defective	Check the continuity of the load circuit; check for loose cable connections. Replace defective components.	
	d. Closed control loop	D12, D14 defective	Perform Regulator Adjust/Check; check for other error messages.	SeSo: Tune-up/QA
	e. Measurement of actual current	SENS and/or D12 defective	If CAV_ERR_X,Y,Z (refer to TSG Chart 12a), replace defective SENS, else replace defective D12.	Test proc. D12 page 314 Repair proc. SENS, GSSU

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01.continued. Dynamic error while running a measurement	f. Power stage	Power stage defective	Check for other error messages; cyclically replace power stages.	
	g. Main power supply, Transformer T1. Main voltage.	Too low output voltage	Measure the output voltage of the main power supply (D32).	Test proc. D32 page 324
02. Error after GPA "Power On"; (in the idle mode)	a. DAC D11	D11 defective/ gradient rise time too fast	Check voltage at D12, X11,12,13 in idle mode: Offset < +/- 0.05mV.	Test proc. D11/12 page 313/ page 314
	b. Regulator D12	D12 defective	Replace defective D12.	
	c. Modulator D14	D 14 defective	Replace defective D14.	
	d. Current sensors (SENS)	Current sensors (SENS) defective	Replace defective current sensors (SENS).	Repair proc.
	e. Monitoring error	D16 defective	Check CAN Bus communication; replace defective D16.	Test proc. D16 page 318

Remarks: The regulation error occurs as a consequence of every other error but will be suppressed in this case; you will see only the causal errors in the error report. The regulation error will only be recorded in the system report if it occurs as a primary error. All modulators will be inhibited by hardware (MOD OFF).

In **Service mode** (Switch S2/2 on D17 in upward position or service mode enabled by the service software) the regulation error is treated as a **warning** and will not be evaluated, i.e. modulators and regulators will be not disabled.

Chart 12a

Error Message: **Current actual value error on X,Y,Z-axis**

Error Code: GPA_CAV_ERR_X/_Y/_Z

Some current sensor signals (current actual value) are not correct. The current signals from the current sensors (in both directions) are monitored and compared on D12 K2217, a possible error signal is forwarded through D18 to D16 and routed via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error after GPA "Power On"; the red LED's V131/A2, V132/A1, V132/A2 for X, Y, Z-axis "CAV_ERR" on D12 light up.	a. Current sensors (SENS) Type LB200-S/SP4 paired or Type IT600S combined with LA 205-S	SENS defective: Difference of measured currents in case of the IT600S/ LA 205-S combination to high (difference > 32A)	Measure the sensor signals at D12, AV_X, AV_Y, AV_Z Replace defective current sensors (SENS).	Repair proc. SENS
	b. Current sensor supply	Break in signal line or power supply disconnected or supply line shorted	Check supply/signal line from SENS to D12/X10 for continuity and shorts.	
	c. D12 Regulator	D12 defective/ defective SENS soft start control (the error is detected, if the sen- sors are switched off)	Replace defective D12.	
	d. Power line from GPA to the gradient coil	Ground short of a gradient power circuits	Check the gradient coil circuit for ground shorts (gradient coil, gradi- ent cables, gradient filters).	
	e.(Type IT600S/LA205-S) Synchronization via FOC	FOC's between D17,U22(X), U24(Y), or U25(Z) defective	Check the FOC's between D17 and SENS.	Repair proc. SENS
02. No error LED lights up.	a. Monitoring error	D16 defective	Check CAN-Bus communication; replace defective D16.	Test proc.D16 page 318

Remarks: All modulators will be inhibited by hardware (MOD OFF).

Chart 13a

Error Message: **Dutycycle/controlling/Temp/OVV/Spare error on X2, Y2, Z2**
-> Overcurrent for X, Y, Z-axis

Error Code: GPA_DUTYCYCLE/_CTRL/_TEMP/
 _HIVOLT/
 _SPARE_ERR_X2/_Y2/_Z2

The actual current exceeds the current limit. The monitoring is done at D12 and an error signal is forwarded through to D16 K2217 and routed via CAN Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Dynamic error while running a measurement; the LED's V130/A1, V130/A2, V131/A1 for X, Y, Z-axis "IMAX" on D12 light up.	a. Measurement sequence	Sporadic overload	First run the duty_cycle sequence: If ok, check the measurement sequence parameters.	
	b. Regulator board D12	Regulator D12 oscillates due to wrong parameters; D12 defective	Run the Regulator Adjust/Check. Replace defective D12.	SeSo: QA/Tune-up: Regulator Adjust/Check
	c. Communication between MMC and D11	Error on MMC (MC4C40/GCTX)	Run MMC Tests.	SeSo: Test Tools MMC
		Gradient data cable and/or connectors defective	Check gradient data cable; Run Gradient tests for DAC.	SeSo: Test Tools Gradient
		D11 defective	Replace defective D11.	Repair proc. GSSU
	d. Modulator board D14	D14 defective	Replace defective D14.	

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01.continued Dynamic error while running a measurement; the LED's V130/A2, V130/A1, V131/A2 for X, Y, Z-axis "IMAX" on D12 light up.	e. Current sensors (SENS)	Measured values incorrect by defective current sensors (SENS)	Replace current sensors (SENS).	TSG chart 12a page 271 Repair proc.
	f. Power stage (only if nominal value and actual value not equal)	Defective Power stage transistors	Replace defective Power stage.	Repair proc.
02. Error after GPA "Power On"	a. Monitoring error	D12 defective	Replace defective D12.	
		D16 defective	Check CAN-Bus communication; replace defective D16.	Test proc. D16 page 318

Remarks: As a consequence of this error the GPA will directly switched in stand-by mode by hardware. (-> K2 off, i.e. primary voltage of transformer T1 off: POW OFF).

The error message will be transmitted five times because of the original meaning of the error code in Numaris 3.5.

Check also for accompanying error messages, e.g. **"Current actual value error"** -> **TSG-Chart 12a**, ([page 271](#)), and refer to the corresponding TSG-Chart.

Chart 14a

Error Message: **Dutycycle/controlling/Temp/OVV/Spare error on X3, Y3, Z3** Error Code: GPA_GPA_DUTYCYCLE/_CTRL/_TEMP/_HIVOLT/_SPARE_ERR_X3/_Y3/_Z3
 -> **Duty cycle for power stage X, Y, Z**

The duty cycle of the actual current is too high. The monitoring is done on D12 and an error signal is forwarded through D18 to D16 K2217 and routed via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Dynamic error while running a measurement; the LED's V133/A1, V133/A2, V134/A1 for X, Y, Z-axis "DuCy" on D12 light up.	a. Measurement sequence	Sporadic overload	Check the measurement sequence parameters Check the system by running a high duty cycle sequence.	
	b. Communication between MMC and D11	Error on MMC (MC4C40/GCTX)	Run MMC Tests.	SeSo: Test Tools
		Gradient data cable and/or connectors defective	Check gradient data cable; Run Gradient Tests for DAC.	SeSo Test Tools: Gradient DAC/ Functional
		D11 defective	Replace defective D11.	
	c. Regulator board D12	Regulator D12 oscillates due to incorrect parameters; D12 defective.	Run Regulator Adjust/Check. Replace defective D2.	
	d. Modulator board D14	Defective Modulator boards	Replace defective D14.	
	e. Current sensors (SENS)	Incorrect values measured by defective current sensors (SENS)	Replace defective current sensors (SENS).	Repair proc.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
02. Error after GPA "Power On"	a. Monitoring error	D14 defective	Replace defective D14.	
		D16 defective	Check CAN-Bus communication; replace defective D16.	Test proc. D16 page 318

Remarks: As a consequence of this error the GPA will directly switched in stand-by mode by hardware. (-> K2 off, i.e. primary voltage of transformer T1 off: POW OFF).

The error message will be transmitted five times because of the original meaning of the error code in Numaris 3.

Check also for accompanying error messages, e.g. "**Current actual value error**" -> **TSG-Chart 12a**, ([page 271](#)), and refer to the corresponding TSG-Chart.

Chart 15a

Error Message: **Controlling frequency error for power stage X1, Y1, Z1**
 -> **Controlling frequency error for power stage X, Y, Z**

Error Code: GPA_CTRL_FREQ_ERR_X1/_Y1/_Z1

The frequency of the control signals for the power transistors of a specific power stage is not ok. The monitoring is done on D14 K2217 (red LED V13 "Drive control Error" lights up) and an error signal is forwarded through D18 to D16 K2217 and routed via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Dynamic error while running a measurement; Red LED V63/A1 on D14 in state 1,2 and/or 3 according to the affected axis X,Y,Z lights up.	a. DAC board D11 (wrong nominal value)	D11 defective	Replace defective D11.	
	b. Regulator board D12	Regulator D12 oscillates or D12 defective	Run the Regulator Adjust/Check. Replace defective D12.	SeSo: QA/Tune-up: Regulator Adjust/Check
02. Error after GPA "Power On"	a. Monitoring error	D14 defective	Replace defective D14.	
03. Error after GPA "Power On"; The red LED V63/A1 on D14 is off during state 1,2 and 3.		D16 defective	Check CAN-Bus communication; replace defective D16.	Test proc. D16 page 318

Remarks: As a consequence of this error the GPA will directly switched in stand-by mode by hardware. (-> K2 off, i.e. primary voltage of transformer T1 off: POW OFF).

Check also for accompanying error messages, e.g. "**Overcurrent for XYZ-axis**" -> **TSG-Chart 13a**, ([page 272](#)), and refer to the corresponding TSG-Chart.

Chart 16a

Error Message: **No Reaction of Modulator**

Error Code: GPA_MOD_SET_ERR/_WARNING

The modulators of the three axes x,y,z can be controlled by the service software or by switches on D17, i.e. they can be switched on and off. If after a control command from the MPCU the modulator is not in the desired condition, this error is generated by the gradient processor, using a combination of several control signals (switch signals from D17 and control signals from D16). The final error signal will be routed via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Normal operation of MR measurement system or Service software tests running (e.g. gradient tests): Red LED V61/A2 on D14 is on in state 1,2 and/or 3 according to the affected axis X, Y, or Z. GPA_Status message: X,Y,Z modulator inhibit = yes	a. I/O board D17	Modulator switches in "off" position (upward position) (means modulator x,y,z-axis disabled)	Set switches in "on" position (downward position). Loading and starting a sequence enables the modulators if no other error condition exists. Eventually do "Reload/All".	SeSo: Test Tools Gradient GPA_Status (On-line help)
	b. Modulator D14	D14 defective	Replace defective D14.	
	c. Service board D16	D16 defective	Check CAN-Bus communication; replace defective D16.	Test proc. D16 page 318

Remarks: In the **service mode** (Switch S1/2 on D17 in upward position or service mode enabled by the service software) this error is treated as a **warning** and will not be evaluated, i.e. modulators and regulators will not be disabled (with the exception of manual disabling via switches on D 17).

Chart 17a

Error Message: **Duty cycle for Y1 -> Modulator defective**

Error Code: GPA_DUTYCYCLE_Y1

This error message is generated, if driver signals for S1/2 and S3/4 want to switch on their switches at the same time. The error is detected on D14, the error signal is routed through D18 to D16 and via CAN-Bus to the MPCU. The red error LED V62/A2 "MOD_DEF" on D14 is on during state 2.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error after GPA "Power On" or Dynamic error while running a measurement Red LED V62/A2 on D14 in state 2 lights up.	a. Modulator board D14	D14 defective	Replace defective D14.	
	b. Monitoring error	D14 defective	Replace defective D14	
		D16 defective	Check CAN-Bus communication; replace defective D16.	Test proc. D16 page 318

Remarks: All modulators will be inhibited by hardware (MOD OFF).

Note:

As a consequence of this error the FOC driver outputs will be inhibited and **overcurrent for Y error** is indicated.
(Refer to TSG-Chart 22a, [page 286](#)).

Chart 18a

Error Message: **Temperature error for power stage X1, Y1, Z1**
-> Temperature error for power stage X, Y, Z

Error Code: GPA_TEMP_ERR_X1/_Y1/_Z1

The power stage temperature exceeds the limit of 56°C. The temperature is measured inside the power stages (in the power stage transistor modul). The signal is evaluated on Driver board D21 (green LED V8 on -> Temp ok) and an error signal is sent via cable to D14 K2217. The error signal is monitored on D14 (red LED V63/A2 "Temp" lights up), forwarded through D18 to D16 K2217 and routed via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Dynamic error while running a measurement; Red LED V63/A2 "Overtemperature" on D14 lights up in state 1,2 or 3 according to the affected axis X, Y or Z.	a. Cooling system	No water flow or water flow in the cooling system is too low or cabinet cooling system (RCA/CCS) defective.	Check for other error messages from RCA or MMC (cabinet temperature control). Check RCA, water pipes, flow rate. Check the cabinet cooling system (RCA/CCS).	TSG Cooling
	b. Power stage	Power stage defective	Replace defective power stage.	Repair proc. PS
	c. Measurement sequence	Load of GPA by measurement parameters are too high or measurement conditions too hard.	Check the sequence parameters; Run an approved high duty cycle sequence.	

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
02. Continuous error, trying GPA "Power On" after wait- ing period of approx. 30 sec; Red LED V63/A2 "Overtemperature" on D14 lights up in state 1,2 or 3 according to the affected axis X,Y or Z.	a. Monitoring error	Temperature sensors (NTC resis- tors) of power stage defective	Replace defective power stage.	Repair proc. PS
		Modulator board D14 defective	Replace defective D14.	
		D16 defective	Check CAN-Bus communication; replace defective D16.	Test proc.D16 page 318
No error LED lights up.				

Remarks: All modulators will be inhibited by hardware (MOD OFF).

If the **duty cycle error for the same axis** (refer to TSG Chart 17a) **occurs at the same time**, check at first the used **measurement parameters**, otherwise check at first the **cooling system**.

Note:

The **Temperature error** is also a consequence of a **driver supply error**. (Refer to TSG-chart 09a, [page 266](#)).

Chart 19a

Error Message: **High voltage by power stage X1, Y1, Z1**
-> High voltage by power stage X, Y, Z

Error Code: GPA_HIVOLT_X1_Y1/_Z1

The output voltage of D32 (rectifier board) exceeds the upper limit. The voltages are measured on D32. The error signal is sent via cable from X1/D32 to X8/D31 and X1/D31 to X3/D17K2217 and is monitored at D14 (dependent on the axis a red LED, V64/A1 "Overvoltage" lights up in state 1,2 or 3). The error signal is forwarded through D18 to D16 and routed via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Dynamic error while running a measurement; Red LED V64/A1 "Overvoltage" on D14 lights up in state 1,2 or 3 according to the affected axis X,Y or Z.	a. Main voltage	Instable main supply voltage, line overvoltage	Measure the output voltages of D32; Provide for stable main supply voltage.	Test proc. D32 page 324
	b. Main power supply Transformer T1	Transformer taps wrong connected	Check the cable connections (taps) at T1.	
	c. Power stage	Power stage defective or a partial stage of a power stage doesn't work	Check if all six green LED's behind the six holes in the front of the power stage are on during normal operation; if not, check fuses between T1 and D32, check D32; replace defective power stage.	Repair proc. PS
	d. Modulator board D14	Modulator D14 defective	Replace defective D14.	Test proc. D14 page 316

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
02. Error after GPA "Power On";	a. Monitoring error	D32 voltage supervision defective	Measure the output voltages of D32; Replace defective D32.	Test proc. D32 page 324
Red LED V64/A1 "Overvoltage" on D14 lights up in state 1,2 or 3 according to the affected axis X,Y or Z.		Cable connections defective between X1/D32 to X8/D31 and X1/D31 to X3/D17.	Check cable connections.	
		Modulator D14 defective	Replace defective D14.	Test proc. D14 page 316
03. No error LED lights up.		D16 defective	Check CAN-Bus communication; Replace defective D16.	Test proc.D16 page 318

Remarks: As a consequence of this error the GPA will directly switched in stand-by mode by hardware. (-> K2 off, i.e. primary voltage of transformer T1 off: POW OFF).

Chart 20a

Error Message: **Duty cycle for X1 -> Pull protection error**

Error Code: GPA_DUTYCYCLE_X1

The cable connections between D14 K2217 and the power stages (X, Y, Z) are not plugged correctly or are defective. The error is monitored on D14. The error signal is routed through D18 to D16 and via CAN_Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Red LED V62/A2 "Cable connected" on D14 lights up in state 1.	a. Cable connection between D14 and power stage	Cable connectors not correctly plugged in or cable broken	Check cable connections.	
	b. Monitoring error	D14 defective	Replace defective D14.	Test proc. D14 page 316
		D16 defective	Check CAN-Bus communication; replace defective D16.	Test proc.D16 page 318

Remarks: All modulators will be inhibited by hardware (MOD OFF).

Note:

As a consequence of this error the FOC driver outputs will be inhibited and overcurrent for Y error is indicated. (Refer to TSG-Chart 22a).

Chart 21a

Error Message: **Overcurrent for X -> Danger of stimulation**

Error Code: GPA_HI_CURR_X

This error occurs only on system equipped with the HW-stimulation monitor D13(1), i.e. Quantum and Sonata. If the stimulation monitor calculates a risk of patient stimulation from the actual gradient pulses (measured by SENS/D12), he sends a signal to D17 K2217 (The red LED "EMMI" on D17 lights up). The signal is forwarded through D18 to D14 (error multiplexer) further to D16 and via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error during imaging: The red LED "EMMI" on D17 lights up; the modulators will be stopped. The red LED V62/A1 "EMMI" on D14 lights up in state 1.	a. Measurement sequence	The measurement parameters generate gradient pulses, which could cause patient stimulation.	Note: This event can occur during routine imaging. Change measurement parameters.	
	b. Regulator board D12/ SENS	D12 and/or current sensors defective.	Run regulator adjustment (General Tune-up) Replace defective components	Repair proc.
	c. Stimulation monitor D13/1	D13(1) defective	Run the function test with two special test sequences. Replace defective D13(1).	Test proc. D13(1) page 319

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01.continued Error during imaging: The red LED "EMMI" on D17 lights up; the modulators will be stopped. The red LED V62/A1 "EMMI" on D14 lights up in state 1.	d. Monitoring error	Defective cable connections between D13(1) and D17	Check /replace defective cable.	
		Wrong jumper setting (D12/D14) or defective D14	Check jumper setting D12/D14; replace defective D14.	Test proc. D12/D14, page 314/ page 316
		Defective D16	Check CAN_Bus communication; replace defective D16.	Test proc. D16

Remarks: All modulators will be inhibited by hardware (MOD OFF).

Note:

This error occurs also during Tune-up/Regulator adjust and QA/Regulator check. During these measurements the stimulation monitor must be manually switched off (S1/D13: right switch in upper position).

After a replacement of the stimulation monitor D13(1) you must perform a function test (refer to test procedures D13(1) [page 319](#)).

Chart 22a

Error Message: **Overcurrent for Y -> FOC outputs to drivers inhibited**

Error Code: GPA_HI_CURR_Y

The FOC driver-IC outputs of the modulator will be inhibited as a consequence of another error. The error signal is generated on D14, routed via D18 to D16 and via CAN-Bus to the MPCU.

Remarks: All modulators will be inhibited by hardware, i.e. no control signals will be sent to the power stages (MOD OFF).

Note:

This error will also be triggered by a "clock error", "supply voltage error", "pull protection error" and the error "modulator defective".

Check for accompanying error messages (e.g. cable connection error, modulator defective, modulator clock error, supply voltages error a.s.o.) and refer to the corresponding TSG-Charts 20a,17a, 07a,.08a.

Proposed remedies:

- a. Replace D14
- b. Check cables from D14 to power stages for short circuit.
- c. Check +/-15V and +5V supply voltages.

Image Quality - Test Strategy

General

The stability of the MR system is critical to the quality of MR imaging. Stability Check and CalcAr are routines used in the tune-up and quality assurance to monitor the stability of the MR imaging system.

If the results of either or both of these routines are greater than a specific tolerance, the source of the instability can be found by investigating the various components making up the MR imaging system.

The service engineer would use the **Gradient Tests (DAC, X,-Y,-Z-Axis, Functional)** in the **Test Tools** service software platform when there is an indication that the source of the problem, e.g. instability, is in the gradient system:

For instance, when the instability measured in the **QA functions** of the service software "**Stability Check**" or "CalcAr" is orientation dependent, it is indicative of instability due to gradient instability and not RF or a magnetic field instability.

Other QA-functions check whether system parameters are adjusted correctly, e.g. **Regulator check/ ECC Check** or **Graddelay**. If the results are not within tolerance, you have to perform the corresponding measurements/adjustments of the Tune-up.

In case of **orientation dependent RF-disturbances in the image**, which may supposed to be caused by the gradient system (switching frequency of the amplifiers), you should interchange the gradient cables from the GPA of the affected axis in order to locate the origin of the problem. If the disturbances disappear with swapped cables, the problem is probably located in the GPA. Otherwise the problem must be related with the gradient/bodycoil system.

Summary of gradient dependent IQ - Parameters

IQ Parameter	Influence Factor	Available Tests
Spatial Resolution Contrast (Symptoms are edge rings and edge blurring)	Non-linear amplitude of gradient pulses Sequence timing related to eddy current fields	Check linearity of gradient pulses (Gradient Loops); eddy current compensation (ECC)
Slice Thickness Slice Profile Slice Position Slice Warp	Eddy currents Linearity of gradient field	ECC Depending on gradient coil design; no improvement possible
Artifacts such as Smearing artifacts Ghosts: discrete/blurred	Phase encoding errors: Any kind of gradient instability; periodic unequal phase encoding steps SENS instability/nonlinearity	DAC: Data loop test Check gradient stability/linearity (Gradient loops) Regulator adjustment
Image Uniformity Spatial Linearity	Non-linear gradient magnetic fields	No improvement possible
Signal to Noise Ratio	Gradient system: Eddy currents; Spikes generated by gradient switching	ECC, Gradient delay; Spike check

Gradient System - Image Quality TSG Chart

Normally there are no specific error messages corresponding to an image quality problem. You have to specify the individual problem according to the column "Symptom" in the table below and to perform the proposed test to localize the problem in the MR-system.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Smearing artifacts in phase encoding direction: - discrete ghosts - blurred ghosts	a. Low phase encoding performance of the gradient system caused by instability	SENS instability and offset at low gradient amplitudes D1/D11 DAC defective/wrong gradient data from MPCU/GCTX Regulator oscillation	Perform the Gradient Tests; check the stability of the DAC/actual value Replace D1/D11 Check regulator adjustments	SeSo: Test Tools Tune-up/QA functions Regulator check
02. N/2 ghosts (especially with EPI; EPI-Bold protocols). For example with sequence: ep2d_fid_16_276b1953; ghosting strength depends on used measurement parameters such as FOV and Matrix.	a. Nonlinearities of gradient pulse ramps during MR-signal sampling.	Note: This error may not be caused by the hardware but by: Instable line voltage; low line voltage (in the range up to minus 10%).	Check and record the line voltage with the help of a main voltage recorder (e.g. Dranetz). Try to establish a correlation between bad image quality in EPI-measurements and line voltage instabilities. Ultimate remedy is the installation of an UPS.	
03. Edge blurring/rings artifacts in phase encoding direction	a. Nonlinearity of gradient pulse steps	Slew rate limitation of the gradient pulses at high amplitudes	Perform the Gradient Tests; check the linearity of the DAC/actual value	Seso: Test Tools
04. Streaks and/or dots in images	a. RF noise caused by rapid gradient switching	Magnet room not rf-tight	Check RF-room shielding, filter-plate, rf-filters	

Image Quality TSG Chart continued

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
05. Low signal to noise ratio and/or spikes	a. Gradient system: Gradient/Bodycoil system	Eddy currents; Spikes caused by mechanical vibrations during gradient switching	Perform - Graddelay - ECC Check - Spike Check	SeSo: Tune-up/QA functions
06. Non-uniformity of images (phantom image)	a. Gradient system	Eddy currents	Perform ECC Check	
07. Image spatially distorted	a. Adjustment of gradient-system	Gradient sensitivity misadjusted	Perform Gradsens	
08. Low contrast/resolution of the image: Edge rings and blurring	a. Gradient system Nonlinear amplitude switching of gradient pulses	Wrong gradient data D1/D11 DAC	Check gradient data transfer: Perform MMC Tests and Gradient Tests (check results of DAC/functional Test, GCTX-DAC) Perform the gradient linearity tests DAC/actual value	SeSo: Test Tools MMC Gradient
	b. Gradient system	Eddy currents	ECC/Check	SeSo: QA/Tune-up functions
09. Wrong slice position and/or warped slice	a. Shim condition of magnet	Distorted magnetic field	Check the shim condition of magnet/ Reshim the magnet	SeSo: Tune-up Shim/Phantom Shim
	b. Gradient coil design	Non-linear gradient field	No improvement	

Image Quality TSG Chart continued

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
10. Broadening of slice thickness/profile	a. Gradient system	Eddy currents	ECC/Check	SeSo: QA/Tune-up functions
11. Insufficient Fat saturation	a. Base magnetic field	Magnetic field inhomogeneities due to bad shim (possible reasons may be: magnet quench, magnetic materials in the magnet bore)	Perform Shim Check. Perform Phantom Shim. Reshim the magnet.	SeSo: QA: Shimcheck Tuneup: Phantom Shim
	b. Gradient offsets	Gradient offset currents have changed (linear gradients)	Perform Shim Check. Perform Phantom Shim. Reshim the magnet.	
	c. Eddy currents	Uncompensated eddy currents	Perform ECC Check/ ECC adjustment Perform CTC Check/ CTC adjustment	SeSo: QA: ECC check CTC check Tune-up: ECC CTC

Procedures

General

This section describes the various tests and checks for field replaceable units (FRU's) of the

Gradient system K2209 and K2217 of the MR systems Harmony/Symphony/Quantum/Sonata.

The tests and checks support error localization (in response to error messages). After replacement of a FRU, the tests and checks may be required to ensure that the switch/jumper setting or other special adjustments are correct.

In a subsequent section the possible **gradient test loops** of the service software, platform **"Test Tools"** are described shortly, i.e.

- Gradient data transfer between **MMC/GCTX and GSSU/D1/D11**
- Test loop DAC-value: **D1/D11(DAC) - D6/D16 - RFSU (ADC)**
- Test loop Actual value: **D2/D12 (Regulator) - D6/16 - RFSU (ADC)**

The gradient test loops verify the stability and linearity of the gradient amplifier.

Required Tools

Prerequisites for successfully troubleshooting the gradient system using this document are:

- Functional Descriptions (FUN)
- Diagrams (Chapter Gradient System)
- Set of standard tools,
- Digital multimeter, Fluke 8060A or equivalent

Summary of FRU's and corresponding tests

FRU's K2209 (page)	Test procedures
DAC board D1 (page 295)	a. Check jumpers and switch position b. Offset measurement/adjustment c. Gradient data test loop GCTX/D1
Regulator board D2 (page 296)	a. Default regulator setting by DIP switches b. SENS offset measurement/adjustment c. Regulator Adjust/Check
Synchronization board D3 (page 297)	a. Check jumper settings b. Measurements of clock signals and supply voltages
Modulator board D4 (page 298)	a. Check of DIP switch settings
Supermodulator board D5 (page 299)	a. Check of DIP switch settings
Service/CAN board D6 (page 300)	a. Check communication to RC b. Manual loop setting c. CAN-Bus communication (Test Tools/Gradient/GPA status).
FOC board D7 (page 301)	a. Board layout b. FOC - Tests c. Supply voltage measurement

FRU's K2209 (page)	Test procedures
Small signal power supply E4A/Driver supply D10 (page 301)	a. Measurement/Adjustment
Supervision board D9 (page 304)	a. DIP switch settings b. Sensor tests
Resonant converter (RC) (page 306)	a. Measurement/adjust. of output volt.

Summary of FRU's and corresponding tests

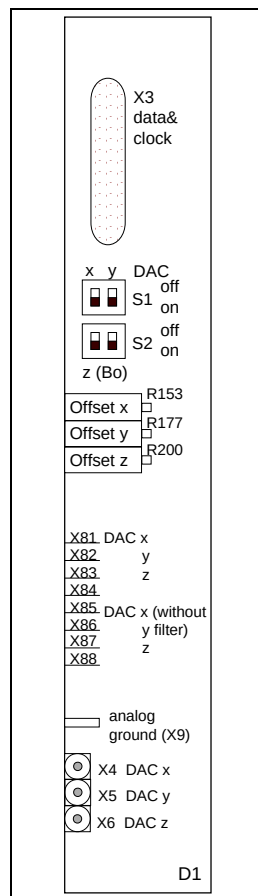
FRU's K2217 (page)	Test procedures
GSSU	
DAC board D11 (57 25 507 K2217) (page 313)	a. Check jumpers and switch position b. Offset measurement/adjustment c. Gradient data test loop GCTX/D11
Regulator board D12 (57 26 349 K2217) (page 314)	a. Default regulator setting by DIP switches b. Jumper settings for maximum current, current regulation mode and current direction dependent switching c. SENS offset measurement/adjustment d. Regulator Adjust/Check
Modulator board D14(2) (57 26 364 K2217) (D142 is used on GPA light, Ultra and Turbo) (page 316)	a. Check of slewrate jumper settings
Service/CAN board D16 (57 25 515 K2217) (page 318)	a. Check of communication to the main power supply. b. Manual loop setting c. CAN-Bus communication (Test Tools/Gradient/GPA status).
Stimulation Monitor D13(1) (59 39 134 K2217 (57 27 313)) (D131 used on 085/Sonata) (page 319)	a. Function test b. Switch on check

FRU's K2217 (page)	Test procedures
I/O board D17 (57 26 430 K2217) (page 320)	a. Switch setting of S1 and S2 b. Jumper settings X9 c. Supply voltage measurement
GPA cabinet	
Small signal power supply E4A (30 59 482) (page 321)	a. Measurement/Adjustment
Current sensor assembly (SENS) (57 25 655 K2217)	a. Offset adjustment of D12 b. Regulator adjustment c. Phantom Shim d. Grad Delay e. Grad Sens f. ECC Check g. Image Orientation h. SN DIP & CalcAr
GPA cabinet Main power supply boards:	
D31 switch on assembly (57 26 414 K2217) (page 322)	a. Check of connection to GSSU b. Supply voltage measurements
D32 rectifier board (57 27 016 K2217) (page 324)	a. Check of PTC temperature sensor b. PS supply voltage measurement

FRU's K2217 (page)	Test procedures
GPA cabinet	
Power stage (59 38 142 K2217)	a. Regulator adjustment b. Phantom Shim c. Grad Delay d. Grad Sense e. Image Orientation
Blowers (30 63 476) (page 326)	Function check

DAC D1 Test procedures

Fig. 50 Front of
DAC D1



Descriptions:

X3:

18 bit parallel gradient data and
4MHz clock signal from MMC

S1/S2:

The DAC output can be switched on/off
for every axis; S2/B₀ is not used.

Potentiometers for offset adjustment

DAC output = nominal value (0-> +/-10V)
10 V corresponds to 300 A

a. Check jumpers and switch positions:

Jumper X16: 2-3 (sign of gradient data, 2-Complement)

Rotary switch S3: position 7

Rotary switch S4: position 2

(this setting results in a clock signal FSYNC = 50 kHz)

b. Offset measurement/adjustment:

Note: Prerequisite is the Regulator adjustment (Tune-up).

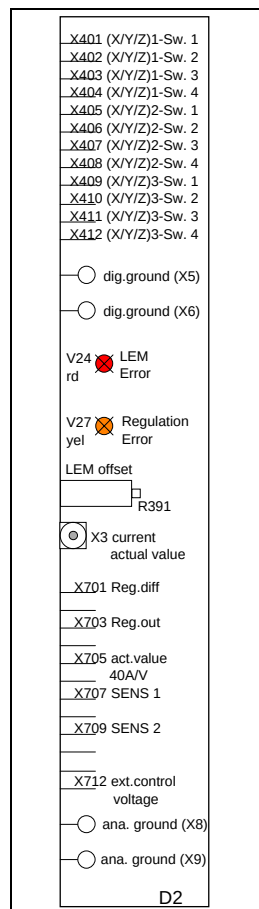
- Gradient cabinet has been on for at least 10 minutes.
- Set switches S1/2 to up position ("DAC output zero")
- The corresponding modulator must be on: S1/2 on D4 in down position.
- Connect a DVM to the desired actual value output X3 on D2. Use the QLA-BNC service cable.
- Check the offset: offset < +/- 0.05 mV
- Use the corresponding R153, 177, 200 on DAC-board D1 to adjust the offset for the X, Y and Z axes if necessary.

c. Gradient data Test loop GCTX/D1:

Use the Test "Gradient/ DAC/Functional" of the Test Tools platform to confirm the correct gradient data transmission between MMC/GCTX and GSSU/D1; (a prerequisite for this test is the correct function of the MMC components, refer to MMC-Tests).

Regulator D2 Test Procedure

Fig. 51 Front of
Regulator D2



Descriptions:

- R391 SENS-Offset adjustment
- X3 Actual current value, QLA (1V=40A)
- X701 Regulator difference signal
- X703 Regulator output signal
- X705 Actual current value (1V=40A)
- X707 SENS1 current value
- X709 SENS2 current value

a. Default regulator setting by DIP switches:

DIP-switch default setting:

Switch	S1 (P)	S2 (I)	S1 corresponds to the adjustment of the proportional portion, S2 corresponds to the integral portion of the regulator.
1	on	on	
2	off	off	
3	off	off	
4	off	off	
5	off	off	
6	off	off	
7	off	off	
8	off	off	

b. SENS offset measurement/adjustment:

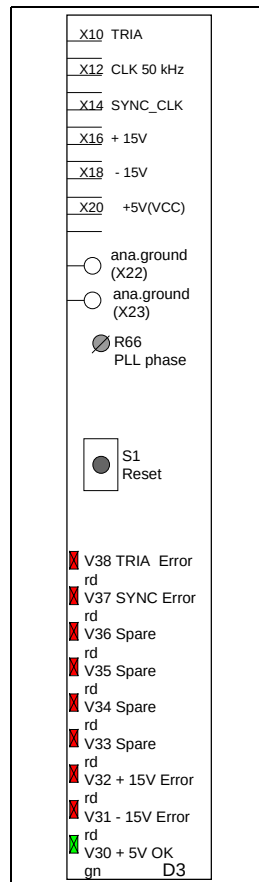
- Gradient cabinet has been on for at least 10 minutes and the regulators are adjusted (see Regulator Adjust/Check below)
- Connect the DVM to X3 ("actual value") on the regulator board. Use the QLA-BNC service cable.
- Set the switch, corresponding to the axis, on the modulator board D4 in upward position ("Modulator off").
- Check the offset: offset < +/- 0.05 mV; use R191 to adjust the offset if necessary.

c. Regulator Adjust/Check:

The regulators are set via the service software function **Tune-up/Regulator Adjust**. The **QA/Regulator Check** function quickly checks the regulator data set.

Synchronization D3 Test Procedures

Fig. 52 Front of
Synchron. D3



Descriptions:

X10 Triangle voltage
X12 Clock signal CLK, 50kHz
X14 SYNC_CLK, 50 kHz
(derived from 4MHz system clock)
X16 +15V (supplied by E4A, slot 3)
X18 -15 V (supplied by E4A, slot 3)
X20 +5V (VCC, supplied by E4A, slot 1)

R66: Adjustment of the phase locked loop
(free running frequency 52 kHz)

S1:
Hardware supervision reset
(Resetting of the error latches)

Error LED's

a. Check jumper settings:

Jumper X3: 3A - 3B, 6A - 6B, 9A - 9B

b. Measurements of clock signals and supply voltages:

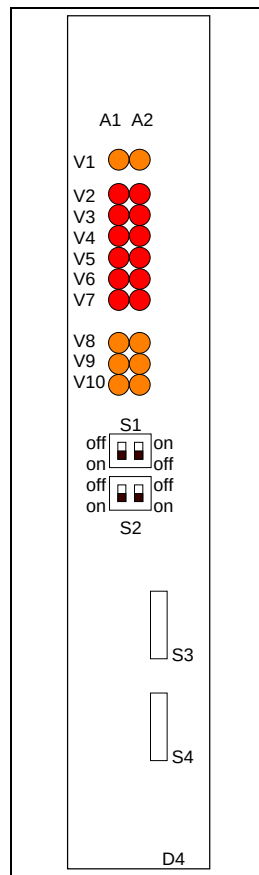
- Triangle voltage:
Amplitude: +/- 9V
Adjustment frequency: 52 kHz
(Adjust with R66; measure with a DMM frequency counter;
during adjustment remove the gradient data cable from MMC to D1/X3)
- Supply voltages:
In case of supply voltage error, you can measure the voltages
supplied by the E4A directly at the power supply terminals, at D3
or D7 (refer also to D7 test procedures).

Tolerance ranges (error status if exceeded):

+/- 15V (+/- 13,65V...+/- 16.35V)
+5V (4,55V...5,45V)

Modulator D4 Test procedures

Fig. 53 Front of Modulator D4



Descriptions:

LED's (left / right):

V1 Service mode / Modulator disabled

V2 Overtemp. / Drive error (X1 or Y1 or Z1)

V3 Spare / Overvoltage X1

V4 Overvoltage Y1 / Overvoltage Z1

V5 Overcurrent X1 / Duty cycle error X1

V6 Overcurrent Y1 / Duty cycle error Y1

V7 Overcurrent Z1 / Duty cycle error Z1

V8 Current X+ / Current X-

V9 Current Y+ / Current Y-

V10 Current Z+ / Current Z-

S1 left: Modulator X on/off

S1 right: Service mode on/off

S2 left: Modulator Z on/off

S2 right: Modulator Y on/off

a. Check of DIP switch settings:

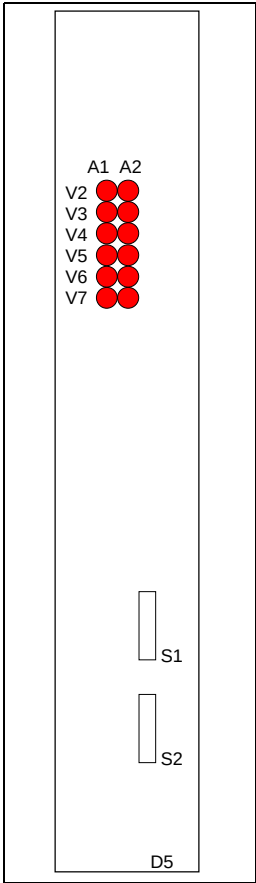
DIP switch default setting:

DIP Switches D4	Setting S3	Function	Setting S4	Function
1	on	deadtime1	on	modsemo x
2	off	deadtime2	on	modsemo y
3	off	mode1	on	modsemo z
4	off	mode3	on	sunuse x
5	off	fqa	on	sunuse y
6	off	mode2	on	sunuse z
7	on	fqb	on	swloss1
8	off	mode4	off	swloss2

(Refer to the functional description for further explanations).

Turbo Modulator D5 Test procedures

Fig. 54 Front of Modulator D5



Descriptions:

- V2 Overtemp. / Drive error (X2 or Y2 or Z2)
- V3 Spare / Overvoltage X2
- V4 Overvoltage Y2 / Overvoltage Z2
- V5 Spare/ Spare
- V6 Spare/ Spare
- V7 Spare/ Spare

a. Check of DIP switch settings:

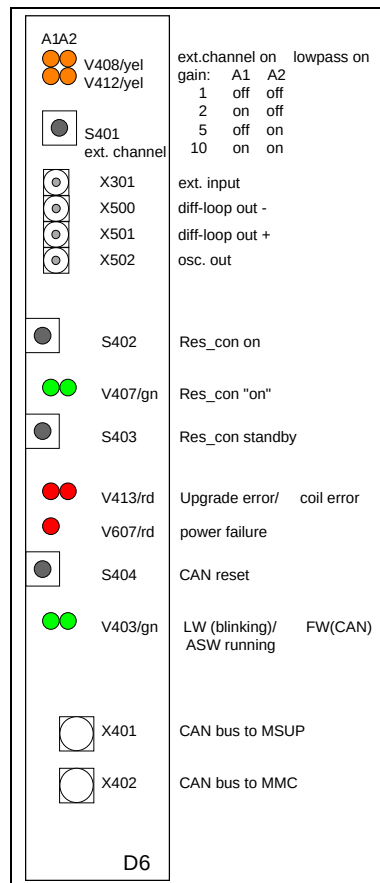
DIP switch default setting:

DIP Switches D5	Setting S1	Function	Setting S2	Function
1	on	deadtime1	on	modsemo x
2	off	deadtime2	on	modsemo y
3	on	mode1	on	modsemo z
4	off	mode3	on	sunuse x
5	on	not used	on	sunuse y
6	on	mode2	on	sunuse z
7	on	not used	on	swloss1
8	on	not used	off	swloss2

(Refer to the functional description for further explanations).

Service/CAN D6 Test Procedures

Fig. 55 Front of Service/CAN D6



Descriptions:

LED's show conditions of external channel settings.

S1: Sets external channel

X301 External signal input

X502 Oscilloscope output

S402: Manual RC "ON"

V407: RC on/off

S403: Manual RC "Standby"

V413 left: Upgrade component is missing or defective

V413 right: Gradient coil error

S404: Warm start of GPA-processor

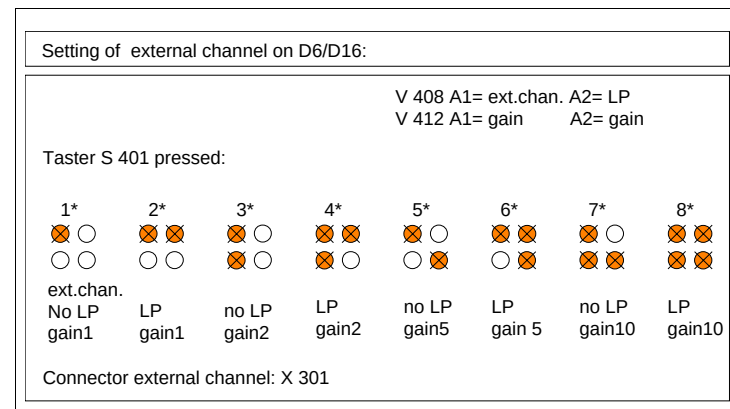
V403 left, blinks during normal conditions

a. Check communication to RC:

The RC receives the switch-on signal via the fiber optic cable of assembly D7 (U51). Test by pressing S403 and then S402. Verify that the RC is switched to stand-by (LED's "EN" and "RDY" off) and on again (switch-on signal is present at D7 and is transmitted to RC).

b. Manual loop setting:

Fig. 56 LED display on D6 showing the possible gain and low pass filter combinations of the external channel



When using the external channel, you can select several combinations of different gain and low pass filter settings by pressing button S401 n-times according to Fig. 56.

Note: If all LED's are off, the automatic loop is activated.

c. CAN-Bus communication/GPA status:

The present status (Status/warning/errors) of the GPA is displayed completely with the help of the service software function "Gradient/Communication" in the platform Test Tools.

FOC D7/Power supply Test Procedures

Assembly D7 is the interface of the GSSU to the power stages, as well as the current measurement and supervision assemblies. Electrical signals are converted into FOC signals and vice versa.

In case of errors, this is an easy method for ruling out "monitoring errors":

The status of the error transmission path is tested by pulling the respective fiber optic cable.

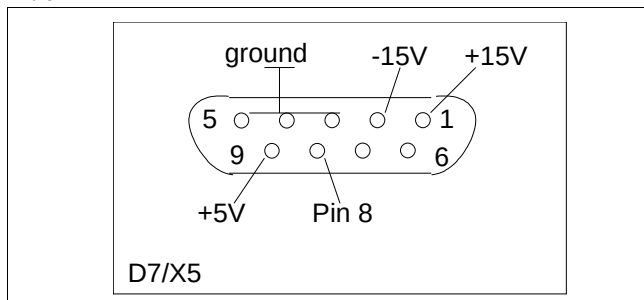
"Light off" means that there is an error and/or one of the corresponding sensors is defective.

Fig. 59 shows the configuration of the connections for the fiber optic cables.

Upgrade interface connector:

If there is no upgrade component installed, the pin 8 must be connected to ground.

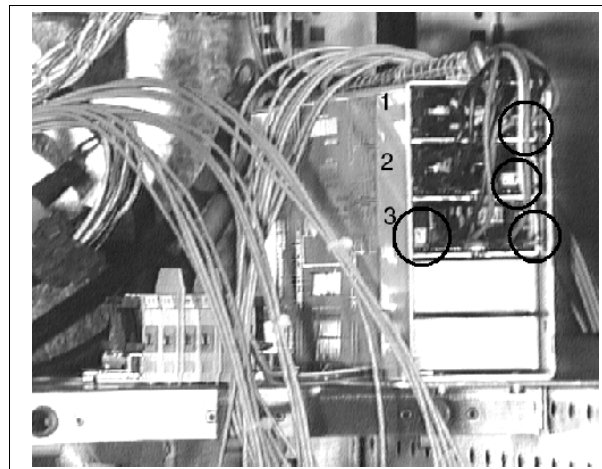
Fig. 57 Upgrade interface connector at D7



Small signal power supply E4A

Adjustment of supply voltages:

Fig. 58 Adjustment of GSSU power supply voltages



Slot 1: +5V Vcc

Measure the voltage at the terminals of E4A or on D7/X5, Pin 9-5

Slot 2: +15V, supply voltage for D10, Driver supply

Measure the voltage at the terminals of E4A

Slot 3: +/- 15V

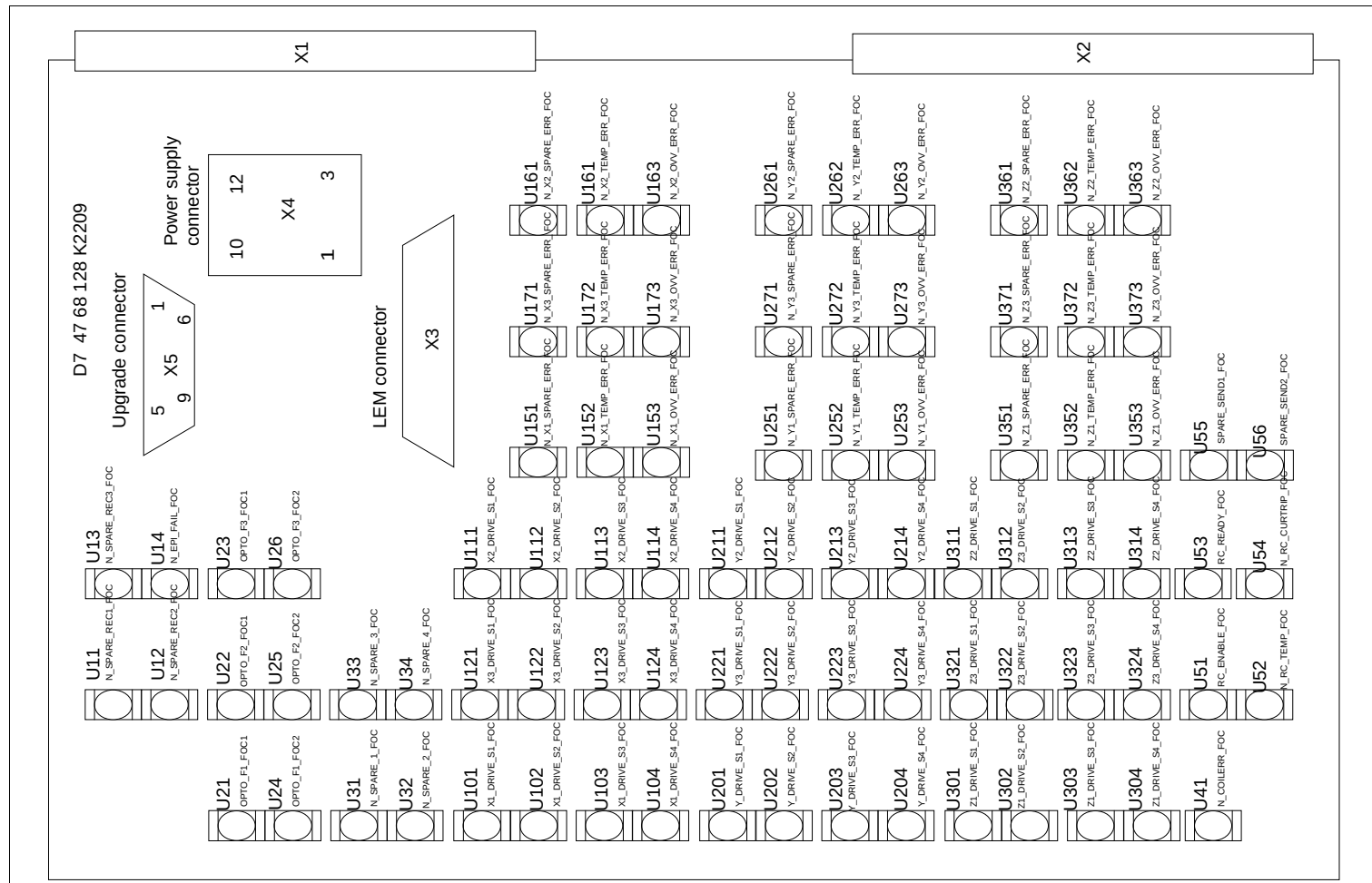
Measure the voltage at the terminals or on D7/X5, Pin 1-5 and 2-5.

Adjust the voltages with the marked potentiometers (Fig. 58).

Driver supply D10

Jumper X4: external clock 1-2 (default); internal clock 2-3.

Fig. 59 D7 K2209 Connector positions



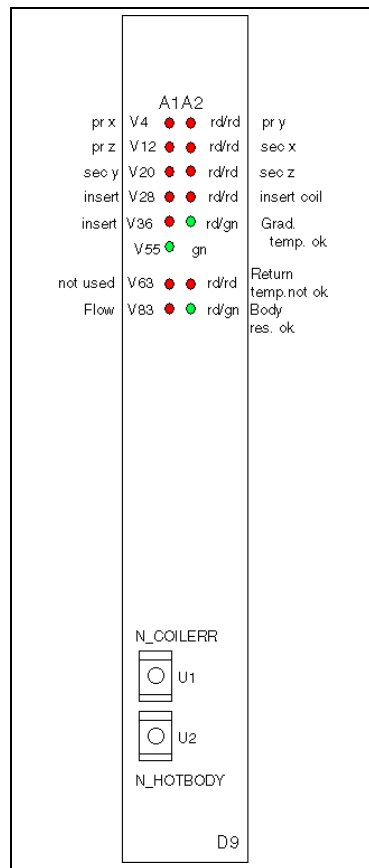
Pin positions of connectors SENS/X1, D7/X3

PIN-No	Signal	Signal function
1	Spare	n.c.
2	C-CON1_X	Actual value X+
3	GND	Ground
4	C-CON2_X	Actual value X-
5	P15_RET	disconnecting protection
6	C-CON1_Y	Actual value Y+
7	GND	Ground
8	C-CON2_Y	Actual value Y-
9	P_15V	Ground
10	C-CON1_Z	Actual value Z+
11	GND	Ground
12	C-CON2_Z	Actual value Z-
13	GND	Ground
14	P15_X	positive supply voltage (X+ SENS)
15	N15_X	negative supply voltage (X+ SENS)
16	P15_X	positive supply voltage (X- SENS)
17	N15_X	negative supply voltage (X- SENS)
18	P15_Y	positive supply voltage (Y+ SENS)
19	N15_Y	negative supply voltage (Y+ SENS)

20	P15_Y	positive supply voltage (Y- SENS)
21	N15_Y	negative supply voltage (Y- SENS)
22	P15_Z	positive supply voltage (Z+ SENS)
23	N15_Z	negative supply voltage (Z+ SENS)
24	P15_Z	positive supply voltage (Z- SENS)
25	N15_Z	negative supply voltage (Z- SENS)

Supervision D9/Grad.coil Test Procedure

Fig. 60 Front of Supervision D9



Descriptions:

LED's:

pr x,y,z:

NTC-sensor status at primary side of the gradient coil

sec x,y,z:

NTC-sensor status at secondary side of the gradient coil

V55gn: 22V supply voltage ok

V63 rd right:

Return temperature status of cooling water for the body resonator (Temperature too high)

V83 rd left:

Water flow supervision for the body resonator (Flow not ok).

V83 gn right: Body coil ok

N_COILERR:

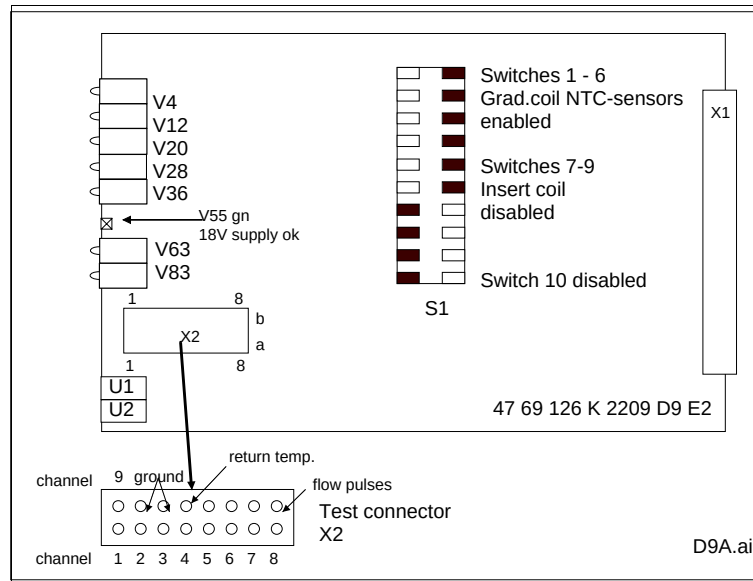
FOC connection to D7/U41
(Light off or glowing weakly:
=> Error)

N_HOTBODY

FOC connection to BTB1/U5
(Light off => Error/ Body coil not ok)

The coil supervision board D9 is located in the left part of the patient table control electronics rack.

Fig. 61 Position of DIP switches at D9



Normal condition: Switches 1-6 on, switches 7-10 off.
(Switch 1 => pr x, switch 2 => pr y, a.s.o.)

An error condition (green LED off) exists if the gradient coil temperature at the location of the NTC-sensor exceeds 83 +/- 7 °C or the cable to the coil is not connected or the NTC is shorted or broken.

In case of overtemperature and/or shorted NTC the corresponding red LED lights up.

At most two defective NTC-sensors may be deactivated by switching the corresponding DIP-switch S1/D9 to the off position. In this case, the corresponding error LED may be permanently on, but no error message is generated.

In case of problems, measure the resistance of the NTC-sensors at the gradient coil connector X2:

Pin 4-5 (x prim), Pin 10-11 (x sec)

Pin 6-7 (y prim), Pin 12-13 (y sec)

Pin 8-9 (z prim), Pin 14-15 (z sec)

or at connector X20/D9

Pin 25 - 13 (x prim), Pin 22 - 10 (x sec)

Pin 24 - 12 (y prim), Pin 21 - 9 (y sec)

Pin 23 - 11 (z prim), Pin 20 - 8 (z sec)

Typical resistance of NTC's:

Room temperature: $R > 15 \text{ k}\Omega$ (approx. $40 \text{ k}\Omega$ at 20 °C)

Temperature limit: $R < 4 \text{ k}\Omega$

In case of a **body coil error**, i.e. cooling water flow too low (refer to **RF-repair instructions** for adjustment) and return cooling water to hot, the red LED's V63 (right) and/or V 83 (left) light up.

Measurement of the cooling water temperature sensor resistance in the **body resonator** (return cooling water flow) can be done at connector X20/D9 between Pins 9 and 10.

Resistance values of the temperature sensor:

$R = 2000 \text{ }\Omega$ at 20 °C

$R > 2140 \text{ }\Omega$ -> overtemperature

$R < 2000 \text{ }\Omega$ -> OK

RC Test procedure

According to the table below check the following FOC's on the display board in order to verify a corresponding error message:

- "RDY" off => RC Output voltage lower than 60% of U_{nom}
- "TEMP" off => RC power stage overheated
- "ITRIP" off => At least one of RC output stages in current trip; if this state occurs when the modulators are switched off, a power stage of the GPA is defective: a lighted LED IX1...IZ2 indicates the defective power stage.
- "V.." off -> Corresponding output stage voltage > 425V; if all six FOC's are off, the RC control electronics are defective.

LED	EN	RDY	Water Temp	Air Temp	Phase	V DIP	V PRI	IX1	IX2	IY1	IY2	IZ1	IZ2
LED's continuing on right side										ILIM	P15V	N15V	HF15V
FOC	EN	RDY	TEMP	ITRIP	VX1	VX2	VY1	VY2	VZ1	VZ2			

Meaning of LED's

LED	Color	Description (LED = ON)
EN	green	RC enabled from GSSU: ON during normal operation, OFF in standby mode of RC.
RDY	green	Output voltage of RC is higher than 60% of the nominal output voltage.
WATER TEMP	red	Cooling water temperature too high.
AIR TEMP	red	Air temperature in the GPA cabinet too high.
PHASE	red	Line voltage phase error.
V DIP	red	Line voltage exceeds 528 V.
V PRI	red	Line supply voltage for RC control is lower than 185 V.
IX1... IZ2 (6*)	red	The current trip threshold of the corresponding channel is exceeded. The corresponding power stage of the GPA is probably defective.
ILIM	yel	Primary power stage of RC is in current limit
P 15	green	Supply voltages (15 VDC) for RC are ok. ON during normal and standby operation of RC.
N 15	green	
HF 15	green	

Measurement /Adjustment of output voltage (RC/old type):

CAUTION Switch off the power supply of the RC by fuse first F9 (3*50A) and then F14 in the CCA before working with removed front cover of the RC or making any connection of test cables to the test points of the output voltages.
Voltages up to 420 VDC may be present at these test points.

CAUTION When you connect test lines or clips to the test points, make sure that they do not touch adjacent circuit lines. There is the danger of destroying the control board of the RC due to attaching high voltages.

With the pot R1, located on the left side of the display board you can adjust the maximum output voltage of the RC: $U_{nom} = 405$ to 410 VDC with modulators switched off. The voltage must be adjusted and measured after installing of a replacement RC (refer to the repair procedures). Through a hole in the front cover you can get access to R1. But in case of six installed power stages you have to remove the front cover in order to measure the output voltage of the RC.

The output voltage of the RC can be measured as described below:

1. Three power stages are installed:

- Remove the protection cover plate right of power stages.
- Connect FOC's, Line voltage and water pipes to the RC.
- Connect the DVM to UB+/UB- in the GPA cabinet (lower fuse holder NH2).
- Switch on the Service mode (S1 right side at D4 in upward position).
- Switch on the RC (first F14 and then F9).
The resonant circuit of the RC can be switched on/off manually by the push-buttons S402 ("on") and S403 ("off") at D6.
- Switch off the modulators (switches S1 left and S2 in upward position).
- Turn pot R1 clockwise to increase the voltage or counter clockwise to decrease the output voltage.
With modulators off, the voltage should be between 405 and 410 VDC.
- After adjustment reassemble the protection cover plate right of power stages.

2. Six power stages installed/ backside of GPA not removable:

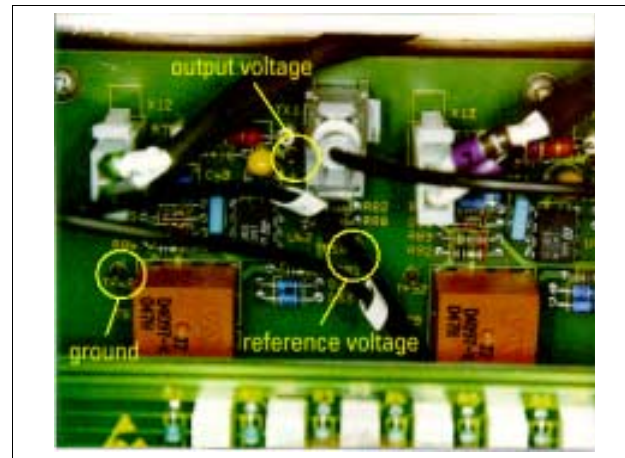
- Remove the power cable, the FOC's and front cover of the RC to get access to the test points which are located above the display board. (refer to the table and figure below).
- Reconnect FOC's, Line voltage and water pipes again.
- Measure and adjust the output voltage as described above. (Measure the voltage of one of the six output testpoints only).

Location of test points: (refer to the the figure below)

Test point	X1	X2	Y1	Y2	Z1	Z2
Ground	43	46	49	52	55	58
U_{out}	42	45	48	51	54	57
U_{ref}	44	47	50	53	56	59

- Optionally you can measure the reference voltage for the output overvoltage supervision: $U_{ref} = 8.6...8.7$ VDC. The resonant circuit of the RC will be switched off, if the output voltage of one output channel exceeds 425 VDC.

Fig. 62 Location of test points on the RC



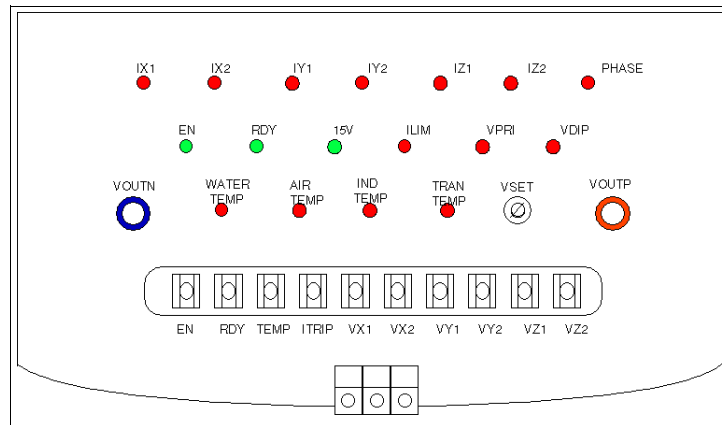
- After adjustment, switch off the RC and remove FOC's, Line voltage and water pipes.
- Reassemble the front cover.
- Finally reconnect FOC's, line voltage and water pipes to the RC again.
- Switch on the gradient system, first fuse F14, then F9.

Measurement /Adjustment of output voltage (RC/MK2):

- Switch off the circuit breakers F9 (3*50A) first, then F14 (4A) in the CCA.
- Connect a DVM (range 1000 VDC) to VOUTN and VOUTP terminals at the RC.
- Switch on the Service mode (S1 right side at D4 in upwards position).
- Switch off the modulators (D4, switches S1 left and S2 in upwards position).
- Switch on the RC (first F14 and then F9, CCA)
The resonant circuit of the RC can be switched on/off manually by the push-buttons S402 ("on") and S403 ("off") at D6.
- Read the voltage at the DVM; The voltage should be between 4,05 and 4,10 VDC.
(4VDC <=> 400VDC at RC output)

- If out of tolerance, adjust the voltage with pot VSET at RC to 4,07 VDC.

Fig. 63 Voltage measurement terminals and adjustment pot of RC MK2

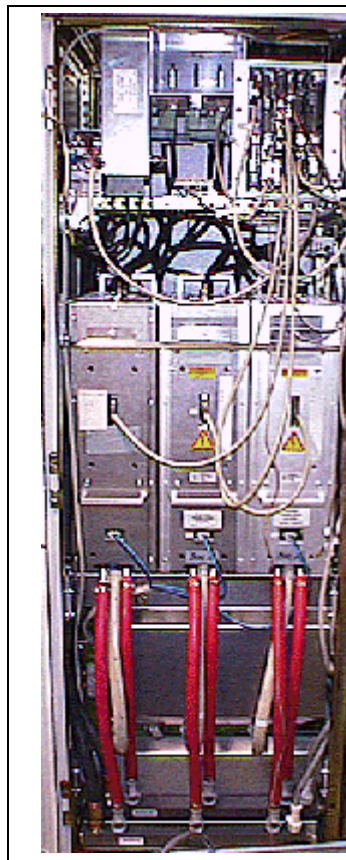


NOTICE

Don't forget to deactivate the "Service mode" for normal operation of the MR-system.

GPA K2217

Fig. 64 Front side of
GPA_059 cabinet



Components:

(from left to right)
 Temperature sensor (above E4A)
 Power supply E4A & Fan
 Line filter
 SENS (in the back)
 GSSU & Fan
 Cabling, FOC's

Power stages X,Y,Z-axis

NF filters (in the back)

5 Blowers

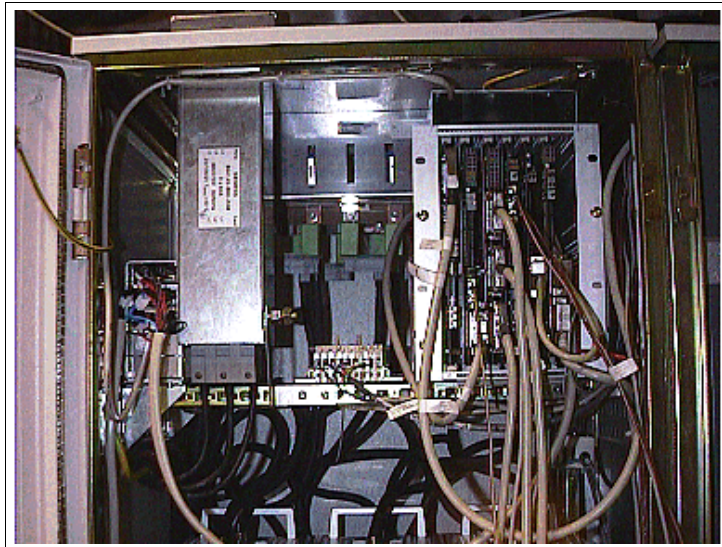
Transformer T1 (in the back)
 Main contactor K2

D31, Soft start unit
 D32, Rectifier board

Cooling water distributor

FRU's in the upper part of the GPA cabinet

Fig. 65 Upper part of GPA cabinet



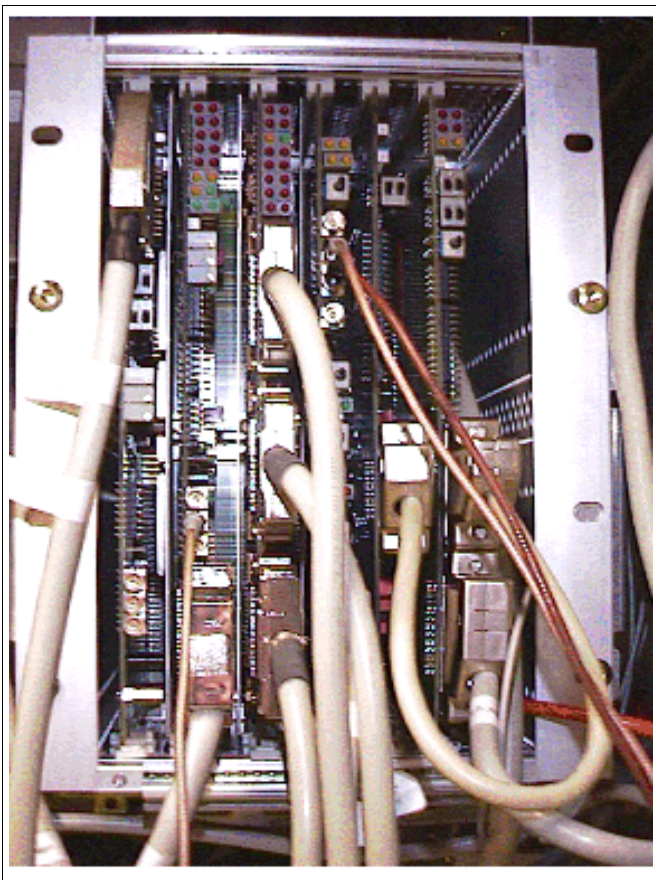
Components from left to right

- Power supply E4A
- Line input filter
- Connector block X100
- GSSU
- In the background: Current sensor assembly

Position of PCB's:

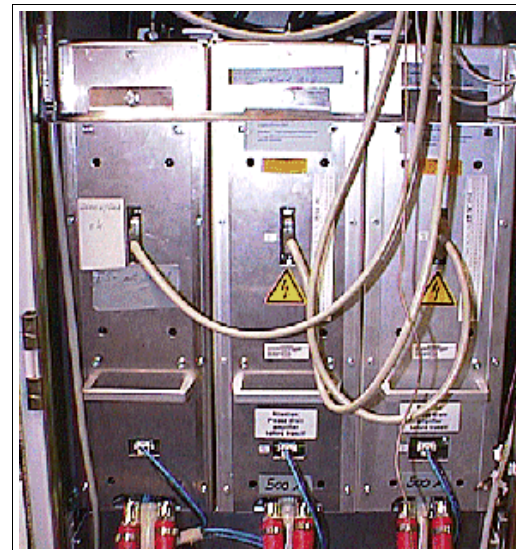
D11 - D12 - D14(2) - D16 - D13(1) - D17

Fig. 66 GSSU of GPA K2217



FRU's in the middle part of the GPA cabinet

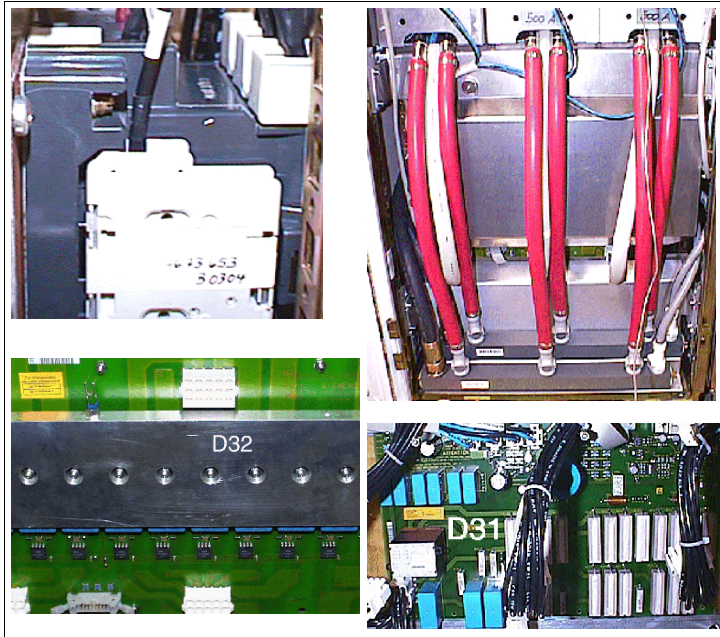
Fig. 67 Middle part of GPA cabinet



X,Y,Z-Power stage with cabling and water pipes

FRU's in the lower part of the GPA cabinet

Fig. 68 Lower part of GPA cabinet (with D32 partially shown left and D31 right and main contactor K2)



Blowers below the power stages

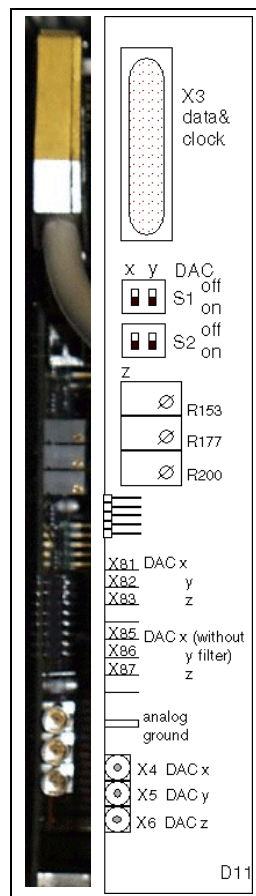
Behind cover plates:

- Rectifier board D32 in front
- Soft start board D31 behind D32
- Main contactor K2

Water distributor with water pipes

DAC D11 Test procedures

Fig. 69 Front of
DAC board D11



Descriptions:

X3:

18 bit parallel gradient data and
4MHz clock signal from MMC

S1/S2:

The DAC output can be switched on/off
for every axis; S2/2 is not used.

Potentiometers for offset adjustment

DAC output = nominal value (0-> +/-10V)
10 V corresponds to the maximum output
current.

a. Check jumpers and switch positions:

Jumper X16: 2-3 (sign of gradient data, 2-Complement)

Rotary switch S3: position 3

Rotary switch S4: position 6

(this setting results in a clock signal FSYNC = 20 kHz)

b. Offset measurement/adjustment:

Note: Prerequisite is the Regulator adjustment (Tune-up).

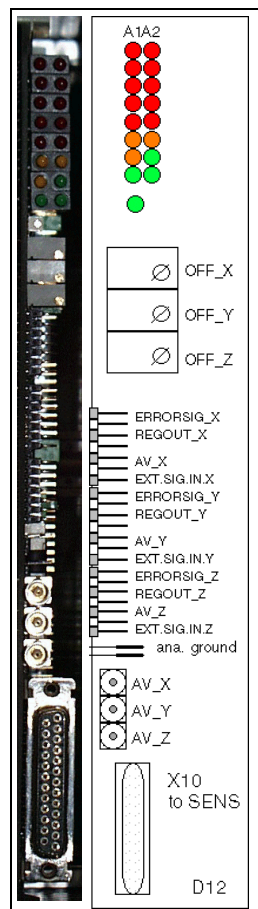
- Gradient cabinet has been on for at least 10 minutes.
- Set switches S1/2 to up position ("DAC output zero")
- The corresponding modulator must be on: S1/2 on D17 in down position.
- Connect a DVM to the desired actual value output X11,12,13 on D12.
Use the QLA-BNC service cable.
- Check the offset: offset < +/- 0.05 mV
- Use the corresponding R153, 177, 200 on DAC-board D11 to adjust the offset for the X, Y and Z axes if necessary.

c. Gradient data Test loop GCTX/D11:

Use the Test "Gradient/ DAC/Functional" of the
Test Tools platform to confirm the correct gradient data transmission.
between MMC/GCTX and GSSU/D11; (a prerequisite for this test is
the correct function of the MMC components, refer to MMC-Tests).

Regulator D12 Test Procedure

Fig. 70 Front of
Regulator D12



Descriptions:

LED's (A1 left / A2 right):

V130: IMAX_X / IMAX_Y
V131: IMAX_Z / CAV_ERR_X
V132: CAV_ERR_Y / CAV_ERR_Z
V133: DUCY_X / DUCY_Y
V134: DUCY_Z / OHMLOSS
V135: LOOP_ERR_X / LOOP_ERR_Y
V136: LOOP_ERR_Z / REG_ON_X
V137: REG_ON_Y / REG_ON_Z

V159: current flow

Offset pots

X: R 155

Y: R 189

Z: R 223

Testpoints

(refer to Functional description)

X11,12,13: Actual current value X,Y,Z

QLA (1V=50A)

X10 SENS connection

a. Default jumper setting by DIP switches (regulator parameters)

- in preparation -

b. Jumper settings:

Jumper	Setting	Function
X16, X18, X20	1-2 2-3	Nominal value X,Y,Z from D11 Nominal value externally
X17, X19, X21	1-2 2-3	Current regulation normally Control from X67,77,87
X22/X23	1-2/1-2	Processor controls DAC gain
	2-3/2-3	DAC maximum range <=> 505A
	open/2-3	DAC maximum range <=> 435A
	2-3/open	DAC maximum range <=> 375A
	without jumper	DAC maximum range <=> 300A
X24	1-2	Switching depending from current direction
	2-3 do not use this position !	Switching not depending from current direction

c. SENS offset measurement/adjustment:

- Gradient cabinet has been on for at least 10 minutes and the regulators are adjusted (see Regulator Adjust/Check below)
- Connect the DVM to X11,12,13 ("actual value") on the regulator board. Use the QLA-BNC service cable.
- Set the switch, corresponding to the axis, on the I/O board D17 in upward position ("Modulator off").
- Check the offset: offset < +/- 0.05 mV; use offset pots to adjust the offset if necessary.

d. Regulator Adjust/Check:

In NUMARIS 3 the regulators are set via DIP-switches S1...S9.

The **QA/Regulator Check** function quickly checks the regulator data set.

Note:

The DIP switch setting of the regulators in NUMARIS 3 depends on the type of gradient coil used in the system.

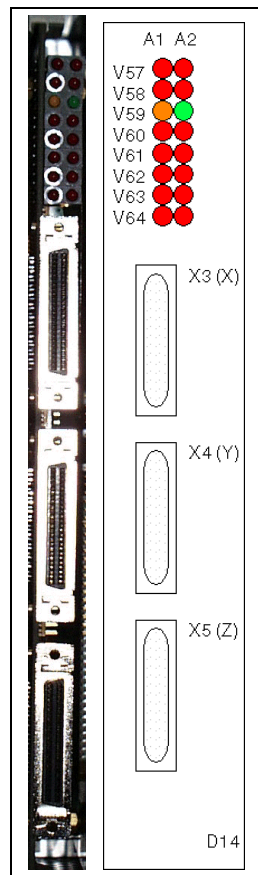
Set the switches depending on the gradient coil used.

Refer to the Diagram Binder and Function description Gradient System.

In Numaris 4 the parameters will be set automatically.

Modulator D14(2) Test procedures

Fig. 71 Front of Modulator D14



Descriptions:

Error LED's

X3,4,5: Connections to power stages X,Y,Z

Clock signals, control of drivers, power stage transistors and temperature supervision (50 pin connectors)

a. Meaning of LED's

LED	LED ON		
V57/A1	N_CLOCK_ERR (error in 82 MHz clock)		
V57/A2	M_MOD_ERR (modulator disabled)		
V58/A1	P_15V_ERR (+15V error)		
V58/A2	N_15V_ERR (-15V error)		
V59/A1	SYNC_ERR_CH (GPA not synchronous to system clock)		
V59/A2	5V_ok		
V60/A1/2	Continuously on: Supervision chip not configured, change D14 (D142)		
V60/A1/A2	State 1: A1 ON / A2 OFF	State 2: A1 OFF / A2 ON	State 3: A1 ON / A2 ON
V61/A1	N_INT_INT Modulator disabled	N_Spare	MOD SPARE 1
V61/A2	OFF_X, N_MODON_X X-modulator disabled	OFF_Y, N_MODON_Y Y-modulator disabled	OFF_Z, N_MODON_Z Z-modulator disabled
V62/A1	EMMI Danger of stimulation	Output disabled FOC control disabled	MOD_SPARE 3
V62/A2	Cable disconnected	MOD_DEF Modulator D14 defective	MOD_SPARE 2

V63/A1	Switching frequency X to high	Frequency Y	Frequency Z
V63/A2	TEMP X Over temperature PS-X	TEMP Y Over temperature PS-Y	TEMP Z Over temperature PS-Z
V64/A1	OVV_X Over voltage PS-Y	OVV_Y Over voltage PS-Y	OVV_Z Over voltage PS-Z
V64/A2	Driver voltage X too low	Driver voltage Y too low	Driver voltage Z too low

After state 3, the LED's V60, A1/2 are off, all 8 LED's below are on.

b. Check of jumper settings:

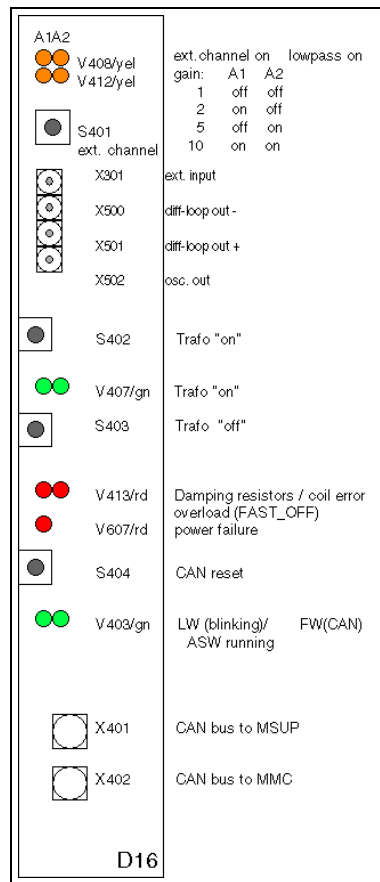
Slew rate settings:

Jumper	Setting	Function
X9/X10	1-2/1-2:	Processor controls slew rate
	2-3/2-3	Maximum slewrate possible
	2-3/open	n/a
	open/2-3	Half slewrate (SR2)
	without jumper SR4 (includes SR2)	Quarter slewrate (SR4)

(Refer to the functional description for further explanations).

Service/CAN D16 Test Procedures

Fig. 72 Front of
Service/CAN board D16



Descriptions:

LED's show conditions of external channel settings.

S1: Sets external channel

X301 External signal input

X502 Oscilloscope output

S402: Manual "ON"

V407: Trafo on/off

S403: Manual "OFF"

V413 left: Damping resistors overload

V413 right: Gradient coil error

S404: Warm start of GPA-processor

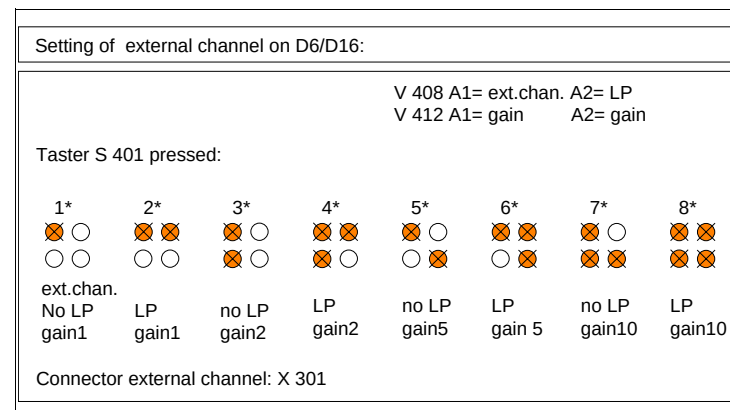
V403 left, blinking during normal conditions

a. Check communication to main power supply:

The main power supply receives the switch-on signal via cable from assembly D17 (X1). Test by pressing S403 and then S402. Verify that the main power supply is switched to stand-by (LED's "EN" and "RDY" off) and on again after approx. 15 seconds (switch-on signal is present at D17 (X1) and is transmitted to the main power supply via cable).

b. Manual loop setting:

Fig. 73 LED display on D6 showing the possible gain and low pass filter combinations of the external channel



When using the external channel, you can select several combinations of different gain and low pass filter settings by pressing button S401 n-times according to Fig. 56. Pressing nine times deselects the external channel (all LED's off).

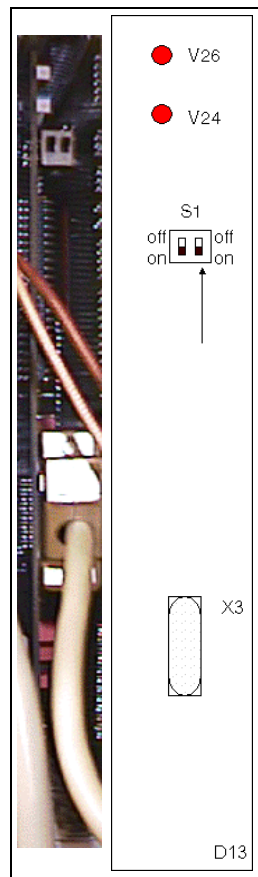
Note: If all LED's are off, the automatic loop is activated.

c. CAN-Bus communication/GPA status:

The present status (Status/warning/errors) of the GPA is displayed completely with the help of the service software function "Gradient/Communication" in the platform Test Tools.

Stimulation Monitor D13(1) Test Procedures

Fig. 74 Front of
Stim.Mon.D13



Descriptions:

LED's

V26: "STIM OFF", stimulation monitor disabled

V24: "Switch OFF", stimulation monitor with switch S1 (right) disabled

a. Function test

(refer to Safety-Related Tests/Binder Installation Volume 2)

- Check the correct jumper position on D12 and D14.
- Position the body phantom with loader in the magnet center. (The adjustments have to be done successfully).
- Select via Main Menu/Measurement/Protocols in the protocol class **safety_test** the following sequences:
 - First measurement: **stimo_function_test** (quantum/sonata); start the sequence by clicking on Measure
-> The measurement will be aborted within 1 second with an error message (popup window message).
 - Second measurement: **stimo_specification_check** (quantum/sonata); start the sequence by clicking on Measure
-> The measurement is performed without an error message.
- In case of any differences in respect to the above described behavior proceed as follows:
 - Perform the Regulator Adjust (General Tune-up)
 - Perform ECC (General Tune-up)
 - Perform Grad Sens (General Tune-up)
- Repeat the **safety_test** measurements.
- In case of further incorrect behavior replace the HW-stimulation monitor once again. Repeat the **safety_test** measurements again.

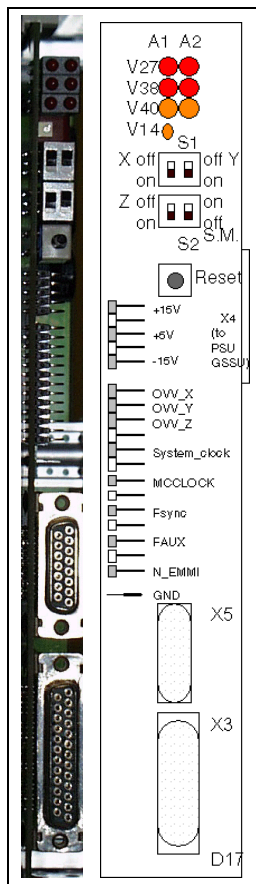
b. Switch on Check:

About 30 minutes after switching off the stimulation monitor with switch S1, the stimulation monitor is enabled automatically (V26 is on again).

To switch the stimulation monitor off again you have to switch S1 to "on" and then to "off" again.

I/O D17/Power supply Test Procedures

Fig. 75 Front of
I/O board D17



Descriptions:

LED's (A1 left / A2 right):

V27: Rectifier -Temp. / Transformer-Temp.

V38: Coilerr / Filterfan

V40: EMMI / DB_WDG_ERR (stim.mon.)

V14: Yellow LED: Service mode

Switches for modulator on/off and
service mode on/off

Error Reset

Test Pins:

+/-15V, +5V

Overvoltage X,Y,Z

System clock 4 MHz

Modulator clock 20 kHz

Synchron.clock 20 kHz

Auxiliary clock 50 kHz (not used)

X5 Connector to stimulation monitor

X3 Connector to switch-on assembly D31

The I/O board D17 is the interface of the GSSU to the switch-on assembly D31, as well as the SENS (synchronization via three FOC's) and supervision assemblies such as D9 (coil supervision) and the stimulation monitor. Electrical signals are converted into FOC signals and vice versa.

Additionally the D17 delivers the supply voltages (from E4A power supply) to the GSSU components. X4 on the board is the connector to the E4A.

The four switches at the front of board allow an individual switch off for every modulators and the selection of the service mode.

A push button allows the reset of an error situation.

In case of errors, this is an easy method for ruling out "monitoring errors":

The status of the error transmission path is tested by pulling the respective fiber optic cable.

"Light off" means that there is an error and/or one of the corresponding sensors is defective.

Check of pin positions jumpers D17/X9

Jumper	Link	Signal function
X9	11-12	Trafo temperature (no supervision)
	19-20	Filterfan (no supervision)

Connector D17/X3 (to softstart unit D31)

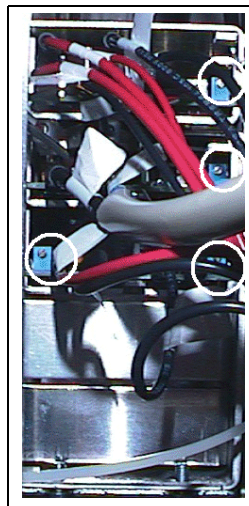
Pin	Meaning	Pin	Meaning
1	K1/K2 switch on (p)	14	message K2 closed
2	K1/K2 switch on (n)	15	+15V
3	Trafo overtemperature (p)	16	+15V
4	Trafo overtemperature (n)	17	+15V
5	Rectifier overtemp (p)	18	0V (GND)
6	Rectifier overtemp (n)	19	0V (GND)
		20	shield (to GND)
		21	-15V
		22	-15V
		23	-15V
		24	Extr. power input (p)
		25	Extr. power input (n)

D17 FOC's

Connector	Meaning
U1,2,3	Synchronization clock to current sensors
U7	Temperature supervision gradient coil

Small signal power supply E4A Procedures

Fig. 76 Adjustment of supply voltages



Slot 1: +5,075 +/- 0,025 VDC

Measure the voltage on D17 (Fig. 75)

Slot 2: +16,5VDC, supply voltage for drivers at D21 on the power stages and blower on top of GSSU.

Measure the voltage at the terminals of E4A (Fig. 76).

Slot 3: +/- 15,05 +/- 0,025VDC

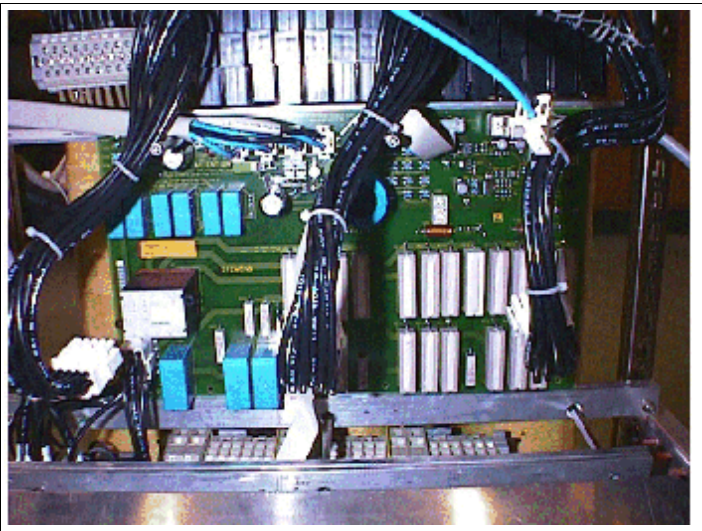
Measure the voltage on D17 (Fig. 75).

Adjust the voltages **with switched off modulators** using the marked potentiometers (Fig. 76).

Slot	Terminal at E4A
Slot 1	A = 5V, B = GND, C = 5V (Sense), D = GND(Sense)
Slot 2	E = +Blower, F = GND Blower, G = +Driver supply, H = GND Driver supply
Slot 3	I = +15V, J = GND, K = GND, L = -15V

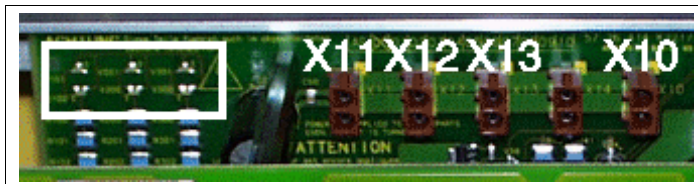
D31 Soft start unit Test Procedure

Fig. 77 D31Soft start unit (D32 rectifier board removed)



Six lighted up LED's at left top of D31 indicate a closed motor protection switch F3/80A in the CCA mainbox, i.e. main voltage is present on D31 (refer to [Fig. 81](#)).

Fig. 78 D31Soft start unit - Voltage indicating LED's and driver supply connectors X11,12,13 (to PS) and X10 (from E4A)



Connector X1 to GSSU, D17/X3

Pin	Meaning	Pin	Meaning
1	K1/K2 switch on (p)	14	Message K2 closed (n)
2	K1/K2 switch on (n)	15	+15V
3	Trafo overtemperature (p)	16	+ !5V
4	Trafo overtemperature (n)	17	+ 15V
5	Rectifier overtemp (p)	18	0V (GND)
6	Rectifier overtemp (n)	19	0V (GND)
7	Overvoltage Z (p)	20	shield (to GND)
8	Overvoltage Z (n)	21	-15V
9	Overvoltage Y (p)	22	-15V
10	Overvoltage Y (n)	23	-15V
11	Overvoltage X (p)	24	Overload (EPI) (p)
12	Overvoltage X(n)	25	Overload (EPI) (n)
13	Message K2 closed (p)	-	Shield

Connectors X11 to X13 and X10

Pin	Meaning	Pin	
1	+16,5V	2	0V

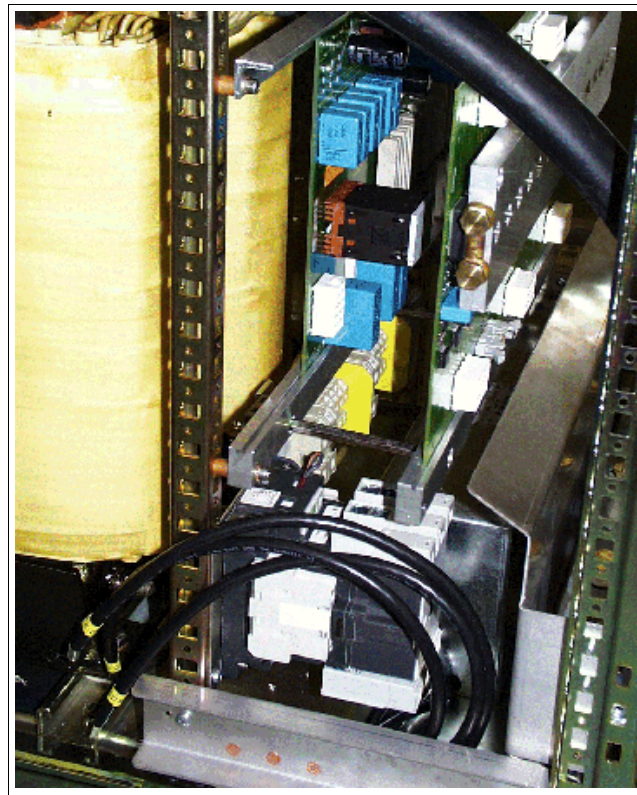
Flat cable connector X8 to D32/Rectifier board

Pin	Meaning	Pin	Meaning
1	Connect. supervision	2	
3	+15V	4	+15V
5	0V (GND)	6	0V (GND)
7	PTC on Rectifier(n)	8	PTC on Rectifier (p)
9	Overvoltage X (p)	10	Overvoltage X (n)
11	Overvoltage Y (p)	12	Overvoltage Y (n)
13	Overvoltage Z (p)	14	Overvoltage Z (n)
15		16	Connect. supervis. (GND)

Connector X2 to K2 main contactor

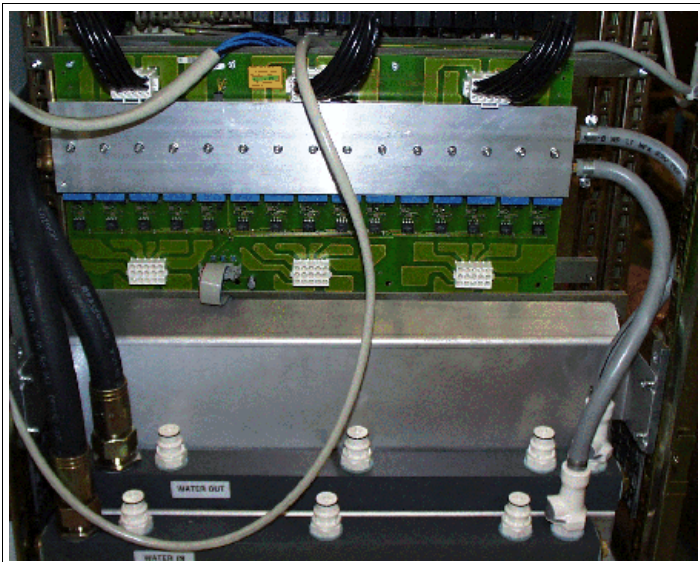
Pin	Meaning	Pin	Meaning	Pin	Meaning
1	K2_A1_old	2	K2_A2	3	K2_A1_new
4	K2 mess. cont.	5	K2 mess.cont.	6	
7		8		9	
10	Transformer L1	11	Transformer L2	12	Transformer L3
13	Line L1	14	Line L2	15	Line L3

Fig. 79 D31 Connector X2 - Main contactor K2



D32 Rectifier unit Test Procedure

Fig. 80 Rectifier board, ACV power connectors from T1 and cooling water pipes plugged in



Connector X1 to D31/Softstart unit

Pin	Meaning	Pin	Meaning
1	Connect. supervision	2	
3	+15V	4	+15V
5	0V (GND)	6	0V (GND)
7	PTC on Rectifier(n)	8	PTC on Rectifier (p)
9	Overvoltage X (p)	10	Overvoltage X (n)
11	Overvoltage Y (p)	12	Overvoltage Y (n)
13	Overvoltage Z (p)	14	Overvoltage Z (n)
15		16	Connect. supervis. (GND)

Connector X11(X), X21(Y), X32(Z) ACV supply of rectifiers

Pin	Meaning	Pin	Meaning	Pin	Meaning
1	ACV X,Y,Z_U11	2	ACV X,Y,Z_U12	3	ACV X,Y,Z_U13
4	ACV X,Y,Z_U21	5	ACV X,Y,Z_U22	6	ACV X,Y,Z_U23
7	ACV X,Y,Z_U31	8	ACV X,Y,Z_U32	9	ACV X,Y,Z_U33

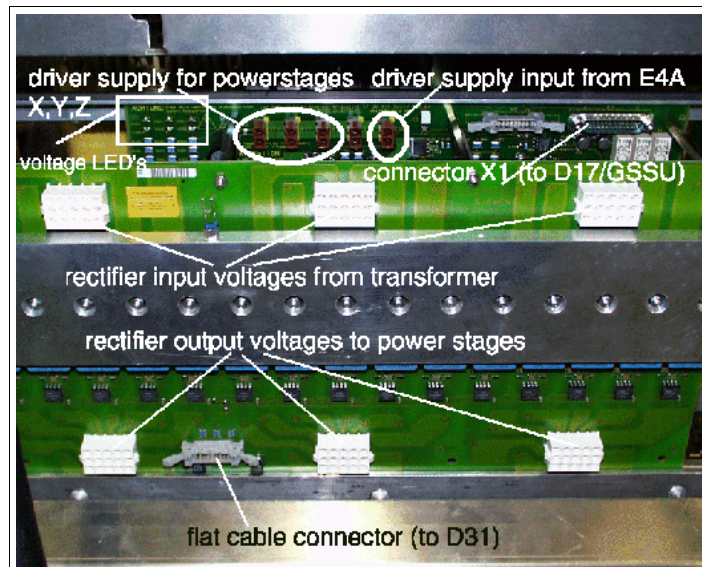
Measurement of supply voltages for power stages

- Switch off K2 with D16/S403.
- Switch off the motor protection switch F3/3*80A in the CCA mainbox.
- Pull one of the output connectors X21, X22, or X 32, (where you want to measure the output voltage).
- Connect a DVM (range 1000 VDC) with appropriate adapter clips to the connector pins at D32 according to the table below.
- Switch off the modulators (D17 S1 and S2 left in upward position).
- Switch on the Service mode (D17 S2 right in upward position).
- Switch on F3/(80A).
- Switch on K2 with D16/S402.
- Read the Output voltage at DVM.
- Switch off F3/3*80A in the CCA mainbox.
- Remove the measurement equipment.
- Plug in the removed connector again.
- Switch on F3/3*80A in the CCA mainbox.
- Switch on the modulators (D17 S1 and S2 left in downward position).
- Switch off the Service mode (D17 S2 right in downward position).

Connector X21(X), X22(Y), X32 (Z) DCV supply of power stages

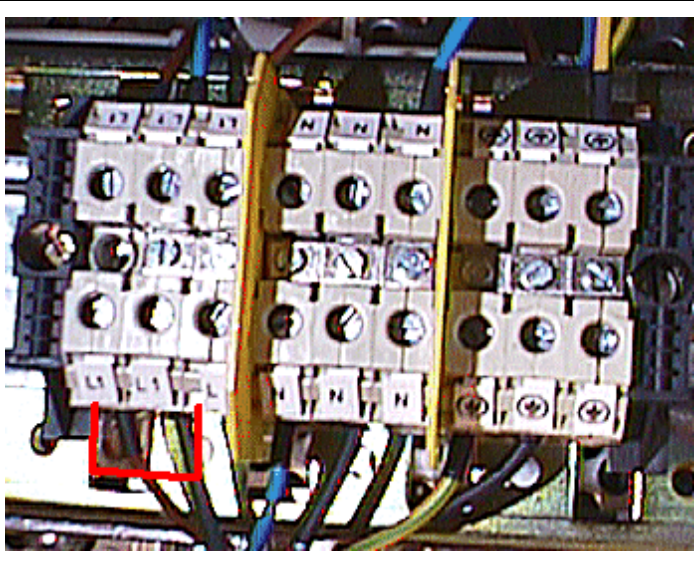
Pin	Meaning	Pin	Meaning
1	DC voltage X,Y,Z_U1+	3	DC voltage X,Y,Z_U1-
4	DC voltage X,Y,Z_U2+	6	DC voltage X,Y,Z_U1-
7	DC voltage X,Y,Z_U3+	9	DC voltage X,Y,Z_U3-
10	DC voltage X,Y,Z_U4+	12	DC voltage X,Y,Z_U4-
13	DC voltage X,Y,Z_U5+	15	DC voltage X,Y,Z_U5-

Fig. 81 Plug positions on D32/D31



GPA Blowers Test Procedure

Fig. 82 X100 Line power connector block in the GPA cabinet



Check of blower function:

A temporary connection between the left L1 with the right L1 terminal at X100 (refer to Fig.60) simulates a closed temperature switch in a filter inductor (switching point 50 °C).

There is just one temperature switch located in all upper three filter inductors behind the power stages. The switches are connected in parallel.

All five blowers must start running.

TestTools - Gradient Loop Tests

Test usage and interpretation of results

The tests are available in the Service software platform **TestTools**.

MMC - DAC Gradient data test (DAC/Functional)

(Loop MMC/GCTX -GSSU/ D1/D11 - MMC/GCTX)

Test of 18 bit parallel data transmission and control signals via cable.

In case of errors, perform the MMC-tests, check the cable connection between GCTX and GPA. In addition check the D1 K2209/D11 K2217.

DAC - nominal value (DAC/Functional)

(Loop MPCU/GCTX - D1/D11/DAC - D6/D16 - RFSU/ADC)

Nominal value for gradient current from DAC for x,y,z axes;
Stability and linearity measurement with several parameter variations, such as amplitude and timing.

If some of the test fail, check D1 K2209/D11 K2217 (DAC), D6 K2209/D16 K2217, cables to the RFSU, receiver board. Perform the MMC tests.

Note:

If the test fails due to a regulation (loop) error, switch on the "service mode" (right switch S1 upward on D4 or right switch S2 on D17). Now you can test the correct function of the DAC's.

Actual current value (X,Y,Z-axis/Functional)

(Loop MPCU/GCTX - D1/D11/DAC - D2/D12 - D14 - ((D3) - D4/5 - D7) - PS - SENS- D2/D12 - D6/D16 - RFSU/ADC)

Actual value for gradient current from D2/D12, actual value amplifier for x,y,z axes;
Stability and linearity measurement.

If some of the test fail, check D2 K2209/D12 K2217 (regulator board) with "Regulator Check", D6 K2209/D16 K2217 (loopmux), cables to the RFSU, receiver board. Perform the MMC tests.

Check the following GPA K2209 components:

D4/5 K2209 (modulator board), D3 K2209 (synchronization board), D7 K2209 (FOC board) and the FOC's to the power stages.
Moreover check for error messages concerning the D10 K2209 (driver supply board), power stages, the current sensors (SENS) and the RC; refer to the corresponding TSG charts.

For further information refer to the online help of the gradient tests.

Normal/Standby Operation

Status of components K2209

Gradient Small Signal Unit Components	
D6 Service-CAN-board Normal Mode:	2 LED's "RC on" (V407,green) on; left LED (V403,green) is blinking (Load ware is ok, Application software (ASW) is running)
Standby:	2 LED's (V407,green) off -> RC in standby; left LED (V403,green) is blinking (Load ware is ok, ASW is running)
D1 DAC	Switches S1/2 in downward position
D2 Regulator	no Error LED on
D3 Synchron.	LED (V30, green) on -> VCC +5V ok
D4/5 Modulator	no red Error LED on; LED's (V8-10, yellow) indicate gradient (+/-) puls- ing during a measurement
	Switches D4 S1/S2 "Modulator on" in downward position; Switch S1 "Service mode off" in downward posi- tion
Standby:	D4, LED V5/A2 "Modulator disabled" (V1/A2,yel- low) is on

D10 Driver supply	Green LED is on -> Output voltage ok
D9 Supervi- sion (located in Pat. tab. control)	LED "Grad. temp.ok" (V36/A2, green) on; LED "Body res. ok" (V83/A2, green) on.
GPA Small Signal Power supply: Fan is running	
CCA cabinet: Circuit breakers F14,15 and F9 are switched on Fan's beside the RC and at the top (inside) of the cabinet are run- ning	
Power stage: FOC "T" is on (red light).	
Resonant Converter (normal mode): Green LED's "EN", "RDY", "P15V", "N15V", "HF15V" are on	
Resonant Converter in standby: Green LED's "P15V", "N15V", "HF15V" are on, "EN" and "RDY" are off	

Status of components K2217

Gradient Small Signal Unit Components	
D16 Service-CAN-board Normal Mode:	2 LED's "Trafo on" (V407,green) on; left LED (V403,green) is blinking (Load ware is ok, Application software (ASW) is running)
Standby:	2 LED's (V407,green) off -> "Trafo off"; left LED (V403,green) is blinking (Load ware is ok, ASW is running)
D11 DAC	Switches S1/2 in downward position
D12 Regulator Normal Mode:	LED's V136/137 green on
Standby:	LED's V136/137 green off
D14 Modulator Normal Mode:	green LED V59 on -> VCC +5V ok
Standby:	D14, LED "Modulator disabled" (V61/A2, red) is on in state 1, 2 an 3.
D17 I/O	Switches S1/S2 "Modulator on" in downward position; Switch S1 "Service mode off" in downward position

D9 Supervision (located in Pat. tab. control)	LED "Grad. temp.ok" (V36/A2, green) on; LED "Body res. ok" (V83/A2, green) on.
GPA Small Signal Power supply: Fan is running	
CCA cabinet: Circuit breakers F14 are switched on; Cutter switch F3/3*80 A in the CCA maijnbx (top of CCA) is switced on.	
Power stages: If the six green LED's behind the six holes in the front of the PS are on, the capacitors are charged.	

Power On - Application software

After "Power On" or Reset) all CAN units run through a self-test and start the firmware. The firmware checks to whether the loadware has been loaded. It also checks the loadware code. The loadware is started if the consistency test is successful.

When the gradient processor is in the loadware mode, a logon command is sent every 500 milliseconds to the MPCU.

The gradient processor is the link between the gradient amplifier and the rest of the system. As a CAN unit, it is connected with the CAN bus to the system.

In loadware mode, the gradient processor is ready for operation with the exception of sending commands to the MPCU.

The gradient processor is fully operational after the MPCU has validated the version of the gradient processor via the login request and the software application has been started via a CAN command.

In other words, after the CAN unit has been booted and the loadware started, the application software is not fully available. However, after the login request by the MPCU, the application software is fully operational.

The application software of the gradient processor allows for the reliable operation of the overall component, namely the gradient amplifier. The most important tasks are:

- Setting the PI regulators for the three gradient axis
- Setting the different test loops for the service software (gradient stability/linearity tests)
- Service mode, switch on/off
- Setting the Power on/Standby mode

GPA Status (example K2209)

Using the test Gradient/GPA status/Comm/Status in the platform Test Tools, you will get the following listing as an example:

```
Starting Test GetGpaStatus ...
Status of GPA is ok.
```

```
...
GPA_STATUS: Resonantconverter current trip    =no
GPA_STATUS: Resonantconverter standby        =no
GPA_STATUS: Resonantconverter defect         =no
GPA_STATUS: Drivercurrentsupply              =ok
...
```

and so on.

In case of errors, the corresponding status is indicated with "!!!". In this case, refer to the "Details" of the Error log, respectively to the corresponding TSG Charts.

Description of Servicemode

Service mode can be activated either by the service software or by setting manual (right) switch S1 on board D4 to the up position.

If service mode is active, the following errors will be handled as **warnings**, e.g. the modulators and regulators will not be disabled:

- K2209: RC not ready ($U_{act} < 60\%$ of U_{nom})
- K2209: Synchronization error (D3, LED V37 red/ D14 LED V3 red)
- K2209/K2217: Regulation error x,y,z (D2, LED V27 yellow/ D12, LED's V6/V7 yellow)
- K2209/K2217: No reaction of modulator

PART 10 Magnet System

In this part you will find the strategies and test procedures for trouble shooting on Magnet, Refrigerator, Magnet Supervision and Magnet Supply

Strategy

This chapter describes the test strategy for the Magnet; Refrigerator; Magnet Supervision and the Magnet Power Supply.

Magnet

In the following table all possible error symptoms or problems are listed.

Symptom	Possible cause	Service action	Comments
High Helium Boil-off	Maintenance interval for displacer exceeded	Replace the displacer	Procedure for replacing: Refer "Maintenance manual"
	Malfunction of the Helium Level-meter	Check correct calibration; Use the second Level probe	
	Malfunction of the screen reset	Check for proper function and setting of the screen reset unit	
	High shield temperature	Check the shield temperature via SeSO	
	Refrigerator does not work correctly	Refer to chapter "" Refrigerator ", page 336"	
	Vacuum leak	Check the vacuum vessel for ice formation (wet areas)	
	Low helium level	Refill helium	
Helium vessel pressure is: • Zero pressure or • Pointer of the pressure gauge is in the red sector	Leakage in the venting system	Search for leaks, using a leakage dedector (Handy)	
	Leakage on the syphon Leakage on the current lead baffle	Replace seals or tighten ring nuts	
	1/3psi valves might be jammed open	Replace the 1/3psi valves	
	Broken bursting disc	Visual inspection of the Burst disc	

Symptom	Possible cause	Service action	Comments
Ice formation on the turret	Helium gas leak on turret/syphon/baffle; Faulty 1/3psia valve or leak through quench valve	Replace the gasket/copper washer Use the leakage dedector (Handy)	Refer to Replacement of Parts, Magnet, "Seal configuration of baffle and syphon"
	High helium boil-off	Refer to symptom:"High helium boil-off"	
	Broken Burst disc	Visual inspection of the burst disc	Refer to Replacement of Parts, Magnet, "Installing a new bursting disk"
Quench	Ramping the magnet	Refer to: What to do after a Quench (page 334)	
	Vacuum leak	Check the vacuum	
Vacuum vessel is wet	Vacuum leakage	Check the vacuum Repump the vacuum	

What to do after a Quench

CAUTION **Disconnect the Magnet Power Supply from the Mains Supply**

The quench valve will open and stay open until the flow subsides. The valve will then re-seat, resealing the venting and thus preventing any air from being sucked back into the vessel.

The quench valve incorporates a 20 psi bursting disc, housed in the middle of the poppet. In normal operation, the disc should remain intact.

The integrity of the disc must be checked however in case the valve has malfunctioned.

- Check the pressure of the Helium vessel on the pressure gauge .

Symtom	Service action
Pressure in the yellow or green sector of the gauge	1) Disc can be assumed o.k. 2) Warm up the ice on the turret with a heat gun 3) Check for leaks with the Handy Detector

Symtom	Service action
Pointer stays on or close to zero	1) A broken disc must be suspected 2) Repressurize the helium vessel Remove the FUSE F4 on D3 of the MSU unit. Energize the quench heater by pressing the magnet stop button on the alarm box for two minutes. Release the magnet stop button after energizing the quench heater. If it does not start to re-pressurise within one minute of energising the quench heater, the bursting disc is almost certainly broken 3) Replace the bursting disc ("Repair Magnet-System") 4) Repressurice the helium vessel. 5) Check for leaks with the Handy Detector 6) Reinsert the Fuse F4

- Refill the magnet with helium

WARNING **If the magnet quenched while unattended, you must check the bursting disc as soon as possible. If the bursting disc is found to be broken, it must be assumed that the system is iced. Call Service department.**

Filling the Helium Vessel

CAUTION What to do if the smart valve of the syphon did not move to the "transfer" position

If the smart valve of the syphon has not moved to the transfer position, the smart valve may be stuck in the closed position.

In this case, the ring on the top of the syphon may be used to open the smart valve.

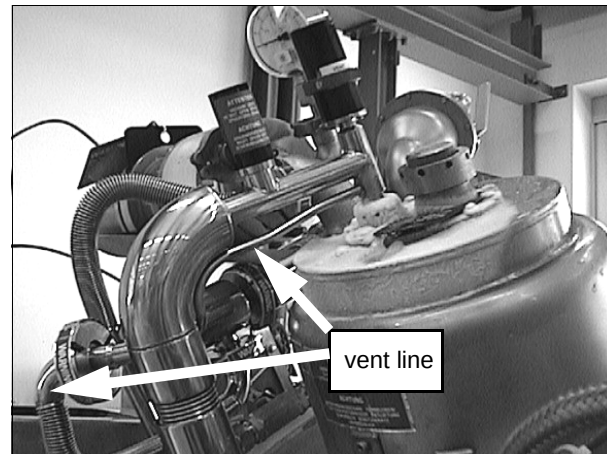
WARNING Do not manually move the valve unless the syphon has had adequate time to cool. If the valve is moved before the syphon is cold the magnet will quench

Before moving manually the valve, check the following:

1. Syphon has been cooled for at least 10 minutes
2. The vent line from the top of the syphon has become very cold (visible vapour falling, or liquid oxygen dripping - this occurs after about 10 minutes under normal circumstances). See Fig.1.
3. The syphon on the dewar side is installed correctly ?
4. The shut-off valve at the syphon on the dewar side is completely open ?
5. The syphon is covered with ice, check the vacuum of the syphon. If vacuum is bad, replace the syphon.

6. The pressure shown at the helium dewar is o.k. (The pressure is given in the filling instructions in the start-up section)
7. Does the helium vessel contain liquid helium?

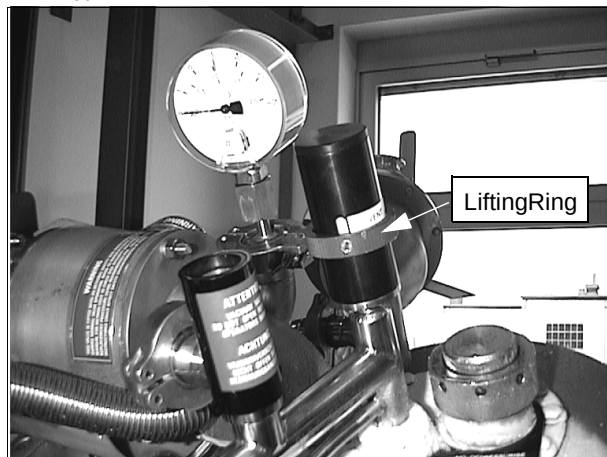
Fig. 83 Smart syphon



After you checked all items, the smart valve of the syphon can be opened manually.

- Apply even pressure to both sides of the ring, lift the ring to the top of the slot in the housing (about 10mm), and hold it in place until the spring force on it reduces. The helium is now flowing into the magnet, and will keep the syphon cold and the valve open (Transfer) See fig.2
- At the end of the transfer, the valve should move to the closed (Vent) position.

Fig. 84 Smart syphon



Symtom	Possible causes
High transfer loses	Ice formation in the magnet Smart valve did not open complete Filling pressure too low Sludge inhibitor on the dewar syphon missing

Refrigerator

The compressor unit shuts down automatically if the helium pressure is too low or the helium temperature is too high.

The compressor display reads:

SYSTEM ERROR

Press the SELECT button to determine the source of thr error.

The compressor unit does not restart automatically.

Locate and remove the cause of the error.

After error removal, press the OFF button,

SYSTEM ERROR

is displayed.

Reset and start the compressor unit by pressing ON

The Following parameters are constantly supervised; an error message is sent to the HOST computer:

- Temperature of the He-gas in the compressor
- He-pressure of the compressor

Actions following a power failure:

Following a temporary line voltage failure, the compressor unit automatically returns to its previous operating status.

The compressor unit will switch back on automatically if it was at normal operation before

Compressor - Error message

Error - message	Possible cause	Service action	Reference
The compressor unit fails to run; a Pressure Alarm message is displayed	Leak in helium circuit	Check helium pressure and top off helium Test for leaks	
The compressor unit fails to run; a Helium temp. fail or Oil cct. fail is displayed, one of the thermal switches has tripped	Primary water circuit not open Insufficient coolant throughput	Insure sufficient water supply	
	Insufficient water flow	Insure sufficient coolant supply	
	Water inlet temperature too high	Check the onsite water supply	
	Water connections reversed	Connect water lines correctly	
	Heat exchanger congested	Flush the heat exchanger	
	Ambient temperature too high		
	Compressor motor too hot		

Compressor display	Possible cause	Service action	Reference
Phase/Fuse ERROR	F1 or F2 has blown Phase sequence of the line voltage is incorrect Cold head motor defective	Replace fuses Check phase sequence, reverse 2 phases of power supply Check cold head motor	"Procedure-Magnet system"
DC Voltage error Compressor unit runs	Line voltage setting is wrong Cold head motor defective	Set compressor to the correct line voltage Check cold head motor	"Installation Volume 1" "Procedure-Magnet system"
MAINS LEVEL !!!	The line voltage setting is wrong	Set compressor to the correct line voltage	
OVERLOAD	Current is too high at switch-on	Check connections; then try to start the compressor unit again. After the compressor has failed 3 times to run with an OVERLOAD; the display will show "Locked rotor"	
Locked rotor	Compressor has failed 3 times to run with an OVERLOAD		
Compressor fail	Thermal switch in compressor capsule (Klixon) has tripped; compressor capsule too hot compressor motor defective	see Error message: Helium temp. fail Wait 2-3 hours to cool down the compressor capsule, then restart	
RAM ERROR 1 RAM ERROR 2 ROM ERROR EEPROM ERROR		Contact the HSC	

Compressor - Symptoms

Symptom	Possible cause	Service action	Reference
The compressor unit fails to run; no error is displayed	Fault in connection to power supply	Check power lead and plug	
	Line voltage has failed, is too low or too high	Check line voltage and line voltage setting of compressor unit	"Installation Volume 1"
	Master switch has tripped	Locate and eliminate cause of overloading	
	Circuit breaker F3 in the RCA has blown	Locate and eliminate cause of overloading.	
	Both fuses in Power Module have blown	Replace fuses. Contact service department	
Helium pressure drops	Leakage	Refill helium gas; Test for leaks	

Compressor - Symptoms

Symptom	Possible cause	Service action	Reference
High helium pressure in the compressor Bypass valve opens up	Pressurized helium hoses are not connected correctly	Check Aeroquip connections	
	Displacer is frozen	Change displacer	
	Cold head motor is defective	Check motor windings Replace motor	"Procedure-Magnet System"
Compressor whistles (sreeching noise)	Bypass valve opens up	Check helium pressure Switch the compressor off and on again Keep compressor running for 1 h Release some helium gas via refill port Check chilled water Check Aeroquip connections Check cold head motor	

Cold head - Symptoms

Symptom	Possible cause	Service action	Reference
Refrigerating capacity is too low	Helium loss in refrigerator system	Refill helium gas at compressor unit (99.999% purity or higher)	"Repair-Magnet System"
	Adsorber has more than 18.000 hours of operation Adsorber is saturated	Replace the adsorber in the compressor	"Maintenance manual"
	Cold-head motor turning in wrong direction	Check wiring	
	Unisock vacuum bad	Pump down vacuum in Unisock	
	Maintenance interval for displacer exceeded	Replace the displacer	"Maintenance manual"
	Control valve is worn	Replace the Control valve	
Cold head becomes irregular or noisy	Helium contaminated because adsorber maintenance exceeded	Replace the adsorber in the compressor	"Maintenance manual"
	Control valve is worn	Replace the control valve	
	External magnetic field exists (>30 mT)	Shield the cold head motor, replace the cold head motor	
	Cold head motor defective	Replace cold head motor	
Unisock iced up	Unisock vacuum is bad	Pump down vacuum in the Unisock	

Cold head - Symptoms

Symptom	Possible cause	Service action	Reference
Displacer does not move	Compressor failure	Switch compressor back on	
	Cold head motor is defective	Coil test of the cold head motor Replace motor	"Procedure-Magnet System"
	Displacer is frozen in place	Replace displacer	"Maintenance manual"
	Rotating valve damaged	Replace rotating valve	

Magnet Supervision

Error messages

Error message: MSU_	Possible Cause	Service Action	Comment
CCATEMPWARN CCATEMPALARM	Temperature in the CCA-cabinet is above 38°C Temperature in the CCA-cabinet is above 43°C	Check the primary water supply for the water cooled Cabinet Cooling system Check the on-site air conditioning for the air-cooled cabinet Cooling system Check the fans in the cabinet Check if the door is closed	Temperature warning repeats every 3 minutes
GPATEMPWARN GPATEMPALARM	Temperature in the GPA-Cabinet is above 38°C Temperature in the GPA-Cabinet is above 43°C	Check the water supply for the water cooled Cabinet Cooling system Check the on-site air conditioning for the air-cooled cabinet Cooling system Check the fans in the cabinet Check if the door is closed	Temperature warning repeats every 3 minutes
HELIUMALARM	Helium level is below 30%	Measure the helium level; refill helium	

Error message: MSU_	Possible Cause	Service Action	Comment
HELEVEL	Helium level is below 50%	Measure the helium level; refill helium Check the warning setting	
REFRIGWARN REFRIGALARM	Shield temperature is above 75K Shield temperature is above 80K	Maintenance interval for displacer exceeded. Replace the displacer Measure the shield temperature; Cooling capacity of the refrigerator is too low; Check the alarm/warning settings	"Maintenance manual" See chapter: "Refrigerator"
Wrong date in the SESO: "1.1.70	Reset of the D4 processor Manual / Software reset or reset due to a power disruption of the supply to the MSUP control crate.	n.a.	

Supervisory indication on D2 (He, EIS & Battery)

LED name	Description	Service action
EIS FAULT	The EIS has detected current flowing when it should not or no current flowing when it should	
BATT. BAD	After the last battery test, the 24V battery voltage was detected as being below the test level	Replace the battery
BATT. TEST	The 24V battery test is in progress	
BATT. LOAD ON	The 24V battery test load is connected	

Supervisory indication on D3 (ERDU)

LED name	Description	Service action
SH	Switch Heater is out of limits	Check jumpering Check cabling
MPS SH	The Switch Heater is energized by the MPS	
QH1	Quench Heater 1 is out of limits	Check jumpering Check cabling
QH2	Quench Heater 2 is out of limits	Check jumpering Check cabling
BUTT. MISSING	One of the Emergency Stop Buttons has an open circuit or is not connected	Check the wiring Check the battery, change the battery
TELL TALE	The ERDU has been activated	Replace fuse F4
BUTT. SHORT	One of the Emergency Stop Buttons has short circuit (or pressed)	Check the wiring

Supervisory indication on D1 (Monitor)

LED name	Description	Service action
ERDU TEST	The ERDU test load is connected	
F.PROBE OC	Circuit in the Shield Temperature Probe is disconnected	Check the wiring
F. PROBE TEST	The Shield Temperature Monitor is in the test position	
F. ALARM	The Shield Temperature is above the alarm level	Check the refrigerator; Check the helium level
F. WARN	The Shield Temperature is above the warning level	Check the refrigerator; Check the helium level
He ALARM	The Helium Level is below the alarm level	Refill helium
He Warn	The Helium Level is below the warning level	Refill helium
He	The Helium Level is/was below the alarm setpoint	Refill helium
ERDU	There is/was a fault with one of the ERDU circuits	Check the wiring
EIS	There is/was a fault with the EIS	
FRIDGE	The Shield Temperature is/was above the alarm level	Check the refrigerator

Alarm Box Indication

LED name	Description	Service action
HE LEVEL	Helium level is below the alarm point	Refill helium
MAGNET STOP	There is a fault in one of the ERDU circuits	Check the ERDU circuits

Magnet Power Supply

Error Messages on the MPS display

NOTICE

In all cases check for error in hardware set up
When corrected , press "off" button to reset fault lamp.

Error - message / symptom	Possible cause	Service action	Comments
Power supply display not on	Main AC input not on Rear circuit breaker not on	Check AC supply is switched on, F5 on CCA Check top box is seated correctly on bottom box Check rear circuit breaker if it is switched on	Display should turn on
Unit will not respond to "on" switch	Main AC input is not on Fault light is on	Check display is on Check circuit breaker is on A fault has been detected, press off button to reset	Fault light should go off
Unit will not start cycle	Current set over limit Main power supply not on	Check current setting for magnet type Did the MPS recognize the correct magnet type? Check if circuit breaker on front of the MPS is on	Unit should run
"....", Magnet Type Error	MPS recognizes the incorrect magnet type	Check the cable connections: MPS - CCA - MSUP - filter plate - magnet - "C" connector Measure the resistance at the "C-connector" on magnet: OR 24; 47 Ohm, PIN G-J OR 70; 100 Ohm, PIN G-J	
Error "002"	Current not at zero	Check for current in leads. Press OFF to reset	Unit should run

Error - message / symptom	Possible cause	Service action	Comments
Error "003"	Switch Heater Switch on	Check switch is off. Press OFF to reset	
PHASE 1, SELF_TEST			
Unit fails self test	MPS not turned on	Check circuit breaker on front of MPS is on	Fans should run
Error "103"	MPS cannot run to 5A	Check top box is seated correctly on bottom box Check 25 way D type lead is connected, turn MPS circuit breaker off and then on.	
Error "104"	MPS does not run back to 0 Amps	Check D type lead	
PHASE 2, CABLE_TEST			
Unit fails cable test	Cables not connected	Check connections at MPS and magnet	
Error "201"	Actual current not found	Check connections at MPS and magnet	
Error "203"	Voltage is detected as positive	Check connections at MPS and magnet	
Error "204"	Power Control cable missing Actual current not equal to demand, power supply is limiting	Check cabling Check MPS voltage limit is not set too low Voltage should set to 9.9V	
Error "205"	Voltage is detected as negative Polarity of the sense lines reversed	Check the Sense-Line. Monitor voltage on rear of the MPS correct the polarity	
Error "206"	Voltage is too high	Check power leads and Current Lead (DCL)	
Error "207"	Actual current does not run back to zero 1 second after the END of the cable test	Check magnet switch heater is off Check Switch Heater on the MPS is off Oscillations of the MPS, replace the MPS	

Error - message / symptom	Possible cause	Service action	Comments
Error "208"	Actual output current is not zero when output short is opened at start of the cable test	Check magnet does not have current flowing in it	
Error "209"	Voltage drop too high Harmony: ≥ 125 mV Symphony: ≥ 175 mV Current Lead (HDCL) not correctly inserted	Check the seating of the current probe Check the connectors Disconnect and reconnect the cables Replace the spring contact on the Current Lead	
PHASE 3, CURRENT SEARCH			
Error "301"	Voltage is detected as positive	Check sense terminals. Monitor voltage on rear of the MPS	
Error " 302"	Sweep stopped at max current, current search failed	Retry, having checked for current in leads	
Error "303"	Actual current does not equal demand	Check power connections and the D type lead	
Error "304"	Unit could not detect current within 4 seconds	Retry, having checked for current in leads	
PHASE 4, RAMP TO SET POINT			
Unit fails Current search	Actual current is not steady	Check water cooling	
Error "401"	Actual current does not equal demand Oscillation of the MPS	Check MPS for correct current setting Change the MPS	
Error "402"	Actual current does not equal demand Oscillation of the MPS	Check magnet voltage for oscillation Change the MPS	

Error - message / symptom	Possible cause	Service action	Comments
Error "403"/"404"/"405"	Timing error in algorithm	Check MPS for correct current setting	
Phase 5, CABLE RUN DOWN			
Error "501"/"502"/"503"/"504"/"505"	Actual current does not equal demand	Check switch heater is operating correctly Check switch heater lamp is off on the MPS Check switch heater LED is off on the MSUP	
Error "506"	Actual current does not run back to zero	Check switch heater is operating correct	
PHASE 6, RAMP TO ZERO			
Unit fails ramp to zero	Unit has overheated or shut down	Check temperatures and fault lights, check cooling water, Bypass; Cooler	
Error "601"	Current is still detected after 1 hour	Check current in leads, lamp and display; if no current flows press off. if current does flow retry a capture and run to zero.	
Error "602"	Voltage still detected after 10 minutes	Check current in leads, lamp and display; if no current flows press off. if current does flow retry a capture and run to zero.	
Error "603"	Switch heaters failed	Check CCA connections	
PHASE 9, ERRORS IN OPERATION			
Error in operation	Poor connections Low cooling water Incorrect mains AC input	Check magnet lead connection Check water flow, cooler Check AC input voltage	
Error "901"	MPS turned off	Check off switch is not damaged	

Error - message / symptom	Possible cause	Service action	Comments
Error "902"	Auto/Manual switch was operated	Switch MPS off and operate the Auto/Manual switch as long as the MPS is off	
Error "910"	Current flow fault	Check for current in leads lamp. Turn unit off and retry	
Error "920"	Current flow in wrong direction	Check orientation of magnet cables	
Error "922"	Actual voltage too high	Check voltage setting and D type lead	
Error "923"	Hardware failure detected	Press off button and retry.	
Error "930"	Cable reversal detected	Check magnet lead connections	
Error "931"	Overcurrent detected	Check maximum current settings	
Error "960"	Internal short failure detected	Press off button and retry	
Error "970"	Water cooling for the MPS is inadequate	Check the cooler in the RCA Check the water fill level of the secondary water circuit Check the flow rate via the service software (gradient coil flow rate) Check the the bypass valve setting Increase the flow rate through the MPS, close bypass valve	Error message appears at 60°C
Error "971"/"972"	Internal short failure detected	Press off button and retry	
Error "973"	Magnet cable reversed The magnet and MPS polarities are different	Check magnet cable connections Slowly de-energize the magnet via the cables. Do not switch the MPS off	

Error - message / symptom	Possible cause	Service action	Comments
Error "974" (Not displayed on the LCD panel)	MPS recognizes the incorrect magnet type	Check the cable connections: MPS - CCA - MSUP - filter plate - magnet - "C" connector Measure the resistance at the "C-connector" on magnet: OR 24; 47 Ohm, PIN G-J OR 70; 100 Ohm, PIN G-J	
Error "987"/"988"/"989"	Software fault detected	Press off button and retry	
MANUAL MODE			
Unit fails during Manual mode	Poor connections Low cooling water Incorrect AC input	Check magnet lead connections Check water flow, Cooler Check the AC input voltage	

Main failures during magnet ramping

WARNING The MPS must remain fully connected to the HDCL during and following a power failure, as potentially hazardous currents/voltages may exist in the magnet, magnet leads and shorting contactor without indication. No action should be taken until the power has been restored and the following procedure carried out.

A mains failure during magnet ramping up or down will cause the magnet power supply to trip out of RUN mode, closing the output shorting contactor. The magnet current will then slowly run down through the magnet leads and shorting contactor.

NOTICE After a power failure, there is **NO** indication on the power supply until mains power is restored.

When power returns, or after a momentary power failure, the red LED "Warning -Current in leads" will be illuminated as a warning that current is still flowing in the magnet.

- Check that the MPS has powered up into the OFF state by verifying that the ON button on the control panel is not illuminated
- Check if the red LED marked "current in leads" is on warning that current remains in the magnet leads
If it is **NOT on**, it is safe to switch off circuit breaker F5 and proceed with removal of the HDCL.
If it is **on**, continue with the following steps:

1. The LCD display will now read:
OXFORD MAGNET
TP154 VX:XX
then
Magnet : Test
ON to start
2. Press the **ON** button on the MPS control panel
3. Then select **Run to Zero** to continue ramping under MPS control. This guarantee sufficient cooling of the HDCL.
4. Wait until the **Cycle Complete** green LED on the MPS control panel illuminates .
Then select **Run to Field** or
press the OFF button.
5. Follow the "Ramping the magnet" procedure.

Procedures

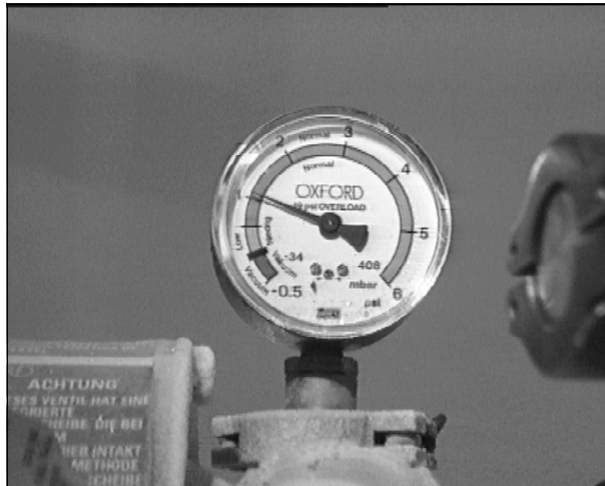
This chapter describes the test procedures for the magnet system.

Magnet

Checking the condition of the bursting disc

- The pressure gauge will indicate if the disk is broken or not..
Yellow/green sector indicates the bursting disc is intact.
If the pointer stays on or close to zero a broken disc **must** be suspected.

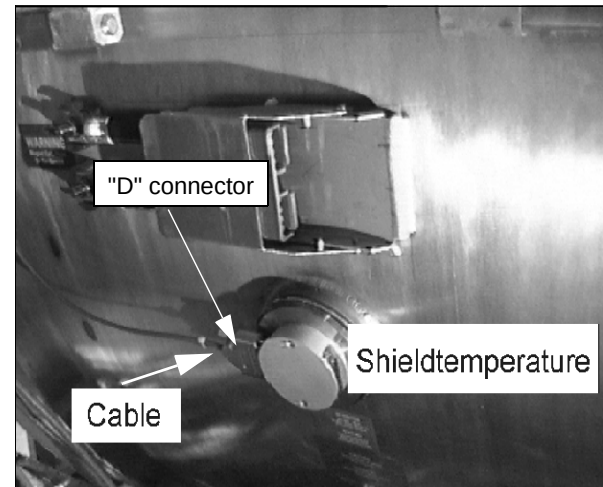
Fig. 85 Manometer



- If you suspect that the bursting disk is broken:
Energize the level probe for a few minutes to increase the rate of boil-off.
If the helium vessel does not start to re-pressurize, the bursting disk is probably broken.
Remove the quench line from the quench valve and check the disc.
- If the pressure remains at or very close to zero there is little chance that the bursting disk is probably not broken.*
Inspect the disc and the sealing O-ring visually. (There are two spare O-rings and one spare burst disc supplied in the magnet spare kit)

Measuring the shield temperature

Fig. 86 Shield temperature



Required Tools

- Temperature Measure Device; Order No: 99 00 317 RE999
(Same as for Impact/Expert/Vision)
- Adapter kit; Order No: 47 65 603 KE999

DVM

Fig. 87 Temperature Measure Device

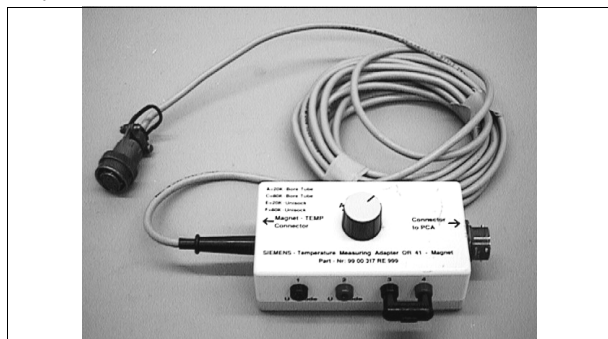


Fig. 88 Adapter kit



Measuring of the temperature

- Disconnect the cable
- Connect the service temperature measuring device
Use the adapter cable set.
- Press "MANUAL SAMPLE" on D1 of the MSUP.
The measurement cycle is for a period of 3 minutes
- Measure the Voltage drop
- Calculate the temperature in Kelvin, use Diode conversion chart.
Refer: Appendix "Magnet system"

Pin configuration on the "D" connector:

- 1 - 9 : 20k bore
- 5 - 9 : 20k neck
- 3 - 9 : 80k bore
- 6 - 9 : 80k neck

Pressurisation of the helium vessel

It is an essential pre-requisite that there is no negative pressure in the system for

- Opening a turret port
- Inserting or removing of the HDCL
- Leak detection

The system could be pressurised by energizing the EIS circuit.

Fig. 89 MSUP - D2



- Insert the shorting plug into D2 "EIS O/R" to energize the EIS circuit
Following LEDs lights up:

1. D2: "EIS FAULT/ON"
2. D1: "EIS"

NOTICE Ignore the error messages
"MSU_SCRUALARM" in the system report

NOTICE The shorting plug should be **removed** when sufficient pressure has been reached

After removing the shorting plug:

1. LED "EIS FAULT/ON" goes out
2. LED "EIS" has to be reseted with the ACKNOWLEDGE button

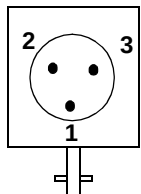
For **leak detection** the pointer of the pressure gauge should be in the yellow sector and closer to the green than to the red sector.

For **opening a turret port or inserting and removing the HDCL** the pressure gauge should read 0 psi.

Refrigerator

Checking the cold head motor

Testing the windings of the cold head motor.



specified values:

PIN 1 - 2: 620 Ohm

PIN 1 - 3: 620 Ohm

PIN 2 - 3: 620 Ohm

- Measure directly at the motor of the cold head

For magnets with cold head shielding:

- Measure at connector X101A on filter plate

Leakage detection

If the operating pressure of the refrigerator system drops continuously, test for leakage.

Check the cold head, the compressor and the pressure lines.

Leakage typically occurs at

- Welded seams; soldered joints
- O-rings
- Aeroquip unions

Required Tools

- helium leakage detector (Handy Detector)
(see Installation and Service Tools)

Working with self-sealing couplings

Fig. 90 Attaching and detaching self-sealing couplings

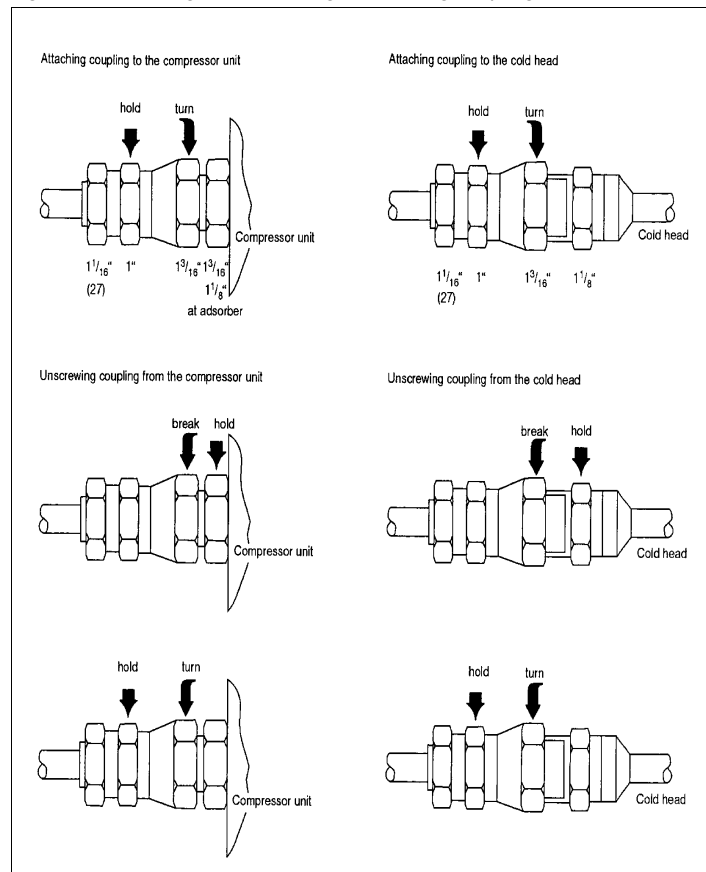
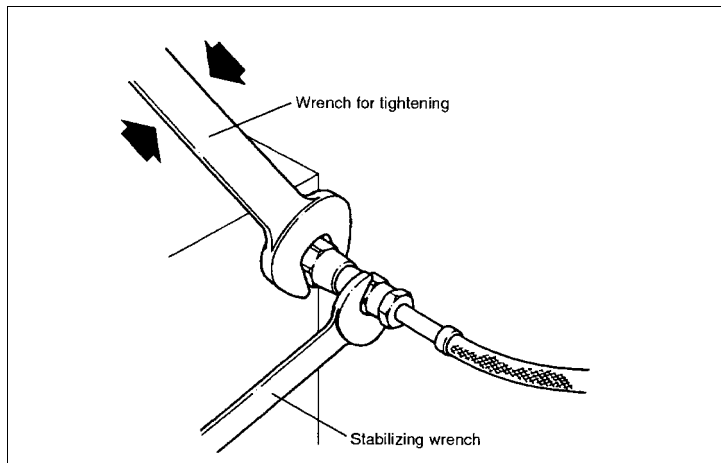


Fig. 91 Mounting the self-sealing couplings



Helium fill pressure of the compressor

If the helium pressure in the compressor drops below the minimum value, you must add helium gas.

With the compressor switched off, the fill pressure should be:

Minimum value

50 Hz; 14 bar

60 Hz; 12.5 bar

Nominal value:

50 Hz; 15 bar

60 Hz; 14 bar

Auxiliary tools:

Helium gas 5.0

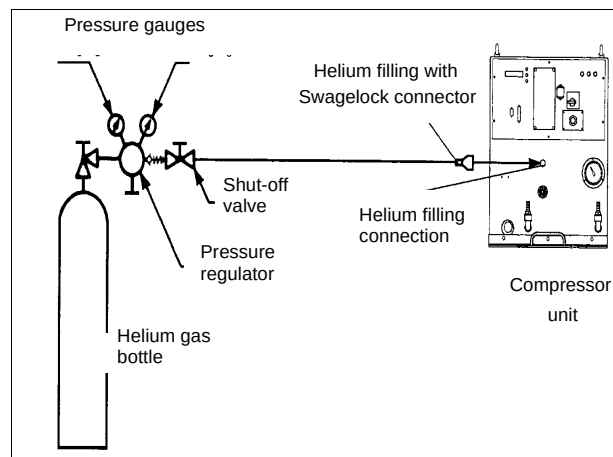
Pressure reducer with at least 16 bar working pressure

High pressure fill hose

Helium filling adapter (delivered with the system)

Helium filling

Fig. 92 Topping off helium gas



CAUTION Use only helium of the highest purity (99.999% or better).

- Switch off the compressor.

- Use the pressure reducer and a suitable high pressure fill hose to connect the helium cylinder to the helium filling adapter; do not tighten the Swagelock connector yet.
- Open the valve at the cylinder.
- Open the pressure reducer and the shut-off valve slightly so that the high pressure fill hose and the filling adapter are purged with helium.
- Tighten down the Swagelock connector after purging the configuration several times.
- Set the pressure reducer to the proper fill pressure.
- Close the shut-off valve of the pressure reducer.
- Connect the coupling at the helium filling adapter to the helium filling nipple.
- Refill the compressor unit up to nominal value and open the shut-off valve slightly.
- Detach the helium filling adapter from the compressor.
- Close the valve at the Helium supply cylinder.
- Seal the helium filling nipple at the compressor with a protective cap.

WARNING The high pressure fill hose at the pressure reducer will still be under high pressure.

Magnet Supervision

Jumper List

D1 Monitor:

LK 1	1.2
S4	Position 1 Fridge Normal Position 2 Fridge Test

D2 Battery; He & EIS Control

LK 1	1.2
------	-----

D3 ERDU

SW 1	Position 1: 1.5T an 1.0T Operation Position 2: not used
SW 2a	ON
SW 2b	OFF

D4 Processor

LK 0	2.3
LK 1	1.2
LK 2	1.2

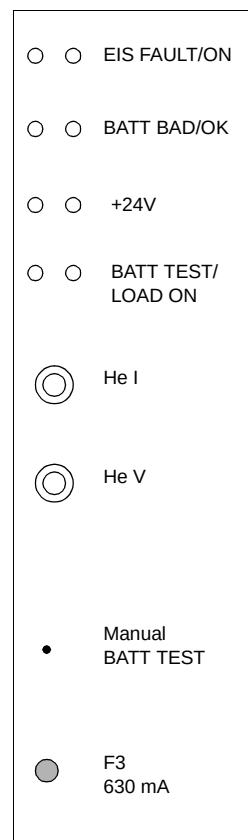
LK 3	1.2
LK 4	1.2
LK 5	Not Fitted

Temperature PCB

LK1	1.2
-----	-----

Helium Level Calculation

Fig. 93 D2; Battery;
He&EIS



- Connect a DVM to the measurement point "He V"
- Press "MANUAL SAMPLE" on D1
- Measure the voltage drop (expected value about 0 20 V)
- Calculate the Helium level

$$\text{Level (\%)} = 100 - (100 \times U / (R \times I))$$

U: Voltage drop

R: Helium probe resistance at 0%

(see: Magnet test certificate)

I: 250 mA probe current; check it at measurement point "He I";
measuring range: "mV"!

Operating parameters - MSU

This chapter provides you with the warning/alarm values for the magnet system.

	nominal value	warning value	alarm value	Reset
Shield temperature	<70K	75 K	80 K	n.a.
Helium level	e.g. 95 %	35% * (default) (35 - 79 %)		warning value +20%
			30 %	40 %

*. adjustable in the SeSo

Conversation list for temperature measuring diodes

Temperature measuring diodes are fitted to:

- 20 K and 80 K radiation shield
- 20 K and 80K stages of the Coldhead

mV	K	mV	K	mV	K
1010	101.93	1029	88.79	1048	75.06
1011	101.25	1030	88.08	1049	74.31
1012	100.57	1031	87.37	1050	73.57
1013	99.89	1032	86.66	1051	72.82
1014	99.21	1033	85.95	1052	72.07
1015	98.52	1034	85.24	1053	71.33
1016	97.84	1035	84.52	1054	70.58
1017	97.15	1036	83.81	1055	69.83
1018	96.46	1037	83.09	1056	69.09
1019	95.77	1038	82.37	1057	68.34
1020	95.08	1039	81.65	1058	67.59
1021	94.39	1040	80.93	1059	66.84
1022	93.69	1041	80.21	1060	66.10
1023	92.99	1042	79.48	1061	65.35
1024	92.30	1043	78.76	1062	64.60
1025	91.60	1044	78.03	1063	63.86
1026	90.90	1045	77.30	1064	63.11
1027	90.19	1046	76.55	1065	62.36
1028	89.49	1047	75.81	1066	61.62

mV	K	mV	K	mV	K	mV	K	mV	K
1067	60.87	1086	46.68	1105	32.49	1124	26.13	1143	23.89
1068	60.12	1087	45.93	1106	31.74	1125	25.92	1144	23.85
1069	59.38	1088	45.18	1107	30.99	1126	25.72	1145	23.80
1070	58.63	1089	44.44	1108	30.25	1127	25.53	1146	23.76
1071	57.88	1090	43.69	1109	29.50	1128	25.33	1147	23.72
1072	57.13	1091	42.94	1110	29.16	1129	25.13	1148	23.68
1073	56.39	1092	42.20	1111	28.93	1130	24.94	1149	23.63
1074	55.64	1093	41.45	1112	28.70	1131	24.75	1150	23.59
1075	54.89	1094	40.70	1113	28.48	1132	24.56	1151	23.55
1076	54.15	1095	39.96	1114	28.25	1133	24.37	1152	23.51
1077	53.40	1096	39.21	1115	28.03	1134	24.18	1153	23.47
1078	52.65	1097	38.46	1116	27.81	1135	23.99	1154	23.42
1079	51.91	1098	37.72	1117	27.60	1136	23.81	1155	23.38
1080	51.16	1099	36.97	1118	27.38	1137	23.63	1156	23.34
1081	50.41	1100	36.22	1119	27.17	1138	23.45	1157	23.30
1082	49.67	1101	35.48	1120	26.96	1139	24.06	1158	23.26
1083	48.92	1102	34.73	1121	26.75	1140	24.01	1159	23.21
1084	48.17	1103	33.98	1122	26.54	1141	23.97	1160	23.17
1085	47.43	1104	33.23	1123	26.33	1142	23.93	1161	23.13

mV	K	mV	K	mV	K	mV	K	mV	K
1162	23.09	1182	22.24	1202	21.40	1222	20.56	1242	19.71
1163	23.04	1183	22.20	1203	21.36	1223	20.51	1243	19.67
1164	23.00	1184	22.16	1204	21.32	1124	20.47	1244	19.63
1165	22.96	1185	22.12	1205	21.27	1225	20.43	1245	19.59
1166	22.92	1186	22.07	1206	21.23	1226	20.39	1246	19.54
1167	22.88	1187	22.03	1207	21.19	1227	20.35	1247	19.50
1168	22.83	1188	21.99	1208	21.15	1228	20.30	1248	19.46
1169	22.79	1189	21.95	1209	21.11	1229	20.26	1249	19.42
1170	22.75	1190	21.91	1210	21.06	1230	20.22	1250	19.38
1171	22.71	1191	21.86	1211	21.02	1231	20.18	1251	19.33
1172	22.67	1192	21.82	1212	20.98	1232	20.14	1252	19.29
1173	22.62	1193	21.78	1213	20.94	1233	20.09	1253	19.25
1174	22.58	1194	21.74	1214	20.89	1234	20.05	1254	19.21
1175	22.54	1195	21.70	1215	20.85	1235	20.01	1255	19.17
1176	22.50	1196	21.65	1216	20.81	1236	19.97	1256	19.12
1177	22.45	1197	21.61	1217	20.77	1237	19.92	1257	19.08
1178	22.41	1198	21.57	1218	20.73	1238	19.88	1258	19.04
1179	22.37	1199	21.53	1219	20.68	1239	19.84	1259	19.00
1180	22.33	1200	21.48	1220	20.64	1240	19.80	1260	18.95
1181	22.29	1201	21.44	1221	20.60	1241	19.76	1261	18.91

mV	K	mV	K	mV	K	mV	K	mV	K
1262	18.87	1282	18.03	1302	17.18	1322	16.34	1342	15.50
1263	18.83	1283	17.98	1303	17.14	1323	16.30	1343	15.46
1264	18.79	1284	17.94	1304	17.10	1324	16.26	1344	15.41
1265	18.74	1285	17.90	1305	17.06	1325	16.21	1345	15.37
1266	18.70	1286	17.86	1306	17.02	1326	16.17	1346	15.33
1267	18.66	1287	17.82	1307	16.97	1327	16.13	1347	15.29
1268	18.62	1288	17.77	1308	16.93	1328	16.09	1348	15.24
1269	18.58	1289	17.73	1309	16.89	1329	16.05	1349	15.20
1270	18.53	1290	17.69	1310	16.85	1330	16.00		
1271	18.49	1291	17.65	1311	16.80	1331	15.96		
1272	18.45	1292	17.61	1312	16.76	1332	15.92		
1273	18.41	1293	17.56	1313	16.72	1333	15.88		
1274	18.36	1294	17.52	1314	16.68	1334	15.83		
1275	18.32	1295	17.48	1315	16.64	1335	15.79		
1276	18.28	1296	17.44	1316	16.59	1336	15.75		
1277	18.24	1297	17.39	1317	16.55	1337	15.71		
1278	18.20	1298	17.35	1318	16.51	1338	15.67		
1279	18.15	1299	17.31	1319	16.47	1339	15.62		
1280	18.11	1300	17.27	1320	16.42	1340	15.58		
1281	18.07	1301	17.23	1321	16.38	1341	15.54		

Conversation list for helium level

Magnet OR 24	
% Level	He Vol (l)
10	93.1
20	232.8
30	454
35	565.1
40	661.7
45	753.6
50	843.1
55	931.3
60	1019
65	1106.8
70	1195.4
75	1285.6
80	1378.9
85	1478.5
90	1595.2
95	1708.8
100	1805.7

Magnet OR 70	
% Level	He Vol (l)
10	88.4
20	222.9
30	436
35	538.6
40	629.2
45	715.7
50	800
55	883.1
60	965.8
65	1048.6
70	1132.1
75	1217
80	1304.7
85	1397.8
90	1505.4
95	1615.9
100	1710

Cooling

In this part you will find the strategies and test procedures for trouble shooting on the Cooler in the RCA and the Cabinet Cooling System (CCS).

Strategy

This chapter describes the test sequence for the cooler in the RCA and Cabinet Cooling System (CCS)

Cooler in the RCA

In the following table all possible error symptoms or problems are listed.

Cooler Error code

Error Code: RCA_.....	Possible cause	Service action
SECWATERTOLOW	Water level in the heater tank is too low	Check the water level via SESO "Magnet&Cooling" Refill the heater tank with distilled water Leakage test Check the position of the drain valves
COOLWATLEVLOW	Water level in the heater tank is low	Refill the heater tank with distilled water Leakage test
PUMPFALT	Motor proximity switch responded Pump is defective	Reset Overload Contactor Check the wiring to the pump Replace pump
	Input voltage out of specification	Check the input voltage
	Pressure switch responded	Check hose connection to coil and electronics cabinet Check valves

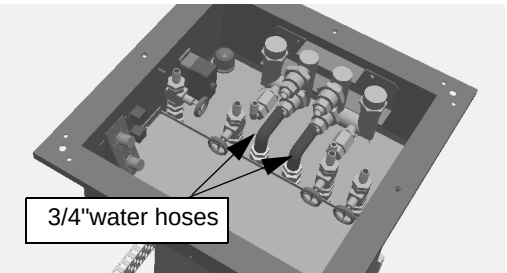
Error Code: RCA_.....	Possible cause	Service action
SECWATTEMPFAULT Secondary water temperature >22°C	Water level in the heater tank is too low	Refill distilled water
	Secondary water temperature < 18°C	
	Flow rate of the secondary circuit is too low	Increase the flow rate by adjusting the flow with the valves Clean the strainer in the secondary circuit Clean the heat exchanger
	The temperature set point is too high	Adjust the set point
	Primary flow rate and the temperature is out of specification	Check the flow rate and the temperature of the primary circuit Clean the strainer in the primary circuit Clean the heat exchanger
	The 3-way valve in the primary circuit is defective	Check the position and operating of the actuator. "0" : heat exchanger is bypassed "10": bypass is closed,
Secondary water temperature < 18°C	The temperature set point is too low	Adjust the set point
	The 3-way valve in the primary circuit is defective	Check the position and operating of the actuator. "0" : heat exchanger is bypassed "10": bypass is closed,
	Temperature of the primary water circuit is out of specification	Check the water circuit
PRIMTEMPFAULT Primary water temperature >13°C Primary water temperature >5°C	Primary water temperature is out of specification	Check the onsite chiller Check the primary water flow rate Clean the strainer in the primary circuit

Error Code: RCA_.....	Possible cause	Service action
SECWATLOWFLOW1 Gradient coil flow low	Water flow through the gradient coil < 28 l/min	Check the position of the valves (in- / outlet of the RCA, valve on the filterplate, quick release connection on the gradient coil) Check the flow rate of the CCA/GPA circuit Check the water hoses for bends and knicks Check the alarm setting Check the direction of rotation for the water pump Check the water level in the secondary water circuit
SECWATLOWFLOW2 CCA/GPA flow low	Water flow through the gradient coil < 43 l/min	Check the position of the valves Check the flow rate of the gradient coil circuit Check the water hoses for bends and knicks Check the alarm setting Check the direction of rotation for the water pump Check the water level in the secondary water circuit
PRIMFLOWRATEFAULT Water flow rate < 28 l/min	The flow rate of the primary water circuit is out of specification	Primary water hoses (3/4") are twisted. See "cooler symptoms) Adjust the flow rate to the desired value Check the position of the valves (in-/outlet) Check the alarm setting Clean the strainer in the primary circuit Check the onsite water chiller

Error Code: RCA_.....	Possible cause	Service action
SECOVERPRESSFAULT1	Overpressure in the secondary water circuit Overpressure in the gradient coil circuit or in the CCA/GPA circuit	Check the position of the valves in the secondary circuit, on top of the RCA (in- / outlet), on the filterplate Check the strainer in the secondary circuit Check the water hoses for bends and knicks Check the quick release connection on the gradient coil NOTICE Switch of the cooler and start the cooler again(Circuit breaker: RCA Control) Restart the Hardware components after cooling down (Switch to standby and switch on again)
LEYOVERTEMP	Insufficient water flow	Check the water supply to the compressor Ensure sufficient cooling water from the chiller
	Cooling water temperature too high	Ensure the right water temperature from the chiller
	Waterpipes connected in reverse	Check the right flow of the water circuit
	Heat exchanger congested	Clean the heat exchanger
LEYLOWPRESS	Helium loss	Top off helium, perform leak test
COMMBREAK	Communication with the cooler has been lost	Check the cabeling (fiber optic)

Cooler Symptoms

Symtom	Probable Cause	Service Action
Yellow LED lights up	Not enough water in the secondary circuit	Refill with water until LED goes out
Pump did not start in "Auto mode"	Pump defect	Check the pump Switch to "Manual mode"
	Check the RCA status via SeSo "Test tool"	Start pump in "manual mode"
	Primary water flow too low	Check the primary water circuit
	Primary water temperature out of specification	Check the primary water circuit
	Values for the warning/alarms level are wrong	Check the values via SeSo; reload the default values Start the pump in "Manual mode"
	Communication failed	Check the wiring
	Fuse F1 in the RCA is off	Switch on
	Fuse F22 is off	Switch on

Symtom	Probable Cause	Service Action
No primary water flow through the cooler, Flow through the Coolpak is o.k.	Primary water hoses (3/4") are twisted. 	Remove isolation and check the water hoses
No communication to the RCA	+15V is missing Fuse F6 on D1 in the cooler electronic box is defect	
No realistic flow rates values in the SESO	+24V supply for the satchwell controller is missing. Fuse F5 on D5 in the cooler electronic box is defect	

Cabinet Cooling System

Water-cooled system

There are two types of cabinet cooling systems, an air-cooled or a water-cooled system.

Error Code: MSU_.....	Possible cause	Service action	Comments
CCATEMPWARN (Temperature in the CCA cabinet is above 38 °C) CCATEMPALARM (Temperature in the CCA cabinet is above 43°C) GPATEMPWARN (Temperature in the GPA cabinet is above 38 °C) GPATEMPALARM (Temperature in the GPA cabinet is above 43°C)	Water temperature or flow rate of the primary water circuit is out of specification	Check primary water circuit/ Chiller	Temperature warning repeats every 3 minutes
	Filter on top of CCS is contaminated	Clean filter	
	Valves in the primary water circuit are closed	Check valve functions on the CCA and RCA	
	Supply voltage (230V) is missing	Check F10 Check voltage 230V at X800 and at X801	
	Fans in air distributor of the CCS are defective (below of the MMC) Fans in the CCS are defective	Check fans	
	Transformer T1 or T2 is defective Regler N1 or N2 is defective Temperatursensor R1 oder R2 defective Regelventil Y1 or Y2 is defektice	Check the components, refer test procedure	
	Cabinet door is open	Check if door is closed	

Air-cooled system

Error Code: MSU_.....	Possible cause	Service action	Comments
CCATEMPWARN (Temperature in the CCA cabinet is above 38 °C)	Supply voltage (230V) is missing	Check voltage 230V at X800 and at X801	Temperature warning repeats every 3 minutes
CCATEMPALARM (Temperature in the CCA cabinet is above 43°C)	Fans in air distributor of the CCS are defective (below of the MMC) Fans in the CCS are defective	Check fans	
	Air filter is dirty	Clean filter	
GPATEMPWARN (Temperature in the GPA cabinet is above 38 °C)	Ambient conditions out of specifications	Check the on-site air conditioning	
GPATEMPALARM (Temperature in the GPA cabinet is above 43°C)	Cabinet door is open	Check if door is closed	

Procedures

This chapter describes the drain and the filling procedures for the cooler in the RCA

Cooler in the RCA

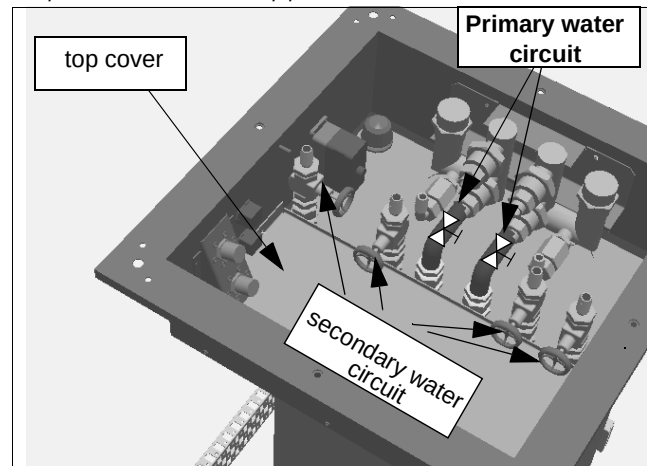
Drain procedure of the cooler

The following instruction cover the procedure to ensure that the cooler is completely empty of water.

Auxiliary tools:

- Spanner: 30 mm
- Screwdrivers
- Adjustable spanner
- Pipe wrench, adjustable
- Bucket

Fig. 94 Top of RCA without water pipes



- Switch off voltage supply to the cooler
Switch off circuit breaker F1 in the RCA
- Turn off primary water, supply and return.
- Turn off secondary water, gradient coil; electronic suit; flow and return
- Remove the front cover.
- Position the bucket in front of the cooler and put the cooler drain hose in it (the cooler holds approximately 30 l of water).

Fig. 95 Drain valves

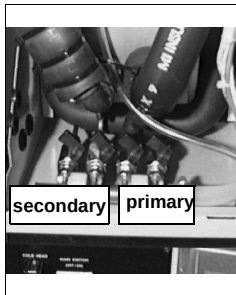
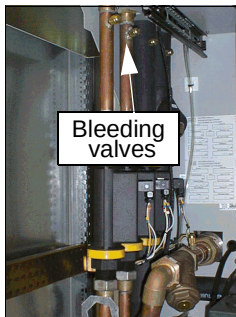


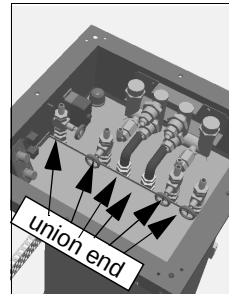
Fig. 96 Bleeding valves



- Open all four drain valves
2 valves for primary
2 valves for secondary

- Open the bleeder valves
The air entering the system will help
drain the unit.

Fig. 97 Union end



Filling the secondary water circuit

Cooler without bleeding valves:

- Loosen the union ends on the primary "flow" and primary "return".
The air entering the system will help
drain the unit.
- Loosen the union ends on the secondary "flow" and secondary "return".
This air entering the system will help
drain the unit..

NOTICE

Fill water only when lamp LP1 lights up
Use distilled water only.
Do not use any inhibitor.

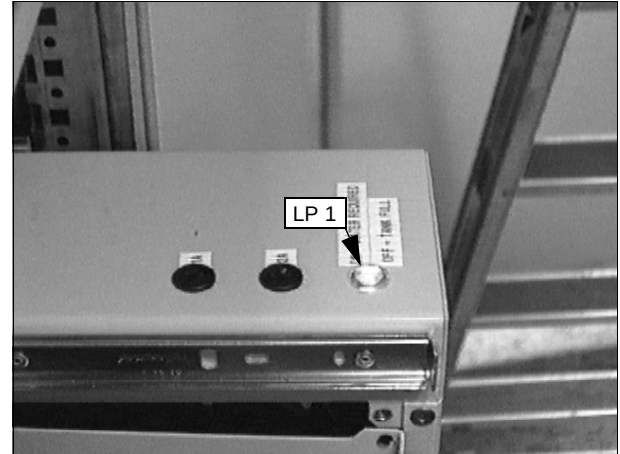
- Switch on the cooler
- Open the cap of the water tank and insert the funnel

Fill the water tank with distilled water
Monitor the Yellow lamp, refer [Fig. 99](#)

Fig. 98 Filling procedure at RCA cabinet



Fig. 99 Water indicator - yellow Lamp LP1



- Refill the water tank, until the yellow lamp goes out.

NOTICE	Stop filling immediately after lamp LP1 goes out. Avoid overfilling!
---------------	---

Temperature Sensor Fault Finding

Required tool: Volt meter

The resistance of the sensors can be measured across the cable ends. Disconnect the wires from the terminal blocks in the Satchwell controller.

	Wire number	Earth number
Primary Temperature Sensor	23	59
Secondary Temperature Sensor	22	58

Use the table below for conversion values.

°C	OHMS	°C	OHMS	°C	OHMS
0	7660	10	6666	20	5573
1	7569	11	6560	21	5469
2	7475	12	6453	22	5353
3	7379	13	6345	23	5243
4	7281	14	6236	24	5134
5	7182	15	6125	25	5024
6	7083	16	6016	26	4917

°C	OHMS	°C	OHMS	°C	OHMS
7	6980	17	5906	27	4809
8	6877	18	5795	28	4703
9	6772	19	5681	29	4597

Flow Meter Fault Finding

Required tool: Volt meter

The flow rates can be measured across the terminal pins within the Satchwell controller. All flow rates are displayed in volts.

1.6V = 0 l/min

7.8V = 60 l/min

Measure across the following pins:

	Pins
Primary flow rate	28 and 46
Gradient coil flow rate	29 and 46
CCA / GPA flow rate	30 and 46

Satchwell testing

The following tables list all input and output devices to the controller

DIGITAL OUTPUTS

Component	Canbuss Alarm	Operated Condition	Fault Condition	Satchwell Terminal No	Comments
FILL LAMP	NO	24 V AC	0 V AC	35 and 1	Disconnect wire from Terminal 35 and measure voltage between the specified Terminals at the Controller
PUMP START SIGNAL	NO	24 V AC	0 V AC	34 and 1	Disconnect wire from Terminal 34 and measure voltage between the specified Terminals at the Controller

DIGITAL INPUTS

Component	Canbuss Alarm	Healthy Condition	Alarm Condition	Satchwell Terminal No	Comments
LEYBOLD PRESSURE	YES	Short circuit	Open circuit	21 and 57	Disconnect wire from Terminal 20 and check for short circuit between wire and Terminal 56
LEYBOLD TEMPERATURE	YES	Short circuit	Open circuit	20 and 56	Disconnect wire from Terminal 21 and check for short circuit between wire and Terminal 57
WATER PUMP OVER-LOAD	YES	Short circuit	Open circuit	19 and 55	Disconnect wire from Terminal 19 and check for short circuit between wire and Terminal 55
LOCAL CONTROL	NO	Open circuit	Short circuit	18 and 54	Disconnect wire from Terminal 18 and check for short circuit between wire and Terminal 54
COOLING WATER PRESSURE	YES	Short circuit	Open circuit	16 and 52	Disconnect wire from Terminal 16 and check for short circuit between wire and Terminal 52

Component	Canbuss Alarm	Healthy Condition	Alarm Condition	Satchwell Terminal No	Comments
FLOAT SWITCH - FILL LEVEL	NO	Open circuit	Short circuit	17 and 53	Amber Lamp lit when tank needs topping up Disconnect wire from Terminal 17 and check for open circuit between wire and Terminal 53
LOAT SWITCH - LOW LEVEL	YES	Short circuit	Open circuit	15 and 51	Disconnect wire from Terminal 15 and check for open circuit between wire and Terminal 51
FLOAT SWITCH - VERY LOW LEVEL	YES	Short circuit	Open circuit	14 and 50	Disconnect wire from Terminal 14 and check for open circuit between wire and Terminal 50

ANALOGUE OUTPUTS

Component	Canbuss Alarm	Normal Condition	Fault Condition	Satchwell Terminal No	Comments
3-WAY VALVE ACTUATOR	NO	Between 0 and 10 V DC	0 V DC	6 and 42	Disconnect wire from Terminal 6 and measure voltage between the specified Terminals at the controller

ANALOGUE INPUTS

Component	Canbuss Alarm	Healthy Condition	Alarm Condition	Satchwell Terminal No	Comments
CCA/GPA WATER FLOW	YES	Voltage greater than 6.1 V	Voltage less than 6.1 V	30	Disconnect wire from Terminal 30 and check voltage between wire and +24VDC. NOTICE The alarm will remain until the voltage has risen above 6.7 V
GRADIENT COIL WATER FLOW	YES	Voltage greater than 4.5 V	Voltage less than 4.5 V	29	Disconnect wire from Terminal 29 and check voltage between wire and +24VDC. NOTICE The alarm will remain until the voltage has risen above 4.9 V
CHILLED WATER FLOW	YES	Voltage greater than 6.3 V	Voltage less than 6.3 V	28	Disconnect wire from Terminal 28 and check voltage between wire and +24VDC. NOTICE The alarm will remain until the voltage has risen above 6.8 V

Operating parameters - RCA / CCS

This chapter provides you with the operating/alarm values for the RCA and CCS

	nominal value	warning value	alarm value	Reset
Primary water flow	50 l/min	28 l/min *	n.a.	30 l/min
Primary temperature	6 - 12 °C	high 14 °C *	n.a.	12 °C
		Low 5 °C *	n.a.	6 °C
Gradient coil flow	30 l/min	27 l/min *	n.a.	30 l/min
CCA / GPA flow	46 l/min	43 l/min *	n.a.	46 l/min
Secondary temperature	20 °C	24 °C	n.a.	22 °C
		16 °C	n.a.	18 °C
Cabinet temperature CCA	<38 %	38 °C *		< 38 °C
			43 °C	< 43 °C
Cabinet temperture GPA	< 38 %	38 °C *		< 38 °C
			43 °C	< 43 °C

*. adjustable in the SeSo

Power Distribution

In this part you will find the strategies for trouble shooting on the power distribution

Strategy

This chapter describes the test strategy for the Power Distribution

CAUTION Voltage continues to be present at the line voltage transformer even after the MR system has been switched off. The line power distributor has to be switched off for service work and the on-site circuit breakers have to be set to OFF.

NOTICE The transformer connections have been made, refer to voltage adaptation in the Start-up section.

Circuit Breakers

Fig. 100 Circuit breakers

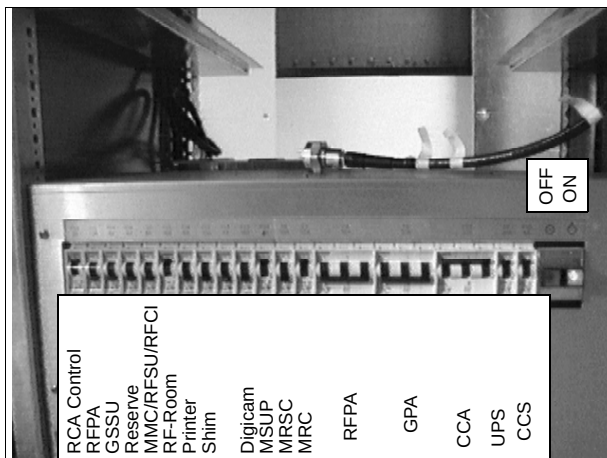
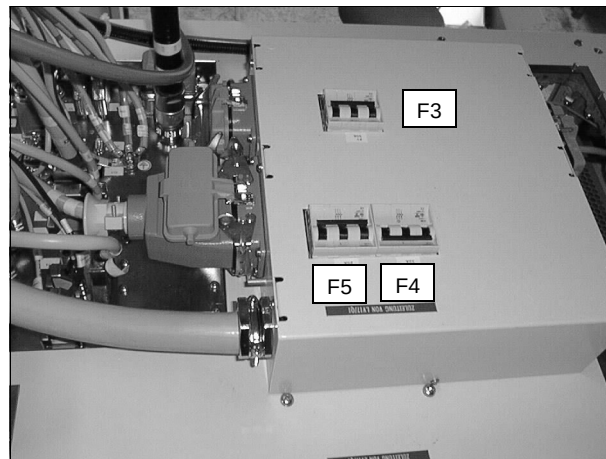


Fig. 101 Mains Input Box



Circuit breaker	Plug (CCA)	Plug (RCA)	Component
F4	X101	X100	RCA(Cooler 1; COOLPACK)
F3	X111		Option
F5	X102		Magnet PSU

Symptoms:MR system cannot be switched on

Symptom	Possible cause	Service action	Comments
Green lamp in the CCA is off	Main Fuse F1 defective	Check for short circuit; replace F1	
	F2 is off	Switch on F2	
	F23 is off	Switch on F23	
	Line filter is defective	Replace the filter	
	If UPS for HOST is not connected: Short circuit adapter is defective	Check the adapter on X103, CCA top panel	
	If UPS for HOST is connected: UPS is defective, open circuit	Check the UPS	
Green lamp in the CCA is on	15 V from MSUP are missing	Check circuit breaker F24 Check the 15V power supply	
	K4 defective	Check K4	
Press the "System ON" key, System or HOST switches off immediately	Relay K2 is defective Relay K3 is defective Relay K4 is defective	Check and replace relay	

Symptom: MR system switches only into Standby after System ON

Symptom	Possible cause	Service action	Comments
Only HOST, MSUP and COOL-PACK is running	"Standby" on the alarm box was pressed	Switch on the system at the alarm box	Temperature inside the cabinet >50°C K7 schaltet K11/ K1/K10
	Temperature switch in CCA or GPA responded	Check cabinet cooling system Check the cabinet temperature	
	Short circuit on cable "W355" K7 is defective	Check cable from MMC X4 to CCA X10 Check K7	
	One of the 3 Thermal Sensor on T1 is defect	Change the 3 Thermal Sensor of T1	contact Tx of diagram 684-945 page 2

Symptom: HOST does not switch correctly

Symptom	Possible cause	Service action	Comments
HOST is not switched on with "Standby"	K9 is defective		
HOST switches off immediately after System ON	K3 is defective		
HOST cannot be switched off	K5 is defective		

Symptom: System does not switch on or off correctly

Symptom	Possible cause	Service action	Comments
System is not switched off with "Standby"	K6 is defective		
System is not switched off with System OFF	K5 is defective		
Single component is not on	respective circuit breaker		
Only HOST, RFPA and GPA (Resonant Converter) are on	K1 is defective		
System switches off immediately after on	K2 is defective		
CCS is not switched on	K13 is defective		

13 Changes

In order to get a quick overview what has changed, here you find the changes to the last version.

Changes to previous version

Part RF-System

RF-trouble shooting flowchart added.

14 Index

In the online version of the index you can click on the specific letter in order to go directly to this topic.

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